Four-Algorithm Active Inverse Optimization

Complete experimental results with convergence figures and all trajectories

Algorithm	Valid at T	Stable joint success	Mean angle (valid)	Mean regret (valid)	Mean runtime
Pedro	38/40	4/40	15.63 deg	0.01387	0.103 s
Genious	15/40	5/40	12.42 deg	0.01738	0.222 s
Score Base	40/40	31/40	3.83 deg	0.00098	13.684 s
Online SAMD	40/40	0/40	42.49 deg	0.07549	0.004 s

Main finding

Score Base is the clear winner at the fixed query horizon. It keeps all 40 estimates valid, reaches mean final angular error 3.83 degrees and mean regret 0.0010, and achieves the sustained joint target in 31 of 40 runs. Pedro is reliable but slower. Genious Pedro can be accurate when its cone survives, but parameter noise collapses 25 of its 40 hard cones. Online SAMD stays structurally valid in all runs but is not query-efficient at T=20 under the frozen one-update-per-observation configuration.

All 160 runs continue to T=20. Invalid estimates remain missing rather than being replaced by 90/180 degree errors or zero regret. Every conditional mean is accompanied by its valid count.

1. Protocol and algorithm definitions

The comparison uses the unchanged eight-scenario protocol: 4D and 6D MIN problems, five paired seeds, 120 fixed candidate queries, 120 fresh hidden test queries, T=20, parameter noise, and no observation noise (Y=X). The objective is F(theta,s,x) = theta dot (s*x). Algorithms never see theta-star or hidden evaluation data.

Algorithm	Next S	Estimate after observing Y	Key property
Pedro	Uniform candidate	Hard-cone incenter	Exact constraints
Genious Pedro	Minimum predicted boundary margin	Hard-cone incenter	Active boundary seeking
Score Base	Maximum sample disagreement	Mean of 16 samples	Noise-tolerant distribution
Online SAMD	Uniform candidate	One signed mirror update	Exact finite ASL oracle

Fairness rules

All algorithms receive one expert response per round and share the same scenario seeds, candidate pools, test queries, horizon, noise settings, and metrics. The SAMD learning rate and L1 radius are fixed before the benchmark and are not tuned per scenario. Wall time is reported separately from query efficiency.

Interpretation boundary

The fourth method is an online finite-oracle specialization using one SAMD update per new observation. It is not a batch rerun with many internal optimization iterations. Therefore poor T=20 query efficiency does not contradict the paper's convergence theory.

2. Final results by scenario

Scenario	Algorithm	Valid	Ever invalid	Angle	Regret	Stable joint
Connecting two groups	Pedro	4/5	1	16.791	0.01556	1
Connecting two groups	Genious	4/5	1	26.959	0.05557	0
Connecting two groups	Score Base	5/5	0	3.588	0.00271	5
Connecting two groups	Online SAMD	5/5	0	41.659	0.13630	0
Several boundaries to locate	Pedro	4/5	1	10.201	0.02063	2
Several boundaries to locate	Genious	0/5	5	-	-	0
Several boundaries to locate	Score Base	5/5	0	1.715	0.00037	5
Several boundaries to locate	Online SAMD	5/5	0	54.919	0.11988	0
Similar queries versus varied queries	Pedro	5/5	0	30.565	0.01878	0
Similar queries versus varied queries	Genious	1/5	4	34.747	0.01946	0
Similar queries versus varied queries	Score Base	5/5	0	5.414	0.00049	2
Similar queries versus varied queries	Online SAMD	5/5	0	58.194	0.02553	0
Ordinary balanced choice	Pedro	5/5	0	10.674	0.01007	0
Ordinary balanced choice	Genious	2/5	3	1.672	0.00049	2
Ordinary balanced choice	Score Base	5/5	0	2.008	0.00037	5
Ordinary balanced choice	Online SAMD	5/5	0	37.238	0.08048	0
Ordinary subset selection	Pedro	5/5	0	9.036	0.00173	1
Ordinary subset selection	Genious	3/5	2	4.752	0.00054	1
Ordinary subset selection	Score Base	5/5	0	3.269	0.00032	4
Ordinary subset selection	Online SAMD	5/5	0	34.375	0.03197	0
Small budget-constrained selection	Pedro	5/5	0	16.254	0.01048	0
Small budget-constrained selection	Genious	3/5	2	3.477	0.00070	2
Small budget-constrained selection	Score Base	5/5	0	5.492	0.00054	2
Small budget-constrained selection	Online SAMD	5/5	0	34.139	0.04127	0
Stronger parameter noise	Pedro	5/5	0	15.320	0.01770	0
Stronger parameter noise	Genious	0/5	5	-	-	0
Stronger parameter noise	Score Base	5/5	0	5.185	0.00205	3
Stronger parameter noise	Online SAMD	5/5	0	39.680	0.08422	0
Query-dependent parameter noise	Pedro	5/5	0	15.320	0.01770	0
Query-dependent parameter noise	Genious	2/5	3	7.854	0.00712	0
Query-dependent parameter noise	Score Base	5/5	0	3.934	0.00097	5
Query-dependent parameter noise	Online SAMD	5/5	0	39.680	0.08422	0

2.1 Recovery and pairwise outcomes

Algorithm	Final valid	Ever invalid	Stable joint	Mean stable step	Mean runtime
Pedro	38/40	2/40	4/40	14.5	0.103 s
Genious	15/40	25/40	5/40	15.8	0.222 s
Score Base	40/40	0/40	31/40	16.5	13.684 s
Online SAMD	40/40	0/40	0/40	-	0.004 s

Paired ordinary-test outcomes at T=20

Left	Right	Both valid	Left invalid	Right invalid	Both invalid	Left angle wins	Right angle wins	Left regret wins	Right regret wins
Pedro	Genious	14	1	24	1	3	11	4	10
Pedro	Score Base	38	2	0	0	2	36	2	36
Pedro	Online SAMD	38	2	0	0	38	0	37	1
Genious	Score Base	15	25	0	0	4	11	3	11
Genious	Online SAMD	15	25	0	0	13	2	15	0
Score Base	Online SAMD	40	0	0	0	40	0	40	0

Wins are counted only when both estimates are valid. Ties are omitted from win columns.

2.2 Convergence checkpoints

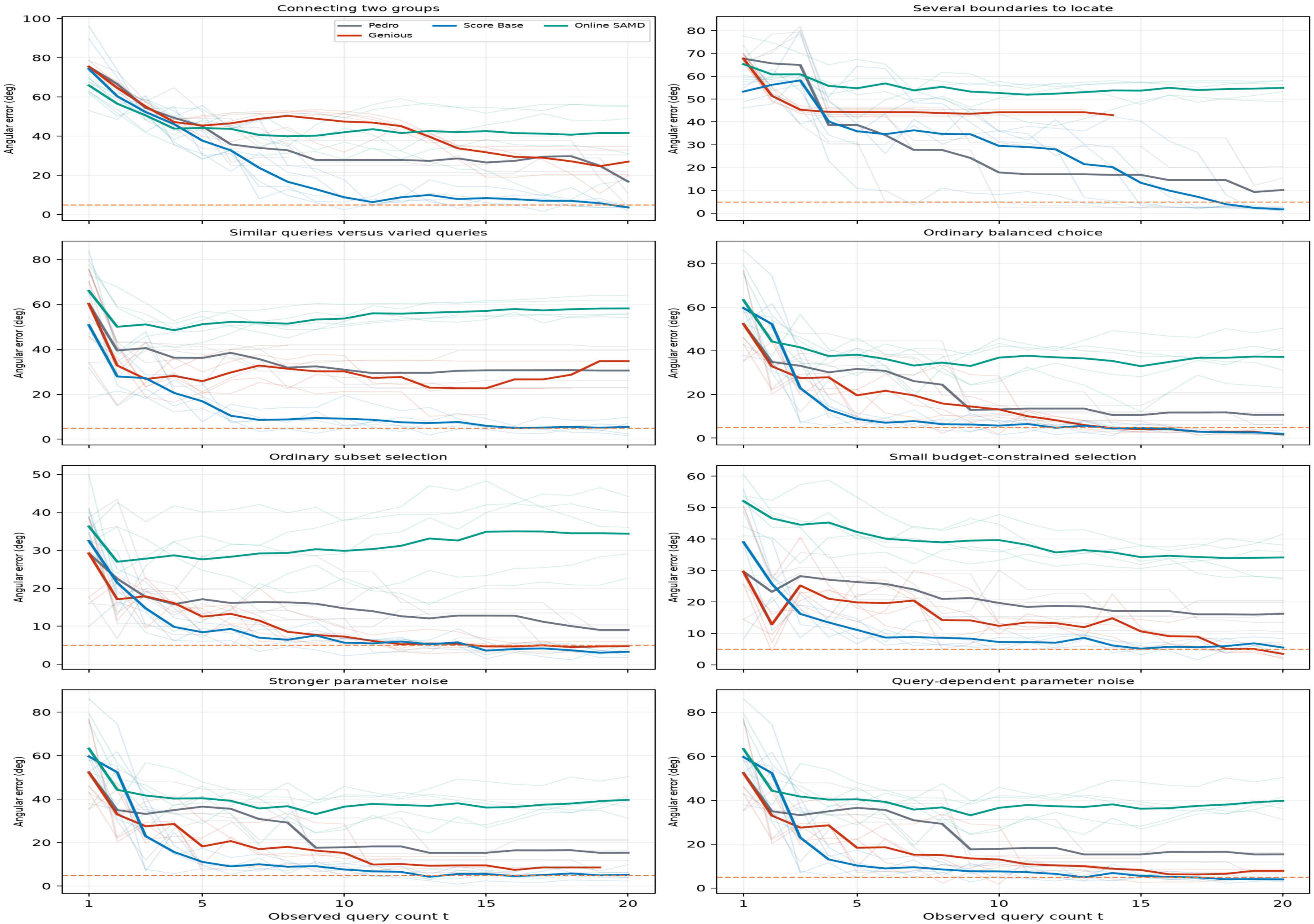
Each cell reports the valid-only mean angular error and normalized regret across all eight scenarios and five seeds.

Algorithm	t=1	t=5	t=10	t=15	t=20
Pedro	52.41 deg / 0.2558	33.53 deg / 0.0784	19.83 deg / 0.0214	17.96 deg / 0.0161	15.63 deg / 0.0139
Genious	52.41 deg / 0.2558	25.51 deg / 0.0434	21.30 deg / 0.0292	13.24 deg / 0.0156	12.42 deg / 0.0174
Score Base	53.61 deg / 0.2483	17.51 deg / 0.0231	10.15 deg / 0.0093	6.53 deg / 0.0039	3.83 deg / 0.0010
Online SAMD	59.45 deg / 0.2789	42.38 deg / 0.1010	40.99 deg / 0.0766	40.97 deg / 0.0758	42.49 deg / 0.0755

Score Base improves steadily from 53.61 degrees at t=1 to 3.83 degrees at t=20. Online SAMD improves early but plateaus near 41-42 degrees. Genious Pedro's late valid-only mean is affected by attrition: only 15 of 40 estimates remain valid at t=20.

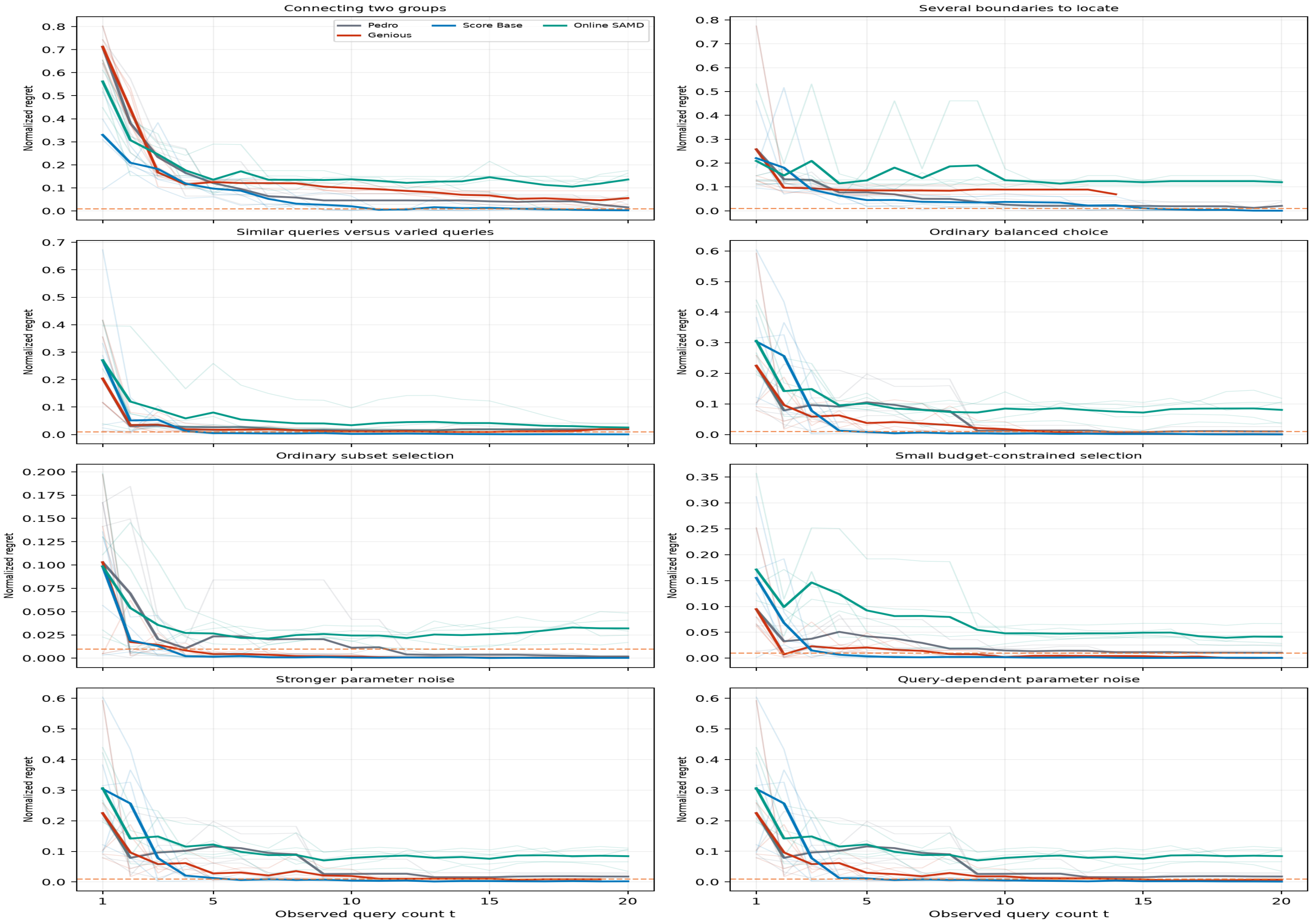
3. Angular-error trajectories

Thin curves are individual seeds; thick curves are valid-only means. The dashed line is 5 degrees.



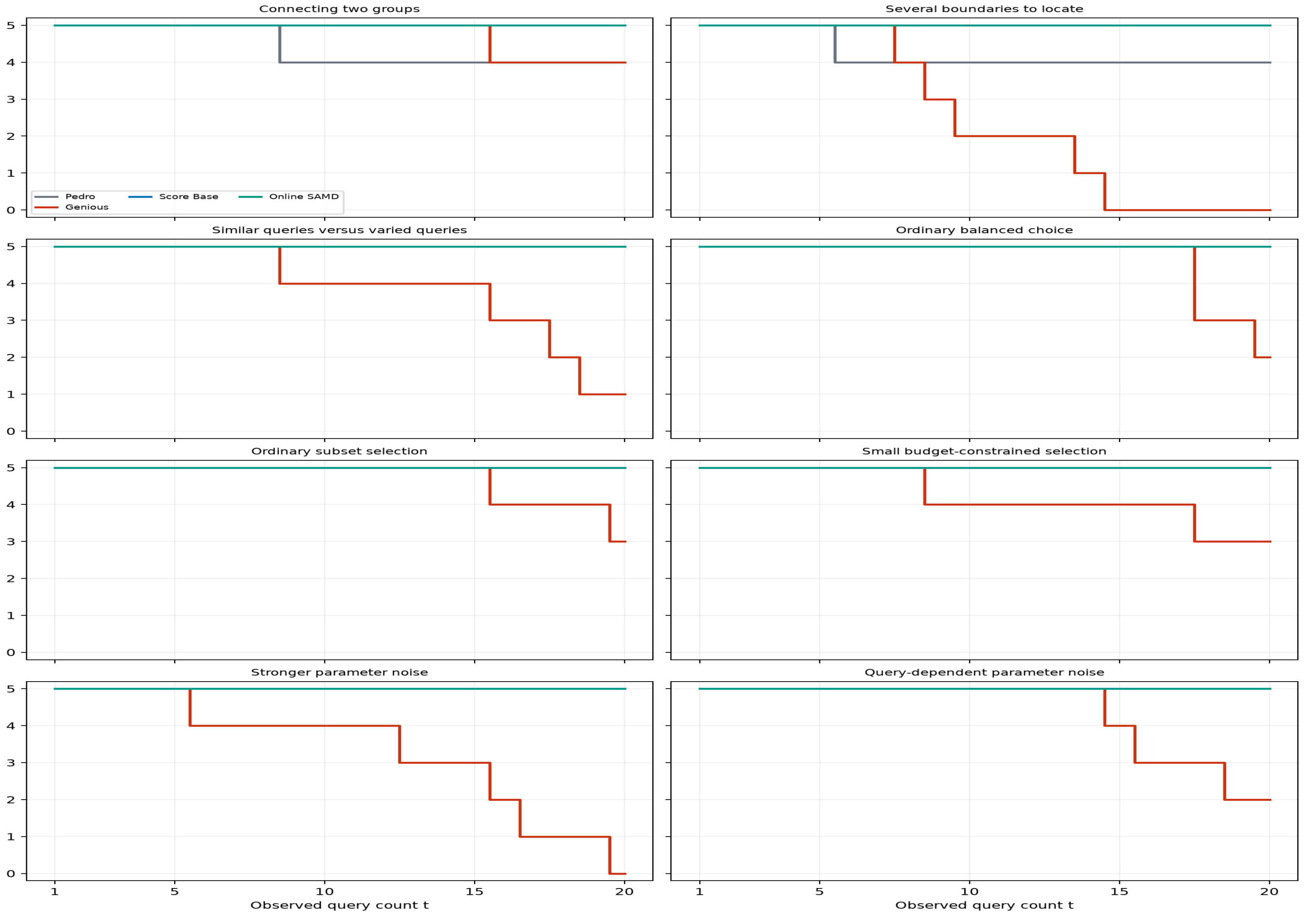
4. Held-out-regret trajectories

Thin curves are individual seeds; thick curves are valid-only means. The dashed line is 0.01.



5. Valid-estimate trajectories

Each curve reports how many of the five estimates remain valid at every time step.



6. Every-step conditional statistics

The next eight pages report all 20 time steps for every algorithm and scenario. Each n value is the number of valid estimates among the five paired seeds.

Connecting two groups

t	Pedro n	angle	regret	Genious n	angle	regret	Score Base n	angle	regret	Online SAMD n	angle	regret
1	5/5	75.453	0.71234	5/5	75.453	0.71234	5/5	74.294	0.33004	5/5	65.908	0.56195
2	5/5	66.578	0.38138	5/5	64.667	0.44271	5/5	60.382	0.20915	5/5	56.555	0.30724
3	5/5	54.411	0.23366	5/5	55.162	0.16731	5/5	52.269	0.18208	5/5	50.577	0.24515
4	5/5	49.345	0.16511	5/5	47.202	0.11498	5/5	46.157	0.11837	5/5	43.853	0.17590
5	5/5	45.062	0.12075	5/5	45.411	0.12504	5/5	37.697	0.09687	5/5	44.124	0.13543
6	5/5	35.792	0.09506	5/5	46.524	0.12080	5/5	32.740	0.08775	5/5	43.704	0.17188
7	5/5	33.976	0.06392	5/5	48.851	0.12055	5/5	23.836	0.05214	5/5	40.610	0.13528
8	5/5	32.797	0.05791	5/5	50.369	0.11981	5/5	16.719	0.03152	5/5	39.893	0.13502
9	4/5	27.815	0.04561	5/5	48.889	0.10518	5/5	12.879	0.02666	5/5	40.214	0.13412
10	4/5	27.815	0.04561	5/5	47.419	0.09925	5/5	8.778	0.02033	5/5	41.981	0.13735
11	4/5	27.815	0.04561	5/5	46.901	0.09417	5/5	6.314	0.00522	5/5	43.541	0.13042
12	4/5	27.815	0.04561	5/5	45.128	0.08674	5/5	8.771	0.00654	5/5	41.572	0.12172
13	4/5	27.393	0.04492	5/5	39.711	0.08044	5/5	9.960	0.01625	5/5	42.630	0.12706
14	4/5	28.627	0.04581	5/5	33.802	0.06988	5/5	7.866	0.01175	5/5	42.001	0.12838
15	4/5	26.552	0.04150	5/5	31.740	0.06709	5/5	8.376	0.01288	5/5	42.585	0.14616
16	4/5	27.368	0.03909	4/5	29.448	0.05223	5/5	7.826	0.00996	5/5	41.570	0.12907
17	4/5	29.513	0.04188	4/5	28.949	0.05468	5/5	7.009	0.00753	5/5	41.244	0.11278
18	4/5	29.747	0.04179	4/5	27.136	0.04955	5/5	6.954	0.00512	5/5	40.719	0.10543
19	4/5	24.930	0.02712	4/5	24.642	0.04649	5/5	5.729	0.00346	5/5	41.624	0.11863
20	4/5	16.791	0.01556	4/5	26.959	0.05557	5/5	3.588	0.00271	5/5	41.659	0.13630

All means are conditional on a valid estimate at that time.

Several boundaries to locate

t	Pedro n	angle	regret	Genious n	angle	regret	Score Base n	angle	regret	Online SAMD n	angle	regret
1	5/5	67.742	0.25773	5/5	67.742	0.25773	5/5	53.305	0.22055	5/5	65.320	0.20906
2	5/5	65.643	0.13230	5/5	51.443	0.09665	5/5	56.241	0.18079	5/5	60.813	0.14793
3	5/5	64.929	0.12960	5/5	45.322	0.09457	5/5	58.224	0.08938	5/5	60.858	0.20906
4	5/5	38.702	0.07674	5/5	44.394	0.08710	5/5	40.168	0.06330	5/5	55.784	0.11483
5	5/5	38.712	0.07733	5/5	44.303	0.08530	5/5	35.930	0.04488	5/5	54.777	0.12735
6	4/5	34.222	0.06898	5/5	44.295	0.08501	5/5	34.649	0.04528	5/5	56.862	0.18092
7	4/5	27.732	0.05022	5/5	44.279	0.08501	5/5	36.340	0.03781	5/5	53.829	0.13705
8	4/5	27.711	0.04998	4/5	43.889	0.08397	5/5	34.743	0.03625	5/5	55.380	0.18621
9	4/5	24.217	0.03716	3/5	43.581	0.08965	5/5	34.616	0.03455	5/5	53.285	0.19020
10	4/5	17.886	0.02569	2/5	44.254	0.08898	5/5	29.490	0.03728	5/5	52.695	0.12811
11	4/5	17.089	0.02086	2/5	44.254	0.08898	5/5	29.040	0.03600	5/5	51.929	0.12261
12	4/5	17.089	0.02086	2/5	44.254	0.08898	5/5	28.000	0.03445	5/5	52.388	0.11413
13	4/5	17.089	0.02086	2/5	44.254	0.08898	5/5	21.546	0.02205	5/5	53.051	0.12415
14	4/5	16.823	0.02054	1/5	43.002	0.06933	5/5	20.235	0.02327	5/5	53.779	0.12415
15	4/5	16.823	0.02054	0/5	-	-	5/5	13.353	0.01039	5/5	53.707	0.11988
16	4/5	14.507	0.01907	0/5	-	-	5/5	9.939	0.00632	5/5	54.944	0.12415
17	4/5	14.507	0.01907	0/5	-	-	5/5	7.222	0.00416	5/5	53.959	0.12415
18	4/5	14.507	0.01907	0/5	-	-	5/5	4.007	0.00390	5/5	54.400	0.12415
19	4/5	9.296	0.01221	0/5	-	-	5/5	2.369	0.00058	5/5	54.581	0.12415
20	4/5	10.201	0.02063	0/5	-	-	5/5	1.715	0.00037	5/5	54.919	0.11988

All means are conditional on a valid estimate at that time.

Similar queries versus varied queries

t	Pedro n	angle	regret	Genious n	angle	regret	Score Base n	angle	regret	Online SAMD n	angle	regret
1	5/5	60.230	0.20373	5/5	60.230	0.20373	5/5	50.769	0.27049	5/5	66.012	0.27068
2	5/5	39.464	0.03076	5/5	32.795	0.03464	5/5	27.931	0.05108	5/5	50.018	0.12007
3	5/5	40.548	0.03238	5/5	26.794	0.03607	5/5	27.168	0.05348	5/5	51.088	0.09031
4	5/5	36.247	0.02747	5/5	28.209	0.01919	5/5	20.622	0.01285	5/5	48.459	0.05866
5	5/5	36.167	0.02619	5/5	25.806	0.01726	5/5	16.870	0.00526	5/5	51.171	0.08006
6	5/5	38.446	0.02731	5/5	29.710	0.01752	5/5	10.444	0.00473	5/5	52.197	0.05468
7	5/5	35.674	0.02250	5/5	32.782	0.01867	5/5	8.609	0.00387	5/5	51.908	0.04730
8	5/5	31.856	0.01578	5/5	31.361	0.01535	5/5	8.782	0.00392	5/5	51.406	0.04066
9	5/5	32.437	0.01697	4/5	30.214	0.01279	5/5	9.444	0.00568	5/5	53.268	0.04028
10	5/5	30.935	0.01456	4/5	30.188	0.01136	5/5	9.133	0.00206	5/5	53.689	0.03380
11	5/5	29.405	0.01427	4/5	27.252	0.01100	5/5	8.661	0.00246	5/5	56.029	0.04164
12	5/5	29.522	0.01448	4/5	27.684	0.01255	5/5	7.547	0.00328	5/5	55.854	0.04512
13	5/5	29.522	0.01448	4/5	22.947	0.00957	5/5	7.158	0.00255	5/5	56.310	0.04611
14	5/5	30.435	0.01899	4/5	22.703	0.00929	5/5	7.695	0.00138	5/5	56.618	0.04162
15	5/5	30.669	0.01905	4/5	22.686	0.00922	5/5	5.991	0.00111	5/5	57.109	0.04162
16	5/5	30.689	0.01878	3/5	26.621	0.01167	5/5	4.939	0.00076	5/5	57.941	0.03678
17	5/5	30.704	0.01878	3/5	26.621	0.01167	5/5	5.221	0.00095	5/5	57.314	0.03174
18	5/5	30.734	0.01878	2/5	28.709	0.01266	5/5	5.444	0.00100	5/5	57.850	0.03027
19	5/5	30.565	0.01878	1/5	34.747	0.01946	5/5	5.159	0.00064	5/5	58.152	0.02707
20	5/5	30.565	0.01878	1/5	34.747	0.01946	5/5	5.414	0.00049	5/5	58.194	0.02553

All means are conditional on a valid estimate at that time.

Ordinary balanced choice

t	Pedro n	angle	regret	Genious n	angle	regret	Score Base n	angle	regret	Online SAMD n	angle	regret
1	5/5	52.344	0.22493	5/5	52.344	0.22493	5/5	59.655	0.30388	5/5	63.304	0.30651
2	5/5	34.957	0.07856	5/5	32.975	0.09635	5/5	52.332	0.25650	5/5	44.317	0.14173
3	5/5	33.170	0.09603	5/5	27.516	0.05815	5/5	22.972	0.07796	5/5	41.650	0.14863
4	5/5	30.191	0.09118	5/5	27.906	0.06261	5/5	13.015	0.01247	5/5	37.652	0.09537
5	5/5	31.754	0.10541	5/5	19.627	0.03740	5/5	8.797	0.00837	5/5	38.292	0.10169
6	5/5	30.860	0.09686	5/5	21.705	0.04047	5/5	7.060	0.00419	5/5	36.303	0.08485
7	5/5	26.155	0.08122	5/5	19.623	0.03604	5/5	7.830	0.00622	5/5	33.291	0.08029
8	5/5	24.541	0.07656	5/5	15.915	0.03071	5/5	6.393	0.00376	5/5	34.666	0.07348
9	5/5	12.944	0.01218	5/5	14.513	0.02110	5/5	6.261	0.00408	5/5	33.091	0.07183
10	5/5	13.162	0.01257	5/5	13.160	0.01761	5/5	5.740	0.00278	5/5	36.943	0.08465
11	5/5	13.558	0.01312	5/5	10.097	0.01123	5/5	6.503	0.00349	5/5	37.788	0.08148
12	5/5	13.558	0.01312	5/5	8.244	0.00790	5/5	4.733	0.00202	5/5	37.076	0.08621
13	5/5	13.558	0.01312	5/5	6.038	0.00364	5/5	5.694	0.00256	5/5	36.589	0.07965
14	5/5	10.578	0.00773	5/5	4.613	0.00278	5/5	4.448	0.00145	5/5	35.398	0.07477
15	5/5	10.578	0.00773	5/5	4.074	0.00220	5/5	4.752	0.00158	5/5	32.999	0.07170
16	5/5	11.755	0.01018	5/5	4.105	0.00151	5/5	4.292	0.00197	5/5	34.964	0.08283
17	5/5	11.754	0.01096	5/5	3.032	0.00115	5/5	2.938	0.00088	5/5	36.843	0.08467
18	5/5	11.820	0.01115	3/5	2.858	0.00088	5/5	2.687	0.00052	5/5	36.856	0.08500
19	5/5	10.640	0.01025	3/5	2.916	0.00085	5/5	2.626	0.00070	5/5	37.426	0.08518
20	5/5	10.674	0.01007	2/5	1.672	0.00049	5/5	2.008	0.00037	5/5	37.238	0.08048

All means are conditional on a valid estimate at that time.

Ordinary subset selection

t	Pedro n	angle	regret	Genious n	angle	regret	Score Base n	angle	regret	Online SAMD n	angle	regret
1	5/5	29.189	0.10299	5/5	29.189	0.10299	5/5	32.505	0.09855	5/5	36.311	0.09849
2	5/5	22.517	0.06944	5/5	17.092	0.01737	5/5	21.464	0.01916	5/5	26.984	0.05392
3	5/5	17.722	0.02047	5/5	17.887	0.01433	5/5	14.739	0.01295	5/5	27.807	0.03569
4	5/5	15.806	0.01065	5/5	16.120	0.00829	5/5	9.840	0.00215	5/5	28.680	0.02708
5	5/5	17.125	0.02333	5/5	12.532	0.00436	5/5	8.414	0.00159	5/5	27.615	0.02656
6	5/5	16.123	0.02337	5/5	13.265	0.00452	5/5	9.302	0.00236	5/5	28.305	0.02183
7	5/5	16.372	0.02013	5/5	11.490	0.00376	5/5	6.987	0.00098	5/5	29.173	0.02114
8	5/5	16.297	0.02054	5/5	8.547	0.00223	5/5	6.431	0.00080	5/5	29.310	0.02481
9	5/5	15.937	0.02033	5/5	7.704	0.00190	5/5	7.561	0.00126	5/5	30.298	0.02596
10	5/5	14.711	0.01121	5/5	7.272	0.00188	5/5	5.664	0.00069	5/5	29.867	0.02434
11	5/5	13.964	0.01176	5/5	6.138	0.00110	5/5	5.483	0.00051	5/5	30.336	0.02431
12	5/5	12.649	0.00398	5/5	5.246	0.00086	5/5	5.970	0.00076	5/5	31.189	0.02166
13	5/5	12.071	0.00347	5/5	5.437	0.00103	5/5	5.245	0.00068	5/5	33.123	0.02541
14	5/5	12.809	0.00367	5/5	5.360	0.00096	5/5	5.769	0.00084	5/5	32.588	0.02475
15	5/5	12.809	0.00367	5/5	4.690	0.00057	5/5	3.572	0.00021	5/5	34.889	0.02574
16	5/5	12.791	0.00367	4/5	4.672	0.00047	5/5	4.003	0.00041	5/5	35.002	0.02687
17	5/5	11.194	0.00302	4/5	5.002	0.00064	5/5	4.147	0.00035	5/5	34.947	0.02971
18	5/5	10.027	0.00249	4/5	4.486	0.00042	5/5	3.670	0.00026	5/5	34.503	0.03300
19	5/5	9.036	0.00173	4/5	4.671	0.00049	5/5	3.028	0.00030	5/5	34.507	0.03207
20	5/5	9.036	0.00173	3/5	4.752	0.00054	5/5	3.269	0.00032	5/5	34.375	0.03197

All means are conditional on a valid estimate at that time.

Small budget-constrained selection

t	Pedro n	angle	regret	Genious n	angle	regret	Score Base n	angle	regret	Online SAMD n	angle	regret
1	5/5	29.639	0.09464	5/5	29.639	0.09464	5/5	39.027	0.15511	5/5	52.098	0.17125
2	5/5	23.210	0.03261	5/5	12.829	0.00737	5/5	25.733	0.06829	5/5	46.599	0.09880
3	5/5	28.230	0.03751	5/5	25.245	0.02310	5/5	16.197	0.01487	5/5	44.559	0.14621
4	5/5	27.069	0.05087	5/5	21.003	0.01875	5/5	13.464	0.00663	5/5	45.273	0.12375
5	5/5	26.338	0.04210	5/5	19.854	0.02061	5/5	11.085	0.00356	5/5	42.266	0.09222
6	5/5	25.737	0.03812	5/5	19.594	0.01634	5/5	8.670	0.00215	5/5	40.144	0.08132
7	5/5	23.980	0.02951	5/5	20.436	0.01403	5/5	8.824	0.00170	5/5	39.449	0.08157
8	5/5	20.934	0.01838	5/5	14.236	0.00797	5/5	8.582	0.00249	5/5	38.965	0.07962
9	5/5	21.267	0.01849	4/5	14.118	0.00737	5/5	8.317	0.00244	5/5	39.520	0.05475
10	5/5	19.678	0.01482	4/5	12.392	0.00211	5/5	7.251	0.00187	5/5	39.673	0.04800
11	5/5	18.394	0.01355	4/5	13.471	0.00459	5/5	7.225	0.00104	5/5	38.176	0.04812
12	5/5	18.772	0.01443	4/5	13.265	0.00480	5/5	7.022	0.00123	5/5	35.752	0.04740
13	5/5	18.556	0.01434	4/5	11.944	0.00430	5/5	8.585	0.00183	5/5	36.493	0.04789
14	5/5	17.132	0.01155	4/5	14.802	0.00420	5/5	6.151	0.00077	5/5	35.776	0.04794
15	5/5	17.132	0.01155	4/5	10.636	0.00398	5/5	5.116	0.00055	5/5	34.269	0.04930
16	5/5	17.080	0.01180	4/5	9.098	0.00244	5/5	5.722	0.00061	5/5	34.673	0.04952
17	5/5	16.063	0.01072	4/5	8.949	0.00315	5/5	5.576	0.00071	5/5	34.306	0.04262
18	5/5	16.063	0.01072	3/5	5.015	0.00044	5/5	5.950	0.00082	5/5	33.969	0.03940
19	5/5	15.970	0.01062	3/5	5.031	0.00026	5/5	6.830	0.00078	5/5	34.038	0.04169
20	5/5	16.254	0.01048	3/5	3.477	0.00070	5/5	5.492	0.00054	5/5	34.139	0.04127

All means are conditional on a valid estimate at that time.

Stronger parameter noise

t	Pedro n	angle	regret	Genious n	angle	regret	Score Base n	angle	regret	Online SAMD n	angle	regret
1	5/5	52.344	0.22493	5/5	52.344	0.22493	5/5	59.655	0.30388	5/5	63.304	0.30651
2	5/5	34.957	0.07856	5/5	32.975	0.09635	5/5	52.332	0.25650	5/5	44.317	0.14173
3	5/5	33.170	0.09603	5/5	27.516	0.05815	5/5	22.972	0.07796	5/5	41.650	0.14863
4	5/5	34.963	0.10166	5/5	28.563	0.06156	5/5	15.679	0.02075	5/5	40.273	0.11549
5	5/5	36.526	0.11588	5/5	18.216	0.02779	5/5	11.084	0.01292	5/5	40.383	0.12243
6	5/5	35.535	0.11042	4/5	20.674	0.03103	5/5	9.021	0.00619	5/5	39.251	0.09830
7	5/5	30.830	0.09478	4/5	16.967	0.02115	5/5	10.011	0.00902	5/5	35.712	0.08777
8	5/5	29.216	0.09012	4/5	18.047	0.03554	5/5	8.881	0.00660	5/5	36.728	0.08835
9	5/5	17.619	0.02574	4/5	16.295	0.02097	5/5	9.115	0.00730	5/5	33.119	0.07021
10	5/5	17.837	0.02613	4/5	15.227	0.01938	5/5	7.598	0.00433	5/5	36.517	0.07837
11	5/5	18.233	0.02668	4/5	9.888	0.01050	5/5	6.786	0.00360	5/5	37.823	0.08332
12	5/5	18.233	0.02668	4/5	10.128	0.01050	5/5	6.494	0.00448	5/5	37.267	0.08621
13	5/5	15.293	0.01534	3/5	9.328	0.01062	5/5	4.239	0.00136	5/5	36.878	0.07876
14	5/5	15.293	0.01534	3/5	9.493	0.01062	5/5	5.547	0.00302	5/5	38.101	0.08156
15	5/5	15.292	0.01534	3/5	9.493	0.01062	5/5	5.549	0.00278	5/5	36.105	0.07581
16	5/5	16.415	0.01778	2/5	7.413	0.00715	5/5	4.525	0.00209	5/5	36.365	0.08643
17	5/5	16.400	0.01859	1/5	8.543	0.00872	5/5	5.102	0.00173	5/5	37.417	0.08720
18	5/5	16.466	0.01877	1/5	8.543	0.00872	5/5	5.753	0.00271	5/5	37.980	0.08428
19	5/5	15.287	0.01787	1/5	8.543	0.00872	5/5	4.935	0.00168	5/5	39.055	0.08574
20	5/5	15.320	0.01770	0/5	-	-	5/5	5.185	0.00205	5/5	39.680	0.08422

All means are conditional on a valid estimate at that time.

Query-dependent parameter noise

t	Pedro n	angle	regret	Genious n	angle	regret	Score Base n	angle	regret	Online SAMD n	angle	regret
1	5/5	52.344	0.22493	5/5	52.344	0.22493	5/5	59.655	0.30388	5/5	63.304	0.30651
2	5/5	34.957	0.07856	5/5	32.975	0.09635	5/5	52.332	0.25650	5/5	44.317	0.14173
3	5/5	33.170	0.09603	5/5	27.516	0.05815	5/5	22.972	0.07796	5/5	41.650	0.14863
4	5/5	34.963	0.10166	5/5	28.563	0.06156	5/5	13.015	0.01247	5/5	40.273	0.11549
5	5/5	36.526	0.11588	5/5	18.335	0.02963	5/5	10.232	0.01130	5/5	40.383	0.12243
6	5/5	35.535	0.11042	5/5	18.554	0.02537	5/5	8.893	0.00610	5/5	39.251	0.09830
7	5/5	30.830	0.09478	5/5	15.175	0.01862	5/5	9.479	0.00832	5/5	35.712	0.08777
8	5/5	29.216	0.09012	5/5	14.991	0.02915	5/5	8.447	0.00643	5/5	36.728	0.08835
9	5/5	17.619	0.02574	5/5	13.455	0.01813	5/5	7.644	0.00650	5/5	33.119	0.07021
10	5/5	17.837	0.02613	5/5	13.016	0.01829	5/5	7.571	0.00504	5/5	36.517	0.07837
11	5/5	18.233	0.02668	5/5	10.830	0.01231	5/5	7.194	0.00385	5/5	37.823	0.08332
12	5/5	18.233	0.02668	5/5	10.327	0.01160	5/5	6.424	0.00299	5/5	37.267	0.08621
13	5/5	15.293	0.01534	5/5	9.933	0.01164	5/5	4.858	0.00176	5/5	36.878	0.07876
14	5/5	15.293	0.01534	5/5	8.797	0.00936	5/5	6.829	0.00435	5/5	38.101	0.08156
15	5/5	15.292	0.01534	4/5	8.207	0.00839	5/5	5.506	0.00195	5/5	36.105	0.07581
16	5/5	16.415	0.01778	3/5	6.233	0.00422	5/5	5.115	0.00202	5/5	36.365	0.08643
17	5/5	16.400	0.01859	3/5	6.152	0.00502	5/5	4.797	0.00205	5/5	37.417	0.08720
18	5/5	16.466	0.01877	3/5	6.473	0.00502	5/5	4.004	0.00147	5/5	37.980	0.08428
19	5/5	15.287	0.01787	2/5	7.854	0.00712	5/5	4.087	0.00136	5/5	39.055	0.08574
20	5/5	15.320	0.01770	2/5	7.854	0.00712	5/5	3.934	0.00097	5/5	39.680	0.08422

All means are conditional on a valid estimate at that time.

7. Complete run summaries

Scenario	Seed	Algorithm	Status	Angle	Regret	First joint	Stable joint	First invalid	Seconds
bridge-6d	0	Pedro	valid	10.731	0.02912	-	-	-	0.671
bridge-6d	0	Genious	valid	19.949	0.04956	-	-	-	0.326
bridge-6d	0	Score Base	valid	3.585	0.00368	14	20	-	44.967
bridge-6d	0	Online SAMD	valid	55.125	0.12584	-	-	-	0.015
bridge-6d	1	Pedro	valid	30.442	0.01819	-	-	-	0.350
bridge-6d	1	Genious	degenerate_cone	-	-	-	-	16	0.730
bridge-6d	1	Score Base	valid	2.226	0.00054	11	20	-	35.703
bridge-6d	1	Online SAMD	valid	49.022	0.15056	-	-	-	0.006
bridge-6d	2	Pedro	valid	21.867	0.01268	-	-	-	0.095
bridge-6d	2	Genious	valid	34.655	0.05874	-	-	-	0.264
bridge-6d	2	Score Base	valid	3.389	0.00159	18	18	-	13.764
bridge-6d	2	Online SAMD	valid	55.482	0.17489	-	-	-	0.004
bridge-6d	3	Pedro	degenerate_cone	-	-	-	-	9	0.055
bridge-6d	3	Genious	valid	20.369	0.02685	-	-	-	0.281
bridge-6d	3	Score Base	valid	4.762	0.00404	11	20	-	15.239
bridge-6d	3	Online SAMD	valid	17.587	0.06637	-	-	-	0.005
bridge-6d	4	Pedro	valid	4.125	0.00226	20	20	-	0.087
bridge-6d	4	Genious	valid	32.864	0.08713	-	-	-	0.276
bridge-6d	4	Score Base	valid	3.979	0.00368	10	18	-	13.607
bridge-6d	4	Online SAMD	valid	31.076	0.16384	-	-	-	0.004
boundaries-4d	0	Pedro	valid	20.498	0.04288	-	-	-	0.093
boundaries-4d	0	Genious	degenerate_cone	-	-	-	-	15	0.188
boundaries-4d	0	Score Base	valid	0.935	0.00002	17	17	-	15.279
boundaries-4d	0	Online SAMD	valid	58.105	0.12929	-	-	-	0.004
boundaries-4d	1	Pedro	valid	15.451	0.03711	-	-	-	0.078
boundaries-4d	1	Genious	degenerate_cone	-	-	-	-	10	0.130

Scenario	Seed	Algorithm	Status	Angle	Regret	First joint	Stable joint	First invalid	Seconds
boundaries-4d	1	Score Base	valid	0.844	0.00030	15	15	-	14.336
boundaries-4d	1	Online SAMD	valid	57.977	0.12480	-	-	-	0.004
boundaries-4d	2	Pedro	valid	2.431	0.00157	11	11	-	0.075
boundaries-4d	2	Genious	degenerate_cone	-	-	-	-	14	0.179
boundaries-4d	2	Score Base	valid	1.877	0.00009	7	17	-	13.707
boundaries-4d	2	Online SAMD	valid	48.824	0.12631	-	-	-	0.005
boundaries-4d	3	Pedro	valid	2.426	0.00096	10	10	-	0.089
boundaries-4d	3	Genious	degenerate_cone	-	-	-	-	9	0.123
boundaries-4d	3	Score Base	valid	2.888	0.00124	15	15	-	14.549
boundaries-4d	3	Online SAMD	valid	56.367	0.11279	-	-	-	0.004
boundaries-4d	4	Pedro	degenerate_cone	-	-	-	-	6	0.039
boundaries-4d	4	Genious	degenerate_cone	-	-	-	-	8	0.106
boundaries-4d	4	Score Base	valid	2.033	0.00019	19	19	-	14.985
boundaries-4d	4	Online SAMD	valid	53.325	0.10620	-	-	-	0.004
redundancy-6d	0	Pedro	valid	39.387	0.04137	-	-	-	0.113
redundancy-6d	0	Genious	degenerate_cone	-	-	-	-	18	0.264
redundancy-6d	0	Score Base	valid	6.185	0.00015	12	-	-	15.383
redundancy-6d	0	Online SAMD	valid	55.780	0.02221	-	-	-	0.004
redundancy-6d	1	Pedro	valid	23.092	0.00912	-	-	-	0.117
redundancy-6d	1	Genious	degenerate_cone	-	-	-	-	9	0.149
redundancy-6d	1	Score Base	valid	7.111	0.00106	13	-	-	14.466
redundancy-6d	1	Online SAMD	valid	55.704	0.03948	-	-	-	0.008
redundancy-6d	2	Pedro	valid	23.155	0.01835	-	-	-	0.104
redundancy-6d	2	Genious	degenerate_cone	-	-	-	-	16	0.245
redundancy-6d	2	Score Base	valid	2.344	0.00032	7	13	-	15.506
redundancy-6d	2	Online SAMD	valid	63.802	0.02213	-	-	-	0.005

Scenario	Seed	Algorithm	Status	Angle	Regret	First joint	Stable joint	First invalid	Seconds
redundancy-6d	3	Pedro	valid	41.148	0.01756	-	-	-	0.109
redundancy-6d	3	Genious	valid	34.747	0.01946	-	-	-	0.297
redundancy-6d	3	Score Base	valid	1.602	0.00001	16	16	-	14.867
redundancy-6d	3	Online SAMD	valid	61.514	0.02599	-	-	-	0.005
redundancy-6d	4	Pedro	valid	26.045	0.00751	-	-	-	0.105
redundancy-6d	4	Genious	degenerate_cone	-	-	-	-	19	0.341
redundancy-6d	4	Score Base	valid	9.831	0.00092	-	-	-	14.017
redundancy-6d	4	Online SAMD	valid	54.169	0.01784	-	-	-	0.004
balanced-choice-4d	0	Pedro	valid	7.923	0.00663	-	-	-	0.075
balanced-choice-4d	0	Genious	degenerate_cone	-	-	17	-	18	0.239
balanced-choice-4d	0	Score Base	valid	1.308	0.00095	5	17	-	14.617
balanced-choice-4d	0	Online SAMD	valid	39.986	0.11841	-	-	-	0.004
balanced-choice-4d	1	Pedro	valid	20.837	0.02571	-	-	-	0.077
balanced-choice-4d	1	Genious	valid	1.686	0.00088	8	14	-	0.269
balanced-choice-4d	1	Score Base	valid	2.043	0.00000	9	15	-	14.078
balanced-choice-4d	1	Online SAMD	valid	41.524	0.10620	-	-	-	0.005
balanced-choice-4d	2	Pedro	valid	6.382	0.00302	-	-	-	0.083
balanced-choice-4d	2	Genious	degenerate_cone	-	-	13	-	18	0.222
balanced-choice-4d	2	Score Base	valid	3.376	0.00077	6	11	-	17.140
balanced-choice-4d	2	Online SAMD	valid	23.170	0.03929	-	-	-	0.004
balanced-choice-4d	3	Pedro	valid	11.865	0.01189	-	-	-	0.077
balanced-choice-4d	3	Genious	valid	1.659	0.00010	13	17	-	0.264
balanced-choice-4d	3	Score Base	valid	1.356	0.00010	17	17	-	12.459
balanced-choice-4d	3	Online SAMD	valid	50.367	0.10357	-	-	-	0.004
balanced-choice-4d	4	Pedro	valid	6.362	0.00313	-	-	-	0.071
balanced-choice-4d	4	Genious	degenerate_cone	-	-	10	-	20	0.234

Scenario	Seed	Algorithm	Status	Angle	Regret	First joint	Stable joint	First invalid	Seconds
balanced-choice-4d	4	Score Base	valid	1.956	0.00003	15	15	-	13.040
balanced-choice-4d	4	Online SAMD	valid	31.145	0.03493	-	-	-	0.004
balanced-subset-6d	0	Pedro	valid	4.880	0.00063	17	17	-	0.099
balanced-subset-6d	0	Genious	degenerate_cone	-	-	9	-	20	0.250
balanced-subset-6d	0	Score Base	valid	5.090	0.00008	13	-	-	13.075
balanced-subset-6d	0	Online SAMD	valid	36.076	0.04840	-	-	-	0.004
balanced-subset-6d	1	Pedro	valid	8.522	0.00132	6	-	-	0.093
balanced-subset-6d	1	Genious	valid	5.956	0.00067	-	-	-	0.254
balanced-subset-6d	1	Score Base	valid	1.740	0.00007	7	16	-	12.305
balanced-subset-6d	1	Online SAMD	valid	29.076	0.02461	-	-	-	0.004
balanced-subset-6d	2	Pedro	valid	14.499	0.00301	-	-	-	0.091
balanced-subset-6d	2	Genious	valid	5.070	0.00048	12	-	-	0.261
balanced-subset-6d	2	Score Base	valid	3.966	0.00063	15	15	-	13.282
balanced-subset-6d	2	Online SAMD	valid	39.848	0.02913	-	-	-	0.004
balanced-subset-6d	3	Pedro	valid	6.912	0.00134	11	-	-	0.100
balanced-subset-6d	3	Genious	valid	3.231	0.00046	12	12	-	0.255
balanced-subset-6d	3	Score Base	valid	3.347	0.00044	15	18	-	13.418
balanced-subset-6d	3	Online SAMD	valid	22.681	0.01691	-	-	-	0.004
balanced-subset-6d	4	Pedro	valid	10.369	0.00237	-	-	-	0.113
balanced-subset-6d	4	Genious	degenerate_cone	-	-	-	-	16	0.224
balanced-subset-6d	4	Score Base	valid	2.203	0.00039	7	7	-	9.368
balanced-subset-6d	4	Online SAMD	valid	44.195	0.04079	-	-	-	0.003
knapsack-6d	0	Pedro	valid	20.440	0.01429	-	-	-	0.105
knapsack-6d	0	Genious	degenerate_cone	-	-	2	-	9	0.133
knapsack-6d	0	Score Base	valid	8.342	0.00081	-	-	-	9.003
knapsack-6d	0	Online SAMD	valid	38.092	0.04387	-	-	-	0.004

Scenario	Seed	Algorithm	Status	Angle	Regret	First joint	Stable joint	First invalid	Seconds
knapsack-6d	1	Pedro	valid	13.507	0.00239	-	-	-	0.104
knapsack-6d	1	Genious	valid	6.435	0.00191	-	-	-	0.248
knapsack-6d	1	Score Base	valid	2.391	0.00024	14	20	-	8.964
knapsack-6d	1	Online SAMD	valid	27.640	0.02353	-	-	-	0.004
knapsack-6d	2	Pedro	valid	19.827	0.02551	-	-	-	0.106
knapsack-6d	2	Genious	valid	1.834	0.00014	18	18	-	0.234
knapsack-6d	2	Score Base	valid	5.715	0.00054	11	-	-	9.187
knapsack-6d	2	Online SAMD	valid	36.055	0.04378	-	-	-	0.004
knapsack-6d	3	Pedro	valid	10.538	0.00192	-	-	-	0.109
knapsack-6d	3	Genious	valid	2.162	0.00005	18	18	-	0.242
knapsack-6d	3	Score Base	valid	7.261	0.00036	16	-	-	8.690
knapsack-6d	3	Online SAMD	valid	27.349	0.02839	-	-	-	0.004
knapsack-6d	4	Pedro	valid	16.956	0.00831	-	-	-	0.111
knapsack-6d	4	Genious	degenerate_cone	-	-	-	-	18	0.220
knapsack-6d	4	Score Base	valid	3.753	0.00075	10	20	-	8.526
knapsack-6d	4	Online SAMD	valid	41.561	0.06679	-	-	-	0.004
strong-noise-4d	0	Pedro	valid	16.253	0.01781	-	-	-	0.053
strong-noise-4d	0	Genious	degenerate_cone	-	-	-	-	17	0.148
strong-noise-4d	0	Score Base	valid	6.183	0.00320	13	-	-	10.794
strong-noise-4d	0	Online SAMD	valid	36.272	0.11053	-	-	-	0.003
strong-noise-4d	1	Pedro	valid	20.837	0.02571	-	-	-	0.053
strong-noise-4d	1	Genious	degenerate_cone	-	-	-	-	6	0.060
strong-noise-4d	1	Score Base	valid	3.767	0.00118	10	20	-	9.640
strong-noise-4d	1	Online SAMD	valid	41.524	0.10620	-	-	-	0.003
strong-noise-4d	2	Pedro	valid	21.282	0.02996	-	-	-	0.053
strong-noise-4d	2	Genious	degenerate_cone	-	-	-	-	20	0.171

Scenario	Seed	Algorithm	Status	Angle	Regret	First joint	Stable joint	First invalid	Seconds
strong-noise-4d	2	Score Base	valid	3.900	0.00058	11	16	-	10.022
strong-noise-4d	2	Online SAMD	valid	39.091	0.06588	-	-	-	0.003
strong-noise-4d	3	Pedro	valid	11.865	0.01189	-	-	-	0.052
strong-noise-4d	3	Genious	degenerate_cone	-	-	-	-	16	0.135
strong-noise-4d	3	Score Base	valid	9.766	0.00389	-	-	-	9.049
strong-noise-4d	3	Online SAMD	valid	50.367	0.10357	-	-	-	0.003
strong-noise-4d	4	Pedro	valid	6.362	0.00313	-	-	-	0.053
strong-noise-4d	4	Genious	degenerate_cone	-	-	-	-	13	0.117
strong-noise-4d	4	Score Base	valid	2.309	0.00140	11	11	-	10.055
strong-noise-4d	4	Online SAMD	valid	31.145	0.03493	-	-	-	0.003
query-noise-4d	0	Pedro	valid	16.253	0.01781	-	-	-	0.056
query-noise-4d	0	Genious	degenerate_cone	-	-	-	-	15	0.127
query-noise-4d	0	Score Base	valid	2.928	0.00093	5	18	-	9.696
query-noise-4d	0	Online SAMD	valid	36.272	0.11053	-	-	-	0.003
query-noise-4d	1	Pedro	valid	20.837	0.02571	-	-	-	0.050
query-noise-4d	1	Genious	valid	7.164	0.00552	8	-	-	0.170
query-noise-4d	1	Score Base	valid	4.384	0.00125	9	19	-	10.366
query-noise-4d	1	Online SAMD	valid	41.524	0.10620	-	-	-	0.004
query-noise-4d	2	Pedro	valid	21.282	0.02996	-	-	-	0.057
query-noise-4d	2	Genious	valid	8.543	0.00872	-	-	-	0.181
query-noise-4d	2	Score Base	valid	3.900	0.00058	11	16	-	10.388
query-noise-4d	2	Online SAMD	valid	39.091	0.06588	-	-	-	0.004
query-noise-4d	3	Pedro	valid	11.865	0.01189	-	-	-	0.057
query-noise-4d	3	Genious	degenerate_cone	-	-	-	-	16	0.146
query-noise-4d	3	Score Base	valid	4.709	0.00111	17	17	-	9.435
query-noise-4d	3	Online SAMD	valid	50.367	0.10357	-	-	-	0.004

Scenario	Seed	Algorithm	Status	Angle	Regret	First joint	Stable joint	First invalid	Seconds
query-noise-4d	4	Pedro	valid	6.362	0.00313	-	-	-	0.058
query-noise-4d	4	Genious	degenerate_cone	-	-	10	-	19	0.182
query-noise-4d	4	Score Base	valid	3.747	0.00096	10	14	-	10.378
query-noise-4d	4	Online SAMD	valid	31.145	0.03493	-	-	-	0.003

8. Complete 3,200-step appendix

Every observed step is included below. Vectors are rounded to three decimals for readability; the versioned JSON checkpoints preserve full precision.

Connecting two groups | seed 0 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	78.62	0.6546	[-0.000,0.938,-0.346,-0.000,-0.000,-0.000]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
2	valid	72.73	0.3239	[-0.000,0.667,-0.246,-0.000,0.667,-0.223]	[0.000,0.000,0.000,0.000,-0.799,-0.602]	[0.000,0.000,0.000,0.000,1.000,0.000]
3	valid	57.69	0.2558	[0.726,0.458,-0.169,-0.000,0.458,-0.153]	[-0.903,-0.431,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
4	valid	49.06	0.2142	[0.232,0.147,-0.054,-0.000,0.688,0.669]	[0.000,0.000,0.000,0.000,-0.580,-0.815]	[0.000,0.000,0.000,0.000,0.000,1.000]
5	valid	49.06	0.2142	[0.232,0.147,-0.054,-0.000,0.688,0.669]	[0.000,-0.896,-0.444,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
6	valid	49.06	0.2142	[0.232,0.147,-0.054,-0.000,0.688,0.669]	[0.000,0.000,0.000,-0.619,-0.785,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
7	valid	36.43	0.0611	[0.202,0.127,-0.047,0.496,0.597,0.581]	[0.000,0.000,0.000,-0.863,-0.504,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
8	valid	36.43	0.0611	[0.202,0.127,-0.047,0.496,0.597,0.581]	[0.000,0.000,-0.699,-0.715,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
9	valid	26.81	0.0741	[0.312,0.391,-0.000,0.443,0.533,0.519]	[-0.619,-0.785,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
10	valid	26.81	0.0741	[0.312,0.391,-0.000,0.443,0.533,0.519]	[0.000,-0.799,-0.602,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
11	valid	26.81	0.0741	[0.312,0.391,-0.000,0.443,0.533,0.519]	[0.000,-0.903,-0.431,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
12	valid	26.81	0.0741	[0.312,0.391,-0.000,0.443,0.533,0.519]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
13	valid	25.13	0.0714	[0.397,0.447,-0.000,0.410,0.494,0.481]	[-0.853,-0.522,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
14	valid	25.13	0.0714	[0.397,0.447,-0.000,0.410,0.494,0.481]	[0.000,0.000,0.000,-0.243,-0.970,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
15	valid	25.60	0.0745	[0.373,0.420,-0.000,0.418,0.519,0.491]	[0.000,0.000,0.000,0.000,-0.781,-0.625]	[0.000,0.000,0.000,0.000,1.000,0.000]
16	valid	28.86	0.0649	[0.247,0.278,-0.000,0.587,0.546,0.469]	[0.000,0.000,0.000,-0.739,-0.674,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
17	valid	28.86	0.0649	[0.247,0.278,-0.000,0.587,0.546,0.469]	[0.000,0.000,0.000,0.000,-0.311,-0.950]	[0.000,0.000,0.000,0.000,0.000,1.000]
18	valid	28.86	0.0649	[0.247,0.278,-0.000,0.587,0.546,0.469]	[0.000,-0.799,-0.602,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
19	valid	28.86	0.0649	[0.247,0.278,-0.000,0.587,0.546,0.469]	[0.000,0.000,0.000,-0.881,-0.472,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
20	valid	10.73	0.0291	[0.324,0.424,0.324,0.494,0.459,0.394]	[0.000,0.000,-0.884,-0.468,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]

Connecting two groups | seed 0 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	78.62	0.6546	[-0.000,0.938,-0.346,-0.000,-0.000,-0.000]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
2	valid	66.43	0.3737	[-0.000,0.674,-0.248,0.674,-0.169,-0.000]	[0.000,0.000,0.000,-0.881,-0.472,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
3	valid	57.07	0.1900	[-0.000,0.557,-0.205,0.557,-0.162,0.557]	[0.000,0.000,0.000,0.000,-0.536,-0.844]	[0.000,0.000,0.000,0.000,0.000,1.000]
4	valid	44.86	0.1220	[0.651,0.423,-0.156,0.423,-0.123,0.423]	[-0.914,-0.406,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
5	valid	43.52	0.1323	[0.603,0.699,0.000,0.267,-0.078,0.267]	[-0.536,-0.844,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
6	valid	47.65	0.1218	[0.738,0.641,0.000,0.146,-0.042,0.146]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
7	valid	51.05	0.1220	[0.798,0.593,-0.000,0.073,-0.021,0.073]	[-0.654,-0.756,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
8	valid	52.36	0.1216	[0.785,0.616,-0.000,0.047,-0.014,0.047]	[-0.580,-0.815,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
9	valid	53.65	0.1218	[0.771,0.637,-0.000,0.023,-0.007,0.023]	[-0.619,-0.785,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
10	valid	52.70	0.1123	[0.770,0.636,0.035,0.023,-0.007,0.023]	[0.000,0.000,-0.914,-0.406,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
11	valid	51.15	0.1056	[0.768,0.635,0.051,0.058,-0.007,0.023]	[0.000,0.000,-0.523,-0.852,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
12	valid	48.02	0.1086	[0.761,0.629,0.083,0.130,-0.007,0.023]	[0.000,0.000,-0.757,-0.653,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
13	valid	41.28	0.0995	[0.728,0.601,0.199,0.263,-0.006,0.022]	[0.000,0.000,-0.836,-0.549,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
14	valid	34.82	0.0794	[0.648,0.536,0.338,0.422,-0.006,0.019]	[0.000,0.000,-0.802,-0.598,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
15	valid	34.23	0.0794	[0.635,0.525,0.355,0.441,-0.005,0.019]	[0.000,-0.536,-0.844,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
16	valid	31.26	0.0382	[0.634,0.524,0.355,0.440,0.042,0.049]	[0.000,0.000,0.000,0.000,-0.914,-0.406]	[0.000,0.000,0.000,0.000,1.000,0.000]
17	valid	27.91	0.0460	[0.630,0.520,0.352,0.437,0.069,0.110]	[0.000,0.000,0.000,0.000,-0.761,-0.649]	[0.000,0.000,0.000,0.000,0.000,1.000]
18	valid	20.66	0.0255	[0.612,0.506,0.342,0.425,0.160,0.216]	[0.000,0.000,0.000,0.000,-0.842,-0.540]	[0.000,0.000,0.000,0.000,1.000,0.000]
19	valid	10.68	0.0132	[0.538,0.445,0.301,0.374,0.332,0.414]	[0.000,0.000,0.000,0.000,-0.799,-0.602]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	valid	19.95	0.0496	[0.405,0.335,0.227,0.281,0.490,0.594]	[0.000,0.000,0.000,0.000,-0.781,-0.625]	[0.000,0.000,0.000,0.000,1.000,0.000]

Connecting two groups | seed 0 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	89.73	0.5171	[0.504,-0.352,0.180,-0.002,-0.344,-0.110]	[-0.654,-0.756,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
2	valid	69.80	0.2831	[0.341,-0.225,-0.304,0.610,0.034,0.038]	[0.000,0.000,-0.757,-0.653,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
3	valid	51.36	0.1362	[0.665,0.253,-0.229,0.556,0.086,-0.036]	[0.000,-0.914,-0.406,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
4	valid	37.09	0.1083	[0.702,0.326,-0.207,0.609,0.195,0.439]	[0.000,0.000,0.000,0.000,-0.243,-0.970]	[0.000,0.000,0.000,0.000,0.000,1.000]
5	valid	28.19	0.0829	[0.551,0.362,0.306,0.627,-0.088,0.596]	[0.000,-0.311,-0.950,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
6	valid	32.63	0.0695	[0.704,0.420,0.460,0.673,-0.217,0.408]	[0.000,0.000,-0.914,-0.406,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
7	valid	14.22	0.0193	[0.730,0.405,0.300,0.447,0.567,0.473]	[0.000,0.000,0.000,0.000,-0.903,-0.431]	[0.000,0.000,0.000,0.000,1.000,0.000]
8	valid	14.44	0.0318	[0.702,0.361,0.366,0.444,0.656,0.466]	[0.000,0.000,-0.802,-0.598,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
9	valid	6.67	0.0039	[0.659,0.357,0.467,0.591,0.535,0.463]	[0.000,0.000,0.000,-0.829,-0.560,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
10	valid	6.34	0.0040	[0.684,0.475,0.484,0.575,0.531,0.431]	[0.000,-0.842,-0.540,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
11	valid	9.27	0.0022	[0.803,0.466,0.486,0.662,0.423,0.451]	[0.000,0.000,0.000,-0.714,-0.700,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
12	valid	15.65	0.0107	[0.764,0.557,0.448,0.543,0.371,0.264]	[-0.487,-0.873,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
13	valid	6.34	0.0023	[0.693,0.494,0.500,0.648,0.402,0.456]	[0.000,0.000,0.000,0.000,-0.487,-0.873]	[0.000,0.000,0.000,0.000,0.000,1.000]
14	valid	4.64	0.0097	[0.761,0.600,0.629,0.757,0.481,0.692]	[0.000,0.000,0.000,0.000,-0.761,-0.649]	[0.000,0.000,0.000,0.000,0.000,1.000]
15	valid	7.93	0.0179	[0.669,0.454,0.498,0.625,0.319,0.545]	[-0.619,-0.785,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
16	valid	9.77	0.0110	[0.747,0.517,0.637,0.750,0.358,0.458]	[0.000,0.000,0.000,0.000,-0.863,-0.504]	[0.000,0.000,0.000,0.000,1.000,0.000]
17	valid	10.11	0.0065	[0.683,0.498,0.489,0.627,0.317,0.390]	[0.000,-0.799,-0.602,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
18	valid	6.53	0.0061	[0.734,0.514,0.479,0.625,0.415,0.563]	[0.000,0.000,0.000,-0.433,-0.901,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
19	valid	8.28	0.0075	[0.780,0.536,0.463,0.583,0.441,0.565]	[0.000,0.000,0.000,0.000,-0.814,-0.580]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	valid	3.59	0.0037	[0.665,0.491,0.513,0.599,0.451,0.556]	[0.000,0.000,0.000,0.000,-0.799,-0.602]	[0.000,0.000,0.000,0.000,1.000,0.000]

Connecting two groups | seed 0 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	69.33	0.6546	[0.000,0.440,0.000,0.000,0.000,0.000]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
2	valid	60.23	0.3239	[0.000,0.423,0.000,0.000,0.276,0.000]	[0.000,0.000,0.000,0.000,-0.799,-0.602]	[0.000,0.000,0.000,0.000,1.000,0.000]
3	valid	48.24	0.2755	[0.220,0.284,0.000,0.000,0.276,0.000]	[-0.903,-0.431,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
4	valid	40.75	0.2415	[0.220,0.284,0.000,0.000,0.129,0.188]	[0.000,0.000,0.000,0.000,-0.580,-0.815]	[0.000,0.000,0.000,0.000,0.000,1.000]
5	valid	48.38	0.2900	[0.210,0.501,0.000,0.000,0.123,0.179]	[0.000,-0.896,-0.444,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
6	valid	45.46	0.2881	[0.203,0.485,0.000,0.000,0.277,0.173]	[0.000,0.000,0.000,-0.619,-0.785,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
7	valid	39.41	0.1465	[0.203,0.485,0.000,0.129,0.188,0.173]	[0.000,0.000,0.000,-0.863,-0.504,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
8	valid	34.76	0.1214	[0.199,0.477,0.000,0.261,0.185,0.170]	[0.000,0.000,-0.699,-0.715,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
9	valid	45.81	0.1214	[0.098,0.658,0.000,0.245,0.174,0.160]	[-0.619,-0.785,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
10	valid	51.60	0.1214	[0.089,0.840,0.000,0.223,0.158,0.145]	[0.000,-0.799,-0.602,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
11	valid	55.97	0.1214	[0.080,1.025,0.000,0.199,0.141,0.130]	[0.000,-0.903,-0.431,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
12	valid	59.21	0.1214	[0.070,1.207,0.000,0.174,0.123,0.114]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
13	valid	55.65	0.1214	[0.137,1.015,0.000,0.174,0.123,0.114]	[-0.853,-0.522,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
14	valid	54.78	0.1380	[0.137,1.015,0.000,0.156,0.196,0.114]	[0.000,0.000,0.000,-0.243,-0.970,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
15	valid	54.59	0.2154	[0.137,1.015,0.000,0.156,0.279,0.065]	[0.000,0.000,0.000,0.000,-0.781,-0.625]	[0.000,0.000,0.000,0.000,1.000,0.000]
16	valid	53.97	0.1566	[0.137,1.015,0.000,0.229,0.204,0.065]	[0.000,0.000,0.000,-0.739,-0.674,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
17	valid	53.05	0.1271	[0.137,1.015,0.000,0.229,0.181,0.123]	[0.000,0.000,0.000,0.000,-0.311,-0.950]	[0.000,0.000,0.000,0.000,0.000,1.000]
18	valid	56.27	0.1271	[0.127,1.200,0.000,0.213,0.169,0.114]	[0.000,-0.799,-0.602,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
19	valid	55.31	0.1313	[0.125,1.181,0.000,0.282,0.135,0.112]	[0.000,0.000,0.000,-0.881,-0.472,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
20	valid	55.13	0.1258	[0.125,1.181,0.045,0.243,0.135,0.112]	[0.000,0.000,-0.884,-0.468,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]

Connecting two groups | seed 1 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	73.28	0.6410	[-0.000,-0.000,-0.000,-0.000,0.978,-0.208]	[0.000,0.000,0.000,0.000,-0.914,-0.406]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	62.91	0.2533	[-0.135,0.693,-0.000,-0.000,0.693,-0.147]	[-0.374,-0.927,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
3	valid	52.44	0.1616	[-0.080,0.414,0.802,-0.000,0.414,-0.088]	[0.000,-0.580,-0.815,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
4	valid	45.64	0.1232	[-0.060,0.308,0.597,0.667,0.308,-0.065]	[0.000,0.000,0.000,-0.761,-0.649,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
5	valid	47.34	0.1232	[-0.106,0.307,0.595,0.665,0.307,-0.065]	[-0.619,-0.785,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
6	valid	44.48	0.0964	[-0.000,0.707,0.648,0.255,0.118,-0.025]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
7	valid	44.48	0.0964	[-0.000,0.707,0.648,0.255,0.118,-0.025]	[0.000,0.000,0.000,-0.908,-0.418,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
8	valid	35.30	0.0630	[0.563,0.585,0.536,0.211,0.097,-0.021]	[-0.799,-0.602,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
9	valid	21.56	0.0043	[0.533,0.554,0.508,0.280,0.187,0.192]	[0.000,0.000,0.000,0.000,-0.433,-0.901]	[0.000,0.000,0.000,0.000,0.000,1.000]
10	valid	21.56	0.0043	[0.533,0.554,0.508,0.280,0.187,0.192]	[0.000,-0.914,-0.406,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
11	valid	21.56	0.0043	[0.533,0.554,0.508,0.280,0.187,0.192]	[0.000,0.000,0.000,-0.243,-0.970,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
12	valid	21.56	0.0043	[0.533,0.554,0.508,0.280,0.187,0.192]	[0.000,-0.580,-0.815,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
13	valid	21.56	0.0043	[0.533,0.554,0.508,0.280,0.187,0.192]	[-0.536,-0.844,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
14	valid	21.56	0.0043	[0.533,0.554,0.508,0.280,0.187,0.192]	[0.000,0.000,0.000,-0.829,-0.560,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
15	valid	21.00	0.0074	[0.527,0.548,0.502,0.316,0.184,0.190]	[0.000,0.000,-0.396,-0.918,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
16	valid	21.00	0.0074	[0.527,0.548,0.502,0.316,0.184,0.190]	[0.000,-0.889,-0.458,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
17	valid	29.58	0.0186	[0.523,0.630,0.495,0.255,0.103,0.092]	[0.000,-0.654,-0.756,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
18	valid	30.52	0.0182	[0.592,0.596,0.468,0.241,0.097,0.087]	[-0.739,-0.674,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
19	valid	30.44	0.0182	[0.592,0.596,0.468,0.241,0.099,0.087]	[0.000,0.000,0.000,0.000,-0.842,-0.540]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	valid	30.44	0.0182	[0.592,0.596,0.468,0.241,0.099,0.087]	[0.000,-0.863,-0.504,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]

Connecting two groups | seed 1 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	73.28	0.6410	[-0.000,-0.000,-0.000,-0.000,0.978,-0.208]	[0.000,0.000,0.000,0.000,-0.914,-0.406]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	64.14	0.3757	[-0.000,-0.135,0.693,-0.000,0.693,-0.147]	[0.000,-0.374,-0.927,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
3	valid	54.26	0.1475	[-0.166,0.479,0.710,-0.000,0.479,-0.102]	[-0.619,-0.785,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
4	valid	48.34	0.1317	[-0.000,0.745,0.603,-0.000,0.280,-0.059]	[0.000,-0.829,-0.560,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
5	valid	49.75	0.1315	[-0.000,0.672,0.724,-0.000,0.152,-0.032]	[0.000,-0.619,-0.785,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
6	valid	50.83	0.1308	[-0.000,0.730,0.679,-0.000,0.081,-0.017]	[0.000,-0.739,-0.674,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
7	valid	51.58	0.1317	[-0.000,0.756,0.652,-0.000,0.044,-0.009]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
8	valid	51.99	0.1312	[-0.000,0.742,0.670,-0.000,0.021,-0.004]	[0.000,-0.654,-0.756,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
9	valid	51.14	0.0884	[-0.000,0.742,0.670,0.033,0.021,-0.004]	[0.000,0.000,0.000,-0.914,-0.406,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
10	valid	49.70	0.0841	[-0.000,0.740,0.668,0.048,0.055,-0.000]	[0.000,0.000,0.000,-0.536,-0.844,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
11	valid	47.21	0.0955	[-0.000,0.734,0.663,0.078,0.124,-0.000]	[0.000,0.000,0.000,-0.761,-0.649,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
12	valid	42.30	0.0906	[-0.000,0.708,0.640,0.178,0.240,-0.000]	[0.000,0.000,0.000,-0.842,-0.540,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
13	valid	36.51	0.0810	[-0.000,0.607,0.548,0.360,0.449,-0.000]	[0.000,0.000,0.000,-0.799,-0.602,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
14	valid	37.92	0.0848	[-0.000,0.440,0.397,0.513,0.621,-0.000]	[0.000,0.000,0.000,-0.781,-0.625,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
15	valid	37.91	0.0848	[-0.000,0.440,0.398,0.512,0.621,-0.000]	[0.000,0.000,-0.802,-0.598,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,0.000,0.000,-0.761,-0.649,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.619,-0.785,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,0.000,0.000,-0.908,-0.418,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.799,-0.602,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]

Connecting two groups | seed 1 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	62.60	0.3092	[-0.011,0.197,0.251,-0.165,-0.144,0.588]	[0.000,0.000,0.000,0.000,-0.619,-0.785]	[0.000,0.000,0.000,0.000,0.000,1.000]
2	valid	53.64	0.1947	[0.070,-0.059,0.100,0.584,-0.032,0.626]	[0.000,0.000,-0.396,-0.918,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
3	valid	42.29	0.1524	[-0.001,0.488,-0.066,0.561,0.195,0.652]	[-0.311,-0.950,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
4	valid	49.93	0.1456	[0.571,0.474,-0.177,0.526,-0.218,0.539]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
5	valid	55.65	0.1379	[0.649,0.655,-0.154,0.512,-0.426,0.420]	[-0.686,-0.728,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
6	valid	44.01	0.1397	[0.573,0.537,-0.126,0.639,-0.130,0.610]	[-0.714,-0.700,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
7	valid	48.61	0.1379	[0.585,0.606,-0.132,0.514,-0.262,0.562]	[-0.686,-0.728,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
8	valid	25.04	0.0650	[0.628,0.624,0.007,0.457,0.482,0.626]	[0.000,0.000,0.000,-0.243,-0.970,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
9	valid	23.36	0.0825	[0.498,0.506,0.080,0.498,0.341,0.687]	[0.000,0.000,0.000,0.000,-0.761,-0.649]	[0.000,0.000,0.000,0.000,0.000,1.000]
10	valid	17.74	0.0644	[0.537,0.554,0.179,0.488,0.452,0.690]	[0.000,0.000,0.000,0.000,-0.896,-0.444]	[0.000,0.000,0.000,0.000,1.000,0.000]
11	valid	4.23	0.0033	[0.519,0.551,0.507,0.446,0.472,0.630]	[0.000,0.000,-0.836,-0.549,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
12	valid	8.29	0.0045	[0.731,0.721,0.537,0.473,0.471,0.611]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
13	valid	17.67	0.0546	[0.625,0.634,0.287,0.215,0.461,0.617]	[0.000,0.000,0.000,-0.714,-0.700,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
14	valid	14.23	0.0285	[0.696,0.693,0.373,0.309,0.535,0.688]	[0.000,0.000,-0.523,-0.852,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
15	valid	14.34	0.0275	[0.638,0.669,0.328,0.341,0.576,0.713]	[0.000,0.000,0.000,0.000,-0.781,-0.625]	[0.000,0.000,0.000,0.000,1.000,0.000]
16	valid	12.68	0.0206	[0.717,0.728,0.438,0.320,0.535,0.633]	[0.000,0.000,-0.699,-0.715,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
17	valid	9.40	0.0130	[0.668,0.648,0.480,0.357,0.479,0.626]	[0.000,0.000,0.000,-0.814,-0.580,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
18	valid	13.69	0.0091	[0.793,0.792,0.489,0.399,0.445,0.533]	[0.000,0.000,0.000,-0.799,-0.602,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
19	valid	6.49	0.0015	[0.757,0.751,0.656,0.489,0.538,0.642]	[0.000,-0.580,-0.815,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
20	valid	2.23	0.0005	[0.624,0.634,0.641,0.507,0.563,0.654]	[0.000,0.000,0.000,-0.714,-0.700,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]

Connecting two groups | seed 1 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	67.36	0.4482	[0.000,0.000,0.000,0.000,0.440,0.000]	[0.000,0.000,0.000,0.000,-0.914,-0.406]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	61.92	0.2533	[-0.104,0.275,0.000,0.000,0.421,0.000]	[-0.374,-0.927,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
3	valid	58.50	0.1704	[-0.104,0.108,0.218,0.000,0.421,0.000]	[0.000,-0.580,-0.815,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
4	valid	50.14	0.1170	[-0.104,0.108,0.218,0.186,0.217,0.000]	[0.000,0.000,0.000,-0.761,-0.649,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
5	valid	45.80	0.0783	[-0.104,0.291,0.218,0.186,0.217,0.000]	[-0.619,-0.785,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
6	valid	51.87	0.1391	[-0.100,0.491,0.080,0.180,0.209,0.000]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
7	valid	51.58	0.1518	[-0.097,0.478,0.078,0.334,0.138,0.000]	[0.000,0.000,0.000,-0.908,-0.418,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
8	valid	42.97	0.1518	[0.023,0.337,0.078,0.334,0.138,0.000]	[-0.799,-0.602,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
9	valid	38.32	0.1355	[0.023,0.337,0.078,0.334,0.081,0.114]	[0.000,0.000,0.000,0.000,-0.433,-0.901]	[0.000,0.000,0.000,0.000,0.000,1.000]
10	valid	44.11	0.1355	[0.023,0.506,0.030,0.333,0.081,0.114]	[0.000,-0.914,-0.406,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
11	valid	40.76	0.1069	[0.023,0.503,0.030,0.297,0.189,0.113]	[0.000,0.000,0.000,-0.243,-0.970,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
12	valid	32.63	0.1012	[0.023,0.389,0.129,0.297,0.189,0.113]	[0.000,-0.580,-0.815,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
13	valid	40.65	0.1069	[-0.035,0.534,0.125,0.289,0.184,0.110]	[-0.536,-0.844,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
14	valid	42.74	0.1142	[-0.034,0.520,0.122,0.406,0.116,0.107]	[0.000,0.000,0.000,-0.829,-0.560,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
15	valid	45.71	0.1330	[-0.033,0.497,0.081,0.530,0.111,0.103]	[0.000,0.000,-0.396,-0.918,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
16	valid	47.95	0.1330	[-0.031,0.625,0.039,0.502,0.105,0.097]	[0.000,-0.889,-0.458,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
17	valid	51.47	0.1330	[-0.029,0.740,-0.028,0.474,0.100,0.092]	[0.000,-0.654,-0.756,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
18	valid	48.11	0.1330	[0.035,0.587,-0.028,0.474,0.100,0.092]	[-0.739,-0.674,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
19	valid	47.76	0.1506	[0.035,0.587,-0.028,0.474,0.169,0.051]	[0.000,0.000,0.000,0.000,-0.842,-0.540]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	valid	49.02	0.1506	[0.034,0.739,-0.028,0.469,0.167,0.050]	[0.000,-0.863,-0.504,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]

Connecting two groups | seed 2 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.06	0.7432	[-0.000,-0.000,-0.000,0.972,-0.235,-0.000]	[0.000,0.000,0.000,-0.889,-0.458,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
2	valid	63.78	0.5740	[-0.000,-0.000,0.850,0.512,-0.124,-0.000]	[0.000,0.000,-0.884,-0.468,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
3	valid	50.96	0.2974	[-0.000,0.870,0.419,0.252,-0.061,-0.000]	[0.000,-0.654,-0.756,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
4	valid	42.76	0.0644	[-0.000,0.808,0.421,0.345,0.211,-0.074]	[0.000,0.000,0.000,0.000,-0.781,-0.625]	[0.000,0.000,0.000,0.000,1.000,0.000]
5	valid	42.76	0.0644	[-0.000,0.808,0.421,0.345,0.211,-0.074]	[-0.311,-0.950,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
6	valid	25.15	0.0289	[-0.000,0.424,0.272,0.361,0.544,0.566]	[0.000,0.000,0.000,0.000,-0.654,-0.756]	[0.000,0.000,0.000,0.000,0.000,1.000]
7	valid	25.15	0.0289	[-0.000,0.424,0.272,0.361,0.544,0.566]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
8	valid	25.15	0.0289	[-0.000,0.424,0.272,0.361,0.544,0.566]	[-0.311,-0.950,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
9	valid	25.15	0.0289	[-0.000,0.424,0.272,0.361,0.544,0.566]	[0.000,0.000,0.000,0.000,-0.619,-0.785]	[0.000,0.000,0.000,0.000,0.000,1.000]
10	valid	25.15	0.0289	[-0.000,0.424,0.272,0.361,0.544,0.566]	[0.000,0.000,0.000,-0.433,-0.901,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
11	valid	25.15	0.0289	[-0.000,0.424,0.272,0.361,0.544,0.566]	[0.000,-0.829,-0.560,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
12	valid	25.15	0.0289	[-0.000,0.424,0.272,0.361,0.544,0.566]	[-0.487,-0.873,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
13	valid	25.15	0.0289	[-0.000,0.424,0.272,0.361,0.544,0.566]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
14	valid	30.12	0.0340	[-0.000,0.682,0.536,0.207,0.313,0.325]	[0.000,-0.580,-0.815,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
15	valid	21.87	0.0127	[0.478,0.613,0.478,0.170,0.257,0.267]	[-0.814,-0.580,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
16	valid	21.87	0.0127	[0.478,0.613,0.478,0.170,0.257,0.267]	[-0.311,-0.950,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
17	valid	21.87	0.0127	[0.478,0.613,0.478,0.170,0.257,0.267]	[0.000,-0.863,-0.504,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
18	valid	21.87	0.0127	[0.478,0.613,0.478,0.170,0.257,0.267]	[0.000,0.000,-0.757,-0.653,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
19	valid	21.87	0.0127	[0.478,0.613,0.478,0.170,0.257,0.267]	[0.000,0.000,0.000,-0.914,-0.406,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
20	valid	21.87	0.0127	[0.478,0.613,0.478,0.170,0.257,0.267]	[0.000,0.000,0.000,-0.889,-0.458,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]

Connecting two groups | seed 2 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.06	0.7432	[-0.000,-0.000,-0.000,0.972,-0.235,-0.000]	[0.000,0.000,0.000,-0.889,-0.458,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
2	valid	59.54	0.5132	[-0.000,-0.000,-0.000,0.846,0.516,-0.134]	[0.000,0.000,0.000,0.000,-0.873,-0.488]	[0.000,0.000,0.000,0.000,1.000,0.000]
3	valid	52.67	0.1047	[-0.145,0.454,-0.000,0.744,0.454,-0.118]	[-0.580,-0.815,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
4	valid	48.31	0.0957	[-0.081,0.252,-0.000,0.650,0.712,-0.000]	[0.000,0.000,0.000,-0.536,-0.844,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
5	valid	50.44	0.0968	[-0.042,0.130,-0.000,0.565,0.813,-0.000]	[0.000,0.000,0.000,-0.739,-0.674,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
6	valid	51.82	0.0950	[-0.019,0.060,-0.000,0.627,0.776,-0.000]	[0.000,0.000,0.000,-0.814,-0.580,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
7	valid	52.61	0.0955	[-0.009,0.028,-0.000,0.602,0.798,-0.000]	[0.000,0.000,0.000,-0.781,-0.625,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
8	valid	52.92	0.0950	[-0.005,0.014,-0.000,0.613,0.790,-0.000]	[0.000,0.000,0.000,-0.799,-0.602,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
9	valid	52.44	0.0904	[-0.005,0.014,0.019,0.613,0.790,-0.000]	[0.000,-0.243,-0.970,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
10	valid	51.38	0.0760	[-0.000,0.038,0.024,0.613,0.789,-0.000]	[0.000,-0.781,-0.625,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
11	valid	49.64	0.0749	[-0.000,0.065,0.058,0.611,0.787,-0.000]	[0.000,-0.536,-0.844,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
12	valid	45.49	0.0749	[-0.000,0.154,0.115,0.602,0.775,-0.000]	[0.000,-0.654,-0.756,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
13	valid	41.81	0.0749	[-0.000,0.229,0.180,0.587,0.756,-0.000]	[0.000,-0.580,-0.815,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
14	valid	35.22	0.0587	[-0.000,0.405,0.304,0.529,0.681,-0.000]	[0.000,-0.619,-0.785,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
15	valid	34.66	0.0587	[-0.000,0.425,0.319,0.520,0.669,-0.000]	[0.000,0.000,-0.863,-0.506,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
16	valid	34.66	0.0587	[-0.000,0.425,0.319,0.520,0.669,-0.000]	[0.000,-0.619,-0.785,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
17	valid	34.66	0.0587	[-0.000,0.425,0.319,0.520,0.669,-0.000]	[0.000,0.000,-0.863,-0.506,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
18	valid	34.66	0.0587	[-0.000,0.425,0.319,0.520,0.669,-0.000]	[0.000,0.000,0.000,-0.799,-0.602,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
19	valid	34.66	0.0587	[-0.000,0.425,0.319,0.520,0.669,-0.000]	[0.000,-0.619,-0.785,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
20	valid	34.66	0.0587	[-0.000,0.425,0.319,0.520,0.669,-0.000]	[0.000,0.000,-0.863,-0.506,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]

Connecting two groups | seed 2 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	52.99	0.0921	[-0.157,0.316,0.125,0.182,0.553,-0.103]	[0.000,0.000,0.000,-0.580,-0.815,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	48.52	0.1730	[0.177,-0.004,0.008,0.570,0.572,-0.000]	[0.000,0.000,-0.523,-0.852,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
3	valid	60.70	0.0997	[0.077,0.473,-0.240,0.335,0.553,-0.295]	[0.000,-0.842,-0.540,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
4	valid	35.69	0.0549	[-0.015,0.641,0.451,0.490,0.720,-0.068]	[0.000,0.000,-0.914,-0.406,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
5	valid	31.03	0.0606	[0.177,0.454,0.503,0.492,0.671,-0.067]	[0.000,-0.536,-0.844,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
6	valid	22.05	0.0240	[0.006,0.600,0.513,0.427,0.606,0.526]	[0.000,0.000,0.000,0.000,-0.243,-0.970]	[0.000,0.000,0.000,0.000,0.000,1.000]
7	valid	31.44	0.0299	[-0.195,0.558,0.604,0.610,0.663,0.471]	[0.000,0.000,0.000,-0.799,-0.602,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
8	valid	21.54	0.0264	[0.029,0.647,0.661,0.595,0.672,0.595]	[0.000,0.000,0.000,-0.739,-0.674,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
9	valid	18.80	0.0224	[0.087,0.721,0.548,0.562,0.624,0.642]	[0.000,-0.714,-0.700,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
10	valid	7.93	0.0162	[0.641,0.533,0.443,0.526,0.603,0.469]	[-0.881,-0.472,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
11	valid	7.80	0.0071	[0.583,0.583,0.472,0.434,0.532,0.618]	[0.000,0.000,-0.802,-0.598,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
12	valid	5.90	0.0034	[0.562,0.584,0.475,0.466,0.508,0.439]	[0.000,0.000,0.000,-0.781,-0.625,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
13	valid	10.85	0.0178	[0.631,0.618,0.553,0.425,0.495,0.679]	[0.000,0.000,0.000,0.000,-0.580,-0.815]	[0.000,0.000,0.000,0.000,0.000,1.000]
14	valid	9.42	0.0177	[0.515,0.726,0.629,0.543,0.551,0.755]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
15	valid	8.26	0.0130	[0.552,0.667,0.586,0.505,0.567,0.726]	[-0.814,-0.580,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
16	valid	6.93	0.0077	[0.563,0.723,0.573,0.511,0.569,0.690]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
17	valid	8.49	0.0160	[0.516,0.645,0.486,0.437,0.494,0.650]	[0.000,-0.619,-0.785,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
18	valid	4.21	0.0015	[0.601,0.762,0.561,0.547,0.617,0.617]	[0.000,0.000,0.000,0.000,-0.781,-0.625]	[0.000,0.000,0.000,0.000,1.000,0.000]
19	valid	3.54	0.0014	[0.593,0.748,0.539,0.538,0.622,0.573]	[0.000,0.000,0.000,0.000,-0.714,-0.700]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	valid	3.39	0.0016	[0.526,0.727,0.571,0.546,0.625,0.530]	[-0.781,-0.625,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]

Connecting two groups | seed 2 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	67.81	0.5613	[0.000,0.000,0.000,0.440,0.000,0.000]	[0.000,0.000,0.000,-0.889,-0.458,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
2	valid	58.71	0.3892	[0.000,0.000,0.287,0.250,0.000,0.000]	[0.000,0.000,-0.884,-0.468,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
3	valid	50.43	0.3347	[0.000,0.195,0.049,0.250,0.000,0.000]	[0.000,-0.654,-0.756,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
4	valid	38.25	0.1008	[0.000,0.195,0.049,0.250,0.195,0.000]	[0.000,0.000,0.000,0.000,-0.781,-0.625]	[0.000,0.000,0.000,0.000,1.000,0.000]
5	valid	44.08	0.1008	[-0.053,0.393,0.047,0.239,0.187,0.000]	[-0.311,-0.950,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
6	valid	43.64	0.1749	[-0.053,0.393,0.047,0.239,0.053,0.150]	[0.000,0.000,0.000,0.000,-0.654,-0.756]	[0.000,0.000,0.000,0.000,0.000,1.000]
7	valid	47.67	0.1749	[-0.049,0.584,0.043,0.223,0.049,0.140]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
8	valid	51.26	0.1749	[-0.044,0.783,0.039,0.202,0.045,0.127]	[-0.311,-0.950,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
9	valid	51.92	0.1749	[-0.043,0.764,0.038,0.197,-0.034,0.240]	[0.000,0.000,0.000,0.000,-0.619,-0.785]	[0.000,0.000,0.000,0.000,0.000,1.000]
10	valid	49.23	0.1495	[-0.043,0.764,0.038,0.145,0.060,0.240]	[0.000,0.000,0.000,-0.433,-0.901,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
11	valid	53.50	0.1495	[-0.039,0.953,-0.020,0.132,0.054,0.218]	[0.000,-0.829,-0.560,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
12	valid	55.36	0.1495	[-0.035,1.135,-0.018,0.117,0.048,0.193]	[-0.487,-0.873,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
13	valid	56.80	0.1495	[-0.030,1.310,-0.015,0.101,0.042,0.168]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
14	valid	54.60	0.1495	[-0.030,1.080,0.040,0.101,0.042,0.168]	[0.000,-0.580,-0.815,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
15	valid	51.69	0.1495	[0.023,0.895,0.040,0.101,0.042,0.168]	[-0.814,-0.580,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
16	valid	54.20	0.1495	[0.006,1.155,0.040,0.101,0.042,0.168]	[-0.311,-0.950,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
17	valid	56.25	0.1495	[0.006,1.385,0.010,0.095,0.039,0.157]	[0.000,-0.863,-0.504,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
18	valid	56.19	0.1132	[0.006,1.385,0.056,0.053,0.039,0.157]	[0.000,0.000,-0.757,-0.653,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
19	valid	55.81	0.1588	[0.006,1.381,0.056,0.101,0.019,0.157]	[0.000,0.000,0.000,-0.914,-0.406,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
20	valid	55.48	0.1749	[0.006,1.365,0.056,0.150,-0.003,0.155]	[0.000,0.000,0.000,-0.889,-0.458,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]

Connecting two groups | seed 3 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.47	0.7203	[-0.000,-0.419,0.908,-0.000,-0.000,-0.000]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
2	valid	65.84	0.3392	[-0.000,-0.303,0.657,-0.000,0.657,-0.210]	[0.000,0.000,0.000,0.000,-0.814,-0.580]	[0.000,0.000,0.000,0.000,1.000,0.000]
3	valid	57.66	0.1533	[0.549,-0.254,0.549,-0.000,0.549,-0.176]	[-0.814,-0.580,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
4	valid	64.68	0.1533	[0.535,-0.247,0.535,-0.225,0.535,-0.171]	[0.000,0.000,-0.699,-0.715,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
5	valid	52.63	0.1323	[0.184,-0.085,0.184,-0.077,0.696,0.660]	[0.000,0.000,0.000,0.000,-0.536,-0.844]	[0.000,0.000,0.000,0.000,0.000,1.000]
6	valid	26.75	0.0662	[0.416,0.422,0.640,-0.000,0.356,0.337]	[0.000,-0.896,-0.444,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
7	valid	29.22	0.0575	[0.428,0.475,0.669,-0.000,0.275,0.260]	[0.000,-0.863,-0.504,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
8	valid	29.22	0.0575	[0.428,0.475,0.669,-0.000,0.275,0.260]	[-0.853,-0.522,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
9	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
10	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,-0.487,-0.873,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
11	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,0.000,0.000,-0.487,-0.873,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
12	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,0.000,0.000,-0.714,-0.700,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
13	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,0.000,0.000,-0.914,-0.406,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
14	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,0.000,0.000,-0.903,-0.431,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,0.000,0.000,-0.881,-0.472,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.686,-0.728,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,-0.889,-0.458,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,0.000,0.000,-0.311,-0.950,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.000,0.000,0.000,0.000,-0.686,-0.728]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.842,-0.540,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]

Connecting two groups | seed 3 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.47	0.7203	[-0.000,-0.419,0.908,-0.000,-0.000,-0.000]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
2	valid	65.05	0.5344	[0.672,-0.310,0.672,-0.000,-0.000,-0.000]	[-0.853,-0.522,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
3	valid	57.66	0.1533	[0.549,-0.254,0.549,-0.000,0.549,-0.176]	[0.000,0.000,0.000,0.000,-0.814,-0.580]	[0.000,0.000,0.000,0.000,1.000,0.000]
4	valid	47.44	0.1115	[0.449,-0.207,0.449,0.575,0.449,-0.144]	[0.000,0.000,-0.243,-0.970,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
5	valid	43.50	0.1353	[0.332,-0.153,0.332,0.425,0.758,-0.000]	[0.000,0.000,0.000,-0.619,-0.785,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
6	valid	45.21	0.1351	[0.188,-0.087,0.188,0.623,0.730,-0.000]	[0.000,0.000,0.000,-0.873,-0.488,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
7	valid	48.31	0.1372	[0.097,-0.045,0.097,0.713,0.686,-0.000]	[0.000,0.000,0.000,-0.761,-0.649,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
8	valid	49.94	0.1349	[0.054,-0.025,0.054,0.687,0.722,-0.000]	[0.000,0.000,0.000,-0.686,-0.728,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
9	valid	50.76	0.1349	[0.035,-0.016,0.035,0.674,0.737,-0.000]	[0.000,0.000,0.000,-0.714,-0.700,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
10	valid	51.49	0.1349	[0.018,-0.008,0.018,0.687,0.726,-0.000]	[0.000,0.000,0.000,-0.739,-0.674,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
11	valid	48.40	0.1060	[0.039,0.028,0.061,0.685,0.724,-0.000]	[-0.243,-0.970,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
12	valid	44.82	0.0702	[0.064,0.071,0.111,0.680,0.719,-0.000]	[0.000,-0.908,-0.418,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
13	valid	38.45	0.0592	[0.110,0.147,0.199,0.662,0.699,-0.000]	[-0.739,-0.674,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
14	valid	28.19	0.0393	[0.218,0.262,0.332,0.604,0.639,-0.000]	[-0.799,-0.602,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
15	valid	19.04	0.0253	[0.354,0.452,0.547,0.419,0.442,-0.000]	[0.000,-0.781,-0.625,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
16	valid	19.01	0.0249	[0.373,0.448,0.543,0.415,0.439,-0.000]	[-0.781,-0.625,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
17	valid	20.37	0.0268	[0.395,0.478,0.576,0.366,0.387,-0.000]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
18	valid	20.37	0.0268	[0.395,0.478,0.576,0.366,0.387,-0.000]	[0.000,0.000,0.000,-0.739,-0.674,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
19	valid	20.37	0.0268	[0.395,0.478,0.576,0.366,0.387,-0.000]	[0.000,0.000,0.000,-0.714,-0.700,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	valid	20.37	0.0268	[0.395,0.478,0.576,0.366,0.387,-0.000]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]

Connecting two groups | seed 3 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	69.94	0.3333	[0.105,0.056,0.047,-0.078,-0.066,0.580]	[0.000,0.000,0.000,0.000,-0.311,-0.950]	[0.000,0.000,0.000,0.000,0.000,1.000]
2	valid	56.25	0.1910	[0.140,0.134,-0.094,0.466,-0.047,0.338]	[0.000,0.000,-0.396,-0.918,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
3	valid	58.81	0.3832	[-0.120,-0.223,0.449,0.488,0.051,0.325]	[0.000,-0.619,-0.785,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
4	valid	52.83	0.1581	[-0.122,0.382,0.626,0.688,-0.337,0.604]	[-0.311,-0.950,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
5	valid	43.42	0.1334	[0.395,0.409,0.655,0.719,-0.338,0.561]	[-0.903,-0.431,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
6	valid	33.97	0.1343	[0.603,0.491,0.494,0.617,-0.117,0.565]	[0.000,-0.829,-0.560,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
7	valid	7.98	0.0076	[0.550,0.519,0.565,0.541,0.587,0.580]	[0.000,0.000,0.000,-0.374,-0.927,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
8	valid	12.83	0.0255	[0.608,0.515,0.480,0.442,0.641,0.609]	[0.000,0.000,0.000,-0.739,-0.674,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
9	valid	9.49	0.0175	[0.540,0.508,0.632,0.448,0.692,0.598]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
10	valid	9.43	0.0156	[0.607,0.549,0.619,0.531,0.739,0.648]	[0.000,-0.781,-0.625,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
11	valid	4.98	0.0015	[0.529,0.636,0.679,0.514,0.659,0.552]	[-0.654,-0.756,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
12	valid	6.70	0.0131	[0.504,0.652,0.705,0.547,0.841,0.480]	[0.000,0.000,0.000,0.000,-0.654,-0.756]	[0.000,0.000,0.000,0.000,1.000,0.000]
13	valid	7.61	0.0040	[0.437,0.545,0.595,0.669,0.758,0.405]	[0.000,0.000,0.000,-0.814,-0.580,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
14	valid	3.19	0.0009	[0.527,0.613,0.666,0.674,0.703,0.444]	[0.000,0.000,0.000,-0.739,-0.674,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
15	valid	5.31	0.0039	[0.412,0.584,0.673,0.606,0.643,0.348]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
16	valid	4.72	0.0056	[0.428,0.631,0.633,0.624,0.715,0.479]	[0.000,0.000,0.000,0.000,-0.433,-0.901]	[0.000,0.000,0.000,0.000,0.000,1.000]
17	valid	1.60	0.0009	[0.517,0.661,0.686,0.616,0.650,0.466]	[-0.829,-0.560,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
18	valid	5.84	0.0038	[0.428,0.583,0.571,0.633,0.651,0.521]	[0.000,0.000,0.000,0.000,-0.580,-0.815]	[0.000,0.000,0.000,0.000,0.000,1.000]
19	valid	6.78	0.0034	[0.447,0.595,0.644,0.679,0.722,0.604]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
20	valid	4.76	0.0040	[0.442,0.625,0.646,0.579,0.631,0.553]	[0.000,0.000,0.000,0.000,-0.619,-0.785]	[0.000,0.000,0.000,0.000,0.000,1.000]

Connecting two groups | seed 3 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	61.89	0.5344	[0.000,0.000,0.440,0.000,0.000,0.000]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
2	valid	50.21	0.1533	[0.000,0.000,0.423,0.000,0.276,0.000]	[0.000,0.000,0.000,0.000,-0.814,-0.580]	[0.000,0.000,0.000,0.000,1.000,0.000]
3	valid	53.97	0.1533	[0.212,-0.147,0.408,0.000,0.266,0.000]	[-0.814,-0.580,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
4	valid	54.49	0.1533	[0.190,-0.132,0.663,0.000,0.239,0.000]	[0.000,0.000,-0.699,-0.715,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
5	valid	53.83	0.1350	[0.190,-0.132,0.663,0.000,0.136,0.144]	[0.000,0.000,0.000,0.000,-0.536,-0.844]	[0.000,0.000,0.000,0.000,0.000,1.000]
6	valid	44.79	0.1350	[0.190,-0.001,0.527,0.000,0.136,0.144]	[0.000,-0.896,-0.444,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
7	valid	34.45	0.1143	[0.190,0.119,0.404,0.000,0.136,0.144]	[0.000,-0.863,-0.504,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
8	valid	39.23	0.1350	[0.333,0.050,0.404,0.000,0.136,0.144]	[-0.853,-0.522,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
9	valid	32.90	0.0934	[0.333,0.159,0.273,0.000,0.136,0.144]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
10	valid	36.88	0.1350	[0.333,0.100,0.420,0.000,0.136,0.144]	[0.000,-0.487,-0.873,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
11	valid	33.43	0.1290	[0.333,0.100,0.420,0.000,0.246,0.144]	[0.000,0.000,0.000,-0.487,-0.873,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
12	valid	32.46	0.1397	[0.333,0.100,0.420,0.000,0.372,0.144]	[0.000,0.000,0.000,-0.714,-0.700,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
13	valid	27.01	0.1363	[0.333,0.100,0.420,0.087,0.315,0.144]	[0.000,0.000,0.000,-0.914,-0.406,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
14	valid	23.21	0.0896	[0.333,0.100,0.420,0.178,0.261,0.144]	[0.000,0.000,0.000,-0.903,-0.431,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
15	valid	22.53	0.0823	[0.333,0.100,0.420,0.278,0.207,0.144]	[0.000,0.000,0.000,-0.881,-0.472,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
16	valid	16.01	0.0557	[0.234,0.186,0.420,0.278,0.207,0.144]	[-0.686,-0.728,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
17	valid	10.53	0.0037	[0.234,0.280,0.358,0.278,0.207,0.144]	[0.000,-0.889,-0.458,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
18	valid	7.59	0.0032	[0.234,0.279,0.357,0.246,0.301,0.143]	[0.000,0.000,0.000,-0.311,-0.950,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
19	valid	15.46	0.0361	[0.229,0.273,0.350,0.241,0.394,0.067]	[0.000,0.000,0.000,0.000,-0.686,-0.728]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	valid	17.59	0.0664	[0.317,0.215,0.346,0.238,0.390,0.066]	[-0.842,-0.540,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]

Connecting two groups | seed 4 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	74.84	0.8026	[-0.000,-0.000,-0.000,-0.000,-0.419,0.908]	[0.000,0.000,0.000,0.000,-0.761,-0.649]	[0.000,0.000,0.000,0.000,0.000,1.000]
2	valid	67.63	0.4165	[-0.000,0.656,-0.219,-0.000,-0.303,0.656]	[0.000,-0.799,-0.602,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
3	valid	53.31	0.3001	[-0.000,0.796,0.397,-0.132,-0.183,0.397]	[0.000,0.000,-0.802,-0.598,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
4	valid	44.58	0.2704	[0.642,0.610,0.304,-0.101,-0.140,0.304]	[-0.896,-0.444,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
5	valid	33.53	0.0696	[0.556,0.617,0.447,0.224,-0.103,0.224]	[0.000,0.000,0.000,-0.842,-0.540,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
6	valid	33.53	0.0696	[0.556,0.617,0.447,0.224,-0.103,0.224]	[0.000,0.000,-0.802,-0.598,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
7	valid	34.60	0.0757	[0.555,0.655,0.411,0.206,-0.095,0.206]	[0.000,-0.739,-0.674,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
8	valid	37.89	0.0790	[0.435,0.666,0.590,0.094,-0.044,0.094]	[0.000,-0.580,-0.815,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
9	valid	37.74	0.0752	[0.471,0.652,0.578,0.093,-0.043,0.093]	[-0.873,-0.488,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
10	valid	37.74	0.0752	[0.471,0.652,0.578,0.093,-0.043,0.093]	[0.000,0.000,0.000,-0.799,-0.602,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
11	valid	37.74	0.0752	[0.471,0.652,0.578,0.093,-0.043,0.093]	[0.000,0.000,-0.623,-0.783,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
12	valid	37.74	0.0752	[0.471,0.652,0.578,0.093,-0.043,0.093]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
13	valid	37.74	0.0752	[0.471,0.652,0.578,0.093,-0.043,0.093]	[0.000,0.000,-0.884,-0.468,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
14	valid	37.71	0.0736	[0.484,0.647,0.573,0.092,-0.042,0.092]	[-0.863,-0.504,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
15	valid	37.74	0.0714	[0.531,0.627,0.555,0.089,-0.041,0.089]	[-0.829,-0.560,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
16	valid	37.74	0.0714	[0.531,0.627,0.555,0.089,-0.041,0.089]	[0.000,0.000,0.000,-0.814,-0.580,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
17	valid	37.74	0.0714	[0.531,0.627,0.555,0.089,-0.041,0.089]	[0.000,0.000,0.000,0.000,-0.433,-0.901]	[0.000,0.000,0.000,0.000,0.000,1.000]
18	valid	37.74	0.0714	[0.531,0.627,0.555,0.089,-0.041,0.089]	[0.000,0.000,0.000,0.000,-0.739,-0.674]	[0.000,0.000,0.000,0.000,0.000,1.000]
19	valid	18.55	0.0128	[0.487,0.576,0.510,0.222,0.158,0.312]	[0.000,0.000,0.000,-0.311,-0.950,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	valid	4.12	0.0023	[0.388,0.487,0.461,0.304,0.321,0.453]	[0.000,0.000,0.000,0.000,-0.863,-0.504]	[0.000,0.000,0.000,0.000,1.000,0.000]

Connecting two groups | seed 4 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	74.84	0.8026	[-0.000,-0.000,-0.000,-0.000,-0.419,0.908]	[0.000,0.000,0.000,0.000,-0.761,-0.649]	[0.000,0.000,0.000,0.000,0.000,1.000]
2	valid	68.17	0.4165	[-0.258,0.649,-0.000,-0.000,-0.300,0.649]	[-0.686,-0.728,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
3	valid	54.15	0.2410	[-0.128,0.324,-0.000,-0.052,0.324,0.878]	[0.000,0.000,0.000,-0.311,-0.950,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
4	valid	47.07	0.1141	[-0.122,0.307,0.307,-0.083,0.307,0.834]	[0.000,0.000,-0.863,-0.506,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
5	valid	39.85	0.1293	[-0.000,0.613,0.257,-0.070,0.257,0.698]	[0.000,-0.714,-0.700,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
6	valid	37.11	0.1214	[-0.000,0.716,0.478,-0.028,0.176,0.477]	[0.000,-0.374,-0.927,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
7	valid	40.72	0.1164	[-0.000,0.738,0.597,-0.017,0.109,0.295]	[0.000,-0.536,-0.844,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
8	valid	44.63	0.1164	[-0.000,0.732,0.657,-0.010,0.063,0.170]	[0.000,-0.619,-0.785,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
9	valid	36.45	0.0905	[-0.000,0.680,0.610,-0.000,0.216,0.343]	[0.000,0.000,0.000,0.000,-0.914,-0.406]	[0.000,0.000,0.000,0.000,1.000,0.000]
10	valid	31.82	0.0889	[-0.000,0.542,0.486,-0.000,0.408,0.550]	[0.000,0.000,0.000,0.000,-0.842,-0.540]	[0.000,0.000,0.000,0.000,1.000,0.000]
11	valid	38.10	0.0888	[-0.000,0.318,0.285,-0.000,0.566,0.705]	[0.000,0.000,0.000,0.000,-0.799,-0.602]	[0.000,0.000,0.000,0.000,1.000,0.000]
12	valid	45.03	0.0894	[-0.000,0.177,0.159,-0.000,0.618,0.749]	[0.000,0.000,0.000,0.000,-0.781,-0.625]	[0.000,0.000,0.000,0.000,1.000,0.000]
13	valid	40.51	0.0877	[-0.000,0.260,0.245,-0.000,0.595,0.720]	[0.000,-0.654,-0.756,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
14	valid	32.86	0.0871	[-0.000,0.444,0.436,-0.000,0.498,0.604]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
15	valid	32.86	0.0871	[-0.000,0.444,0.436,-0.000,0.498,0.604]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
16	valid	32.86	0.0871	[-0.000,0.444,0.436,-0.000,0.498,0.604]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
17	valid	32.86	0.0871	[-0.000,0.444,0.436,-0.000,0.498,0.604]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
18	valid	32.86	0.0871	[-0.000,0.444,0.436,-0.000,0.498,0.604]	[0.000,-0.686,-0.728,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
19	valid	32.86	0.0871	[-0.000,0.444,0.436,-0.000,0.498,0.604]	[0.000,-0.714,-0.700,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
20	valid	32.86	0.0871	[-0.000,0.444,0.436,-0.000,0.498,0.604]	[0.000,0.000,0.000,0.000,-0.761,-0.649]	[0.000,0.000,0.000,0.000,0.000,1.000]

Connecting two groups | seed 4 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	96.20	0.3985	[0.114,-0.122,-0.052,0.064,0.484,-0.538]	[0.000,0.000,0.000,0.000,-0.842,-0.540]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	73.70	0.2039	[-0.180,0.561,-0.371,0.074,0.490,0.067]	[0.000,-0.842,-0.540,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
3	valid	48.18	0.1390	[0.038,0.617,0.564,-0.167,0.549,0.001]	[0.000,0.000,-0.914,-0.406,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
4	valid	55.25	0.1248	[0.609,0.456,0.301,-0.265,0.411,-0.174]	[-0.889,-0.458,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
5	valid	30.19	0.0696	[0.615,0.624,0.473,-0.170,0.625,0.462]	[0.000,0.000,0.000,0.000,-0.311,-0.950]	[0.000,0.000,0.000,0.000,0.000,1.000]
6	valid	31.05	0.0713	[0.638,0.718,0.428,-0.197,0.605,0.535]	[-0.686,-0.728,0.000,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
7	valid	16.93	0.0661	[0.435,0.597,0.622,0.126,0.588,0.468]	[0.000,-0.619,-0.785,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
8	valid	9.75	0.0090	[0.494,0.667,0.706,0.503,0.499,0.424]	[0.000,0.000,0.000,-0.914,-0.406,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
9	valid	6.07	0.0070	[0.566,0.636,0.728,0.459,0.513,0.530]	[0.000,0.000,0.000,0.000,-0.580,-0.815]	[0.000,0.000,0.000,0.000,0.000,1.000]
10	valid	2.45	0.0015	[0.576,0.668,0.683,0.565,0.566,0.651]	[0.000,0.000,0.000,0.000,-0.714,-0.700]	[0.000,0.000,0.000,0.000,0.000,1.000]
11	valid	5.29	0.0120	[0.601,0.638,0.667,0.417,0.608,0.729]	[0.000,0.000,0.000,-0.714,-0.700,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
12	valid	7.31	0.0010	[0.466,0.584,0.576,0.483,0.591,0.745]	[0.000,0.000,-0.523,-0.852,0.000,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
13	valid	7.33	0.0025	[0.610,0.621,0.666,0.595,0.719,0.836]	[-0.799,-0.602,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
14	valid	7.85	0.0020	[0.493,0.584,0.638,0.601,0.649,0.759]	[0.000,0.000,0.000,-0.761,-0.649,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
15	valid	6.04	0.0021	[0.618,0.620,0.622,0.586,0.663,0.769]	[-0.761,-0.649,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
16	valid	5.02	0.0050	[0.734,0.733,0.653,0.557,0.605,0.658]	[0.000,0.000,0.000,0.000,-0.761,-0.649]	[0.000,0.000,0.000,0.000,1.000,0.000]
17	valid	5.43	0.0013	[0.687,0.754,0.739,0.506,0.533,0.605]	[0.000,0.000,-0.623,-0.783,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
18	valid	4.50	0.0050	[0.699,0.718,0.849,0.571,0.606,0.680]	[0.000,-0.714,-0.700,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
19	valid	3.56	0.0034	[0.559,0.585,0.684,0.470,0.550,0.613]	[0.000,0.000,0.000,-0.714,-0.700,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	valid	3.98	0.0037	[0.590,0.613,0.663,0.471,0.535,0.560]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]

Connecting two groups | seed 4 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	63.15	0.6111	[0.000,0.000,0.000,0.000,0.000,0.440]	[0.000,0.000,0.000,0.000,-0.761,-0.649]	[0.000,0.000,0.000,0.000,0.000,1.000]
2	valid	51.71	0.4165	[0.000,0.276,0.000,0.000,0.000,0.423]	[0.000,-0.799,-0.602,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
3	valid	41.74	0.2919	[0.000,0.270,0.214,0.000,0.000,0.413]	[0.000,0.000,-0.802,-0.598,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
4	valid	35.64	0.2669	[0.183,0.167,0.214,0.000,0.000,0.413]	[-0.896,-0.444,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
5	valid	28.54	0.0731	[0.181,0.165,0.212,0.161,0.000,0.408]	[0.000,0.000,0.000,-0.842,-0.540,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
6	valid	32.76	0.1223	[0.175,0.159,0.386,0.048,0.000,0.394]	[0.000,0.000,-0.802,-0.598,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
7	valid	29.94	0.0888	[0.175,0.314,0.230,0.048,0.000,0.394]	[0.000,-0.739,-0.674,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
8	valid	31.24	0.0920	[0.173,0.205,0.386,0.047,0.000,0.389]	[0.000,-0.580,-0.815,0.000,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
9	valid	32.12	0.1454	[0.303,0.133,0.379,0.047,0.000,0.382]	[-0.873,-0.488,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
10	valid	28.08	0.1454	[0.299,0.132,0.374,0.152,0.000,0.377]	[0.000,0.000,0.000,-0.799,-0.602,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
11	valid	34.04	0.1454	[0.291,0.128,0.490,0.049,0.000,0.366]	[0.000,0.000,-0.623,-0.783,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
12	valid	28.19	0.0968	[0.291,0.232,0.360,0.049,0.000,0.366]	[0.000,-0.761,-0.649,0.000,0.000,0.000]	[0.000,1.000,0.000,0.000,0.000,0.000]
13	valid	33.03	0.1212	[0.283,0.226,0.495,0.003,0.000,0.357]	[0.000,0.000,-0.884,-0.468,0.000,0.000]	[0.000,0.000,1.000,0.000,0.000,0.000]
14	valid	34.68	0.1506	[0.396,0.165,0.483,0.003,0.000,0.348]	[-0.863,-0.504,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
15	valid	38.40	0.1506	[0.518,0.104,0.465,0.003,0.000,0.335]	[-0.829,-0.560,0.000,0.000,0.000,0.000]	[1.000,0.000,0.000,0.000,0.000,0.000]
16	valid	35.72	0.1506	[0.516,0.104,0.463,0.075,0.000,0.334]	[0.000,0.000,0.000,-0.814,-0.580,0.000]	[0.000,0.000,0.000,1.000,0.000,0.000]
17	valid	34.92	0.1506	[0.497,0.100,0.446,0.072,0.000,0.434]	[0.000,0.000,0.000,0.000,-0.433,-0.901]	[0.000,0.000,0.000,0.000,0.000,1.000]
18	valid	35.44	0.1506	[0.474,0.095,0.425,0.069,0.000,0.542]	[0.000,0.000,0.000,0.000,-0.739,-0.674]	[0.000,0.000,0.000,0.000,0.000,1.000]
19	valid	33.78	0.1164	[0.473,0.095,0.425,0.049,0.060,0.541]	[0.000,0.000,0.000,-0.311,-0.950,0.000]	[0.000,0.000,0.000,0.000,1.000,0.000]
20	valid	31.08	0.1638	[0.473,0.095,0.425,0.049,0.122,0.467]	[0.000,0.000,0.000,0.000,-0.863,-0.504]	[0.000,0.000,0.000,0.000,1.000,0.000]

Several boundaries to locate | seed 0 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	66.58	0.1293	[0.977,-0.000,-0.000,-0.212]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
2	valid	70.24	0.1293	[0.951,-0.000,-0.000,-0.309]	[-0.809,0.000,0.000,-0.587]	[1.000,0.000,0.000,0.000]
3	valid	82.00	0.1293	[0.894,-0.000,-0.341,-0.290]	[-0.746,0.000,-0.665,0.000]	[1.000,0.000,0.000,0.000]
4	valid	46.23	0.1053	[0.801,-0.000,-0.000,0.598]	[-0.333,0.000,0.000,-0.943]	[0.000,0.000,0.000,1.000]
5	valid	46.23	0.1053	[0.801,-0.000,-0.000,0.598]	[-0.761,-0.649,0.000,0.000]	[1.000,0.000,0.000,0.000]
6	valid	46.23	0.1053	[0.801,-0.000,-0.000,0.598]	[-0.903,0.000,-0.431,0.000]	[1.000,0.000,0.000,0.000]
7	valid	46.23	0.1053	[0.801,-0.000,-0.000,0.598]	[-0.774,-0.633,0.000,0.000]	[1.000,0.000,0.000,0.000]
8	valid	46.14	0.1043	[0.666,-0.000,-0.000,0.746]	[-0.676,0.000,0.000,-0.737]	[0.000,0.000,0.000,1.000]
9	valid	31.00	0.0505	[0.590,-0.000,0.463,0.661]	[-0.520,0.000,-0.854,0.000]	[0.000,0.000,1.000,0.000]
10	valid	31.00	0.0505	[0.590,-0.000,0.463,0.661]	[-0.787,0.000,-0.617,0.000]	[1.000,0.000,0.000,0.000]
11	valid	29.37	0.0433	[0.570,-0.000,0.610,0.550]	[-0.714,0.000,-0.700,0.000]	[0.000,0.000,1.000,0.000]
12	valid	29.37	0.0433	[0.570,-0.000,0.610,0.550]	[-0.809,-0.587,0.000,0.000]	[1.000,0.000,0.000,0.000]
13	valid	29.37	0.0433	[0.570,-0.000,0.610,0.550]	[-0.860,-0.510,0.000,0.000]	[1.000,0.000,0.000,0.000]
14	valid	29.37	0.0433	[0.570,-0.000,0.610,0.550]	[-0.879,0.000,0.000,-0.478]	[1.000,0.000,0.000,0.000]
15	valid	29.37	0.0433	[0.570,-0.000,0.610,0.550]	[-0.654,0.000,-0.756,0.000]	[0.000,0.000,1.000,0.000]
16	valid	20.11	0.0374	[0.563,0.161,0.602,0.543]	[-0.243,-0.970,0.000,0.000]	[0.000,1.000,0.000,0.000]
17	valid	20.11	0.0374	[0.563,0.161,0.602,0.543]	[-0.520,0.000,0.000,-0.854]	[0.000,0.000,0.000,1.000]
18	valid	20.11	0.0374	[0.563,0.161,0.602,0.543]	[-0.787,0.000,-0.617,0.000]	[1.000,0.000,0.000,0.000]
19	valid	20.11	0.0374	[0.563,0.161,0.602,0.543]	[-0.860,0.000,-0.510,0.000]	[1.000,0.000,0.000,0.000]
20	valid	20.50	0.0429	[0.572,0.154,0.599,0.539]	[-0.695,0.000,0.000,-0.719]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 0 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	66.58	0.1293	[0.977,-0.000,-0.000,-0.212]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
2	valid	48.85	0.0752	[0.610,-0.000,0.781,-0.132]	[-0.243,0.000,-0.970,0.000]	[0.000,0.000,1.000,0.000]
3	valid	44.53	0.0807	[0.839,-0.000,0.544,-0.000]	[-0.787,0.000,-0.617,0.000]	[1.000,0.000,0.000,0.000]
4	valid	43.10	0.0701	[0.735,-0.000,0.678,-0.000]	[-0.551,0.000,-0.835,0.000]	[0.000,0.000,1.000,0.000]
5	valid	43.02	0.0696	[0.679,-0.000,0.734,-0.000]	[-0.676,0.000,-0.737,0.000]	[0.000,0.000,1.000,0.000]
6	valid	43.01	0.0693	[0.710,-0.000,0.704,-0.000]	[-0.731,0.000,-0.683,0.000]	[1.000,0.000,0.000,0.000]
7	valid	43.00	0.0693	[0.701,-0.000,0.713,-0.000]	[-0.695,0.000,-0.719,0.000]	[0.000,0.000,1.000,0.000]
8	valid	43.00	0.0693	[0.692,-0.000,0.722,-0.000]	[-0.714,0.000,-0.700,0.000]	[0.000,0.000,1.000,0.000]
9	valid	43.00	0.0693	[0.692,-0.000,0.722,-0.000]	[-0.714,0.000,-0.700,0.000]	[0.000,0.000,1.000,0.000]
10	valid	43.00	0.0693	[0.692,-0.000,0.722,-0.000]	[-0.731,0.000,-0.683,0.000]	[1.000,0.000,0.000,0.000]
11	valid	43.00	0.0693	[0.692,-0.000,0.722,-0.000]	[-0.714,0.000,-0.700,0.000]	[0.000,0.000,1.000,0.000]
12	valid	43.00	0.0693	[0.692,-0.000,0.722,-0.000]	[-0.731,0.000,-0.683,0.000]	[1.000,0.000,0.000,0.000]
13	valid	43.00	0.0693	[0.692,-0.000,0.722,-0.000]	[-0.731,0.000,-0.683,0.000]	[1.000,0.000,0.000,0.000]
14	valid	43.00	0.0693	[0.692,-0.000,0.722,-0.000]	[-0.731,0.000,-0.683,0.000]	[1.000,0.000,0.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.714,0.000,-0.700,0.000]	[1.000,0.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.520,0.000,-0.854,0.000]	[0.000,0.000,1.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.787,0.000,-0.617,0.000]	[1.000,0.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.714,0.000,-0.700,0.000]	[0.000,0.000,1.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.809,-0.587,0.000,0.000]	[1.000,0.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.860,-0.510,0.000,0.000]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 0 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	48.83	0.1809	[0.145,0.558,-0.002,0.076]	[-0.551,-0.835,0.000,0.000]	[0.000,1.000,0.000,0.000]
2	valid	52.15	0.5168	[-0.017,0.370,-0.051,0.508]	[-0.452,0.000,0.000,-0.892]	[0.000,0.000,0.000,1.000]
3	valid	39.12	0.0980	[0.450,0.669,-0.110,0.642]	[-0.714,0.000,-0.700,0.000]	[1.000,0.000,0.000,0.000]
4	valid	43.10	0.0610	[0.604,0.549,-0.203,0.594]	[-0.809,-0.587,0.000,0.000]	[1.000,0.000,0.000,0.000]
5	valid	35.22	0.0629	[0.623,0.511,-0.051,0.460]	[-0.714,0.000,0.000,-0.700]	[1.000,0.000,0.000,0.000]
6	valid	43.68	0.0601	[0.626,0.590,-0.218,0.559]	[-0.580,0.000,0.000,-0.815]	[0.000,0.000,0.000,1.000]
7	valid	50.04	0.0617	[0.604,0.512,-0.309,0.496]	[-0.714,-0.700,0.000,0.000]	[1.000,0.000,0.000,0.000]
8	valid	50.43	0.0601	[0.725,0.704,-0.418,0.700]	[-0.654,-0.756,0.000,0.000]	[0.000,1.000,0.000,0.000]
9	valid	44.12	0.0605	[0.754,0.664,-0.266,0.667]	[-0.695,-0.719,0.000,0.000]	[1.000,0.000,0.000,0.000]
10	valid	36.04	0.0613	[0.654,0.600,-0.077,0.528]	[-0.676,0.000,0.000,-0.737]	[1.000,0.000,0.000,0.000]
11	valid	32.00	0.0613	[0.685,0.662,-0.003,0.555]	[-0.695,-0.719,0.000,0.000]	[0.000,1.000,0.000,0.000]
12	valid	41.01	0.0601	[0.792,0.756,-0.216,0.724]	[-0.654,0.000,0.000,-0.756]	[0.000,0.000,0.000,1.000]
13	valid	30.40	0.0605	[0.541,0.529,0.021,0.536]	[-0.676,0.000,0.000,-0.737]	[0.000,0.000,0.000,1.000]
14	valid	28.87	0.0600	[0.693,0.649,0.056,0.647]	[-0.695,0.000,0.000,-0.719]	[1.000,0.000,0.000,0.000]
15	valid	16.44	0.0310	[0.728,0.716,0.331,0.658]	[-0.243,0.000,-0.970,0.000]	[0.000,0.000,1.000,0.000]
16	valid	7.26	0.0063	[0.796,0.763,0.599,0.728]	[-0.333,0.000,-0.943,0.000]	[0.000,0.000,1.000,0.000]
17	valid	2.87	0.0029	[0.588,0.542,0.530,0.528]	[-0.580,0.000,-0.815,0.000]	[0.000,0.000,1.000,0.000]
18	valid	2.34	0.0009	[0.654,0.627,0.610,0.585]	[-0.654,0.000,-0.756,0.000]	[0.000,0.000,1.000,0.000]
19	valid	1.87	0.0006	[0.579,0.519,0.582,0.550]	[-0.676,0.000,-0.737,0.000]	[0.000,0.000,1.000,0.000]
20	valid	0.93	0.0000	[0.698,0.689,0.704,0.651]	[-0.654,-0.756,0.000,0.000]	[0.000,1.000,0.000,0.000]

Several boundaries to locate | seed 0 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	59.38	0.1293	[0.517,0.000,0.000,0.000]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
2	valid	59.38	0.1293	[0.912,0.000,0.000,0.000]	[-0.809,0.000,0.000,-0.587]	[1.000,0.000,0.000,0.000]
3	valid	59.38	0.1293	[1.221,0.000,0.000,0.000]	[-0.746,0.000,-0.665,0.000]	[1.000,0.000,0.000,0.000]
4	valid	55.55	0.1293	[1.019,0.000,0.000,0.131]	[-0.333,0.000,0.000,-0.943]	[0.000,0.000,0.000,1.000]
5	valid	58.97	0.1293	[1.322,-0.081,0.000,0.108]	[-0.761,-0.649,0.000,0.000]	[1.000,0.000,0.000,0.000]
6	valid	59.07	0.1293	[1.496,-0.060,0.000,0.081]	[-0.903,0.000,-0.431,0.000]	[1.000,0.000,0.000,0.000]
7	valid	59.15	0.1293	[1.627,-0.045,0.000,0.060]	[-0.774,-0.633,0.000,0.000]	[1.000,0.000,0.000,0.000]
8	valid	57.82	0.1293	[1.175,-0.045,0.000,0.111]	[-0.676,0.000,0.000,-0.737]	[0.000,0.000,0.000,1.000]
9	valid	56.10	0.1293	[0.958,-0.045,0.039,0.111]	[-0.520,0.000,-0.854,0.000]	[0.000,0.000,1.000,0.000]
10	valid	57.71	0.1293	[1.317,-0.045,0.010,0.111]	[-0.787,0.000,-0.617,0.000]	[1.000,0.000,0.000,0.000]
11	valid	55.93	0.1293	[0.972,-0.045,0.045,0.111]	[-0.714,0.000,-0.700,0.000]	[0.000,0.000,1.000,0.000]
12	valid	56.75	0.1293	[1.300,-0.045,0.045,0.111]	[-0.809,-0.587,0.000,0.000]	[1.000,0.000,0.000,0.000]
13	valid	57.36	0.1293	[1.614,-0.042,0.042,0.104]	[-0.860,-0.510,0.000,0.000]	[1.000,0.000,0.000,0.000]
14	valid	58.12	0.1293	[1.700,-0.034,0.034,0.067]	[-0.879,0.000,0.000,-0.478]	[1.000,0.000,0.000,0.000]
15	valid	57.17	0.1293	[1.359,-0.034,0.060,0.067]	[-0.654,0.000,-0.756,0.000]	[0.000,0.000,1.000,0.000]
16	valid	56.42	0.1293	[1.277,-0.011,0.060,0.067]	[-0.243,-0.970,0.000,0.000]	[0.000,1.000,0.000,0.000]
17	valid	55.17	0.1293	[1.101,-0.011,0.060,0.096]	[-0.520,0.000,0.000,-0.854]	[0.000,0.000,0.000,1.000]
18	valid	56.50	0.1293	[1.394,-0.011,0.041,0.096]	[-0.787,0.000,-0.617,0.000]	[1.000,0.000,0.000,0.000]
19	valid	57.34	0.1293	[1.698,-0.011,0.027,0.093]	[-0.860,0.000,-0.510,0.000]	[1.000,0.000,0.000,0.000]
20	valid	58.10	0.1293	[1.763,-0.009,0.023,0.056]	[-0.695,0.000,0.000,-0.719]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 1 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	67.56	0.1248	[0.972,-0.000,-0.235,-0.000]	[-0.889,0.000,-0.458,0.000]	[1.000,0.000,0.000,0.000]
2	valid	78.46	0.1248	[0.924,-0.000,-0.224,-0.309]	[-0.799,0.000,0.000,-0.602]	[1.000,0.000,0.000,0.000]
3	valid	79.39	0.1248	[0.931,-0.227,-0.143,-0.248]	[-0.819,-0.573,0.000,0.000]	[1.000,0.000,0.000,0.000]
4	valid	42.24	0.0673	[0.668,-0.000,0.744,-0.000]	[-0.551,0.000,-0.835,0.000]	[0.000,0.000,1.000,0.000]
5	valid	42.24	0.0673	[0.668,-0.000,0.744,-0.000]	[-0.520,0.000,-0.854,0.000]	[0.000,0.000,1.000,0.000]
6	valid	42.24	0.0673	[0.668,-0.000,0.744,-0.000]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
7	valid	31.90	0.0338	[0.591,0.534,0.604,-0.000]	[-0.487,-0.873,0.000,0.000]	[0.000,1.000,0.000,0.000]
8	valid	31.90	0.0338	[0.591,0.534,0.604,-0.000]	[-0.746,0.000,0.000,-0.665]	[1.000,0.000,0.000,0.000]
9	valid	33.06	0.0363	[0.580,0.624,0.524,-0.000]	[-0.631,-0.776,0.000,0.000]	[0.000,1.000,0.000,0.000]
10	valid	33.06	0.0363	[0.580,0.624,0.524,-0.000]	[-0.898,-0.439,0.000,0.000]	[1.000,0.000,0.000,0.000]
11	valid	33.06	0.0363	[0.580,0.624,0.524,-0.000]	[-0.879,0.000,0.000,-0.478]	[1.000,0.000,0.000,0.000]
12	valid	33.06	0.0363	[0.580,0.624,0.524,-0.000]	[-0.819,-0.573,0.000,0.000]	[1.000,0.000,0.000,0.000]
13	valid	33.06	0.0363	[0.580,0.624,0.524,-0.000]	[-0.829,0.000,-0.560,0.000]	[1.000,0.000,0.000,0.000]
14	valid	33.06	0.0363	[0.580,0.624,0.524,-0.000]	[-0.879,-0.478,0.000,0.000]	[1.000,0.000,0.000,0.000]
15	valid	33.06	0.0363	[0.580,0.624,0.524,-0.000]	[-0.837,-0.547,0.000,0.000]	[1.000,0.000,0.000,0.000]
16	valid	33.06	0.0363	[0.580,0.624,0.524,-0.000]	[-0.853,0.000,-0.522,0.000]	[1.000,0.000,0.000,0.000]
17	valid	33.06	0.0363	[0.580,0.624,0.524,-0.000]	[-0.889,0.000,0.000,-0.458]	[1.000,0.000,0.000,0.000]
18	valid	33.06	0.0363	[0.580,0.624,0.524,-0.000]	[-0.774,0.000,0.000,-0.633]	[1.000,0.000,0.000,0.000]
19	valid	12.22	0.0089	[0.534,0.574,0.482,0.392]	[-0.452,0.000,0.000,-0.892]	[0.000,0.000,0.000,1.000]
20	valid	15.45	0.0371	[0.639,0.536,0.437,0.337]	[-0.654,-0.756,0.000,0.000]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 1 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	67.56	0.1248	[0.972,-0.000,-0.235,-0.000]	[-0.889,0.000,-0.458,0.000]	[1.000,0.000,0.000,0.000]
2	valid	49.49	0.0958	[0.609,-0.000,-0.147,0.780]	[-0.243,0.000,0.000,-0.970]	[0.000,0.000,0.000,1.000]
3	valid	43.67	0.1052	[0.839,-0.000,-0.000,0.544]	[-0.787,0.000,0.000,-0.617]	[1.000,0.000,0.000,0.000]
4	valid	42.30	0.0924	[0.735,-0.000,-0.000,0.678]	[-0.551,0.000,0.000,-0.835]	[0.000,0.000,0.000,1.000]
5	valid	42.26	0.0913	[0.679,-0.000,-0.000,0.734]	[-0.676,0.000,0.000,-0.737]	[0.000,0.000,0.000,1.000]
6	valid	42.23	0.0910	[0.710,-0.000,-0.000,0.704]	[-0.731,0.000,0.000,-0.683]	[1.000,0.000,0.000,0.000]
7	valid	42.23	0.0910	[0.701,-0.000,-0.000,0.713]	[-0.695,0.000,0.000,-0.719]	[0.000,0.000,0.000,1.000]
8	valid	42.23	0.0910	[0.709,-0.000,-0.000,0.705]	[-0.714,0.000,0.000,-0.700]	[1.000,0.000,0.000,0.000]
9	valid	42.23	0.0910	[0.709,-0.000,-0.000,0.705]	[-0.695,0.000,0.000,-0.719]	[0.000,0.000,0.000,1.000]
10	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.714,0.000,0.000,-0.700]	[0.000,0.000,0.000,1.000]
11	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.520,0.000,-0.854,0.000]	[0.000,0.000,1.000,0.000]
12	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
13	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.487,-0.873,0.000,0.000]	[0.000,1.000,0.000,0.000]
14	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.746,0.000,0.000,-0.665]	[1.000,0.000,0.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.631,-0.776,0.000,0.000]	[1.000,0.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.898,-0.439,0.000,0.000]	[1.000,0.000,0.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.879,0.000,0.000,-0.478]	[1.000,0.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.819,-0.573,0.000,0.000]	[1.000,0.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.829,0.000,-0.560,0.000]	[1.000,0.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.879,-0.478,0.000,0.000]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 1 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	46.19	0.2592	[0.001,0.460,0.228,0.093]	[-0.374,-0.927,0.000,0.000]	[0.000,1.000,0.000,0.000]
2	valid	47.82	0.0697	[0.464,0.565,0.201,-0.126]	[-0.903,0.000,0.000,-0.431]	[1.000,0.000,0.000,0.000]
3	valid	60.37	0.0924	[0.492,0.395,0.030,-0.205]	[-0.799,-0.602,0.000,0.000]	[1.000,0.000,0.000,0.000]
4	valid	50.94	0.0986	[0.404,0.491,0.082,-0.094]	[-0.580,-0.815,0.000,0.000]	[0.000,1.000,0.000,0.000]
5	valid	64.39	0.0924	[0.626,0.545,-0.024,-0.301]	[-0.746,-0.665,0.000,0.000]	[1.000,0.000,0.000,0.000]
6	valid	63.36	0.0934	[0.592,0.569,-0.311,0.025]	[-0.654,-0.756,0.000,0.000]	[0.000,1.000,0.000,0.000]
7	valid	48.35	0.0660	[0.656,0.625,-0.288,0.504]	[-0.333,0.000,0.000,-0.943]	[0.000,0.000,0.000,1.000]
8	valid	48.02	0.0596	[0.645,0.619,-0.316,0.651]	[-0.551,0.000,0.000,-0.835]	[0.000,0.000,0.000,1.000]
9	valid	39.74	0.0586	[0.625,0.538,-0.146,0.655]	[-0.676,-0.737,0.000,0.000]	[1.000,0.000,0.000,0.000]
10	valid	31.35	0.0719	[0.519,0.480,0.021,0.697]	[-0.731,0.000,0.000,-0.683]	[0.000,0.000,0.000,1.000]
11	valid	30.71	0.0627	[0.541,0.496,0.025,0.661]	[-0.809,0.000,0.000,-0.587]	[1.000,0.000,0.000,0.000]
12	valid	28.71	0.0589	[0.647,0.570,0.062,0.694]	[-0.774,0.000,0.000,-0.633]	[1.000,0.000,0.000,0.000]
13	valid	6.86	0.0038	[0.572,0.523,0.479,0.635]	[-0.243,0.000,-0.970,0.000]	[0.000,0.000,1.000,0.000]
14	valid	5.41	0.0069	[0.541,0.489,0.707,0.612]	[-0.551,0.000,-0.835,0.000]	[0.000,0.000,1.000,0.000]
15	valid	4.31	0.0013	[0.651,0.596,0.609,0.700]	[-0.761,0.000,-0.649,0.000]	[1.000,0.000,0.000,0.000]
16	valid	2.92	0.0024	[0.556,0.510,0.630,0.621]	[-0.676,0.000,-0.737,0.000]	[0.000,0.000,1.000,0.000]
17	valid	2.24	0.0003	[0.541,0.482,0.563,0.587]	[-0.654,-0.756,0.000,0.000]	[1.000,0.000,0.000,0.000]
18	valid	3.08	0.0007	[0.564,0.468,0.602,0.656]	[-0.714,0.000,-0.700,0.000]	[1.000,0.000,0.000,0.000]
19	valid	3.41	0.0010	[0.588,0.481,0.541,0.632]	[-0.746,0.000,0.000,-0.665]	[1.000,0.000,0.000,0.000]
20	valid	0.84	0.0003	[0.691,0.549,0.692,0.673]	[-0.731,0.000,0.000,-0.683]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 1 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	58.63	0.1248	[0.517,0.000,0.000,0.000]	[-0.889,0.000,-0.458,0.000]	[1.000,0.000,0.000,0.000]
2	valid	58.63	0.1248	[0.912,0.000,0.000,0.000]	[-0.799,0.000,0.000,-0.602]	[1.000,0.000,0.000,0.000]
3	valid	61.10	0.1248	[1.208,-0.103,0.000,0.000]	[-0.819,-0.573,0.000,0.000]	[1.000,0.000,0.000,0.000]
4	valid	57.30	0.1248	[0.860,-0.103,0.130,0.000]	[-0.551,0.000,-0.835,0.000]	[0.000,0.000,1.000,0.000]
5	valid	51.92	0.0959	[0.645,-0.103,0.272,0.000]	[-0.520,0.000,-0.854,0.000]	[0.000,0.000,1.000,0.000]
6	valid	53.34	0.1124	[0.990,-0.103,0.272,0.000]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
7	valid	48.68	0.1064	[0.795,-0.006,0.272,0.000]	[-0.487,-0.873,0.000,0.000]	[0.000,1.000,0.000,0.000]
8	valid	51.23	0.1248	[1.124,-0.006,0.267,0.000]	[-0.746,0.000,0.000,-0.665]	[1.000,0.000,0.000,0.000]
9	valid	46.54	0.1064	[0.850,0.076,0.267,0.000]	[-0.631,-0.776,0.000,0.000]	[0.000,1.000,0.000,0.000]
10	valid	50.43	0.1248	[1.146,0.036,0.260,0.000]	[-0.898,-0.439,0.000,0.000]	[1.000,0.000,0.000,0.000]
11	valid	52.42	0.1248	[1.295,0.030,0.216,0.000]	[-0.879,0.000,0.000,-0.478]	[1.000,0.000,0.000,0.000]
12	valid	54.58	0.1248	[1.425,-0.008,0.178,0.000]	[-0.819,-0.573,0.000,0.000]	[1.000,0.000,0.000,0.000]
13	valid	56.26	0.1248	[1.552,-0.007,0.112,0.000]	[-0.829,0.000,-0.560,0.000]	[1.000,0.000,0.000,0.000]
14	valid	56.81	0.1248	[1.639,-0.005,0.090,0.000]	[-0.879,-0.478,0.000,0.000]	[1.000,0.000,0.000,0.000]
15	valid	57.22	0.1248	[1.710,-0.004,0.073,0.000]	[-0.837,-0.547,0.000,0.000]	[1.000,0.000,0.000,0.000]
16	valid	57.79	0.1248	[1.774,-0.003,0.045,0.000]	[-0.853,0.000,-0.522,0.000]	[1.000,0.000,0.000,0.000]
17	valid	57.97	0.1248	[1.819,-0.003,0.036,0.000]	[-0.889,0.000,0.000,-0.458]	[1.000,0.000,0.000,0.000]
18	valid	58.10	0.1248	[1.854,-0.002,0.029,0.000]	[-0.774,0.000,0.000,-0.633]	[1.000,0.000,0.000,0.000]
19	valid	57.81	0.1248	[1.651,-0.002,0.029,0.011]	[-0.452,0.000,0.000,-0.892]	[0.000,0.000,0.000,1.000]
20	valid	57.98	0.1248	[1.867,-0.002,0.027,0.010]	[-0.654,-0.756,0.000,0.000]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 2 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	69.97	0.1470	[0.971,-0.241,-0.000,-0.000]	[-0.884,-0.467,0.000,0.000]	[1.000,0.000,0.000,0.000]
2	valid	52.77	0.1755	[0.388,-0.096,-0.000,0.916]	[-0.695,0.000,0.000,-0.719]	[0.000,0.000,0.000,1.000]
3	valid	45.88	0.1182	[0.596,-0.000,-0.000,0.803]	[-0.889,0.000,0.000,-0.458]	[1.000,0.000,0.000,0.000]
4	valid	31.28	0.0609	[0.481,-0.000,0.590,0.648]	[-0.654,0.000,-0.756,0.000]	[0.000,0.000,1.000,0.000]
5	valid	31.33	0.0638	[0.473,-0.000,0.608,0.637]	[-0.676,0.000,-0.737,0.000]	[0.000,0.000,1.000,0.000]
6	valid	5.06	0.0150	[0.404,0.552,0.504,0.528]	[-0.714,-0.700,0.000,0.000]	[0.000,1.000,0.000,0.000]
7	valid	5.06	0.0150	[0.404,0.552,0.504,0.528]	[-0.903,-0.431,0.000,0.000]	[1.000,0.000,0.000,0.000]
8	valid	5.06	0.0150	[0.404,0.552,0.504,0.528]	[-0.676,0.000,-0.737,0.000]	[0.000,0.000,1.000,0.000]
9	valid	5.06	0.0150	[0.404,0.552,0.504,0.528]	[-0.333,0.000,-0.943,0.000]	[0.000,0.000,1.000,0.000]
10	valid	5.06	0.0150	[0.404,0.552,0.504,0.528]	[-0.910,0.000,-0.414,0.000]	[1.000,0.000,0.000,0.000]
11	valid	3.50	0.0029	[0.430,0.546,0.496,0.521]	[-0.837,0.000,0.000,-0.547]	[1.000,0.000,0.000,0.000]
12	valid	3.50	0.0029	[0.430,0.546,0.496,0.521]	[-0.243,0.000,-0.970,0.000]	[0.000,0.000,1.000,0.000]
13	valid	3.50	0.0029	[0.430,0.546,0.496,0.521]	[-0.903,-0.431,0.000,0.000]	[1.000,0.000,0.000,0.000]
14	valid	2.43	0.0016	[0.449,0.541,0.490,0.515]	[-0.819,-0.573,0.000,0.000]	[1.000,0.000,0.000,0.000]
15	valid	2.43	0.0016	[0.449,0.541,0.490,0.515]	[-0.867,0.000,-0.499,0.000]	[1.000,0.000,0.000,0.000]
16	valid	2.43	0.0016	[0.449,0.541,0.490,0.515]	[-0.676,0.000,-0.737,0.000]	[0.000,0.000,1.000,0.000]
17	valid	2.43	0.0016	[0.449,0.541,0.490,0.515]	[-0.654,-0.756,0.000,0.000]	[0.000,1.000,0.000,0.000]
18	valid	2.43	0.0016	[0.449,0.541,0.490,0.515]	[-0.907,0.000,0.000,-0.422]	[1.000,0.000,0.000,0.000]
19	valid	2.43	0.0016	[0.449,0.541,0.490,0.515]	[-0.873,0.000,0.000,-0.488]	[1.000,0.000,0.000,0.000]
20	valid	2.43	0.0016	[0.449,0.541,0.490,0.515]	[-0.884,-0.467,0.000,0.000]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 2 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	69.97	0.1470	[0.971,-0.241,-0.000,-0.000]	[-0.884,-0.467,0.000,0.000]	[1.000,0.000,0.000,0.000]
2	valid	52.22	0.1120	[0.608,-0.151,-0.000,0.779]	[-0.243,0.000,0.000,-0.970]	[0.000,0.000,0.000,1.000]
3	valid	47.10	0.1263	[0.839,-0.000,-0.000,0.544]	[-0.787,0.000,0.000,-0.617]	[1.000,0.000,0.000,0.000]
4	valid	45.64	0.1113	[0.735,-0.000,-0.000,0.678]	[-0.551,0.000,0.000,-0.835]	[0.000,0.000,0.000,1.000]
5	valid	45.50	0.1086	[0.679,-0.000,-0.000,0.734]	[-0.676,0.000,0.000,-0.737]	[0.000,0.000,0.000,1.000]
6	valid	45.56	0.1086	[0.650,-0.000,-0.000,0.760]	[-0.731,0.000,0.000,-0.683]	[0.000,0.000,0.000,1.000]
7	valid	45.52	0.1086	[0.666,-0.000,-0.000,0.746]	[-0.761,0.000,0.000,-0.649]	[1.000,0.000,0.000,0.000]
8	valid	45.51	0.1086	[0.674,-0.000,-0.000,0.739]	[-0.746,0.000,0.000,-0.665]	[1.000,0.000,0.000,0.000]
9	valid	45.51	0.1086	[0.674,-0.000,-0.000,0.739]	[-0.731,0.000,0.000,-0.683]	[0.000,0.000,0.000,1.000]
10	valid	45.51	0.1086	[0.674,-0.000,-0.000,0.739]	[-0.746,0.000,0.000,-0.665]	[1.000,0.000,0.000,0.000]
11	valid	45.51	0.1086	[0.674,-0.000,-0.000,0.739]	[-0.731,0.000,0.000,-0.683]	[0.000,0.000,0.000,1.000]
12	valid	45.51	0.1086	[0.674,-0.000,-0.000,0.739]	[-0.746,0.000,0.000,-0.665]	[1.000,0.000,0.000,0.000]
13	valid	45.51	0.1086	[0.674,-0.000,-0.000,0.739]	[-0.746,0.000,0.000,-0.665]	[1.000,0.000,0.000,0.000]
14	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.731,0.000,0.000,-0.683]	[1.000,0.000,0.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.676,0.000,-0.737,0.000]	[0.000,0.000,1.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.333,0.000,-0.943,0.000]	[0.000,0.000,1.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.910,0.000,-0.414,0.000]	[1.000,0.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.837,0.000,0.000,-0.547]	[1.000,0.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.243,0.000,-0.970,0.000]	[0.000,0.000,1.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.903,-0.431,0.000,0.000]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 2 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	60.46	0.4610	[-0.061,-0.030,0.082,0.522]	[-0.288,0.000,0.000,-0.957]	[0.000,0.000,0.000,1.000]
2	valid	58.41	0.1149	[0.451,0.052,-0.264,0.593]	[-0.714,0.000,-0.700,0.000]	[1.000,0.000,0.000,0.000]
3	valid	57.64	0.1022	[0.397,0.653,-0.498,0.598]	[-0.243,-0.970,0.000,0.000]	[0.000,1.000,0.000,0.000]
4	valid	22.92	0.0652	[0.675,0.512,0.146,0.598]	[-0.853,-0.522,0.000,0.000]	[1.000,0.000,0.000,0.000]
5	valid	10.85	0.0197	[0.610,0.498,0.358,0.543]	[-0.333,0.000,-0.943,0.000]	[0.000,0.000,1.000,0.000]
6	valid	9.87	0.0219	[0.700,0.465,0.526,0.598]	[-0.414,0.000,-0.910,0.000]	[0.000,0.000,1.000,0.000]
7	valid	3.66	0.0045	[0.627,0.650,0.551,0.595]	[-0.520,-0.854,0.000,0.000]	[0.000,1.000,0.000,0.000]
8	valid	11.09	0.0162	[0.528,0.630,0.378,0.685]	[-0.654,0.000,0.000,-0.756]	[0.000,0.000,0.000,1.000]
9	valid	8.86	0.0085	[0.585,0.560,0.435,0.693]	[-0.799,-0.602,0.000,0.000]	[1.000,0.000,0.000,0.000]
10	valid	8.15	0.0085	[0.593,0.626,0.438,0.676]	[-0.676,-0.737,0.000,0.000]	[0.000,1.000,0.000,0.000]
11	valid	9.52	0.0125	[0.552,0.591,0.436,0.727]	[-0.580,0.000,-0.815,0.000]	[0.000,0.000,1.000,0.000]
12	valid	7.82	0.0086	[0.660,0.737,0.487,0.680]	[-0.809,0.000,0.000,-0.587]	[1.000,0.000,0.000,0.000]
13	valid	4.45	0.0028	[0.678,0.694,0.576,0.726]	[-0.746,-0.665,0.000,0.000]	[1.000,0.000,0.000,0.000]
14	valid	4.26	0.0055	[0.611,0.618,0.518,0.565]	[-0.731,0.000,0.000,-0.683]	[1.000,0.000,0.000,0.000]
15	valid	2.28	0.0009	[0.628,0.620,0.580,0.635]	[-0.676,0.000,0.000,-0.737]	[0.000,0.000,0.000,1.000]
16	valid	5.17	0.0045	[0.679,0.670,0.560,0.732]	[-0.695,0.000,0.000,-0.719]	[0.000,0.000,0.000,1.000]
17	valid	1.79	0.0001	[0.662,0.697,0.632,0.664]	[-0.714,-0.700,0.000,0.000]	[0.000,1.000,0.000,0.000]
18	valid	3.46	0.0029	[0.623,0.667,0.553,0.631]	[-0.695,0.000,0.000,-0.719]	[0.000,0.000,0.000,1.000]
19	valid	2.08	0.0001	[0.617,0.650,0.578,0.638]	[-0.654,0.000,-0.756,0.000]	[0.000,0.000,1.000,0.000]
20	valid	1.88	0.0001	[0.594,0.634,0.566,0.609]	[-0.714,0.000,0.000,-0.700]	[0.000,0.000,0.000,1.000]

Several boundaries to locate | seed 2 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	77.54	0.1470	[0.502,-0.236,0.000,0.000]	[-0.884,-0.467,0.000,0.000]	[1.000,0.000,0.000,0.000]
2	valid	74.80	0.1755	[0.137,-0.236,0.000,0.328]	[-0.695,0.000,0.000,-0.719]	[0.000,0.000,0.000,1.000]
3	valid	70.10	0.1470	[0.434,-0.236,0.000,0.180]	[-0.889,0.000,0.000,-0.458]	[1.000,0.000,0.000,0.000]
4	valid	65.03	0.0526	[0.203,-0.236,0.222,0.180]	[-0.654,0.000,-0.756,0.000]	[0.000,0.000,1.000,0.000]
5	valid	67.43	0.1762	[0.021,-0.235,0.465,0.179]	[-0.676,0.000,-0.737,0.000]	[0.000,0.000,1.000,0.000]
6	valid	65.29	0.4610	[-0.155,-0.055,0.465,0.179]	[-0.714,-0.700,0.000,0.000]	[0.000,1.000,0.000,0.000]
7	valid	53.53	0.1762	[0.008,-0.055,0.465,0.179]	[-0.903,-0.431,0.000,0.000]	[1.000,0.000,0.000,0.000]
8	valid	61.76	0.4610	[-0.121,-0.050,0.665,0.165]	[-0.676,0.000,-0.737,0.000]	[0.000,0.000,1.000,0.000]
9	valid	61.00	0.4610	[-0.106,-0.044,0.850,0.145]	[-0.333,0.000,-0.943,0.000]	[0.000,0.000,1.000,0.000]
10	valid	56.13	0.1762	[0.004,-0.044,0.726,0.145]	[-0.910,0.000,-0.414,0.000]	[1.000,0.000,0.000,0.000]
11	valid	54.65	0.1270	[0.110,-0.044,0.726,0.074]	[-0.837,0.000,0.000,-0.547]	[1.000,0.000,0.000,0.000]
12	valid	56.79	0.1146	[0.080,-0.042,0.942,0.070]	[-0.243,0.000,-0.970,0.000]	[0.000,0.000,1.000,0.000]
13	valid	53.92	0.1477	[0.175,-0.041,0.926,0.069]	[-0.903,-0.431,0.000,0.000]	[1.000,0.000,0.000,0.000]
14	valid	51.25	0.1477	[0.272,-0.040,0.899,0.067]	[-0.819,-0.573,0.000,0.000]	[1.000,0.000,0.000,0.000]
15	valid	47.74	0.1263	[0.389,-0.040,0.767,0.067]	[-0.867,0.000,-0.499,0.000]	[1.000,0.000,0.000,0.000]
16	valid	51.72	0.1477	[0.270,-0.038,0.951,0.064]	[-0.676,0.000,-0.737,0.000]	[0.000,0.000,1.000,0.000]
17	valid	50.92	0.1477	[0.191,0.034,0.951,0.064]	[-0.654,-0.756,0.000,0.000]	[0.000,1.000,0.000,0.000]
18	valid	49.90	0.1477	[0.280,0.034,0.950,0.031]	[-0.907,0.000,0.000,-0.422]	[1.000,0.000,0.000,0.000]
19	valid	49.30	0.1477	[0.368,0.033,0.916,-0.007]	[-0.873,0.000,0.000,-0.488]	[1.000,0.000,0.000,0.000]
20	valid	48.82	0.1263	[0.461,-0.001,0.875,-0.007]	[-0.884,-0.467,0.000,0.000]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 3 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	65.16	0.1128	[0.977,-0.000,-0.000,-0.212]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
2	valid	71.42	0.1128	[0.930,-0.000,-0.000,-0.367]	[-0.731,0.000,0.000,-0.683]	[1.000,0.000,0.000,0.000]
3	valid	78.63	0.1128	[0.903,-0.000,-0.241,-0.356]	[-0.867,0.000,-0.499,0.000]	[1.000,0.000,0.000,0.000]
4	valid	43.36	0.0884	[0.710,-0.000,-0.000,0.704]	[-0.676,0.000,0.000,-0.737]	[0.000,0.000,0.000,1.000]
5	valid	43.36	0.0884	[0.710,-0.000,-0.000,0.704]	[-0.845,0.000,-0.534,0.000]	[1.000,0.000,0.000,0.000]
6	valid	43.36	0.0884	[0.710,-0.000,-0.000,0.704]	[-0.761,-0.649,0.000,0.000]	[1.000,0.000,0.000,0.000]
7	valid	27.75	0.0468	[0.591,0.554,-0.000,0.586]	[-0.654,-0.756,0.000,0.000]	[0.000,1.000,0.000,0.000]
8	valid	27.75	0.0468	[0.591,0.554,-0.000,0.586]	[-0.860,-0.510,0.000,0.000]	[1.000,0.000,0.000,0.000]
9	valid	27.75	0.0468	[0.591,0.554,-0.000,0.586]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
10	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.607,0.000,-0.795,0.000]	[0.000,0.000,1.000,0.000]
11	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.860,0.000,0.000,-0.510]	[1.000,0.000,0.000,0.000]
12	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.607,0.000,0.000,-0.795]	[0.000,0.000,0.000,1.000]
13	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.873,0.000,0.000,-0.488]	[1.000,0.000,0.000,0.000]
14	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.695,0.000,-0.719,0.000]	[1.000,0.000,0.000,0.000]
15	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.860,0.000,-0.510,0.000]	[1.000,0.000,0.000,0.000]
16	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.551,-0.835,0.000,0.000]	[0.000,1.000,0.000,0.000]
17	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.853,0.000,-0.522,0.000]	[1.000,0.000,0.000,0.000]
18	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.873,0.000,-0.488,0.000]	[1.000,0.000,0.000,0.000]
19	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.374,-0.927,0.000,0.000]	[0.000,1.000,0.000,0.000]
20	valid	2.43	0.0010	[0.531,0.497,0.441,0.526]	[-0.551,0.000,0.000,-0.835]	[0.000,0.000,0.000,1.000]

Several boundaries to locate | seed 3 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	65.16	0.1128	[0.977,-0.000,-0.000,-0.212]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
2	valid	51.31	0.0811	[0.610,-0.000,0.781,-0.132]	[-0.243,0.000,-0.970,0.000]	[0.000,0.000,1.000,0.000]
3	valid	45.35	0.0708	[0.839,-0.000,0.544,-0.000]	[-0.787,0.000,-0.617,0.000]	[1.000,0.000,0.000,0.000]
4	valid	44.83	0.0676	[0.735,-0.000,0.678,-0.000]	[-0.551,0.000,-0.835,0.000]	[0.000,0.000,1.000,0.000]
5	valid	44.88	0.0677	[0.788,-0.000,0.615,-0.000]	[-0.676,0.000,-0.737,0.000]	[1.000,0.000,0.000,0.000]
6	valid	44.81	0.0669	[0.767,-0.000,0.642,-0.000]	[-0.607,0.000,-0.795,0.000]	[0.000,0.000,1.000,0.000]
7	valid	44.81	0.0669	[0.757,-0.000,0.654,-0.000]	[-0.631,0.000,-0.776,0.000]	[0.000,0.000,1.000,0.000]
8	valid	44.81	0.0669	[0.766,-0.000,0.643,-0.000]	[-0.654,0.000,-0.756,0.000]	[1.000,0.000,0.000,0.000]
9	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.631,0.000,-0.776,0.000]	[1.000,0.000,0.000,0.000]
10	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.867,0.000,-0.499,0.000]	[1.000,0.000,0.000,0.000]
11	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.676,0.000,0.000,-0.737]	[1.000,0.000,0.000,0.000]
12	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.845,0.000,-0.534,0.000]	[1.000,0.000,0.000,0.000]
13	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.761,-0.649,0.000,0.000]	[1.000,0.000,0.000,0.000]
14	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.654,-0.756,0.000,0.000]	[0.000,1.000,0.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.860,-0.510,0.000,0.000]	[1.000,0.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.607,0.000,-0.795,0.000]	[0.000,0.000,1.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.860,0.000,0.000,-0.510]	[1.000,0.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.607,0.000,0.000,-0.795]	[0.000,0.000,0.000,1.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.873,0.000,0.000,-0.488]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 3 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	54.84	0.0955	[0.501,0.094,0.166,-0.148]	[-0.774,0.000,0.000,-0.633]	[1.000,0.000,0.000,0.000]
2	valid	63.11	0.0847	[0.432,0.640,-0.113,-0.264]	[-0.631,-0.776,0.000,0.000]	[0.000,1.000,0.000,0.000]
3	valid	53.75	0.0744	[0.347,0.430,-0.026,-0.116]	[-0.845,-0.534,0.000,0.000]	[1.000,0.000,0.000,0.000]
4	valid	34.31	0.0282	[0.580,0.657,0.628,-0.084]	[-0.288,0.000,-0.957,0.000]	[0.000,0.000,1.000,0.000]
5	valid	30.12	0.0314	[0.680,0.707,0.415,0.014]	[-0.761,-0.649,0.000,0.000]	[1.000,0.000,0.000,0.000]
6	valid	30.29	0.0320	[0.495,0.519,0.595,0.002]	[-0.487,0.000,-0.873,0.000]	[0.000,0.000,1.000,0.000]
7	valid	38.52	0.0303	[0.509,0.423,0.570,-0.124]	[-0.714,-0.700,0.000,0.000]	[1.000,0.000,0.000,0.000]
8	valid	30.29	0.0273	[0.655,0.596,0.484,-0.011]	[-0.695,0.000,-0.719,0.000]	[1.000,0.000,0.000,0.000]
9	valid	34.57	0.0265	[0.753,0.621,0.583,-0.099]	[-0.676,-0.737,0.000,0.000]	[1.000,0.000,0.000,0.000]
10	valid	36.15	0.0276	[0.635,0.550,0.460,-0.109]	[-0.631,-0.776,0.000,0.000]	[0.000,1.000,0.000,0.000]
11	valid	34.94	0.0256	[0.717,0.607,0.631,-0.107]	[-0.654,-0.756,0.000,0.000]	[1.000,0.000,0.000,0.000]
12	valid	25.21	0.0276	[0.651,0.556,0.468,0.078]	[-0.631,-0.776,0.000,0.000]	[0.000,1.000,0.000,0.000]
13	valid	30.00	0.0260	[0.635,0.561,0.576,-0.008]	[-0.607,0.000,-0.795,0.000]	[0.000,0.000,1.000,0.000]
14	valid	20.26	0.0270	[0.697,0.582,0.661,0.191]	[-0.654,0.000,-0.756,0.000]	[0.000,0.000,1.000,0.000]
15	valid	3.42	0.0016	[0.587,0.496,0.529,0.493]	[-0.414,0.000,0.000,-0.910]	[0.000,0.000,0.000,1.000]
16	valid	2.69	0.0012	[0.710,0.619,0.618,0.586]	[-0.654,0.000,-0.756,0.000]	[1.000,0.000,0.000,0.000]
17	valid	2.59	0.0003	[0.697,0.596,0.617,0.641]	[-0.580,0.000,0.000,-0.815]	[0.000,0.000,0.000,1.000]
18	valid	3.04	0.0012	[0.707,0.599,0.616,0.585]	[-0.676,0.000,0.000,-0.737]	[1.000,0.000,0.000,0.000]
19	valid	2.67	0.0012	[0.620,0.553,0.538,0.507]	[-0.654,-0.756,0.000,0.000]	[0.000,1.000,0.000,0.000]
20	valid	2.89	0.0012	[0.663,0.582,0.591,0.550]	[-0.654,0.000,-0.756,0.000]	[0.000,0.000,1.000,0.000]

Several boundaries to locate | seed 3 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	57.56	0.1128	[0.517,0.000,0.000,0.000]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
2	valid	57.56	0.1128	[0.912,0.000,0.000,0.000]	[-0.731,0.000,0.000,-0.683]	[1.000,0.000,0.000,0.000]
3	valid	57.56	0.1128	[1.221,0.000,0.000,0.000]	[-0.867,0.000,-0.499,0.000]	[1.000,0.000,0.000,0.000]
4	valid	52.26	0.1128	[0.760,0.000,0.000,0.131]	[-0.676,0.000,0.000,-0.737]	[0.000,0.000,0.000,1.000]
5	valid	54.11	0.1128	[1.196,0.000,0.000,0.130]	[-0.845,0.000,-0.534,0.000]	[1.000,0.000,0.000,0.000]
6	valid	56.98	0.1128	[1.381,-0.067,0.000,0.099]	[-0.761,-0.649,0.000,0.000]	[1.000,0.000,0.000,0.000]
7	valid	54.17	0.1128	[0.992,0.006,0.000,0.099]	[-0.654,-0.756,0.000,0.000]	[0.000,1.000,0.000,0.000]
8	valid	56.15	0.1128	[1.399,-0.034,0.000,0.098]	[-0.860,-0.510,0.000,0.000]	[1.000,0.000,0.000,0.000]
9	valid	57.04	0.1128	[1.539,-0.027,0.000,0.052]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
10	valid	55.69	0.1128	[1.208,-0.027,0.047,0.052]	[-0.607,0.000,-0.795,0.000]	[0.000,0.000,1.000,0.000]
11	valid	56.71	0.1128	[1.560,-0.025,0.045,0.024]	[-0.860,0.000,0.000,-0.510]	[1.000,0.000,0.000,0.000]
12	valid	55.40	0.1128	[1.251,-0.025,0.045,0.067]	[-0.607,0.000,0.000,-0.795]	[0.000,0.000,0.000,1.000]
13	valid	56.38	0.1128	[1.577,-0.024,0.043,0.041]	[-0.873,0.000,0.000,-0.488]	[1.000,0.000,0.000,0.000]
14	valid	57.20	0.1128	[1.662,-0.020,0.005,0.034]	[-0.695,0.000,-0.719,0.000]	[1.000,0.000,0.000,0.000]
15	valid	57.53	0.1128	[1.729,-0.016,-0.009,0.027]	[-0.860,0.000,-0.510,0.000]	[1.000,0.000,0.000,0.000]
16	valid	57.01	0.1128	[1.465,0.006,-0.009,0.027]	[-0.551,-0.835,0.000,0.000]	[0.000,1.000,0.000,0.000]
17	valid	57.12	0.1128	[1.748,0.006,-0.009,0.025]	[-0.853,0.000,-0.522,0.000]	[1.000,0.000,0.000,0.000]
18	valid	57.22	0.1128	[1.796,0.005,-0.007,0.020]	[-0.873,0.000,-0.488,0.000]	[1.000,0.000,0.000,0.000]
19	valid	56.86	0.1128	[1.637,0.020,-0.007,0.020]	[-0.374,-0.927,0.000,0.000]	[0.000,1.000,0.000,0.000]
20	valid	56.37	0.1128	[1.412,0.020,-0.007,0.037]	[-0.551,0.000,0.000,-0.835]	[0.000,0.000,0.000,1.000]

Several boundaries to locate | seed 4 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	69.44	0.7748	[-0.146,-0.000,-0.000,0.989]	[-0.288,0.000,0.000,-0.957]	[0.000,0.000,0.000,1.000]
2	valid	55.33	0.1192	[0.584,-0.000,-0.202,0.786]	[-0.787,0.000,-0.617,0.000]	[1.000,0.000,0.000,0.000]
3	valid	38.74	0.1629	[0.327,0.829,-0.113,0.440]	[-0.731,-0.683,0.000,0.000]	[0.000,1.000,0.000,0.000]
4	valid	30.40	0.0619	[0.588,0.762,-0.000,0.272]	[-0.845,-0.534,0.000,0.000]	[1.000,0.000,0.000,0.000]
5	valid	30.40	0.0619	[0.588,0.762,-0.000,0.272]	[-0.731,0.000,-0.683,0.000]	[1.000,0.000,0.000,0.000]
6	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.731,-0.683,0.000,0.000]	[1.000,0.000,0.000,0.000]
7	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.654,0.000,0.000,-0.756]	[0.000,0.000,0.000,1.000]
8	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.819,-0.573,0.000,0.000]	[1.000,0.000,0.000,0.000]
9	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.580,-0.815,0.000,0.000]	[0.000,1.000,0.000,0.000]
10	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.774,0.000,-0.633,0.000]	[1.000,0.000,0.000,0.000]
11	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.487,0.000,-0.873,0.000]	[0.000,0.000,1.000,0.000]
12	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
13	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.695,0.000,0.000,-0.719]	[1.000,0.000,0.000,0.000]
14	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.907,0.000,-0.422,0.000]	[1.000,0.000,0.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.853,0.000,0.000,-0.522]	[1.000,0.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.631,0.000,-0.776,0.000]	[1.000,0.000,0.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.631,-0.776,0.000,0.000]	[0.000,1.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.799,0.000,-0.602,0.000]	[1.000,0.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.873,0.000,-0.488,0.000]	[1.000,0.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.829,-0.560,0.000,0.000]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 4 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	69.44	0.7748	[-0.146,-0.000,-0.000,0.989]	[-0.288,0.000,0.000,-0.957]	[0.000,0.000,0.000,1.000]
2	valid	55.33	0.1192	[0.584,-0.000,-0.202,0.786]	[-0.787,0.000,-0.617,0.000]	[1.000,0.000,0.000,0.000]
3	valid	45.96	0.0899	[0.820,-0.000,-0.000,0.572]	[-0.799,0.000,0.000,-0.602]	[1.000,0.000,0.000,0.000]
4	valid	46.10	0.0940	[0.717,-0.000,-0.000,0.697]	[-0.580,0.000,0.000,-0.815]	[0.000,0.000,0.000,1.000]
5	valid	45.85	0.0892	[0.769,-0.000,-0.000,0.639]	[-0.695,0.000,0.000,-0.719]	[1.000,0.000,0.000,0.000]
6	valid	45.85	0.0892	[0.796,-0.000,-0.000,0.606]	[-0.631,0.000,0.000,-0.776]	[1.000,0.000,0.000,0.000]
7	valid	45.84	0.0892	[0.785,-0.000,-0.000,0.619]	[-0.607,0.000,0.000,-0.795]	[0.000,0.000,0.000,1.000]
8	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.631,0.000,0.000,-0.776]	[0.000,0.000,0.000,1.000]
9	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.845,-0.534,0.000,0.000]	[1.000,0.000,0.000,0.000]
10	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.731,0.000,-0.683,0.000]	[1.000,0.000,0.000,0.000]
11	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.731,-0.683,0.000,0.000]	[1.000,0.000,0.000,0.000]
12	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.654,0.000,0.000,-0.756]	[1.000,0.000,0.000,0.000]
13	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.819,-0.573,0.000,0.000]	[1.000,0.000,0.000,0.000]
14	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.580,-0.815,0.000,0.000]	[0.000,1.000,0.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.774,0.000,-0.633,0.000]	[1.000,0.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.487,0.000,-0.873,0.000]	[0.000,0.000,1.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.695,0.000,0.000,-0.719]	[1.000,0.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.907,0.000,-0.422,0.000]	[1.000,0.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.853,0.000,0.000,-0.522]	[1.000,0.000,0.000,0.000]

Several boundaries to locate | seed 4 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	56.19	0.1062	[0.566,-0.081,0.084,0.051]	[-0.761,-0.649,0.000,0.000]	[1.000,0.000,0.000,0.000]
2	valid	59.71	0.1177	[0.393,-0.254,0.681,0.089]	[-0.520,0.000,-0.854,0.000]	[0.000,0.000,1.000,0.000]
3	valid	80.23	0.0800	[0.460,-0.536,0.604,-0.149]	[-0.654,0.000,0.000,-0.756]	[1.000,0.000,0.000,0.000]
4	valid	49.56	0.0635	[0.654,0.056,0.528,-0.160]	[-0.761,0.000,-0.649,0.000]	[1.000,0.000,0.000,0.000]
5	valid	39.06	0.0180	[0.666,0.472,0.572,-0.212]	[-0.243,-0.970,0.000,0.000]	[0.000,1.000,0.000,0.000]
6	valid	26.04	0.0190	[0.712,0.496,0.514,0.024]	[-0.654,0.000,-0.756,0.000]	[1.000,0.000,0.000,0.000]
7	valid	41.13	0.0265	[0.662,0.425,0.556,-0.234]	[-0.607,0.000,-0.795,0.000]	[0.000,0.000,1.000,0.000]
8	valid	33.89	0.0180	[0.689,0.492,0.598,-0.122]	[-0.631,0.000,-0.776,0.000]	[0.000,0.000,1.000,0.000]
9	valid	45.79	0.0186	[0.601,0.540,0.543,-0.349]	[-0.580,-0.815,0.000,0.000]	[0.000,1.000,0.000,0.000]
10	valid	35.76	0.0170	[0.624,0.650,0.527,-0.183]	[-0.676,-0.737,0.000,0.000]	[0.000,1.000,0.000,0.000]
11	valid	38.03	0.0179	[0.514,0.587,0.426,-0.191]	[-0.631,0.000,-0.776,0.000]	[1.000,0.000,0.000,0.000]
12	valid	37.25	0.0170	[0.602,0.595,0.492,-0.198]	[-0.761,-0.649,0.000,0.000]	[1.000,0.000,0.000,0.000]
13	valid	36.02	0.0172	[0.653,0.607,0.531,-0.186]	[-0.714,-0.700,0.000,0.000]	[1.000,0.000,0.000,0.000]
14	valid	42.36	0.0170	[0.588,0.597,0.497,-0.290]	[-0.676,-0.737,0.000,0.000]	[0.000,1.000,0.000,0.000]
15	valid	40.31	0.0172	[0.611,0.603,0.483,-0.255]	[-0.631,0.000,-0.776,0.000]	[1.000,0.000,0.000,0.000]
16	valid	31.66	0.0172	[0.703,0.689,0.561,-0.116]	[-0.714,-0.700,0.000,0.000]	[1.000,0.000,0.000,0.000]
17	valid	26.63	0.0172	[0.657,0.666,0.526,-0.015]	[-0.631,0.000,-0.776,0.000]	[0.000,0.000,1.000,0.000]
18	valid	8.12	0.0138	[0.765,0.735,0.619,0.394]	[-0.333,0.000,0.000,-0.943]	[0.000,0.000,0.000,1.000]
19	valid	1.82	0.0000	[0.742,0.728,0.609,0.543]	[-0.452,0.000,0.000,-0.892]	[0.000,0.000,0.000,1.000]
20	valid	2.03	0.0002	[0.608,0.582,0.481,0.497]	[-0.607,0.000,0.000,-0.795]	[0.000,0.000,0.000,1.000]

Several boundaries to locate | seed 4 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	73.50	0.5315	[-0.133,0.000,0.000,0.512]	[-0.288,0.000,0.000,-0.957]	[0.000,0.000,0.000,1.000]
2	valid	53.70	0.1973	[0.180,0.000,0.000,0.508]	[-0.787,0.000,-0.617,0.000]	[1.000,0.000,0.000,0.000]
3	valid	56.15	0.5315	[-0.073,0.240,0.000,0.499]	[-0.731,-0.683,0.000,0.000]	[0.000,1.000,0.000,0.000]
4	valid	48.77	0.1547	[0.142,0.096,0.000,0.499]	[-0.845,-0.534,0.000,0.000]	[1.000,0.000,0.000,0.000]
5	valid	41.46	0.1226	[0.350,0.092,0.000,0.481]	[-0.731,0.000,-0.683,0.000]	[1.000,0.000,0.000,0.000]
6	valid	49.63	0.0892	[0.554,-0.059,0.000,0.440]	[-0.731,-0.683,0.000,0.000]	[1.000,0.000,0.000,0.000]
7	valid	53.62	0.1606	[0.350,-0.058,0.000,0.674]	[-0.654,0.000,0.000,-0.756]	[0.000,0.000,0.000,1.000]
8	valid	49.94	0.1032	[0.514,-0.053,0.000,0.624]	[-0.819,-0.573,0.000,0.000]	[1.000,0.000,0.000,0.000]
9	valid	45.75	0.1416	[0.381,0.059,0.000,0.624]	[-0.580,-0.815,0.000,0.000]	[0.000,1.000,0.000,0.000]
10	valid	43.52	0.0975	[0.552,0.058,0.000,0.611]	[-0.774,0.000,-0.633,0.000]	[1.000,0.000,0.000,0.000]
11	valid	39.93	0.1192	[0.451,0.058,0.101,0.611]	[-0.487,0.000,-0.873,0.000]	[0.000,0.000,1.000,0.000]
12	valid	38.42	0.0892	[0.630,0.058,0.100,0.522]	[-0.910,0.000,0.000,-0.414]	[1.000,0.000,0.000,0.000]
13	valid	41.34	0.1062	[0.815,0.056,0.097,0.356]	[-0.695,0.000,0.000,-0.719]	[1.000,0.000,0.000,0.000]
14	valid	45.52	0.1062	[0.964,0.050,0.050,0.317]	[-0.907,0.000,-0.422,0.000]	[1.000,0.000,0.000,0.000]
15	valid	48.88	0.1062	[1.124,0.044,0.045,0.227]	[-0.853,0.000,0.000,-0.522]	[1.000,0.000,0.000,0.000]
16	valid	51.78	0.1062	[1.229,0.039,-0.016,0.202]	[-0.631,0.000,-0.776,0.000]	[1.000,0.000,0.000,0.000]
17	valid	48.62	0.1062	[1.005,0.096,-0.016,0.202]	[-0.631,-0.776,0.000,0.000]	[0.000,1.000,0.000,0.000]
18	valid	50.27	0.1062	[1.238,0.093,-0.016,0.195]	[-0.799,0.000,-0.602,0.000]	[1.000,0.000,0.000,0.000]
19	valid	51.60	0.1062	[1.346,0.080,-0.014,0.168]	[-0.873,0.000,-0.488,0.000]	[1.000,0.000,0.000,0.000]
20	valid	53.32	0.1062	[1.449,0.043,-0.012,0.145]	[-0.829,-0.560,0.000,0.000]	[1.000,0.000,0.000,0.000]

Similar queries versus varied queries | seed 0 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.59	0.1175	[0.185,0.677,0.201,-0.191,-0.642,-0.136]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
2	valid	43.30	0.0273	[-0.220,0.727,-0.000,0.000,-0.625,0.182]	[-0.656,-0.317,-0.180,0.228,0.294,0.547]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	43.30	0.0299	[-0.220,0.727,-0.000,-0.000,-0.625,0.182]	[-0.652,-0.289,0.060,0.145,0.433,0.528]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	40.86	0.0299	[-0.243,0.711,-0.000,-0.000,-0.607,0.259]	[-0.593,-0.407,-0.053,0.035,0.207,0.659]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	41.29	0.0299	[-0.235,0.732,-0.000,-0.000,-0.588,0.251]	[0.289,-0.352,0.325,-0.427,0.703,-0.102]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	37.79	0.0236	[-0.417,0.715,-0.000,0.089,-0.538,0.133]	[0.400,-0.263,-0.417,0.680,0.090,0.356]	[1.000,1.000,0.000,0.000,0.000,0.000]
7	valid	31.97	0.0128	[-0.313,0.685,-0.446,0.048,-0.477,0.060]	[0.855,-0.083,0.241,0.093,-0.240,0.370]	[1.000,0.000,1.000,0.000,0.000,0.000]
8	valid	31.97	0.0128	[-0.313,0.685,-0.446,0.048,-0.477,0.060]	[-0.634,-0.155,0.300,0.471,-0.459,0.224]	[0.000,1.000,1.000,0.000,0.000,0.000]
9	valid	34.87	0.0188	[-0.291,0.694,-0.421,0.029,-0.505,0.022]	[0.274,-0.432,0.455,-0.277,0.650,-0.178]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	34.87	0.0188	[-0.291,0.694,-0.421,0.029,-0.505,0.022]	[-0.564,-0.480,-0.069,0.006,0.446,0.498]	[0.000,1.000,0.000,0.000,1.000,0.000]
11	valid	34.87	0.0188	[-0.291,0.694,-0.421,0.029,-0.505,0.022]	[0.062,-0.678,0.348,-0.298,0.572,0.004]	[0.000,1.000,0.000,0.000,1.000,0.000]
12	valid	34.87	0.0188	[-0.291,0.694,-0.421,0.029,-0.505,0.022]	[0.081,-0.520,0.402,-0.458,0.586,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
13	valid	34.87	0.0188	[-0.291,0.694,-0.421,0.029,-0.505,0.022]	[0.206,-0.469,0.419,-0.414,0.625,-0.009]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	39.39	0.0414	[-0.164,0.672,-0.557,-0.000,-0.461,-0.000]	[0.229,-0.294,0.407,-0.410,0.725,0.047]	[0.000,0.000,1.000,0.000,1.000,0.000]
15	valid	39.39	0.0414	[-0.164,0.672,-0.557,-0.000,-0.461,-0.000]	[-0.505,-0.404,-0.177,0.053,0.305,0.674]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	39.39	0.0414	[-0.164,0.672,-0.557,-0.000,-0.461,-0.000]	[0.281,-0.615,0.349,-0.360,0.536,-0.058]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	39.39	0.0414	[-0.164,0.672,-0.557,-0.000,-0.461,-0.000]	[-0.584,-0.413,-0.236,0.104,0.435,0.482]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	39.39	0.0414	[-0.164,0.672,-0.557,-0.000,-0.461,-0.000]	[-0.564,-0.480,-0.069,0.006,0.446,0.498]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	39.39	0.0414	[-0.164,0.672,-0.557,-0.000,-0.461,-0.000]	[0.232,-0.604,0.358,-0.323,0.559,-0.192]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	39.39	0.0414	[-0.164,0.672,-0.557,-0.000,-0.461,-0.000]	[-0.623,-0.367,-0.015,0.129,0.426,0.529]	[0.000,1.000,0.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 0 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.59	0.1175	[0.185,0.677,0.201,-0.191,-0.642,-0.136]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
2	valid	41.22	0.0299	[-0.256,0.716,-0.000,-0.000,-0.609,0.228]	[-0.713,-0.184,-0.201,0.087,0.222,0.600]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	23.19	0.0081	[-0.279,0.651,-0.245,0.267,-0.544,0.268]	[-0.634,-0.155,0.300,0.471,-0.459,0.224]	[0.000,1.000,1.000,0.000,0.000,0.000]
4	valid	29.55	0.0162	[-0.103,0.620,-0.384,0.055,-0.494,0.458]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
5	valid	20.20	0.0111	[-0.343,0.585,-0.362,0.067,-0.467,0.432]	[0.400,-0.263,-0.417,0.680,0.090,0.356]	[1.000,1.000,0.000,0.000,0.000,0.000]
6	valid	26.33	0.0173	[-0.235,0.593,-0.533,0.014,-0.447,0.329]	[0.156,-0.322,0.506,-0.267,0.733,-0.087]	[0.000,0.000,1.000,0.000,1.000,0.000]
7	valid	27.34	0.0109	[-0.245,0.548,-0.487,-0.000,-0.560,0.299]	[0.208,-0.553,0.453,-0.393,0.539,-0.023]	[0.000,1.000,0.000,0.000,1.000,0.000]
8	valid	22.54	0.0022	[-0.243,0.544,-0.483,0.123,-0.556,0.296]	[0.503,0.301,-0.511,-0.616,-0.123,-0.016]	[1.000,0.000,0.000,1.000,0.000,0.000]
9	valid	28.80	0.0069	[-0.181,0.536,-0.567,0.062,-0.549,0.231]	[0.255,-0.591,0.488,-0.385,0.438,-0.088]	[0.000,1.000,1.000,0.000,0.000,0.000]
10	valid	27.45	0.0076	[-0.178,0.528,-0.559,0.062,-0.540,0.286]	[0.109,-0.193,0.075,0.548,0.665,-0.451]	[0.000,0.000,0.000,0.000,1.000,1.000]
11	valid	27.60	0.0106	[-0.169,0.568,-0.534,0.058,-0.520,0.299]	[0.228,-0.499,0.467,-0.408,0.549,-0.114]	[0.000,1.000,0.000,0.000,1.000,0.000]
12	valid	30.06	0.0141	[-0.144,0.565,-0.565,0.034,-0.516,0.272]	[0.179,-0.633,0.477,-0.282,0.487,-0.151]	[0.000,1.000,1.000,0.000,0.000,0.000]
13	valid	21.44	0.0096	[-0.395,0.518,-0.517,0.032,-0.472,0.292]	[0.450,-0.010,0.299,0.450,0.346,-0.621]	[1.000,0.000,0.000,0.000,0.000,1.000]
14	valid	22.30	0.0101	[-0.393,0.501,-0.519,0.019,-0.492,0.288]	[-0.151,-0.889,0.286,0.076,0.311,0.042]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	22.37	0.0099	[-0.391,0.510,-0.515,0.019,-0.489,0.286]	[0.118,-0.502,0.481,-0.306,0.627,-0.128]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	22.45	0.0097	[-0.389,0.518,-0.513,0.019,-0.487,0.285]	[0.276,-0.436,0.427,-0.450,0.582,-0.100]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	22.45	0.0097	[-0.389,0.518,-0.513,0.019,-0.487,0.285]	[0.450,-0.010,0.299,0.450,0.346,-0.621]	[1.000,0.000,0.000,0.000,0.000,1.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.151,-0.889,0.286,0.076,0.311,0.042]	[0.000,1.000,1.000,0.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.593,-0.407,-0.053,0.035,0.207,0.659]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.289,-0.352,0.325,-0.427,0.703,-0.102]	[0.000,1.000,0.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 0 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	31.86	0.0241	[-0.338,0.456,-0.358,-0.110,-0.651,0.324]	[-0.604,-0.525,-0.221,0.151,0.292,0.449]	[0.000,1.000,0.000,0.000,1.000,0.000]
2	valid	14.80	0.0070	[-0.437,0.626,-0.271,0.361,-0.339,0.344]	[0.400,-0.263,-0.417,0.680,0.090,0.356]	[1.000,1.000,0.000,0.000,0.000,0.000]
3	valid	22.53	0.1043	[-0.501,0.440,-0.647,0.634,-0.097,0.413]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,1.000,0.000,0.000]
4	valid	14.79	0.0093	[-0.462,0.559,-0.651,0.586,-0.502,0.285]	[0.284,-0.618,0.197,-0.304,0.622,-0.140]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	18.80	0.0026	[-0.414,0.706,-0.597,0.519,-0.523,0.195]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	7.45	0.0079	[-0.571,0.698,-0.595,0.514,-0.465,0.463]	[0.450,-0.010,0.299,0.450,0.346,-0.621]	[1.000,0.000,0.000,0.000,0.000,1.000]
7	valid	8.25	0.0019	[-0.565,0.714,-0.510,0.436,-0.448,0.456]	[0.276,-0.436,0.427,-0.450,0.582,-0.100]	[0.000,1.000,0.000,0.000,1.000,0.000]
8	valid	9.33	0.0029	[-0.549,0.746,-0.621,0.453,-0.533,0.432]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,0.000,1.000,0.000]
9	valid	9.78	0.0015	[-0.488,0.743,-0.580,0.477,-0.538,0.459]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,0.000,1.000,1.000,0.000]
10	valid	9.37	0.0015	[-0.520,0.689,-0.584,0.520,-0.542,0.402]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,0.000,1.000,1.000,0.000]
11	valid	5.38	0.0011	[-0.630,0.696,-0.716,0.591,-0.700,0.618]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,1.000,0.000,1.000,0.000]
12	valid	3.35	0.0008	[-0.660,0.633,-0.651,0.542,-0.640,0.570]	[0.450,-0.010,0.299,0.450,0.346,-0.621]	[1.000,0.000,0.000,0.000,0.000,1.000]
13	valid	7.31	0.0003	[-0.477,0.626,-0.608,0.473,-0.556,0.489]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,0.000,1.000,0.000]
14	valid	7.07	0.0002	[-0.538,0.618,-0.610,0.508,-0.560,0.434]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,0.000,1.000,1.000,0.000]
15	valid	5.45	0.0005	[-0.668,0.744,-0.666,0.547,-0.638,0.526]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,0.000,1.000,0.000]
16	valid	5.67	0.0000	[-0.698,0.695,-0.656,0.532,-0.613,0.482]	[0.289,-0.352,0.325,-0.427,0.703,-0.102]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	4.45	0.0004	[-0.648,0.726,-0.684,0.554,-0.672,0.596]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,0.000,1.000,1.000,0.000]
18	valid	5.71	0.0005	[-0.649,0.702,-0.657,0.539,-0.650,0.506]	[0.240,-0.442,0.478,-0.329,0.632,-0.110]	[0.000,0.000,1.000,0.000,1.000,0.000]
19	valid	6.76	0.0006	[-0.568,0.688,-0.652,0.532,-0.648,0.517]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	6.18	0.0001	[-0.629,0.722,-0.680,0.565,-0.646,0.518]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,0.000,1.000,1.000,0.000]

Similar queries versus varied queries | seed 0 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	79.66	0.2636	[0.185,0.261,0.202,0.000,-0.406,0.000]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
2	valid	58.46	0.0766	[-0.063,0.395,0.118,0.000,-0.553,0.000]	[-0.656,-0.317,-0.180,0.228,0.294,0.547]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	52.46	0.0508	[-0.051,0.471,0.096,0.041,-0.757,0.159]	[-0.652,-0.289,0.060,0.145,0.433,0.528]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	51.60	0.0508	[-0.042,0.666,0.064,0.044,-0.812,0.130]	[-0.593,-0.407,-0.053,0.035,0.207,0.659]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	54.70	0.0659	[-0.033,0.672,0.051,-0.012,-1.025,0.091]	[0.289,-0.352,0.325,-0.427,0.703,-0.102]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	50.14	0.0289	[-0.100,0.830,-0.018,-0.011,-0.889,0.087]	[0.400,-0.263,-0.417,0.680,0.090,0.356]	[1.000,1.000,0.000,0.000,0.000,0.000]
7	valid	47.60	0.0289	[-0.180,0.791,-0.035,-0.004,-0.880,0.086]	[0.855,-0.083,0.241,0.093,-0.240,0.370]	[1.000,0.000,1.000,0.000,0.000,0.000]
8	valid	44.98	0.0175	[-0.170,0.844,-0.070,0.050,-0.834,0.112]	[-0.634,-0.155,0.300,0.471,-0.459,0.224]	[0.000,1.000,1.000,0.000,0.000,0.000]
9	valid	48.83	0.0175	[-0.116,0.875,-0.027,0.042,-0.968,0.093]	[0.274,-0.432,0.455,-0.277,0.650,-0.178]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	50.27	0.0175	[-0.091,0.949,-0.026,0.033,-1.025,0.073]	[-0.564,-0.480,-0.069,0.006,0.446,0.498]	[0.000,1.000,0.000,0.000,1.000,0.000]
11	valid	51.75	0.0272	[-0.073,1.023,-0.008,0.016,-1.055,0.058]	[0.062,-0.678,0.348,-0.298,0.572,0.004]	[0.000,1.000,0.000,0.000,1.000,0.000]
12	valid	52.83	0.0348	[-0.054,1.046,-0.006,-0.002,-1.114,0.046]	[0.081,-0.520,0.402,-0.458,0.586,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
13	valid	53.72	0.0398	[-0.038,1.037,0.004,-0.001,-1.184,0.037]	[0.206,-0.469,0.419,-0.414,0.625,-0.009]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	55.34	0.0352	[-0.029,0.843,-0.003,-0.001,-1.402,0.034]	[0.229,-0.294,0.407,-0.410,0.725,0.047]	[0.000,0.000,1.000,0.000,1.000,0.000]
15	valid	55.15	0.0303	[-0.024,0.892,-0.007,0.000,-1.391,0.027]	[-0.505,-0.404,-0.177,0.053,0.305,0.674]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	55.35	0.0334	[-0.015,0.923,-0.001,0.000,-1.394,0.022]	[0.281,-0.615,0.349,-0.360,0.536,-0.058]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	55.48	0.0303	[-0.012,0.927,-0.004,0.002,-1.417,0.018]	[-0.584,-0.413,-0.236,0.104,0.435,0.482]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	55.51	0.0303	[-0.010,0.945,-0.004,0.001,-1.420,0.014]	[-0.564,-0.480,-0.069,0.006,0.446,0.498]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	55.58	0.0287	[-0.007,0.962,-0.003,0.001,-1.420,0.010]	[0.232,-0.604,0.358,-0.323,0.559,-0.192]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	55.78	0.0222	[-0.005,0.949,-0.003,0.002,-1.446,0.008]	[-0.623,-0.367,-0.015,0.129,0.426,0.529]	[0.000,1.000,0.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 1 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	70.12	0.4169	[0.079,-0.187,-0.687,-0.155,-0.678,-0.053]	[0.156,-0.322,0.506,-0.267,0.733,-0.087]	[0.000,0.000,1.000,0.000,1.000,0.000]
2	valid	43.05	0.0608	[-0.000,0.677,-0.630,-0.000,-0.381,-0.000]	[0.240,-0.442,0.478,-0.329,0.632,-0.110]	[0.000,1.000,1.000,0.000,0.000,0.000]
3	valid	34.00	0.0330	[-0.000,0.649,-0.603,-0.000,-0.365,0.286]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
4	valid	35.41	0.0303	[-0.000,0.642,-0.595,-0.000,-0.429,0.225]	[0.231,-0.512,0.447,-0.198,0.651,-0.143]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	35.73	0.0303	[-0.000,0.630,-0.610,-0.000,-0.430,0.213]	[0.274,-0.432,0.455,-0.277,0.650,-0.178]	[0.000,0.000,1.000,0.000,1.000,0.000]
6	valid	35.73	0.0303	[-0.000,0.630,-0.610,-0.000,-0.430,0.213]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
7	valid	35.73	0.0303	[-0.000,0.630,-0.610,-0.000,-0.430,0.213]	[-0.505,-0.222,-0.114,-0.099,0.320,0.755]	[0.000,1.000,0.000,0.000,1.000,0.000]
8	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,1.000,0.000,0.000]
9	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[0.130,-0.498,0.414,-0.392,0.634,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[0.205,-0.495,0.374,-0.387,0.611,-0.225]	[0.000,1.000,0.000,0.000,1.000,0.000]
11	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[0.229,-0.294,0.407,-0.410,0.725,0.047]	[0.000,0.000,1.000,0.000,1.000,0.000]
12	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
13	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[-0.285,0.562,0.767,-0.101,-0.058,0.030]	[0.000,0.000,1.000,1.000,0.000,0.000]
14	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[-0.577,-0.450,-0.106,0.104,0.360,0.560]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[-0.595,-0.392,0.049,0.122,0.321,0.610]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[0.174,-0.538,0.395,-0.320,0.620,-0.194]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[0.038,-0.654,0.370,-0.421,0.507,-0.018]	[0.000,1.000,1.000,0.000,0.000,0.000]
18	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[0.173,-0.522,0.196,-0.262,0.768,0.035]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[-0.520,-0.415,-0.209,0.088,0.389,0.595]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	23.09	0.0091	[-0.000,0.555,-0.538,0.358,-0.379,0.362]	[-0.619,-0.365,-0.121,0.133,0.380,0.554]	[0.000,1.000,0.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 1 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	70.12	0.4169	[0.079,-0.187,-0.687,-0.155,-0.678,-0.053]	[0.156,-0.322,0.506,-0.267,0.733,-0.087]	[0.000,0.000,1.000,0.000,1.000,0.000]
2	valid	29.23	0.0807	[-0.200,0.360,-0.672,0.138,-0.567,0.198]	[-0.557,-0.407,-0.009,0.282,0.388,0.542]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	43.17	0.0669	[-0.000,0.582,-0.624,-0.000,-0.522,-0.000]	[0.145,-0.553,0.300,-0.374,0.653,-0.132]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	31.30	0.0166	[-0.199,0.565,-0.606,0.089,-0.507,0.091]	[-0.371,0.456,-0.600,-0.331,-0.262,-0.341]	[0.000,0.000,0.000,1.000,0.000,1.000]
5	valid	36.88	0.0157	[-0.103,0.573,-0.684,0.046,-0.433,0.047]	[0.208,-0.553,0.453,-0.393,0.539,-0.023]	[0.000,1.000,1.000,0.000,0.000,0.000]
6	valid	40.00	0.0187	[-0.052,0.567,-0.663,0.023,-0.486,0.024]	[0.139,-0.538,0.405,-0.349,0.636,-0.040]	[0.000,1.000,0.000,0.000,1.000,0.000]
7	valid	41.74	0.0211	[-0.025,0.552,-0.689,0.011,-0.469,0.012]	[0.038,-0.654,0.370,-0.421,0.507,-0.018]	[0.000,1.000,1.000,0.000,0.000,0.000]
8	valid	41.60	0.0169	[-0.025,0.573,-0.677,0.011,-0.461,0.011]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
9	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.118,-0.502,0.481,-0.306,0.627,-0.128]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.240,-0.442,0.478,-0.329,0.632,-0.110]	[0.000,0.000,1.000,0.000,1.000,0.000]
11	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
12	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.231,-0.512,0.447,-0.198,0.651,-0.143]	[0.000,1.000,0.000,0.000,1.000,0.000]
13	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.274,-0.432,0.455,-0.277,0.650,-0.178]	[0.000,1.000,1.000,0.000,0.000,0.000]
14	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.505,-0.222,-0.114,-0.099,0.320,0.755]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,0.000,0.000,1.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.130,-0.498,0.414,-0.392,0.634,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.205,-0.495,0.374,-0.387,0.611,-0.225]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.229,-0.294,0.407,-0.410,0.725,0.047]	[0.000,0.000,1.000,0.000,1.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]

Similar queries versus varied queries | seed 1 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	54.45	0.3318	[0.033,-0.061,-0.598,-0.148,-0.245,0.490]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
2	valid	40.99	0.0521	[-0.140,0.592,-0.548,-0.346,-0.465,0.630]	[0.265,-0.542,0.262,-0.446,0.577,-0.186]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	28.60	0.0286	[-0.688,0.614,-0.440,-0.118,-0.535,0.582]	[0.450,-0.010,0.299,0.450,0.346,-0.621]	[1.000,0.000,0.000,0.000,0.000,1.000]
4	valid	28.76	0.0273	[-0.617,0.750,-0.579,-0.169,-0.363,0.630]	[0.276,-0.436,0.427,-0.450,0.582,-0.100]	[0.000,1.000,1.000,0.000,0.000,0.000]
5	valid	25.01	0.0144	[-0.522,0.809,-0.611,-0.078,-0.417,0.565]	[0.289,-0.352,0.325,-0.427,0.703,-0.102]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	7.61	0.0034	[-0.509,0.793,-0.592,0.446,-0.384,0.550]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,1.000,0.000,0.000]
7	valid	8.44	0.0023	[-0.519,0.789,-0.593,0.421,-0.397,0.508]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,1.000,0.000,0.000]
8	valid	8.45	0.0030	[-0.498,0.748,-0.623,0.446,-0.388,0.456]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,1.000,0.000,0.000]
9	valid	8.36	0.0019	[-0.561,0.803,-0.630,0.519,-0.420,0.495]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]
10	valid	5.90	0.0009	[-0.581,0.703,-0.674,0.531,-0.452,0.515]	[0.229,-0.294,0.407,-0.410,0.725,0.047]	[0.000,0.000,1.000,0.000,1.000,0.000]
11	valid	6.48	0.0030	[-0.475,0.740,-0.636,0.479,-0.401,0.508]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,1.000,0.000,0.000,0.000]
12	valid	6.00	0.0075	[-0.563,0.777,-0.715,0.499,-0.402,0.574]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
13	valid	4.51	0.0030	[-0.588,0.832,-0.773,0.596,-0.483,0.643]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]
14	valid	5.91	0.0017	[-0.480,0.738,-0.666,0.497,-0.450,0.519]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	5.76	0.0021	[-0.559,0.871,-0.754,0.562,-0.499,0.620]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,1.000,0.000,0.000,0.000]
16	valid	5.57	0.0011	[-0.448,0.700,-0.614,0.458,-0.426,0.502]	[0.081,-0.520,0.402,-0.458,0.586,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	4.80	0.0011	[-0.467,0.747,-0.658,0.489,-0.449,0.574]	[0.274,-0.432,0.455,-0.277,0.650,-0.178]	[0.000,1.000,1.000,0.000,0.000,0.000]
18	valid	5.41	0.0013	[-0.530,0.780,-0.700,0.519,-0.476,0.558]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]
19	valid	3.65	0.0011	[-0.502,0.702,-0.722,0.522,-0.479,0.594]	[0.274,-0.432,0.455,-0.277,0.650,-0.178]	[0.000,0.000,1.000,0.000,1.000,0.000]
20	valid	7.11	0.0011	[-0.360,0.642,-0.591,0.437,-0.400,0.455]	[0.289,-0.352,0.325,-0.427,0.703,-0.102]	[0.000,1.000,0.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 1 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	73.00	0.3958	[0.076,-0.161,-0.265,0.000,-0.416,0.000]	[0.156,-0.322,0.506,-0.267,0.733,-0.087]	[0.000,0.000,1.000,0.000,1.000,0.000]
2	valid	67.73	0.3944	[0.075,0.019,-0.546,-0.132,-0.104,0.000]	[0.240,-0.442,0.478,-0.329,0.632,-0.110]	[0.000,1.000,1.000,0.000,0.000,0.000]
3	valid	60.59	0.2802	[-0.019,-0.020,-0.871,-0.111,-0.088,0.116]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
4	valid	52.69	0.1667	[-0.019,0.097,-0.601,-0.111,-0.259,0.082]	[0.231,-0.512,0.447,-0.198,0.651,-0.143]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	56.95	0.2588	[-0.017,0.008,-0.780,-0.103,-0.432,0.041]	[0.274,-0.432,0.455,-0.277,0.650,-0.178]	[0.000,0.000,1.000,0.000,1.000,0.000]
6	valid	55.92	0.1800	[-0.017,0.084,-0.592,-0.099,-0.661,0.012]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
7	valid	51.22	0.1494	[-0.080,0.108,-0.555,-0.093,-0.736,0.108]	[-0.505,-0.222,-0.114,-0.099,0.320,0.755]	[0.000,1.000,0.000,0.000,1.000,0.000]
8	valid	48.72	0.1271	[-0.080,0.108,-0.755,-0.004,-0.518,0.040]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,1.000,0.000,0.000]
9	valid	48.77	0.1247	[-0.061,0.180,-0.589,-0.004,-0.730,0.039]	[0.130,-0.498,0.414,-0.392,0.634,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	52.44	0.0967	[-0.032,0.242,-0.436,-0.003,-0.929,0.036]	[0.205,-0.495,0.374,-0.387,0.611,-0.225]	[0.000,1.000,0.000,0.000,1.000,0.000]
11	valid	56.32	0.1272	[-0.009,0.176,-0.455,-0.003,-1.094,0.031]	[0.229,-0.294,0.407,-0.410,0.725,0.047]	[0.000,0.000,1.000,0.000,1.000,0.000]
12	valid	54.55	0.1418	[-0.009,0.151,-0.582,-0.037,-1.029,0.065]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
13	valid	52.49	0.1422	[-0.025,0.140,-0.720,-0.028,-0.955,0.062]	[-0.285,0.562,0.767,-0.101,-0.058,0.030]	[0.000,0.000,1.000,1.000,0.000,0.000]
14	valid	52.02	0.1279	[-0.066,0.174,-0.661,-0.018,-1.038,0.057]	[-0.577,-0.450,-0.106,0.104,0.360,0.560]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	53.20	0.1223	[-0.060,0.216,-0.574,-0.005,-1.155,0.051]	[-0.595,-0.392,0.049,0.122,0.321,0.610]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	56.47	0.0967	[-0.044,0.248,-0.435,-0.005,-1.326,0.046]	[0.174,-0.538,0.395,-0.320,0.620,-0.194]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	51.76	0.0689	[-0.042,0.323,-0.501,-0.005,-1.097,0.046]	[0.038,-0.654,0.370,-0.421,0.507,-0.018]	[0.000,1.000,1.000,0.000,0.000,0.000]
18	valid	54.51	0.0448	[-0.032,0.351,-0.432,-0.004,-1.275,0.042]	[0.173,-0.522,0.196,-0.262,0.768,0.035]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	55.14	0.0389	[-0.054,0.376,-0.382,0.000,-1.340,0.037]	[-0.520,-0.415,-0.209,0.088,0.389,0.595]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	55.70	0.0395	[-0.048,0.390,-0.341,0.005,-1.396,0.056]	[-0.619,-0.365,-0.121,0.133,0.380,0.554]	[0.000,1.000,0.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 2 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	45.81	0.3558	[-0.656,-0.050,-0.237,0.261,0.050,0.663]	[0.766,-0.085,-0.331,0.374,0.087,-0.386]	[1.000,0.000,0.000,0.000,0.000,1.000]
2	valid	33.24	0.0265	[-0.509,0.453,-0.035,0.080,-0.453,0.568]	[-0.623,-0.367,-0.015,0.129,0.426,0.529]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	39.31	0.0423	[-0.457,0.455,0.004,-0.029,-0.494,0.582]	[0.038,-0.654,0.370,-0.421,0.507,-0.018]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	39.31	0.0423	[-0.457,0.455,0.004,-0.029,-0.494,0.582]	[-0.505,-0.404,-0.177,0.053,0.305,0.674]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	37.64	0.0359	[-0.457,0.473,-0.000,0.004,-0.518,0.547]	[-0.631,-0.344,-0.096,0.230,0.329,0.559]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	36.45	0.0317	[-0.412,0.442,-0.062,-0.000,-0.484,0.630]	[0.150,0.018,-0.505,-0.063,0.802,-0.273]	[0.000,0.000,0.000,0.000,1.000,1.000]
7	valid	30.31	0.0198	[-0.228,0.483,-0.529,-0.000,-0.584,0.308]	[0.150,-0.419,0.606,-0.240,0.593,-0.161]	[0.000,0.000,1.000,0.000,1.000,0.000]
8	valid	30.31	0.0198	[-0.228,0.483,-0.529,-0.000,-0.584,0.308]	[-0.631,-0.344,-0.096,0.230,0.329,0.559]	[0.000,1.000,0.000,0.000,1.000,0.000]
9	valid	30.31	0.0198	[-0.228,0.483,-0.529,-0.000,-0.584,0.308]	[0.237,-0.595,0.306,-0.233,0.645,-0.158]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	30.31	0.0198	[-0.228,0.483,-0.529,-0.000,-0.584,0.308]	[-0.371,0.456,-0.600,-0.331,-0.262,-0.341]	[0.000,0.000,0.000,1.000,0.000,1.000]
11	valid	22.65	0.0183	[-0.148,0.439,-0.407,0.526,-0.546,0.211]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,0.000,1.000,1.000,0.000]
12	valid	22.65	0.0183	[-0.148,0.439,-0.407,0.526,-0.546,0.211]	[0.503,0.301,-0.511,-0.616,-0.123,-0.016]	[1.000,0.000,0.000,1.000,0.000,0.000]
13	valid	22.65	0.0183	[-0.148,0.439,-0.407,0.526,-0.546,0.211]	[0.150,-0.419,0.606,-0.240,0.593,-0.161]	[0.000,0.000,1.000,0.000,1.000,0.000]
14	valid	22.65	0.0183	[-0.148,0.439,-0.407,0.526,-0.546,0.211]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,1.000,0.000,0.000]
15	valid	22.65	0.0183	[-0.148,0.439,-0.407,0.526,-0.546,0.211]	[-0.597,-0.267,-0.012,0.195,0.330,0.652]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	22.65	0.0183	[-0.148,0.439,-0.407,0.526,-0.546,0.211]	[-0.631,-0.344,-0.096,0.230,0.329,0.559]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	22.65	0.0183	[-0.148,0.439,-0.407,0.526,-0.546,0.211]	[-0.619,-0.365,-0.121,0.133,0.380,0.554]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	23.15	0.0183	[-0.141,0.447,-0.404,0.523,-0.549,0.204]	[0.324,-0.585,0.399,-0.395,0.463,-0.149]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	23.15	0.0183	[-0.141,0.447,-0.404,0.523,-0.549,0.204]	[0.197,-0.631,0.392,-0.331,0.543,-0.069]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	23.15	0.0183	[-0.141,0.447,-0.404,0.523,-0.549,0.204]	[0.766,-0.085,-0.331,0.374,0.087,-0.386]	[1.000,0.000,0.000,0.000,0.000,1.000]

Similar queries versus varied queries | seed 2 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	45.81	0.3558	[-0.656,-0.050,-0.237,0.261,0.050,0.663]	[0.766,-0.085,-0.331,0.374,0.087,-0.386]	[1.000,0.000,0.000,0.000,0.000,1.000]
2	valid	36.24	0.0297	[-0.454,0.523,-0.000,0.030,-0.501,0.518]	[0.139,-0.538,0.405,-0.349,0.636,-0.040]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	18.95	0.0159	[-0.260,0.549,-0.229,0.484,-0.490,0.323]	[0.075,0.205,-0.944,0.120,-0.196,0.088]	[1.000,0.000,0.000,0.000,0.000,1.000]
4	valid	16.46	0.0112	[-0.217,0.518,-0.373,0.508,-0.459,0.276]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,1.000,0.000,0.000]
5	valid	20.87	0.0181	[-0.188,0.617,-0.313,0.432,-0.484,0.257]	[0.276,-0.436,0.427,-0.450,0.582,-0.100]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	20.25	0.0197	[-0.145,0.588,-0.320,0.459,-0.454,0.339]	[0.109,-0.193,0.075,0.548,0.665,-0.451]	[0.000,0.000,0.000,0.000,1.000,1.000]
7	valid	25.27	0.0192	[-0.104,0.598,-0.242,0.527,-0.454,0.298]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]
8	valid	23.73	0.0192	[-0.140,0.595,-0.241,0.524,-0.452,0.296]	[0.400,-0.263,-0.417,0.680,0.090,0.356]	[1.000,1.000,0.000,0.000,0.000,0.000]
9	valid	20.18	0.0077	[-0.118,0.535,-0.509,0.459,-0.405,0.258]	[0.324,-0.585,0.399,-0.395,0.463,-0.149]	[0.000,1.000,1.000,0.000,0.000,0.000]
10	valid	21.23	0.0084	[-0.106,0.538,-0.529,0.441,-0.405,0.246]	[0.274,-0.432,0.455,-0.277,0.650,-0.178]	[0.000,0.000,1.000,0.000,1.000,0.000]
11	valid	21.14	0.0081	[-0.108,0.529,-0.521,0.453,-0.417,0.242]	[0.118,-0.502,0.481,-0.306,0.627,-0.128]	[0.000,1.000,0.000,0.000,1.000,0.000]
12	valid	21.16	0.0084	[-0.109,0.521,-0.514,0.462,-0.426,0.238]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,1.000,0.000,1.000,0.000]
13	valid	10.18	0.0010	[-0.336,0.491,-0.484,0.435,-0.401,0.247]	[0.450,-0.010,0.299,0.450,0.346,-0.621]	[1.000,0.000,0.000,0.000,0.000,1.000]
14	valid	10.88	0.0017	[-0.329,0.483,-0.487,0.438,-0.415,0.240]	[0.208,-0.553,0.453,-0.393,0.539,-0.023]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	10.96	0.0017	[-0.329,0.483,-0.491,0.434,-0.415,0.239]	[0.228,-0.499,0.467,-0.408,0.549,-0.114]	[0.000,1.000,1.000,0.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.208,-0.553,0.453,-0.393,0.539,-0.023]	[0.000,1.000,1.000,0.000,0.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.038,-0.654,0.370,-0.421,0.507,-0.018]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.505,-0.404,-0.177,0.053,0.305,0.674]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.631,-0.344,-0.096,0.230,0.329,0.559]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.150,0.018,-0.505,-0.063,0.802,-0.273]	[0.000,0.000,0.000,0.000,1.000,1.000]

Similar queries versus varied queries | seed 2 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	18.29	0.0346	[-0.435,0.336,-0.463,0.463,-0.450,0.108]	[-0.634,-0.155,0.300,0.471,-0.459,0.224]	[0.000,1.000,1.000,0.000,0.000,0.000]
2	valid	21.50	0.0511	[-0.332,0.667,-0.737,0.321,-0.226,0.263]	[0.324,-0.585,0.399,-0.395,0.463,-0.149]	[0.000,1.000,1.000,0.000,0.000,0.000]
3	valid	18.81	0.0445	[-0.348,0.620,-0.725,0.257,-0.289,0.368]	[-0.575,-0.148,0.067,-0.036,0.390,0.700]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	12.10	0.0092	[-0.329,0.652,-0.669,0.663,-0.446,0.442]	[0.289,-0.352,0.325,-0.427,0.703,-0.102]	[0.000,0.000,0.000,1.000,1.000,0.000]
5	valid	13.63	0.0052	[-0.410,0.654,-0.590,0.581,-0.561,0.254]	[0.276,-0.436,0.427,-0.450,0.582,-0.100]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	8.47	0.0031	[-0.549,0.656,-0.548,0.545,-0.478,0.727]	[0.450,-0.010,0.299,0.450,0.346,-0.621]	[1.000,0.000,0.000,0.000,0.000,1.000]
7	valid	3.39	0.0033	[-0.592,0.717,-0.593,0.616,-0.526,0.573]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]
8	valid	4.56	0.0042	[-0.524,0.691,-0.561,0.574,-0.525,0.543]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,1.000,0.000,0.000,1.000,0.000]
9	valid	2.61	0.0030	[-0.672,0.706,-0.674,0.619,-0.606,0.638]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,1.000,0.000,1.000,0.000]
10	valid	4.77	0.0001	[-0.591,0.700,-0.678,0.623,-0.561,0.462]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,1.000,1.000,0.000,0.000]
11	valid	8.01	0.0004	[-0.579,0.701,-0.686,0.647,-0.572,0.378]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,0.000,1.000,1.000,0.000]
12	valid	5.85	0.0022	[-0.493,0.689,-0.686,0.631,-0.583,0.608]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,1.000,0.000,1.000,0.000]
13	valid	1.82	0.0017	[-0.618,0.664,-0.643,0.635,-0.567,0.578]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,0.000,1.000,1.000,0.000]
14	valid	4.54	0.0023	[-0.533,0.684,-0.668,0.650,-0.591,0.583]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,1.000,0.000,1.000,0.000]
15	valid	4.44	0.0010	[-0.487,0.614,-0.611,0.600,-0.530,0.513]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,1.000,1.000,0.000,0.000]
16	valid	2.69	0.0010	[-0.645,0.741,-0.766,0.710,-0.631,0.629]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,1.000,0.000,1.000,0.000]
17	valid	2.47	0.0010	[-0.624,0.724,-0.722,0.688,-0.607,0.609]	[0.218,-0.470,0.459,-0.428,0.567,-0.131]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	4.50	0.0012	[-0.600,0.753,-0.739,0.702,-0.627,0.551]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	4.08	0.0004	[-0.616,0.705,-0.684,0.656,-0.575,0.489]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,1.000,1.000,0.000,0.000,0.000]
20	valid	2.34	0.0003	[-0.537,0.636,-0.604,0.580,-0.509,0.506]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,1.000,1.000,0.000,0.000,0.000]

Similar queries versus varied queries | seed 2 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	52.28	0.4156	[-0.425,-0.040,-0.161,0.000,0.000,0.190]	[0.766,-0.085,-0.331,0.374,0.087,-0.386]	[1.000,0.000,0.000,0.000,0.000,1.000]
2	valid	27.12	0.0353	[-0.400,0.177,-0.161,0.073,-0.262,0.179]	[-0.623,-0.367,-0.015,0.129,0.426,0.529]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	40.93	0.0373	[-0.356,0.393,-0.040,-0.048,-0.434,0.160]	[0.038,-0.654,0.370,-0.421,0.507,-0.018]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	41.13	0.0310	[-0.297,0.585,-0.090,-0.023,-0.558,0.133]	[-0.505,-0.404,-0.177,0.053,0.305,0.674]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	42.95	0.0277	[-0.235,0.741,-0.098,0.042,-0.694,0.105]	[-0.631,-0.344,-0.096,0.230,0.329,0.559]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	47.12	0.0288	[-0.185,0.648,-0.086,0.031,-0.924,0.120]	[0.150,0.018,-0.505,-0.063,0.802,-0.273]	[0.000,0.000,0.000,0.000,1.000,1.000]
7	valid	52.35	0.0232	[-0.145,0.440,-0.149,0.027,-1.198,0.107]	[0.150,-0.419,0.606,-0.240,0.593,-0.161]	[0.000,0.000,1.000,0.000,1.000,0.000]
8	valid	53.73	0.0218	[-0.113,0.495,-0.133,0.051,-1.311,0.083]	[-0.631,-0.344,-0.096,0.230,0.329,0.559]	[0.000,1.000,0.000,0.000,1.000,0.000]
9	valid	56.54	0.0226	[-0.073,0.533,-0.083,0.040,-1.439,0.066]	[0.237,-0.595,0.306,-0.233,0.645,-0.158]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	54.62	0.0218	[-0.095,0.517,-0.125,0.057,-1.398,0.085]	[-0.371,0.456,-0.600,-0.331,-0.262,-0.341]	[0.000,0.000,0.000,1.000,0.000,1.000]
11	valid	59.36	0.0179	[-0.083,0.352,-0.078,0.076,-1.640,0.074]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,0.000,1.000,1.000,0.000]
12	valid	57.90	0.0179	[-0.109,0.342,-0.104,0.108,-1.594,0.071]	[0.503,0.301,-0.511,-0.616,-0.123,-0.016]	[1.000,0.000,0.000,1.000,0.000,0.000]
13	valid	60.90	0.0180	[-0.092,0.236,-0.121,0.079,-1.760,0.060]	[0.150,-0.419,0.606,-0.240,0.593,-0.161]	[0.000,0.000,1.000,0.000,1.000,0.000]
14	valid	60.41	0.0159	[-0.091,0.220,-0.161,0.097,-1.739,0.044]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,1.000,0.000,0.000]
15	valid	61.61	0.0213	[-0.073,0.221,-0.131,0.094,-1.819,0.036]	[-0.597,-0.267,-0.012,0.195,0.330,0.652]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	61.95	0.0180	[-0.064,0.227,-0.121,0.082,-1.850,0.046]	[-0.631,-0.344,-0.096,0.230,0.329,0.559]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	62.70	0.0213	[-0.052,0.232,-0.106,0.073,-1.903,0.037]	[-0.619,-0.365,-0.121,0.133,0.380,0.554]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	63.48	0.0213	[-0.044,0.251,-0.076,0.052,-1.954,0.032]	[0.324,-0.585,0.399,-0.395,0.463,-0.149]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	64.06	0.0225	[-0.037,0.264,-0.054,0.038,-1.994,0.027]	[0.197,-0.631,0.392,-0.331,0.543,-0.069]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	63.80	0.0221	[-0.048,0.258,-0.054,0.038,-1.944,0.031]	[0.766,-0.085,-0.331,0.374,0.087,-0.386]	[1.000,0.000,0.000,0.000,0.000,1.000]

Similar queries versus varied queries | seed 3 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.58	0.1156	[0.185,0.677,0.201,-0.191,-0.642,-0.136]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
2	valid	43.14	0.0263	[-0.201,0.721,-0.000,0.000,-0.622,0.228]	[-0.575,-0.310,-0.153,0.168,0.295,0.659]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	43.14	0.0263	[-0.201,0.721,-0.000,0.000,-0.622,0.228]	[-0.597,-0.267,-0.012,0.195,0.330,0.652]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	23.43	0.0069	[-0.279,0.651,-0.245,0.267,-0.544,0.268]	[-0.634,-0.155,0.300,0.471,-0.459,0.224]	[0.000,1.000,1.000,0.000,0.000,0.000]
5	valid	23.97	0.0069	[-0.274,0.638,-0.239,0.263,-0.567,0.265]	[0.139,-0.538,0.405,-0.349,0.636,-0.040]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	40.06	0.0231	[-0.069,0.607,-0.093,0.633,-0.463,0.045]	[0.289,-0.352,0.325,-0.427,0.703,-0.102]	[0.000,0.000,0.000,1.000,1.000,0.000]
7	valid	40.06	0.0231	[-0.069,0.607,-0.093,0.633,-0.463,0.045]	[-0.619,-0.365,-0.121,0.133,0.380,0.554]	[0.000,1.000,0.000,0.000,1.000,0.000]
8	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[0.206,-0.469,0.419,-0.414,0.625,-0.009]	[0.000,1.000,0.000,0.000,1.000,0.000]
9	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[-0.661,-0.373,-0.032,0.114,0.306,0.563]	[0.000,1.000,0.000,0.000,1.000,0.000]
11	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[-0.674,-0.271,0.005,0.033,0.300,0.617]	[0.000,1.000,0.000,0.000,1.000,0.000]
12	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[-0.577,-0.224,-0.182,0.179,0.271,0.691]	[0.000,1.000,0.000,0.000,1.000,0.000]
13	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[0.197,-0.631,0.392,-0.331,0.543,-0.069]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[-0.592,-0.238,-0.269,0.099,0.367,0.613]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[0.232,-0.604,0.358,-0.323,0.559,-0.192]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[-0.621,-0.337,-0.250,0.063,0.374,0.544]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[0.174,-0.538,0.395,-0.320,0.620,-0.194]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	41.99	0.0176	[-0.046,0.626,-0.063,0.606,-0.484,0.031]	[0.328,-0.520,0.348,-0.340,0.619,-0.032]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	41.15	0.0176	[-0.058,0.625,-0.037,0.605,-0.483,0.065]	[-0.384,0.203,-0.042,-0.808,-0.077,-0.388]	[0.000,0.000,0.000,1.000,0.000,1.000]
20	valid	41.15	0.0176	[-0.058,0.625,-0.037,0.605,-0.483,0.065]	[0.271,-0.568,0.441,-0.291,0.540,-0.182]	[0.000,1.000,0.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 3 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.58	0.1156	[0.185,0.677,0.201,-0.191,-0.642,-0.136]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
2	valid	42.09	0.0293	[-0.256,0.716,-0.000,-0.000,-0.609,0.228]	[-0.713,-0.184,-0.201,0.087,0.222,0.600]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	23.43	0.0069	[-0.279,0.651,-0.245,0.267,-0.544,0.268]	[-0.634,-0.155,0.300,0.471,-0.459,0.224]	[0.000,1.000,1.000,0.000,0.000,0.000]
4	valid	27.48	0.0115	[-0.103,0.620,-0.384,0.055,-0.494,0.458]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
5	valid	21.19	0.0102	[-0.343,0.585,-0.362,0.067,-0.467,0.432]	[0.400,-0.263,-0.417,0.680,0.090,0.356]	[1.000,1.000,0.000,0.000,0.000,0.000]
6	valid	27.33	0.0169	[-0.235,0.593,-0.533,0.014,-0.447,0.329]	[0.156,-0.322,0.506,-0.267,0.733,-0.087]	[0.000,0.000,1.000,0.000,1.000,0.000]
7	valid	31.96	0.0310	[-0.169,0.584,-0.617,0.000,-0.425,0.263]	[0.208,-0.553,0.453,-0.393,0.539,-0.023]	[0.000,1.000,1.000,0.000,0.000,0.000]
8	valid	33.57	0.0247	[-0.143,0.582,-0.611,0.000,-0.465,0.227]	[0.231,-0.512,0.447,-0.198,0.651,-0.143]	[0.000,1.000,0.000,0.000,1.000,0.000]
9	valid	34.02	0.0227	[-0.131,0.602,-0.595,-0.000,-0.467,0.220]	[0.264,-0.475,0.453,-0.334,0.619,-0.074]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	34.85	0.0186	[-0.122,0.588,-0.600,-0.000,-0.489,0.201]	[0.260,-0.499,0.442,-0.411,0.559,-0.078]	[0.000,1.000,0.000,0.000,1.000,0.000]
11	valid	35.23	0.0195	[-0.118,0.582,-0.602,-0.000,-0.498,0.193]	[0.271,-0.568,0.441,-0.291,0.540,-0.182]	[0.000,1.000,0.000,0.000,1.000,0.000]
12	valid	35.16	0.0195	[-0.117,0.590,-0.597,-0.000,-0.494,0.196]	[0.218,-0.470,0.459,-0.428,0.567,-0.131]	[0.000,1.000,0.000,0.000,1.000,0.000]
13	valid	35.14	0.0195	[-0.117,0.591,-0.596,-0.000,-0.493,0.196]	[0.276,-0.436,0.427,-0.450,0.582,-0.100]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	34.75	0.0195	[-0.117,0.591,-0.596,0.010,-0.493,0.196]	[0.503,0.301,-0.511,-0.616,-0.123,-0.016]	[1.000,0.000,0.000,1.000,0.000,0.000]
15	valid	34.75	0.0195	[-0.117,0.591,-0.596,0.010,-0.493,0.196]	[0.503,0.301,-0.511,-0.616,-0.123,-0.016]	[1.000,0.000,0.000,1.000,0.000,0.000]
16	valid	34.75	0.0195	[-0.117,0.591,-0.596,0.010,-0.493,0.196]	[0.503,0.301,-0.511,-0.616,-0.123,-0.016]	[1.000,0.000,0.000,1.000,0.000,0.000]
17	valid	34.75	0.0195	[-0.117,0.591,-0.596,0.010,-0.493,0.196]	[0.503,0.301,-0.511,-0.616,-0.123,-0.016]	[1.000,0.000,0.000,1.000,0.000,0.000]
18	valid	34.75	0.0195	[-0.117,0.591,-0.596,0.010,-0.493,0.196]	[0.400,-0.263,-0.417,0.680,0.090,0.356]	[1.000,1.000,0.000,0.000,0.000,0.000]
19	valid	34.75	0.0195	[-0.117,0.591,-0.596,0.010,-0.493,0.196]	[0.276,-0.436,0.427,-0.450,0.582,-0.100]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	34.75	0.0195	[-0.117,0.591,-0.596,0.010,-0.493,0.196]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]

Similar queries versus varied queries | seed 3 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	84.05	0.6734	[0.384,0.005,0.240,0.441,0.232,0.499]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,0.000,1.000,0.000,1.000]
2	valid	38.56	0.1351	[-0.299,0.482,0.209,0.691,-0.225,0.725]	[-0.151,-0.889,0.286,0.076,0.311,0.042]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	47.83	0.0532	[0.052,0.603,0.355,0.597,-0.452,0.691]	[0.289,-0.528,0.354,-0.292,0.606,-0.245]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	21.07	0.0088	[-0.246,0.636,-0.147,0.615,-0.490,0.733]	[0.383,0.212,0.659,0.175,-0.547,0.208]	[1.000,0.000,1.000,0.000,0.000,0.000]
5	valid	7.72	0.0016	[-0.428,0.736,-0.499,0.545,-0.460,0.747]	[0.255,-0.591,0.488,-0.385,0.438,-0.088]	[0.000,1.000,1.000,0.000,0.000,0.000]
6	valid	15.53	0.0063	[-0.256,0.702,-0.347,0.360,-0.343,0.574]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,1.000,0.000,0.000,1.000,0.000]
7	valid	13.38	0.0074	[-0.270,0.708,-0.401,0.475,-0.374,0.660]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]
8	valid	14.13	0.0079	[-0.251,0.760,-0.419,0.521,-0.482,0.737]	[0.228,-0.499,0.467,-0.408,0.549,-0.114]	[0.000,1.000,0.000,0.000,1.000,0.000]
9	valid	13.98	0.0061	[-0.231,0.739,-0.478,0.540,-0.513,0.799]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	19.52	0.0067	[-0.085,0.727,-0.481,0.539,-0.486,0.822]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,0.000,1.000,0.000,1.000]
11	valid	15.78	0.0064	[-0.138,0.626,-0.485,0.555,-0.506,0.708]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,0.000,1.000,1.000,0.000]
12	valid	12.65	0.0042	[-0.255,0.687,-0.485,0.611,-0.567,0.803]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,1.000,0.000,0.000,1.000,0.000]
13	valid	14.17	0.0064	[-0.205,0.699,-0.496,0.571,-0.509,0.785]	[0.255,-0.591,0.488,-0.385,0.438,-0.088]	[0.000,1.000,1.000,0.000,0.000,0.000]
14	valid	9.31	0.0019	[-0.349,0.656,-0.498,0.550,-0.507,0.812]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	5.17	0.0007	[-0.625,0.630,-0.510,0.552,-0.512,0.740]	[0.450,-0.010,0.299,0.450,0.346,-0.621]	[1.000,0.000,0.000,0.000,0.000,1.000]
16	valid	4.75	0.0012	[-0.484,0.621,-0.435,0.485,-0.460,0.596]	[0.279,-0.040,0.446,-0.548,0.502,-0.411]	[0.000,0.000,0.000,1.000,1.000,0.000]
17	valid	4.47	0.0014	[-0.497,0.726,-0.542,0.591,-0.557,0.771]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	3.14	0.0012	[-0.535,0.692,-0.531,0.567,-0.558,0.727]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	2.82	0.0003	[-0.699,0.745,-0.658,0.660,-0.632,0.796]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,1.000,0.000,1.000,0.000]
20	valid	1.60	0.0000	[-0.588,0.652,-0.605,0.579,-0.572,0.704]	[0.149,-0.447,0.472,-0.480,0.569,-0.027]	[0.000,0.000,1.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 3 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	77.79	0.2390	[0.185,0.261,0.202,0.000,-0.406,0.000]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
2	valid	59.11	0.0751	[-0.061,0.408,0.118,0.000,-0.573,0.000]	[-0.575,-0.310,-0.153,0.168,0.295,0.659]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	55.73	0.0582	[-0.047,0.540,0.085,0.084,-0.823,0.000]	[-0.597,-0.267,-0.012,0.195,0.330,0.652]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	49.21	0.0253	[-0.173,0.589,0.025,0.080,-0.786,0.041]	[-0.634,-0.155,0.300,0.471,-0.459,0.224]	[0.000,1.000,1.000,0.000,0.000,0.000]
5	valid	53.07	0.0356	[-0.114,0.683,0.020,0.017,-0.970,0.032]	[0.139,-0.538,0.405,-0.349,0.636,-0.040]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	57.20	0.0232	[-0.069,0.482,0.017,0.055,-1.279,0.028]	[0.289,-0.352,0.325,-0.427,0.703,-0.102]	[0.000,0.000,0.000,1.000,1.000,0.000]
7	valid	57.47	0.0232	[-0.052,0.527,-0.001,0.058,-1.406,0.021]	[-0.619,-0.365,-0.121,0.133,0.380,0.554]	[0.000,1.000,0.000,0.000,1.000,0.000]
8	valid	59.14	0.0232	[-0.029,0.530,-0.001,0.022,-1.538,0.016]	[0.206,-0.469,0.419,-0.414,0.625,-0.009]	[0.000,1.000,0.000,0.000,1.000,0.000]
9	valid	60.37	0.0237	[-0.010,0.515,-0.001,0.004,-1.648,0.012]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	60.03	0.0201	[-0.008,0.554,-0.002,0.009,-1.672,0.010]	[-0.661,-0.373,-0.032,0.114,0.306,0.563]	[0.000,1.000,0.000,0.000,1.000,0.000]
11	valid	60.29	0.0230	[-0.006,0.555,-0.001,0.008,-1.724,0.007]	[-0.674,-0.271,0.005,0.033,0.300,0.617]	[0.000,1.000,0.000,0.000,1.000,0.000]
12	valid	60.48	0.0176	[-0.005,0.543,-0.007,0.013,-1.773,0.006]	[-0.577,-0.224,-0.182,0.179,0.271,0.691]	[0.000,1.000,0.000,0.000,1.000,0.000]
13	valid	60.47	0.0180	[-0.004,0.567,-0.001,0.006,-1.779,0.004]	[0.197,-0.631,0.392,-0.331,0.543,-0.069]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	60.97	0.0166	[-0.003,0.532,-0.005,0.006,-1.835,0.004]	[-0.592,-0.238,-0.269,0.099,0.367,0.613]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	60.98	0.0177	[-0.002,0.545,-0.002,0.003,-1.841,0.003]	[0.232,-0.604,0.358,-0.323,0.559,-0.192]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	61.11	0.0180	[-0.002,0.537,-0.003,0.003,-1.862,0.002]	[-0.621,-0.337,-0.250,0.063,0.374,0.544]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	61.35	0.0182	[-0.001,0.526,-0.001,0.001,-1.883,0.002]	[0.174,-0.538,0.395,-0.320,0.620,-0.194]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	61.59	0.0348	[-0.000,0.513,0.000,0.001,-1.905,0.001]	[0.328,-0.520,0.348,-0.340,0.619,-0.032]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	61.54	0.0275	[-0.001,0.513,0.000,0.003,-1.905,0.002]	[-0.384,0.203,-0.042,-0.808,-0.077,-0.388]	[0.000,0.000,0.000,1.000,0.000,1.000]
20	valid	61.51	0.0260	[-0.001,0.519,0.000,0.001,-1.905,0.001]	[0.271,-0.568,0.441,-0.291,0.540,-0.182]	[0.000,1.000,0.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 4 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	34.05	0.0129	[-0.333,0.613,-0.068,0.033,-0.615,0.359]	[-0.548,-0.406,-0.085,0.026,0.355,0.633]	[0.000,1.000,0.000,0.000,1.000,0.000]
2	valid	34.59	0.0129	[-0.332,0.614,-0.065,0.024,-0.616,0.359]	[-0.564,-0.480,-0.069,0.006,0.446,0.498]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	42.99	0.0304	[-0.187,0.691,-0.000,-0.000,-0.662,0.221]	[0.289,-0.528,0.354,-0.292,0.606,-0.245]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	42.23	0.0279	[-0.208,0.686,-0.000,-0.000,-0.656,0.236]	[-0.575,-0.348,-0.130,0.054,0.322,0.651]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	42.20	0.0279	[-0.216,0.686,-0.000,-0.000,-0.655,0.232]	[-0.626,-0.310,-0.154,0.270,0.423,0.486]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	42.20	0.0279	[-0.216,0.686,-0.000,-0.000,-0.655,0.232]	[0.289,-0.528,0.354,-0.292,0.606,-0.245]	[0.000,1.000,0.000,0.000,1.000,0.000]
7	valid	40.30	0.0265	[-0.289,0.668,-0.000,-0.000,-0.633,0.263]	[-0.713,-0.184,-0.201,0.087,0.222,0.600]	[0.000,1.000,0.000,0.000,1.000,0.000]
8	valid	31.92	0.0196	[-0.121,0.578,-0.392,-0.000,-0.533,0.462]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
9	valid	31.92	0.0196	[-0.121,0.578,-0.392,-0.000,-0.533,0.462]	[0.178,-0.508,0.283,-0.378,0.633,-0.296]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	24.41	0.0075	[-0.238,0.563,-0.382,0.094,-0.519,0.450]	[0.383,0.212,0.659,0.175,-0.547,0.208]	[1.000,0.000,1.000,0.000,0.000,0.000]
11	valid	24.41	0.0075	[-0.238,0.563,-0.382,0.094,-0.519,0.450]	[-0.575,-0.148,0.067,-0.036,0.390,0.700]	[0.000,1.000,0.000,0.000,1.000,0.000]
12	valid	25.00	0.0086	[-0.217,0.602,-0.373,0.099,-0.494,0.445]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
13	valid	25.00	0.0086	[-0.217,0.602,-0.373,0.099,-0.494,0.445]	[-0.623,-0.367,-0.015,0.129,0.426,0.529]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	25.05	0.0086	[-0.215,0.598,-0.370,0.099,-0.504,0.442]	[0.162,-0.581,0.359,-0.329,0.589,-0.230]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	26.22	0.0088	[-0.200,0.569,-0.347,0.092,-0.579,0.415]	[0.208,-0.553,0.453,-0.393,0.539,-0.023]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	26.32	0.0075	[-0.180,0.550,-0.360,0.082,-0.560,0.464]	[-0.668,-0.158,0.564,0.052,-0.292,-0.351]	[0.000,0.000,1.000,0.000,0.000,1.000]
17	valid	26.39	0.0075	[-0.179,0.554,-0.359,0.082,-0.558,0.464]	[0.130,-0.498,0.414,-0.392,0.634,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	26.04	0.0075	[-0.175,0.554,-0.359,0.090,-0.558,0.464]	[0.173,-0.482,-0.072,0.774,-0.314,-0.186]	[0.000,1.000,0.000,0.000,0.000,1.000]
19	valid	26.04	0.0075	[-0.175,0.554,-0.359,0.090,-0.558,0.464]	[0.328,-0.520,0.348,-0.340,0.619,-0.032]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	26.04	0.0075	[-0.175,0.554,-0.359,0.090,-0.558,0.464]	[-0.608,-0.363,-0.039,0.183,0.250,0.633]	[0.000,1.000,0.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 4 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	34.05	0.0129	[-0.333,0.613,-0.068,0.033,-0.615,0.359]	[-0.548,-0.406,-0.085,0.026,0.355,0.633]	[0.000,1.000,0.000,0.000,1.000,0.000]
2	valid	15.20	0.0036	[-0.488,0.449,-0.191,0.486,-0.446,0.298]	[0.503,0.301,-0.511,-0.616,-0.123,-0.016]	[1.000,0.000,0.000,1.000,0.000,0.000]
3	valid	25.23	0.0826	[-0.281,0.618,-0.223,0.634,-0.229,0.187]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]
4	valid	36.25	0.0404	[-0.158,0.627,-0.066,0.646,-0.401,-0.003]	[0.237,-0.595,0.306,-0.233,0.645,-0.158]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	29.89	0.0312	[-0.161,0.619,-0.061,0.638,-0.396,0.154]	[-0.064,0.151,0.038,0.230,0.457,0.842]	[0.000,0.000,1.000,0.000,1.000,0.000]
6	valid	34.63	0.0150	[-0.086,0.603,-0.032,0.623,-0.483,0.082]	[0.130,-0.498,0.414,-0.392,0.634,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
7	valid	37.61	0.0111	[-0.045,0.589,-0.017,0.610,-0.526,0.043]	[0.265,-0.542,0.262,-0.446,0.577,-0.186]	[0.000,1.000,0.000,0.000,1.000,0.000]
8	valid	35.36	0.0138	[-0.046,0.587,-0.015,0.607,-0.524,0.102]	[0.173,-0.482,-0.072,0.774,-0.314,-0.186]	[0.000,1.000,0.000,0.000,0.000,1.000]
9	valid	37.86	0.0138	[-0.026,0.580,-0.008,0.602,-0.545,0.056]	[0.324,-0.585,0.399,-0.395,0.463,-0.149]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	37.22	0.0108	[-0.026,0.580,-0.029,0.601,-0.545,0.056]	[0.075,0.205,-0.944,0.120,-0.196,0.088]	[1.000,0.000,0.000,0.000,0.000,1.000]
11	valid	25.04	0.0058	[-0.031,0.550,-0.177,0.570,-0.516,0.272]	[-0.668,-0.158,0.564,0.052,-0.292,-0.351]	[0.000,0.000,1.000,0.000,0.000,1.000]
12	valid	24.37	0.0082	[-0.031,0.547,-0.206,0.567,-0.514,0.270]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,1.000,0.000,0.000]
13	valid	25.02	0.0083	[-0.026,0.547,-0.196,0.567,-0.521,0.264]	[0.255,-0.591,0.488,-0.385,0.438,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	22.89	0.0059	[-0.078,0.545,-0.196,0.565,-0.519,0.264]	[0.766,-0.085,-0.331,0.374,0.087,-0.386]	[1.000,0.000,0.000,0.000,0.000,1.000]
15	valid	22.67	0.0059	[-0.078,0.544,-0.205,0.564,-0.518,0.263]	[0.855,-0.083,0.241,0.093,-0.240,0.370]	[1.000,0.000,1.000,0.000,0.000,0.000]
16	valid	22.67	0.0059	[-0.078,0.544,-0.205,0.564,-0.518,0.263]	[0.255,-0.591,0.488,-0.385,0.438,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	22.67	0.0059	[-0.078,0.544,-0.205,0.564,-0.518,0.263]	[0.855,-0.083,0.241,0.093,-0.240,0.370]	[1.000,0.000,1.000,0.000,0.000,0.000]
18	valid	22.67	0.0059	[-0.078,0.544,-0.205,0.564,-0.518,0.263]	[0.766,-0.085,-0.331,0.374,0.087,-0.386]	[1.000,0.000,0.000,0.000,0.000,1.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.255,-0.591,0.488,-0.385,0.438,-0.088]	[0.000,1.000,0.000,1.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.289,-0.528,0.354,-0.292,0.606,-0.245]	[0.000,1.000,0.000,0.000,1.000,0.000]

Similar queries versus varied queries | seed 4 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	65.20	0.2885	[-0.205,0.382,-0.318,0.291,-0.242,-0.504]	[0.075,0.205,-0.944,0.120,-0.196,0.088]	[1.000,0.000,0.000,0.000,0.000,1.000]
2	valid	23.82	0.0101	[-0.398,0.640,-0.518,0.474,-0.562,0.092]	[-0.634,-0.155,0.300,0.471,-0.459,0.224]	[0.000,1.000,1.000,0.000,0.000,0.000]
3	valid	18.08	0.0367	[-0.337,0.379,-0.306,0.714,-0.549,0.252]	[0.229,-0.294,0.407,-0.410,0.725,0.047]	[0.000,0.000,0.000,1.000,1.000,0.000]
4	valid	26.39	0.0097	[-0.383,0.606,-0.434,0.708,-0.781,0.041]	[0.179,-0.633,0.477,-0.282,0.487,-0.151]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	19.18	0.0026	[-0.315,0.686,-0.345,0.757,-0.679,0.254]	[0.855,-0.083,0.241,0.093,-0.240,0.370]	[1.000,0.000,1.000,0.000,0.000,0.000]
6	valid	13.17	0.0029	[-0.734,0.603,-0.349,0.556,-0.714,0.570]	[0.766,-0.085,-0.331,0.374,0.087,-0.386]	[1.000,0.000,0.000,0.000,0.000,1.000]
7	valid	9.58	0.0044	[-0.407,0.623,-0.424,0.695,-0.531,0.408]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]
8	valid	7.45	0.0016	[-0.340,0.647,-0.397,0.714,-0.634,0.610]	[0.450,-0.010,0.299,0.450,0.346,-0.621]	[0.000,0.000,0.000,0.000,1.000,1.000]
9	valid	12.49	0.0159	[-0.248,0.526,-0.348,0.766,-0.505,0.504]	[0.218,-0.470,0.459,-0.428,0.567,-0.131]	[0.000,0.000,0.000,1.000,1.000,0.000]
10	valid	6.11	0.0011	[-0.333,0.619,-0.489,0.697,-0.594,0.624]	[0.150,-0.419,0.606,-0.240,0.593,-0.161]	[0.000,0.000,1.000,0.000,1.000,0.000]
11	valid	7.65	0.0014	[-0.319,0.639,-0.494,0.734,-0.689,0.650]	[0.228,-0.499,0.467,-0.408,0.549,-0.114]	[0.000,1.000,0.000,0.000,1.000,0.000]
12	valid	9.88	0.0017	[-0.299,0.627,-0.501,0.770,-0.631,0.524]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,1.000,0.000,0.000]
13	valid	7.99	0.0013	[-0.353,0.626,-0.559,0.736,-0.610,0.534]	[0.260,-0.499,0.442,-0.411,0.559,-0.078]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	11.64	0.0008	[-0.342,0.679,-0.515,0.751,-0.736,0.454]	[0.255,-0.591,0.488,-0.385,0.438,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	9.13	0.0012	[-0.332,0.682,-0.566,0.726,-0.708,0.555]	[-0.650,-0.072,0.025,-0.064,0.745,-0.111]	[0.000,0.000,0.000,0.000,1.000,1.000]
16	valid	6.02	0.0006	[-0.415,0.688,-0.536,0.758,-0.658,0.582]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]
17	valid	9.92	0.0009	[-0.332,0.683,-0.529,0.752,-0.687,0.508]	[0.255,-0.591,0.488,-0.385,0.438,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	8.45	0.0009	[-0.338,0.676,-0.527,0.741,-0.698,0.564]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,0.000,0.000,1.000,1.000,0.000]
19	valid	8.49	0.0008	[-0.369,0.690,-0.522,0.764,-0.706,0.543]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]
20	valid	9.83	0.0009	[-0.361,0.742,-0.573,0.803,-0.716,0.540]	[0.363,-0.494,0.354,-0.476,0.495,-0.162]	[0.000,1.000,0.000,1.000,0.000,0.000]

Similar queries versus varied queries | seed 4 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	47.33	0.0394	[-0.408,0.281,-0.054,0.000,-0.241,0.000]	[-0.548,-0.406,-0.085,0.026,0.355,0.633]	[0.000,1.000,0.000,0.000,1.000,0.000]
2	valid	37.67	0.0190	[-0.322,0.535,-0.043,0.002,-0.455,0.210]	[-0.564,-0.480,-0.069,0.006,0.446,0.498]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	45.72	0.0249	[-0.173,0.735,-0.034,0.002,-0.685,0.108]	[0.289,-0.528,0.354,-0.292,0.606,-0.245]	[0.000,1.000,0.000,0.000,1.000,0.000]
4	valid	47.67	0.0195	[-0.126,0.899,-0.056,0.014,-0.808,0.079]	[-0.575,-0.348,-0.130,0.054,0.322,0.651]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	48.18	0.0124	[-0.094,0.936,-0.063,0.046,-0.949,0.059]	[-0.626,-0.310,-0.154,0.270,0.423,0.486]	[0.000,1.000,0.000,0.000,1.000,0.000]
6	valid	50.60	0.0124	[-0.049,0.993,-0.024,0.034,-1.061,0.044]	[0.289,-0.528,0.354,-0.292,0.606,-0.245]	[0.000,1.000,0.000,0.000,1.000,0.000]
7	valid	50.91	0.0117	[-0.036,1.003,-0.042,0.035,-1.143,0.032]	[-0.713,-0.184,-0.201,0.087,0.222,0.600]	[0.000,1.000,0.000,0.000,1.000,0.000]
8	valid	50.45	0.0137	[-0.036,0.930,-0.073,0.019,-1.143,0.050]	[-0.308,-0.138,0.644,-0.419,-0.337,-0.427]	[0.000,0.000,1.000,0.000,0.000,1.000]
9	valid	51.83	0.0129	[-0.028,0.954,-0.046,0.015,-1.252,0.030]	[0.178,-0.508,0.283,-0.378,0.633,-0.296]	[0.000,1.000,0.000,0.000,1.000,0.000]
10	valid	51.08	0.0129	[-0.039,0.943,-0.070,0.019,-1.238,0.035]	[0.383,0.212,0.659,0.175,-0.547,0.208]	[1.000,0.000,1.000,0.000,0.000,0.000]
11	valid	52.42	0.0129	[-0.032,0.864,-0.054,0.015,-1.368,0.029]	[-0.575,-0.148,0.067,-0.036,0.390,0.700]	[0.000,1.000,0.000,0.000,1.000,0.000]
12	valid	53.50	0.0135	[-0.020,0.839,-0.035,0.012,-1.449,0.023]	[0.331,-0.450,0.359,-0.341,0.645,-0.166]	[0.000,1.000,0.000,0.000,1.000,0.000]
13	valid	53.96	0.0126	[-0.015,0.831,-0.027,0.011,-1.491,0.018]	[-0.623,-0.367,-0.015,0.129,0.426,0.529]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	54.35	0.0125	[-0.012,0.841,-0.017,0.009,-1.514,0.011]	[0.162,-0.581,0.359,-0.329,0.589,-0.230]	[0.000,1.000,0.000,0.000,1.000,0.000]
15	valid	54.60	0.0165	[-0.009,0.853,-0.009,0.004,-1.526,0.009]	[0.208,-0.553,0.453,-0.393,0.539,-0.023]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	54.82	0.0178	[-0.009,0.795,-0.014,0.004,-1.526,0.012]	[-0.668,-0.158,0.564,0.052,-0.292,-0.351]	[0.000,0.000,1.000,0.000,0.000,1.000]
17	valid	55.28	0.0200	[-0.008,0.791,-0.009,0.002,-1.599,0.010]	[0.130,-0.498,0.414,-0.392,0.634,-0.088]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	valid	54.16	0.0202	[-0.006,0.922,-0.009,0.001,-1.472,0.010]	[0.173,-0.482,-0.072,0.774,-0.314,-0.186]	[0.000,1.000,0.000,0.000,0.000,1.000]
19	valid	54.43	0.0178	[-0.004,0.905,-0.006,0.001,-1.501,0.008]	[0.328,-0.520,0.348,-0.340,0.619,-0.032]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	54.17	0.0178	[-0.003,0.947,-0.005,0.002,-1.465,0.007]	[-0.608,-0.363,-0.039,0.183,0.250,0.633]	[0.000,1.000,0.000,0.000,1.000,0.000]

Ordinary balanced choice | seed 0 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	43.02	0.0789	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	35.20	0.0659	[-0.134,0.235,-0.485,0.832]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
3	valid	35.20	0.0659	[-0.134,0.235,-0.485,0.832]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
4	valid	9.63	0.0089	[-0.448,0.536,-0.503,0.509]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]
5	valid	9.63	0.0089	[-0.448,0.536,-0.503,0.509]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
6	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
7	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[0.863,0.029,0.095,0.495]	[1.000,0.000,0.000,0.000]
8	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
9	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
10	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
11	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[-0.805,0.381,-0.140,-0.432]	[0.000,0.000,0.000,1.000]
12	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[-0.261,0.493,-0.095,-0.824]	[0.000,0.000,0.000,1.000]
13	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
14	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
15	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
16	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[0.872,0.150,0.014,-0.466]	[1.000,0.000,0.000,0.000]
17	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[-0.602,-0.667,0.317,0.303]	[0.000,1.000,0.000,0.000]
18	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
19	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
20	valid	7.92	0.0066	[-0.507,0.504,-0.472,0.515]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]

Ordinary balanced choice | seed 0 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	43.02	0.0789	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	22.12	0.0432	[-0.550,0.068,-0.629,0.545]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
3	valid	29.36	0.0666	[-0.811,0.016,-0.462,0.360]	[0.405,0.823,-0.005,-0.399]	[1.000,0.000,0.000,0.000]
4	valid	29.80	0.0605	[-0.658,-0.023,-0.697,0.283]	[0.473,-0.294,0.808,0.193]	[0.000,0.000,1.000,0.000]
5	valid	7.00	0.0041	[-0.621,0.382,-0.580,0.363]	[-0.237,-0.863,0.169,-0.413]	[0.000,1.000,0.000,0.000]
6	valid	7.58	0.0027	[-0.630,0.353,-0.496,0.483]	[-0.007,-0.683,-0.056,-0.728]	[0.000,0.000,0.000,1.000]
7	valid	7.66	0.0051	[-0.610,0.413,-0.453,0.502]	[0.259,-0.495,-0.732,0.389]	[0.000,1.000,0.000,0.000]
8	valid	10.68	0.0148	[-0.581,0.484,-0.397,0.521]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
9	valid	9.50	0.0144	[-0.617,0.455,-0.414,0.490]	[0.563,-0.674,0.433,-0.203]	[1.000,0.000,0.000,0.000]
10	valid	10.95	0.0184	[-0.616,0.481,-0.390,0.486]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
11	valid	9.56	0.0148	[-0.610,0.476,-0.412,0.481]	[0.434,0.565,0.692,-0.118]	[0.000,0.000,1.000,0.000]
12	valid	9.86	0.0157	[-0.603,0.471,-0.407,0.499]	[-0.794,-0.426,0.031,-0.432]	[0.000,0.000,0.000,1.000]
13	valid	7.96	0.0077	[-0.594,0.463,-0.437,0.492]	[-0.700,0.090,0.547,-0.451]	[0.000,0.000,1.000,0.000]
14	valid	8.33	0.0077	[-0.600,0.464,-0.431,0.489]	[0.540,0.560,0.062,-0.625]	[1.000,0.000,0.000,0.000]
15	valid	7.72	0.0077	[-0.596,0.462,-0.441,0.486]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
16	valid	5.43	0.0029	[-0.584,0.452,-0.476,0.476]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
17	valid	3.34	0.0010	[-0.572,0.443,-0.508,0.467]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.509,-0.697,-0.096,-0.496]	[1.000,0.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]

Ordinary balanced choice | seed 0 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	57.86	0.3149	[-0.001,0.056,-0.505,-0.074]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
2	valid	55.47	0.3261	[-0.529,0.062,-0.516,-0.327]	[0.787,0.491,0.145,-0.344]	[1.000,0.000,0.000,0.000]
3	valid	6.85	0.0043	[-0.500,0.391,-0.590,0.332]	[-0.376,0.796,-0.152,-0.450]	[0.000,0.000,0.000,1.000]
4	valid	11.54	0.0115	[-0.425,0.538,-0.434,0.453]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
5	valid	4.71	0.0029	[-0.607,0.594,-0.637,0.544]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
6	valid	9.42	0.0057	[-0.502,0.487,-0.781,0.529]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
7	valid	5.07	0.0032	[-0.613,0.508,-0.716,0.448]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
8	valid	6.92	0.0038	[-0.654,0.411,-0.723,0.445]	[0.563,-0.674,0.433,-0.203]	[1.000,0.000,0.000,0.000]
9	valid	5.38	0.0032	[-0.526,0.447,-0.623,0.385]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
10	valid	1.36	0.0003	[-0.705,0.521,-0.698,0.542]	[0.563,-0.674,0.433,-0.203]	[1.000,0.000,0.000,0.000]
11	valid	2.95	0.0020	[-0.635,0.505,-0.691,0.477]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
12	valid	5.57	0.0036	[-0.641,0.487,-0.683,0.411]	[0.563,-0.674,0.433,-0.203]	[1.000,0.000,0.000,0.000]
13	valid	2.54	0.0004	[-0.684,0.520,-0.650,0.490]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
14	valid	1.20	0.0000	[-0.675,0.534,-0.686,0.522]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
15	valid	6.59	0.0033	[-0.621,0.470,-0.626,0.363]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
16	valid	5.33	0.0036	[-0.655,0.503,-0.697,0.424]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
17	valid	1.16	0.0001	[-0.572,0.451,-0.601,0.487]	[-0.162,-0.596,-0.390,-0.683]	[0.000,0.000,0.000,1.000]
18	valid	2.51	0.0009	[-0.685,0.505,-0.692,0.497]	[0.509,-0.697,-0.096,-0.496]	[1.000,0.000,0.000,0.000]
19	valid	1.86	0.0000	[-0.580,0.449,-0.590,0.434]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
20	valid	1.31	0.0009	[-0.567,0.438,-0.595,0.480]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]

Ordinary balanced choice | seed 0 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	50.40	0.2082	[-0.082,0.000,-0.515,0.000]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	37.73	0.0841	[-0.082,0.000,-0.289,0.337]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
3	valid	56.76	0.2107	[0.086,0.000,-0.255,0.646]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
4	valid	40.78	0.0887	[-0.110,0.000,-0.255,0.467]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]
5	valid	37.43	0.0816	[-0.110,0.000,-0.495,0.438]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
6	valid	29.75	0.0644	[-0.288,0.000,-0.326,0.438]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
7	valid	28.73	0.0730	[-0.473,0.000,-0.292,0.419]	[0.863,0.029,0.095,0.495]	[1.000,0.000,0.000,0.000]
8	valid	37.21	0.0866	[-0.432,-0.065,-0.266,0.590]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
9	valid	39.55	0.1043	[-0.603,-0.061,-0.166,0.554]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
10	valid	43.46	0.1393	[-0.775,-0.055,-0.084,0.504]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
11	valid	40.62	0.1041	[-0.712,0.020,-0.077,0.647]	[-0.805,0.381,-0.140,-0.432]	[0.000,0.000,0.000,1.000]
12	valid	42.13	0.1091	[-0.641,0.018,-0.078,0.792]	[-0.261,0.493,-0.095,-0.824]	[0.000,0.000,0.000,1.000]
13	valid	39.97	0.0865	[-0.623,0.086,-0.078,0.792]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
14	valid	39.29	0.0844	[-0.758,0.079,-0.071,0.716]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
15	valid	37.30	0.0921	[-0.685,0.142,-0.071,0.716]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
16	valid	37.61	0.0956	[-0.860,0.138,-0.069,0.602]	[0.872,0.150,0.014,-0.466]	[1.000,0.000,0.000,0.000]
17	valid	36.99	0.1093	[-0.846,0.200,-0.043,0.592]	[-0.602,-0.667,0.317,0.303]	[0.000,1.000,0.000,0.000]
18	valid	40.33	0.1403	[-0.969,0.180,-0.008,0.533]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
19	valid	42.17	0.1448	[-1.087,0.160,-0.014,0.473]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
20	valid	39.99	0.1184	[-1.063,0.211,-0.036,0.463]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]

Ordinary balanced choice | seed 1 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	76.92	0.5929	[-0.947,-0.172,0.128,-0.239]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	22.52	0.0207	[-0.796,0.424,-0.336,0.273]	[-0.520,-0.406,-0.727,0.188]	[0.000,1.000,0.000,0.000]
3	valid	34.13	0.1197	[-0.720,0.479,-0.498,-0.065]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
4	valid	34.13	0.1197	[-0.720,0.479,-0.498,-0.065]	[0.660,-0.284,-0.415,0.558]	[1.000,0.000,0.000,0.000]
5	valid	41.22	0.1977	[-0.738,0.524,-0.388,-0.174]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
6	valid	38.45	0.1572	[-0.751,0.492,-0.421,-0.126]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
7	valid	38.45	0.1572	[-0.751,0.492,-0.421,-0.126]	[-0.704,-0.560,-0.219,-0.377]	[0.000,1.000,0.000,0.000]
8	valid	38.51	0.1596	[-0.652,0.711,-0.253,-0.076]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
9	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
10	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.501,0.494,0.669,-0.241]	[0.000,0.000,1.000,0.000]
11	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
12	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
13	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.613,-0.730,0.233,-0.193]	[0.000,1.000,0.000,0.000]
14	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.794,-0.426,0.031,-0.432]	[0.000,1.000,0.000,0.000]
15	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.238,-0.766,0.324,0.502]	[0.000,1.000,0.000,0.000]
16	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
17	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
18	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[-0.462,0.422,0.201,0.754]	[0.000,0.000,1.000,0.000]
19	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.787,0.491,0.145,-0.344]	[1.000,0.000,0.000,0.000]
20	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]

Ordinary balanced choice | seed 1 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	76.92	0.5929	[-0.947,-0.172,0.128,-0.239]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	20.22	0.0170	[-0.757,0.423,-0.432,0.247]	[-0.686,0.426,-0.583,-0.092]	[0.000,0.000,0.000,1.000]
3	valid	28.37	0.0521	[-0.767,0.190,-0.220,0.572]	[0.291,-0.106,0.323,-0.894]	[0.000,0.000,0.000,1.000]
4	valid	19.94	0.0223	[-0.694,0.459,-0.199,0.518]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
5	valid	22.33	0.0452	[-0.602,0.657,-0.154,0.426]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
6	valid	11.50	0.0129	[-0.568,0.619,-0.364,0.402]	[-0.501,0.494,0.669,-0.241]	[0.000,0.000,1.000,0.000]
7	valid	9.57	0.0131	[-0.548,0.523,-0.345,0.554]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
8	valid	2.18	0.0005	[-0.510,0.503,-0.471,0.515]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]
9	valid	7.30	0.0097	[-0.598,0.476,-0.398,0.507]	[0.405,0.823,-0.005,-0.399]	[1.000,0.000,0.000,0.000]
10	valid	9.49	0.0109	[-0.595,0.430,-0.386,0.558]	[0.540,0.560,0.062,-0.625]	[0.000,0.000,0.000,1.000]
11	valid	8.55	0.0108	[-0.584,0.466,-0.379,0.547]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
12	valid	7.42	0.0070	[-0.578,0.461,-0.400,0.541]	[0.434,0.565,0.692,-0.118]	[0.000,0.000,1.000,0.000]
13	valid	5.84	0.0040	[-0.568,0.454,-0.433,0.532]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
14	valid	4.58	0.0006	[-0.557,0.444,-0.470,0.521]	[-0.700,0.090,0.547,-0.451]	[0.000,0.000,1.000,0.000]
15	valid	4.30	0.0010	[-0.546,0.436,-0.499,0.512]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
16	valid	4.52	0.0023	[-0.540,0.431,-0.516,0.506]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
17	valid	1.56	0.0005	[-0.526,0.477,-0.503,0.493]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
18	valid	1.57	0.0007	[-0.510,0.524,-0.487,0.477]	[-0.794,-0.426,0.031,-0.432]	[0.000,1.000,0.000,0.000]
19	valid	1.69	0.0009	[-0.503,0.518,-0.480,0.498]	[-0.007,-0.683,-0.056,-0.728]	[0.000,0.000,0.000,1.000]
20	valid	1.69	0.0009	[-0.503,0.518,-0.480,0.498]	[-0.007,-0.683,-0.056,-0.728]	[0.000,0.000,0.000,1.000]

Ordinary balanced choice | seed 1 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	63.66	0.3826	[0.076,0.556,-0.142,-0.100]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
2	valid	34.47	0.0771	[-0.011,0.656,-0.295,0.485]	[-0.099,-0.037,-0.437,-0.893]	[0.000,0.000,0.000,1.000]
3	valid	22.61	0.0579	[-0.617,0.614,-0.127,0.528]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
4	valid	14.90	0.0154	[-0.612,0.692,-0.296,0.544]	[0.198,-0.025,-0.905,-0.376]	[0.000,0.000,0.000,1.000]
5	valid	9.64	0.0080	[-0.575,0.665,-0.691,0.427]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
6	valid	5.44	0.0042	[-0.570,0.585,-0.693,0.625]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
7	valid	7.92	0.0042	[-0.506,0.673,-0.634,0.498]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
8	valid	7.20	0.0050	[-0.536,0.569,-0.438,0.398]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
9	valid	2.70	0.0006	[-0.636,0.645,-0.648,0.563]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
10	valid	6.59	0.0047	[-0.671,0.753,-0.587,0.537]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
11	valid	8.62	0.0055	[-0.593,0.684,-0.538,0.428]	[0.540,0.560,0.062,-0.625]	[1.000,0.000,0.000,0.000]
12	valid	4.70	0.0008	[-0.640,0.602,-0.578,0.480]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
13	valid	8.21	0.0060	[-0.634,0.677,-0.602,0.431]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
14	valid	5.51	0.0011	[-0.629,0.614,-0.573,0.461]	[-0.543,0.615,0.311,-0.480]	[0.000,0.000,0.000,1.000]
15	valid	4.27	0.0012	[-0.714,0.734,-0.664,0.575]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
16	valid	2.30	0.0003	[-0.618,0.589,-0.555,0.522]	[0.540,0.560,0.062,-0.625]	[0.000,0.000,0.000,1.000]
17	valid	3.37	0.0011	[-0.663,0.701,-0.604,0.587]	[0.540,0.560,0.062,-0.625]	[0.000,0.000,0.000,1.000]
18	valid	2.84	0.0007	[-0.640,0.664,-0.624,0.560]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
19	valid	2.55	0.0010	[-0.691,0.615,-0.686,0.653]	[-0.162,-0.596,-0.390,-0.683]	[0.000,0.000,0.000,1.000]
20	valid	2.04	0.0000	[-0.578,0.565,-0.589,0.525]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]

Ordinary balanced choice | seed 1 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.48	0.4395	[-0.501,-0.241,0.000,0.000]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	49.51	0.2497	[-0.488,-0.060,-0.319,0.000]	[-0.520,-0.406,-0.727,0.188]	[0.000,1.000,0.000,0.000]
3	valid	46.48	0.2320	[-0.428,-0.020,-0.608,0.000]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
4	valid	39.57	0.0942	[-0.667,-0.017,-0.522,0.136]	[0.660,-0.284,-0.415,0.558]	[1.000,0.000,0.000,0.000]
5	valid	45.95	0.1211	[-0.954,-0.015,-0.322,0.121]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
6	valid	38.48	0.0841	[-0.879,0.092,-0.469,0.111]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
7	valid	37.18	0.1016	[-0.863,0.201,-0.460,0.043]	[-0.704,-0.560,-0.219,-0.377]	[0.000,1.000,0.000,0.000]
8	valid	30.10	0.0713	[-0.676,0.330,-0.460,0.043]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
9	valid	24.55	0.0415	[-0.652,0.330,-0.460,0.131]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
10	valid	27.42	0.0572	[-0.626,0.316,-0.626,0.096]	[-0.501,0.494,0.669,-0.241]	[0.000,0.000,1.000,0.000]
11	valid	31.04	0.0692	[-0.780,0.267,-0.568,0.087]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
12	valid	34.42	0.1031	[-0.717,0.245,-0.708,0.023]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
13	valid	31.42	0.0993	[-0.710,0.340,-0.638,0.023]	[-0.613,-0.730,0.233,-0.193]	[0.000,1.000,0.000,0.000]
14	valid	29.19	0.0984	[-0.678,0.438,-0.598,0.022]	[-0.794,-0.426,0.031,-0.432]	[0.000,1.000,0.000,0.000]
15	valid	28.20	0.0897	[-0.655,0.558,-0.515,0.021]	[-0.238,-0.766,0.324,0.502]	[0.000,1.000,0.000,0.000]
16	valid	31.44	0.1022	[-0.814,0.429,-0.494,0.021]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
17	valid	35.70	0.1109	[-0.960,0.396,-0.403,0.019]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
18	valid	33.60	0.0792	[-0.943,0.389,-0.424,0.057]	[-0.462,0.422,0.201,0.754]	[0.000,0.000,1.000,0.000]
19	valid	37.82	0.0860	[-1.069,0.349,-0.364,0.052]	[0.787,0.491,0.145,-0.344]	[1.000,0.000,0.000,0.000]
20	valid	41.52	0.1062	[-1.185,0.301,-0.322,0.046]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]

Ordinary balanced choice | seed 2 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	60.45	0.2583	[0.286,0.002,-0.914,0.286]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
2	valid	33.13	0.0438	[-0.367,0.447,-0.808,0.114]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
3	valid	36.69	0.0718	[-0.623,0.229,-0.740,0.106]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
4	valid	32.93	0.0774	[-0.570,0.460,-0.677,0.067]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
5	valid	33.37	0.0774	[-0.579,0.455,-0.674,0.060]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
6	valid	33.37	0.0774	[-0.579,0.455,-0.674,0.060]	[-0.686,0.426,-0.583,-0.092]	[0.000,0.000,0.000,1.000]
7	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
8	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
9	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.376,0.796,-0.152,-0.450]	[0.000,0.000,0.000,1.000]
10	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.670,-0.109,0.690,-0.251]	[0.000,0.000,1.000,0.000]
11	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.300,0.937,0.153,-0.090]	[0.000,0.000,1.000,0.000]
12	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.237,-0.863,0.169,-0.413]	[0.000,1.000,0.000,0.000]
13	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
14	valid	6.38	0.0030	[-0.393,0.415,-0.492,0.657]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
15	valid	6.38	0.0030	[-0.393,0.415,-0.492,0.657]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
16	valid	6.38	0.0030	[-0.393,0.415,-0.492,0.657]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
17	valid	6.38	0.0030	[-0.393,0.415,-0.492,0.657]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
18	valid	6.38	0.0030	[-0.393,0.415,-0.492,0.657]	[0.194,0.637,0.742,0.079]	[0.000,0.000,1.000,0.000]
19	valid	6.38	0.0030	[-0.393,0.415,-0.492,0.657]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
20	valid	6.38	0.0030	[-0.393,0.415,-0.492,0.657]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]

Ordinary balanced choice | seed 2 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	60.45	0.2583	[0.286,0.002,-0.914,0.286]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
2	valid	46.23	0.1513	[0.225,0.677,-0.664,0.225]	[-0.005,-0.305,-0.938,0.166]	[0.000,1.000,0.000,0.000]
3	valid	21.37	0.0416	[-0.094,0.482,-0.654,0.575]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
4	valid	22.78	0.0417	[-0.443,0.360,-0.756,0.320]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
5	valid	22.21	0.0300	[-0.407,0.615,-0.630,0.245]	[-0.602,-0.667,0.317,0.303]	[0.000,1.000,0.000,0.000]
6	valid	29.09	0.0463	[-0.601,0.484,-0.623,0.130]	[0.769,-0.202,0.494,0.352]	[1.000,0.000,0.000,0.000]
7	valid	29.17	0.0456	[-0.530,0.636,-0.549,0.114]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
8	valid	33.60	0.0932	[-0.605,0.606,-0.515,0.044]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
9	valid	29.15	0.0462	[-0.601,0.602,-0.511,0.123]	[-0.272,0.576,0.069,-0.768]	[0.000,0.000,0.000,1.000]
10	valid	28.53	0.0419	[-0.571,0.572,-0.575,0.125]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
11	valid	7.19	0.0074	[-0.393,0.394,-0.550,0.623]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
12	valid	5.63	0.0040	[-0.434,0.405,-0.533,0.603]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
13	valid	3.63	0.0019	[-0.426,0.438,-0.524,0.593]	[-0.704,-0.560,-0.219,-0.377]	[0.000,1.000,0.000,0.000]
14	valid	3.98	0.0033	[-0.453,0.432,-0.516,0.585]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
15	valid	2.88	0.0007	[-0.448,0.450,-0.511,0.579]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
16	valid	2.77	0.0004	[-0.436,0.448,-0.510,0.591]	[0.377,-0.706,-0.186,-0.571]	[0.000,0.000,0.000,1.000]
17	valid	3.03	0.0017	[-0.439,0.445,-0.514,0.587]	[-0.700,0.090,0.547,-0.451]	[0.000,0.000,1.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]

Ordinary balanced choice | seed 2 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	86.15	0.6042	[-0.558,-0.287,0.049,-0.054]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	74.60	0.4336	[-0.627,-0.123,-0.311,-0.271]	[-0.100,-0.189,0.951,0.225]	[0.000,0.000,1.000,0.000]
3	valid	36.73	0.1208	[-0.711,0.520,-0.456,0.023]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
4	valid	14.40	0.0125	[-0.690,0.553,-0.585,0.461]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
5	valid	10.06	0.0088	[-0.596,0.440,-0.446,0.696]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
6	valid	3.96	0.0007	[-0.539,0.548,-0.618,0.632]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
7	valid	6.45	0.0062	[-0.566,0.461,-0.577,0.688]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
8	valid	3.47	0.0021	[-0.628,0.607,-0.680,0.759]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
9	valid	3.18	0.0009	[-0.576,0.580,-0.596,0.676]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
10	valid	5.59	0.0014	[-0.475,0.666,-0.534,0.631]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
11	valid	4.34	0.0022	[-0.620,0.619,-0.610,0.684]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
12	valid	1.58	0.0003	[-0.552,0.641,-0.608,0.713]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
13	valid	2.77	0.0006	[-0.638,0.659,-0.648,0.766]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
14	valid	2.97	0.0008	[-0.596,0.626,-0.594,0.715]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]
15	valid	3.03	0.0010	[-0.583,0.633,-0.575,0.712]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,0.000,1.000]
16	valid	4.29	0.0023	[-0.718,0.691,-0.668,0.840]	[-0.700,0.090,0.547,-0.451]	[0.000,0.000,0.000,1.000]
17	valid	3.19	0.0010	[-0.558,0.623,-0.552,0.687]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
18	valid	2.63	0.0008	[-0.550,0.592,-0.557,0.670]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
19	valid	3.93	0.0015	[-0.576,0.626,-0.542,0.697]	[-0.700,0.090,0.547,-0.451]	[0.000,0.000,0.000,1.000]
20	valid	3.38	0.0008	[-0.553,0.593,-0.540,0.659]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]

Ordinary balanced choice | seed 2 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	79.48	0.4224	[0.314,0.000,-0.490,0.000]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
2	valid	56.39	0.1684	[-0.005,0.064,-0.490,0.000]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
3	valid	44.38	0.1426	[-0.260,0.064,-0.333,0.000]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
4	valid	36.95	0.1091	[-0.172,0.292,-0.333,0.000]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
5	valid	41.30	0.1352	[-0.398,0.148,-0.333,0.000]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
6	valid	33.03	0.0811	[-0.385,0.335,-0.321,0.036]	[-0.686,0.426,-0.583,-0.092]	[0.000,0.000,0.000,1.000]
7	valid	19.26	0.0242	[-0.380,0.330,-0.289,0.192]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
8	valid	32.25	0.0399	[-0.572,0.203,-0.275,0.182]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
9	valid	24.28	0.0392	[-0.540,0.192,-0.311,0.316]	[-0.376,0.796,-0.152,-0.450]	[0.000,0.000,0.000,1.000]
10	valid	25.95	0.0399	[-0.516,0.183,-0.458,0.252]	[-0.670,-0.109,0.690,-0.251]	[0.000,0.000,1.000,0.000]
11	valid	32.52	0.0434	[-0.485,0.172,-0.614,0.169]	[-0.300,0.937,0.153,-0.090]	[0.000,0.000,1.000,0.000]
12	valid	27.86	0.0452	[-0.477,0.282,-0.567,0.167]	[-0.237,-0.863,0.169,-0.413]	[0.000,1.000,0.000,0.000]
13	valid	24.32	0.0324	[-0.474,0.409,-0.468,0.166]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
14	valid	18.33	0.0233	[-0.474,0.409,-0.332,0.261]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
15	valid	12.10	0.0132	[-0.466,0.402,-0.334,0.369]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
16	valid	21.20	0.0313	[-0.600,0.309,-0.322,0.356]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
17	valid	28.91	0.0408	[-0.724,0.276,-0.298,0.330]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
18	valid	25.05	0.0388	[-0.665,0.271,-0.396,0.324]	[0.194,0.637,0.742,0.079]	[0.000,0.000,1.000,0.000]
19	valid	24.05	0.0339	[-0.630,0.257,-0.490,0.315]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
20	valid	23.17	0.0393	[-0.522,0.253,-0.616,0.310]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]

Ordinary balanced choice | seed 3 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	46.07	0.0913	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	29.99	0.0323	[-0.134,0.694,-0.683,0.181]	[-0.005,-0.305,-0.938,0.166]	[0.000,1.000,0.000,0.000]
3	valid	11.94	0.0127	[-0.266,0.578,-0.542,0.549]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
4	valid	26.37	0.0398	[-0.098,0.455,-0.447,0.764]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
5	valid	30.01	0.0615	[-0.014,0.426,-0.555,0.714]	[0.434,0.565,0.692,-0.118]	[0.000,0.000,1.000,0.000]
6	valid	30.01	0.0615	[-0.014,0.426,-0.555,0.714]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
7	valid	18.58	0.0308	[-0.381,0.337,-0.644,0.571]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
8	valid	10.45	0.0051	[-0.333,0.505,-0.537,0.588]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
9	valid	10.45	0.0051	[-0.333,0.505,-0.537,0.588]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
10	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.263,0.146,0.859,-0.414]	[0.000,0.000,1.000,0.000]
11	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[0.077,0.882,-0.212,0.415]	[1.000,0.000,0.000,0.000]
12	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.018,0.197,-0.819,-0.539]	[0.000,0.000,0.000,1.000]
13	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
14	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
15	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
16	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[0.328,0.107,-0.455,-0.821]	[0.000,0.000,0.000,1.000]
17	valid	11.53	0.0110	[-0.263,0.621,-0.485,0.557]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
18	valid	11.86	0.0119	[-0.258,0.610,-0.512,0.547]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
19	valid	11.86	0.0119	[-0.258,0.610,-0.512,0.547]	[-0.013,0.056,-0.998,-0.022]	[0.000,0.000,0.000,1.000]
20	valid	11.86	0.0119	[-0.258,0.610,-0.512,0.547]	[-0.457,0.529,0.535,0.474]	[0.000,0.000,1.000,0.000]

Ordinary balanced choice | seed 3 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	46.07	0.0913	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	32.05	0.0826	[-0.550,0.068,-0.629,0.545]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
3	valid	38.85	0.1016	[-0.349,-0.000,-0.472,0.810]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
4	valid	38.24	0.1108	[-0.493,0.032,-0.294,0.818]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
5	valid	36.81	0.1009	[-0.512,-0.000,-0.433,0.742]	[-0.501,0.494,0.669,-0.241]	[0.000,0.000,1.000,0.000]
6	valid	37.78	0.1135	[-0.598,-0.000,-0.347,0.723]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
7	valid	36.52	0.1025	[-0.577,-0.000,-0.423,0.698]	[-0.263,0.146,0.859,-0.414]	[0.000,0.000,1.000,0.000]
8	valid	19.77	0.0318	[-0.551,0.301,-0.404,0.666]	[0.302,-0.755,-0.526,-0.250]	[0.000,1.000,0.000,0.000]
9	valid	19.18	0.0324	[-0.533,0.291,-0.464,0.645]	[-0.300,0.937,0.153,-0.090]	[0.000,0.000,1.000,0.000]
10	valid	15.08	0.0168	[-0.520,0.361,-0.452,0.629]	[-0.100,-0.640,0.394,0.652]	[0.000,1.000,0.000,0.000]
11	valid	12.23	0.0136	[-0.482,0.397,-0.519,0.583]	[0.223,0.325,0.735,-0.551]	[0.000,0.000,1.000,0.000]
12	valid	8.31	0.0062	[-0.467,0.459,-0.502,0.565]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]
13	valid	4.57	0.0010	[-0.449,0.518,-0.484,0.544]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
14	valid	2.18	0.0009	[-0.419,0.587,-0.477,0.503]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
15	valid	2.87	0.0008	[-0.422,0.567,-0.455,0.542]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
16	valid	5.02	0.0018	[-0.412,0.532,-0.489,0.555]	[-0.700,0.090,0.547,-0.451]	[0.000,0.000,1.000,0.000]
17	valid	4.49	0.0018	[-0.417,0.537,-0.485,0.551]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
18	valid	3.29	0.0012	[-0.404,0.574,-0.470,0.534]	[-0.794,-0.426,0.031,-0.432]	[0.000,1.000,0.000,0.000]
19	valid	3.35	0.0009	[-0.398,0.593,-0.463,0.525]	[-0.007,-0.683,-0.056,-0.728]	[0.000,1.000,0.000,0.000]
20	valid	1.66	0.0001	[-0.426,0.585,-0.456,0.518]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]

Ordinary balanced choice | seed 3 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	52.60	0.0960	[-0.051,0.089,-0.611,0.048]	[-0.220,0.017,0.903,-0.368]	[0.000,0.000,1.000,0.000]
2	valid	61.78	0.3659	[-0.657,0.113,-0.518,-0.334]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
3	valid	41.01	0.2047	[-0.574,0.538,-0.488,-0.166]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
4	valid	18.51	0.0196	[-0.587,0.598,-0.604,0.234]	[-0.444,0.444,-0.465,-0.624]	[0.000,0.000,0.000,1.000]
5	valid	11.46	0.0150	[-0.700,0.591,-0.452,0.489]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
6	valid	9.03	0.0046	[-0.637,0.685,-0.530,0.442]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]
7	valid	12.89	0.0138	[-0.758,0.575,-0.597,0.501]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
8	valid	5.64	0.0030	[-0.527,0.604,-0.620,0.582]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
9	valid	11.28	0.0113	[-0.650,0.638,-0.817,0.593]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
10	valid	7.21	0.0041	[-0.595,0.650,-0.678,0.551]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
11	valid	9.74	0.0047	[-0.474,0.661,-0.738,0.561]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
12	valid	5.73	0.0027	[-0.560,0.722,-0.725,0.665]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
13	valid	7.50	0.0031	[-0.485,0.676,-0.661,0.513]	[-0.007,-0.683,-0.056,-0.728]	[0.000,1.000,0.000,0.000]
14	valid	7.32	0.0031	[-0.455,0.605,-0.610,0.481]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
15	valid	6.08	0.0019	[-0.598,0.797,-0.755,0.619]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
16	valid	6.00	0.0028	[-0.473,0.591,-0.583,0.487]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
17	valid	4.04	0.0016	[-0.658,0.828,-0.757,0.688]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
18	valid	3.08	0.0003	[-0.540,0.716,-0.591,0.558]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
19	valid	2.83	0.0010	[-0.509,0.734,-0.588,0.593]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
20	valid	1.36	0.0001	[-0.543,0.746,-0.586,0.638]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]

Ordinary balanced choice | seed 3 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	58.59	0.2668	[-0.082,0.000,-0.515,0.000]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	30.67	0.0596	[-0.077,0.314,-0.480,0.161]	[-0.005,-0.305,-0.938,0.166]	[0.000,1.000,0.000,0.000]
3	valid	26.12	0.0557	[-0.070,0.286,-0.460,0.417]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
4	valid	37.37	0.0934	[-0.063,0.151,-0.413,0.684]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
5	valid	38.13	0.0896	[-0.057,0.138,-0.638,0.572]	[0.434,0.565,0.692,-0.118]	[0.000,0.000,1.000,0.000]
6	valid	43.18	0.1017	[-0.050,0.120,-0.858,0.473]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
7	valid	40.20	0.1002	[-0.155,0.106,-0.844,0.465]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
8	valid	32.83	0.0577	[-0.155,0.214,-0.731,0.465]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
9	valid	38.95	0.0751	[-0.155,0.191,-0.920,0.413]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
10	valid	45.85	0.0878	[-0.135,0.167,-1.113,0.302]	[-0.263,0.146,0.859,-0.414]	[0.000,0.000,1.000,0.000]
11	valid	43.07	0.0918	[-0.141,0.157,-1.051,0.404]	[0.077,0.882,-0.212,0.415]	[1.000,0.000,0.000,0.000]
12	valid	40.84	0.0922	[-0.134,0.147,-0.983,0.516]	[-0.018,0.197,-0.819,-0.539]	[0.000,0.000,0.000,1.000]
13	valid	44.89	0.1019	[-0.115,0.127,-1.118,0.455]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
14	valid	49.25	0.1103	[-0.100,0.110,-1.273,0.356]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
15	valid	48.17	0.1047	[-0.145,0.109,-1.252,0.351]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
16	valid	46.75	0.1184	[-0.121,0.105,-1.200,0.437]	[0.328,0.107,-0.455,-0.821]	[0.000,0.000,0.000,1.000]
17	valid	46.22	0.0969	[-0.121,0.148,-1.200,0.389]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
18	valid	50.23	0.1029	[-0.107,0.131,-1.347,0.301]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
19	valid	48.25	0.1052	[-0.123,0.125,-1.287,0.366]	[-0.013,0.056,-0.998,-0.022]	[0.000,0.000,0.000,1.000]
20	valid	50.37	0.1036	[-0.133,0.107,-1.371,0.312]	[-0.457,0.529,0.535,0.474]	[0.000,0.000,1.000,0.000]

Ordinary balanced choice | seed 4 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	35.26	0.1033	[-0.421,0.527,0.098,0.732]	[-0.477,0.831,0.084,-0.272]	[0.000,0.000,0.000,1.000]
2	valid	53.94	0.2301	[-0.487,0.511,0.449,0.548]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
3	valid	47.90	0.2100	[-0.303,0.830,0.311,0.350]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
4	valid	47.90	0.2100	[-0.303,0.830,0.311,0.350]	[-0.259,-0.531,-0.093,0.801]	[0.000,1.000,0.000,0.000]
5	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[0.291,-0.106,0.323,-0.894]	[0.000,0.000,0.000,1.000]
6	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
7	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[-0.406,-0.902,-0.036,0.142]	[0.000,1.000,0.000,0.000]
8	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
9	valid	10.11	0.0057	[-0.402,0.662,-0.299,0.558]	[-0.100,-0.189,0.951,0.225]	[0.000,0.000,1.000,0.000]
10	valid	10.11	0.0057	[-0.402,0.662,-0.299,0.558]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
11	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
12	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
13	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
14	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.266,0.572,0.740,0.233]	[0.000,0.000,1.000,0.000]
15	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.489,0.191,-0.234,0.818]	[1.000,0.000,0.000,0.000]
16	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.503,0.277,-0.818,-0.031]	[0.000,0.000,0.000,1.000]
17	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
18	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.157,0.765,-0.245,-0.575]	[0.000,0.000,0.000,1.000]
19	valid	6.20	0.0040	[-0.394,0.584,-0.414,0.576]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
20	valid	6.36	0.0031	[-0.392,0.570,-0.434,0.578]	[-0.064,-0.505,0.844,0.167]	[0.000,0.000,1.000,0.000]

Ordinary balanced choice | seed 4 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	35.26	0.1033	[-0.421,0.527,0.098,0.732]	[-0.477,0.831,0.084,-0.272]	[0.000,0.000,0.000,1.000]
2	valid	44.26	0.1876	[-0.294,0.833,0.252,0.394]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
3	valid	19.63	0.0289	[-0.400,0.784,-0.159,0.447]	[-0.457,0.529,0.535,0.474]	[0.000,0.000,1.000,0.000]
4	valid	28.77	0.0777	[-0.627,0.733,-0.072,0.255]	[0.814,-0.418,0.255,0.313]	[1.000,0.000,0.000,0.000]
5	valid	9.79	0.0069	[-0.520,0.609,-0.490,0.345]	[0.223,0.325,0.735,-0.551]	[0.000,0.000,1.000,0.000]
6	valid	22.58	0.0269	[-0.520,0.786,-0.275,0.193]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
7	valid	15.20	0.0140	[-0.493,0.745,-0.342,0.292]	[0.340,0.096,-0.269,-0.896]	[0.000,0.000,0.000,1.000]
8	valid	13.35	0.0133	[-0.471,0.711,-0.442,0.279]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
9	valid	7.43	0.0027	[-0.453,0.684,-0.425,0.382]	[-0.543,0.615,0.311,-0.480]	[0.000,0.000,0.000,1.000]
10	valid	1.74	0.0000	[-0.417,0.630,-0.456,0.470]	[0.540,0.560,0.062,-0.625]	[0.000,0.000,0.000,1.000]
11	valid	12.95	0.0095	[-0.311,0.499,-0.524,0.617]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
12	valid	10.00	0.0065	[-0.329,0.539,-0.502,0.591]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
13	valid	8.18	0.0035	[-0.380,0.528,-0.491,0.579]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
14	valid	3.99	0.0014	[-0.419,0.570,-0.457,0.539]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
15	valid	2.60	0.0008	[-0.404,0.613,-0.440,0.518]	[-0.007,-0.683,-0.056,-0.728]	[0.000,1.000,0.000,0.000]
16	valid	2.78	0.0002	[-0.403,0.599,-0.456,0.520]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
17	valid	2.75	0.0008	[-0.392,0.616,-0.457,0.508]	[-0.162,-0.596,-0.390,-0.683]	[0.000,1.000,0.000,0.000]
18	valid	3.71	0.0008	[-0.392,0.592,-0.485,0.511]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
19	valid	3.71	0.0008	[-0.392,0.592,-0.485,0.511]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]

Ordinary balanced choice | seed 4 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	38.00	0.1218	[0.061,0.514,-0.083,0.417]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
2	valid	35.34	0.0798	[0.133,0.531,-0.517,0.418]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
3	valid	7.66	0.0021	[-0.387,0.662,-0.597,0.483]	[0.850,0.267,-0.190,0.413]	[1.000,0.000,0.000,0.000]
4	valid	5.72	0.0033	[-0.470,0.619,-0.412,0.594]	[-0.612,0.267,-0.382,0.639]	[0.000,0.000,1.000,0.000]
5	valid	8.11	0.0072	[-0.527,0.528,-0.385,0.502]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
6	valid	7.46	0.0058	[-0.494,0.535,-0.362,0.402]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
7	valid	6.83	0.0038	[-0.616,0.658,-0.462,0.545]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
8	valid	8.74	0.0049	[-0.624,0.609,-0.502,0.481]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
9	valid	8.76	0.0045	[-0.386,0.577,-0.554,0.392]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
10	valid	7.95	0.0035	[-0.437,0.592,-0.592,0.450]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
11	valid	6.87	0.0030	[-0.480,0.543,-0.538,0.472]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
12	valid	6.10	0.0027	[-0.450,0.566,-0.489,0.393]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
13	valid	7.44	0.0027	[-0.451,0.660,-0.615,0.464]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
14	valid	5.25	0.0023	[-0.462,0.612,-0.511,0.423]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
15	valid	3.78	0.0006	[-0.478,0.602,-0.524,0.524]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
16	valid	3.55	0.0008	[-0.375,0.581,-0.472,0.471]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
17	valid	2.93	0.0007	[-0.476,0.693,-0.554,0.518]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
18	valid	2.37	0.0000	[-0.435,0.612,-0.488,0.469]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
19	valid	1.95	0.0000	[-0.448,0.592,-0.473,0.485]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
20	valid	1.96	0.0000	[-0.538,0.728,-0.548,0.546]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]

Ordinary balanced choice | seed 4 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	52.58	0.1957	[-0.499,0.000,0.000,0.256]	[-0.477,0.831,0.084,-0.272]	[0.000,0.000,0.000,1.000]
2	valid	47.29	0.1468	[-0.844,0.246,0.000,0.193]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
3	valid	34.52	0.1021	[-0.601,0.534,0.000,0.193]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
4	valid	33.59	0.0914	[-0.510,0.786,-0.024,0.164]	[-0.259,-0.531,-0.093,0.801]	[0.000,1.000,0.000,0.000]
5	valid	28.65	0.0810	[-0.423,0.771,-0.023,0.321]	[0.291,-0.106,0.323,-0.894]	[0.000,0.000,0.000,1.000]
6	valid	37.08	0.0930	[-0.367,1.022,-0.020,0.172]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
7	valid	41.09	0.1025	[-0.297,1.214,-0.019,0.139]	[-0.406,-0.902,-0.036,0.142]	[0.000,1.000,0.000,0.000]
8	valid	40.94	0.1119	[-0.287,1.176,0.046,0.218]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
9	valid	38.13	0.0991	[-0.287,1.100,-0.015,0.218]	[-0.100,-0.189,0.951,0.225]	[0.000,0.000,1.000,0.000]
10	valid	42.02	0.0991	[-0.226,1.312,-0.013,0.189]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
11	valid	41.69	0.0989	[-0.152,1.305,-0.013,0.270]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
12	valid	40.13	0.0815	[-0.162,1.292,-0.059,0.268]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
13	valid	42.35	0.0781	[-0.134,1.408,-0.072,0.221]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
14	valid	40.93	0.0575	[-0.149,1.382,-0.112,0.216]	[-0.266,0.572,0.740,0.233]	[0.000,0.000,1.000,0.000]
15	valid	39.23	0.0588	[-0.199,1.341,-0.130,0.210]	[0.489,0.191,-0.234,0.818]	[1.000,0.000,0.000,0.000]
16	valid	37.82	0.0667	[-0.256,1.301,-0.126,0.207]	[-0.503,0.277,-0.818,-0.031]	[0.000,0.000,0.000,1.000]
17	valid	36.40	0.0655	[-0.243,1.269,-0.123,0.265]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
18	valid	35.07	0.0638	[-0.220,1.234,-0.120,0.332]	[0.157,0.765,-0.245,-0.575]	[0.000,0.000,0.000,1.000]
19	valid	34.84	0.0560	[-0.220,1.234,-0.161,0.291]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
20	valid	31.14	0.0349	[-0.220,1.079,-0.209,0.291]	[-0.064,-0.505,0.844,0.167]	[0.000,0.000,1.000,0.000]

Ordinary subset selection | seed 0 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	38.70	0.1669	[-0.475,-0.235,-0.447,0.405,-0.471,0.366]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
2	valid	43.45	0.1844	[-0.555,-0.286,-0.279,0.422,-0.522,0.288]	[0.288,0.142,0.145,-0.867,0.339,0.096]	[1.000,0.000,0.000,1.000,1.000,0.000]
3	valid	22.43	0.0438	[-0.550,0.064,-0.378,0.388,-0.510,0.373]	[-0.538,-0.418,0.274,0.420,0.256,0.467]	[0.000,1.000,1.000,0.000,1.000,0.000]
4	valid	20.66	0.0243	[-0.550,0.297,-0.446,0.369,-0.506,0.136]	[0.032,-0.460,-0.470,-0.621,-0.424,-0.023]	[1.000,1.000,0.000,1.000,0.000,0.000]
5	valid	13.54	0.0041	[-0.581,0.292,-0.386,0.305,-0.423,0.394]	[0.311,-0.695,-0.538,0.240,0.088,-0.257]	[1.000,1.000,0.000,0.000,0.000,1.000]
6	valid	13.54	0.0041	[-0.581,0.292,-0.386,0.305,-0.423,0.394]	[0.422,0.621,0.029,-0.427,-0.478,-0.156]	[1.000,0.000,0.000,1.000,0.000,1.000]
7	valid	13.54	0.0041	[-0.581,0.292,-0.386,0.305,-0.423,0.394]	[0.113,0.689,0.259,0.465,0.477,-0.026]	[1.000,0.000,1.000,0.000,1.000,0.000]
8	valid	15.36	0.0049	[-0.482,0.310,-0.349,0.557,-0.362,0.330]	[0.796,0.423,-0.128,0.149,-0.334,-0.194]	[1.000,0.000,1.000,0.000,0.000,1.000]
9	valid	15.36	0.0049	[-0.482,0.310,-0.349,0.557,-0.362,0.330]	[-0.149,-0.313,-0.726,0.471,0.329,-0.154]	[0.000,1.000,0.000,0.000,1.000,1.000]
10	valid	15.36	0.0049	[-0.482,0.310,-0.349,0.557,-0.362,0.330]	[-0.462,-0.494,0.424,0.128,0.199,0.554]	[0.000,1.000,1.000,0.000,1.000,0.000]
11	valid	20.49	0.0118	[-0.577,0.234,-0.263,0.421,-0.273,0.541]	[0.110,-0.860,0.078,-0.437,0.148,-0.173]	[0.000,1.000,0.000,1.000,0.000,1.000]
12	valid	17.36	0.0084	[-0.531,0.243,-0.333,0.465,-0.290,0.497]	[-0.123,-0.104,0.472,-0.427,0.733,-0.179]	[0.000,0.000,1.000,1.000,1.000,0.000]
13	valid	12.95	0.0039	[-0.403,0.405,-0.432,0.504,-0.294,0.383]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
14	valid	12.95	0.0039	[-0.403,0.405,-0.432,0.504,-0.294,0.383]	[0.087,0.348,0.805,-0.110,-0.091,0.451]	[1.000,0.000,1.000,1.000,0.000,0.000]
15	valid	12.95	0.0039	[-0.403,0.405,-0.432,0.504,-0.294,0.383]	[-0.234,-0.246,-0.668,-0.642,0.156,0.050]	[0.000,1.000,0.000,1.000,1.000,0.000]
16	valid	12.86	0.0039	[-0.406,0.404,-0.432,0.497,-0.289,0.394]	[0.671,0.475,-0.338,0.128,0.378,0.224]	[1.000,0.000,0.000,1.000,1.000,0.000]
17	valid	4.88	0.0006	[-0.361,0.359,-0.388,0.364,-0.499,0.456]	[0.815,-0.108,-0.232,0.190,-0.150,0.460]	[1.000,1.000,0.000,1.000,0.000,0.000]
18	valid	4.88	0.0006	[-0.361,0.359,-0.388,0.364,-0.499,0.456]	[-0.462,-0.494,0.424,0.128,0.199,0.554]	[0.000,1.000,1.000,0.000,1.000,0.000]
19	valid	4.88	0.0006	[-0.361,0.359,-0.388,0.364,-0.499,0.456]	[0.573,0.327,0.513,0.058,0.477,-0.266]	[1.000,0.000,1.000,0.000,1.000,0.000]
20	valid	4.88	0.0006	[-0.361,0.359,-0.388,0.364,-0.499,0.456]	[0.127,-0.020,-0.673,0.298,-0.642,0.172]	[1.000,1.000,0.000,0.000,0.000,1.000]

Ordinary subset selection | seed 0 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	38.70	0.1669	[-0.475,-0.235,-0.447,0.405,-0.471,0.366]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
2	valid	15.66	0.0062	[-0.448,0.287,-0.564,0.269,-0.456,0.343]	[0.399,0.619,0.220,0.492,-0.304,-0.274]	[1.000,0.000,1.000,0.000,0.000,1.000]
3	valid	22.82	0.0270	[-0.310,0.141,-0.540,0.518,-0.315,0.475]	[0.054,0.210,0.474,-0.512,0.301,-0.612]	[0.000,0.000,1.000,1.000,0.000,1.000]
4	valid	15.43	0.0039	[-0.279,0.371,-0.447,0.466,-0.289,0.533]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[0.000,1.000,0.000,1.000,0.000,1.000]
5	valid	14.62	0.0078	[-0.215,0.329,-0.492,0.466,-0.465,0.414]	[-0.228,0.368,0.217,-0.793,-0.218,0.299]	[1.000,0.000,1.000,1.000,0.000,0.000]
6	valid	10.54	0.0027	[-0.329,0.313,-0.502,0.388,-0.477,0.404]	[-0.270,0.085,-0.159,-0.836,-0.128,0.423]	[0.000,1.000,0.000,1.000,1.000,0.000]
7	valid	8.24	0.0017	[-0.305,0.388,-0.465,0.360,-0.442,0.464]	[0.382,0.461,-0.285,0.068,-0.646,0.372]	[1.000,0.000,1.000,1.000,0.000,0.000]
8	valid	7.15	0.0018	[-0.336,0.356,-0.447,0.418,-0.449,0.428]	[-0.507,0.208,-0.667,0.392,-0.094,0.304]	[0.000,1.000,0.000,0.000,1.000,1.000]
9	valid	5.00	0.0002	[-0.349,0.409,-0.436,0.390,-0.450,0.408]	[-0.459,0.384,-0.073,0.327,0.227,0.692]	[0.000,0.000,1.000,1.000,1.000,0.000]
10	valid	6.92	0.0014	[-0.325,0.381,-0.453,0.375,-0.488,0.407]	[0.590,-0.332,0.430,0.510,0.303,0.075]	[1.000,0.000,1.000,0.000,1.000,0.000]
11	valid	5.28	0.0009	[-0.381,0.373,-0.442,0.366,-0.477,0.397]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
12	valid	3.76	0.0002	[-0.380,0.371,-0.412,0.383,-0.459,0.437]	[0.110,-0.860,0.078,-0.437,0.148,-0.173]	[0.000,1.000,0.000,1.000,0.000,1.000]
13	valid	4.17	0.0002	[-0.405,0.355,-0.409,0.373,-0.464,0.434]	[0.250,-0.124,0.112,-0.245,0.649,-0.654]	[1.000,0.000,0.000,0.000,1.000,1.000]
14	valid	4.43	0.0004	[-0.403,0.352,-0.384,0.385,-0.449,0.465]	[0.440,-0.213,0.638,0.046,0.434,-0.405]	[0.000,0.000,1.000,0.000,1.000,1.000]
15	valid	4.33	0.0004	[-0.406,0.355,-0.377,0.394,-0.446,0.462]	[0.077,0.627,-0.555,0.059,0.536,0.048]	[1.000,0.000,0.000,0.000,1.000,1.000]
16	valid	4.44	0.0004	[-0.404,0.353,-0.387,0.392,-0.444,0.460]	[-0.013,0.610,-0.185,0.173,-0.744,0.105]	[1.000,0.000,0.000,1.000,0.000,1.000]
17	valid	4.62	0.0004	[-0.403,0.352,-0.386,0.393,-0.441,0.464]	[-0.446,-0.656,0.134,-0.312,-0.491,-0.117]	[0.000,1.000,0.000,1.000,0.000,1.000]
18	valid	4.58	0.0004	[-0.403,0.353,-0.386,0.393,-0.441,0.464]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
19	valid	4.58	0.0004	[-0.403,0.353,-0.386,0.393,-0.441,0.464]	[0.250,-0.124,0.112,-0.245,0.649,-0.654]	[1.000,0.000,0.000,0.000,1.000,1.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.013,0.610,-0.185,0.173,-0.744,0.105]	[1.000,0.000,1.000,0.000,0.000,1.000]

Ordinary subset selection | seed 0 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	9.92	0.0032	[-0.462,0.473,-0.251,0.385,-0.372,0.407]	[-0.538,-0.418,0.274,0.420,0.256,0.467]	[0.000,1.000,1.000,0.000,1.000,0.000]
2	valid	16.44	0.0068	[-0.609,0.271,-0.256,0.382,-0.479,0.575]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
3	valid	13.80	0.0049	[-0.525,0.624,-0.735,0.474,-0.501,0.401]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
4	valid	6.05	0.0003	[-0.555,0.645,-0.658,0.556,-0.610,0.564]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
5	valid	9.49	0.0017	[-0.470,0.545,-0.719,0.569,-0.619,0.545]	[0.440,-0.213,0.638,0.046,0.434,-0.405]	[0.000,0.000,1.000,0.000,1.000,1.000]
6	valid	8.11	0.0011	[-0.513,0.487,-0.676,0.550,-0.648,0.545]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
7	valid	9.10	0.0014	[-0.540,0.597,-0.753,0.505,-0.628,0.562]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
8	valid	9.20	0.0018	[-0.420,0.467,-0.619,0.501,-0.595,0.458]	[0.440,-0.213,0.638,0.046,0.434,-0.405]	[1.000,0.000,1.000,0.000,1.000,0.000]
9	valid	11.21	0.0022	[-0.466,0.534,-0.726,0.471,-0.581,0.478]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[0.000,1.000,0.000,0.000,1.000,1.000]
10	valid	6.41	0.0001	[-0.478,0.517,-0.602,0.494,-0.558,0.506]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
11	valid	6.47	0.0003	[-0.565,0.646,-0.708,0.526,-0.674,0.587]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
12	valid	7.07	0.0007	[-0.536,0.624,-0.667,0.520,-0.692,0.509]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
13	valid	4.19	0.0004	[-0.557,0.587,-0.611,0.542,-0.728,0.570]	[0.137,-0.573,0.372,-0.529,0.421,0.239]	[0.000,1.000,0.000,1.000,1.000,0.000]
14	valid	3.82	0.0004	[-0.621,0.695,-0.638,0.641,-0.714,0.608]	[0.570,0.398,-0.261,-0.403,0.287,-0.451]	[1.000,0.000,0.000,1.000,0.000,1.000]
15	valid	3.56	0.0003	[-0.680,0.705,-0.691,0.651,-0.708,0.688]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
16	valid	3.86	0.0001	[-0.600,0.630,-0.650,0.590,-0.673,0.608]	[0.440,-0.213,0.638,0.046,0.434,-0.405]	[1.000,0.000,1.000,0.000,1.000,0.000]
17	valid	5.79	0.0003	[-0.570,0.586,-0.676,0.577,-0.729,0.567]	[-0.308,0.274,0.274,0.341,-0.084,0.795]	[1.000,0.000,1.000,0.000,1.000,0.000]
18	valid	5.35	0.0004	[-0.555,0.647,-0.687,0.596,-0.722,0.599]	[-0.410,-0.016,-0.600,0.533,-0.107,0.420]	[1.000,1.000,0.000,0.000,1.000,0.000]
19	valid	3.45	0.0001	[-0.595,0.637,-0.646,0.587,-0.702,0.610]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
20	valid	5.09	0.0001	[-0.598,0.648,-0.693,0.579,-0.704,0.594]	[0.440,-0.213,0.638,0.046,0.434,-0.405]	[1.000,0.000,1.000,0.000,1.000,0.000]

Ordinary subset selection | seed 0 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	35.34	0.1302	[-0.287,-0.038,-0.087,0.391,-0.239,0.236]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
2	valid	42.66	0.0959	[-0.320,0.001,-0.038,0.711,-0.292,0.215]	[0.288,0.142,0.145,-0.867,0.339,0.096]	[1.000,0.000,0.000,1.000,1.000,0.000]
3	valid	37.54	0.0423	[-0.449,0.089,-0.086,0.813,-0.295,0.301]	[-0.538,-0.418,0.274,0.420,0.256,0.467]	[0.000,1.000,1.000,0.000,1.000,0.000]
4	valid	41.77	0.0227	[-0.353,0.134,-0.132,1.034,-0.332,0.225]	[0.032,-0.460,-0.470,-0.621,-0.424,-0.023]	[1.000,1.000,0.000,1.000,0.000,0.000]
5	valid	40.10	0.0153	[-0.374,0.204,-0.179,1.033,-0.267,0.233]	[0.311,-0.695,-0.538,0.240,0.088,-0.257]	[1.000,1.000,0.000,0.000,0.000,1.000]
6	valid	40.68	0.0143	[-0.390,0.251,-0.137,1.071,-0.292,0.205]	[0.422,0.621,0.029,-0.427,-0.478,-0.156]	[1.000,0.000,0.000,1.000,0.000,1.000]
7	valid	42.10	0.0167	[-0.338,0.304,-0.132,1.126,-0.313,0.164]	[0.113,0.689,0.259,0.465,0.477,-0.026]	[1.000,0.000,1.000,0.000,1.000,0.000]
8	valid	39.88	0.0273	[-0.426,0.324,-0.108,1.056,-0.319,0.158]	[0.796,0.423,-0.128,0.149,-0.334,-0.194]	[1.000,0.000,1.000,0.000,0.000,1.000]
9	valid	41.33	0.0234	[-0.386,0.318,-0.134,1.111,-0.315,0.145]	[-0.149,-0.313,-0.726,0.471,0.329,-0.154]	[0.000,1.000,0.000,0.000,1.000,1.000]
10	valid	37.68	0.0178	[-0.428,0.359,-0.147,1.015,-0.301,0.172]	[-0.462,-0.494,0.424,0.128,0.199,0.554]	[0.000,1.000,1.000,0.000,1.000,0.000]
11	valid	39.88	0.0247	[-0.371,0.438,-0.129,1.066,-0.257,0.165]	[0.110,-0.860,0.078,-0.437,0.148,-0.173]	[0.000,1.000,0.000,1.000,0.000,1.000]
12	valid	41.49	0.0255	[-0.347,0.375,-0.140,1.125,-0.307,0.137]	[-0.123,-0.104,0.472,-0.427,0.733,-0.179]	[0.000,0.000,1.000,1.000,1.000,0.000]
13	valid	47.02	0.0399	[-0.253,0.413,-0.152,1.283,-0.234,0.098]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
14	valid	45.77	0.0334	[-0.243,0.433,-0.186,1.241,-0.225,0.107]	[0.087,0.348,0.805,-0.110,-0.091,0.451]	[1.000,0.000,1.000,1.000,0.000,0.000]
15	valid	48.33	0.0383	[-0.222,0.397,-0.202,1.327,-0.199,0.091]	[-0.234,-0.246,-0.668,-0.642,0.156,0.050]	[0.000,1.000,0.000,1.000,1.000,0.000]
16	valid	44.45	0.0445	[-0.270,0.448,-0.216,1.195,-0.217,0.094]	[0.671,0.475,-0.338,0.128,0.378,0.224]	[1.000,0.000,0.000,1.000,1.000,0.000]
17	valid	41.31	0.0395	[-0.336,0.451,-0.226,1.102,-0.221,0.106]	[0.815,-0.108,-0.232,0.190,-0.150,0.460]	[1.000,1.000,0.000,1.000,0.000,0.000]
18	valid	38.83	0.0395	[-0.360,0.491,-0.239,1.024,-0.212,0.118]	[-0.462,-0.494,0.424,0.128,0.199,0.554]	[0.000,1.000,1.000,0.000,1.000,0.000]
19	valid	36.29	0.0501	[-0.407,0.503,-0.264,0.943,-0.231,0.095]	[0.573,0.327,0.513,0.058,0.477,-0.266]	[1.000,0.000,1.000,0.000,1.000,0.000]
20	valid	36.08	0.0484	[-0.388,0.462,-0.302,0.950,-0.261,0.082]	[0.127,-0.020,-0.673,0.298,-0.642,0.172]	[1.000,1.000,0.000,0.000,0.000,1.000]

Ordinary subset selection | seed 1 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	38.96	0.1971	[-0.519,0.422,0.113,0.519,-0.029,0.519]	[-0.054,-0.261,-0.025,0.815,0.006,0.514]	[0.000,1.000,1.000,0.000,1.000,0.000]
2	valid	10.68	0.0031	[-0.583,0.261,-0.379,0.386,-0.381,0.393]	[-0.354,-0.292,-0.741,0.008,-0.489,-0.024]	[0.000,1.000,0.000,1.000,0.000,1.000]
3	valid	10.68	0.0031	[-0.583,0.261,-0.379,0.386,-0.381,0.393]	[-0.462,0.037,-0.577,-0.165,0.568,-0.320]	[0.000,0.000,0.000,1.000,1.000,1.000]
4	valid	7.82	0.0010	[-0.498,0.405,-0.325,0.512,-0.328,0.336]	[-0.611,0.612,-0.061,0.298,0.374,0.141]	[0.000,0.000,1.000,0.000,1.000,1.000]
5	valid	7.82	0.0010	[-0.498,0.405,-0.325,0.512,-0.328,0.336]	[-0.149,-0.313,-0.726,0.471,0.329,-0.154]	[0.000,1.000,0.000,0.000,1.000,1.000]
6	valid	4.72	0.0008	[-0.511,0.353,-0.456,0.447,-0.341,0.299]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
7	valid	11.65	0.0045	[-0.580,0.290,-0.452,0.485,-0.280,0.249]	[-0.410,-0.016,-0.600,0.533,-0.107,0.420]	[0.000,1.000,0.000,0.000,1.000,1.000]
8	valid	11.65	0.0045	[-0.580,0.290,-0.452,0.485,-0.280,0.249]	[-0.438,-0.103,-0.054,-0.001,0.801,-0.391]	[0.000,1.000,0.000,0.000,1.000,1.000]
9	valid	11.65	0.0045	[-0.580,0.290,-0.452,0.485,-0.280,0.249]	[0.490,0.520,-0.154,-0.215,0.644,-0.073]	[1.000,0.000,0.000,1.000,1.000,0.000]
10	valid	11.96	0.0031	[-0.584,0.278,-0.450,0.485,-0.259,0.278]	[-0.676,0.066,-0.050,0.440,0.480,0.334]	[0.000,1.000,1.000,0.000,1.000,0.000]
11	valid	11.96	0.0031	[-0.584,0.278,-0.450,0.485,-0.259,0.278]	[0.087,0.348,0.805,-0.110,-0.091,0.451]	[1.000,0.000,1.000,1.000,0.000,0.000]
12	valid	11.96	0.0031	[-0.584,0.278,-0.450,0.485,-0.259,0.278]	[-0.462,0.037,-0.577,-0.165,0.568,-0.320]	[0.000,0.000,0.000,1.000,1.000,1.000]
13	valid	13.48	0.0051	[-0.630,0.222,-0.400,0.436,-0.320,0.318]	[0.250,-0.124,0.112,-0.245,0.649,-0.654]	[1.000,0.000,0.000,0.000,1.000,1.000]
14	valid	13.48	0.0051	[-0.630,0.222,-0.400,0.436,-0.320,0.318]	[-0.106,0.510,0.045,0.750,0.123,-0.386]	[0.000,0.000,1.000,0.000,1.000,1.000]
15	valid	13.48	0.0051	[-0.630,0.222,-0.400,0.436,-0.320,0.318]	[0.124,-0.072,-0.090,-0.854,-0.201,-0.450]	[1.000,0.000,0.000,1.000,0.000,1.000]
16	valid	13.48	0.0051	[-0.630,0.222,-0.400,0.436,-0.320,0.318]	[-0.600,-0.024,-0.134,0.484,0.360,0.507]	[0.000,1.000,1.000,0.000,1.000,0.000]
17	valid	13.48	0.0051	[-0.630,0.222,-0.400,0.436,-0.320,0.318]	[-0.507,0.208,-0.667,0.392,-0.094,0.304]	[0.000,1.000,0.000,0.000,1.000,1.000]
18	valid	13.48	0.0051	[-0.630,0.222,-0.400,0.436,-0.320,0.318]	[-0.566,-0.227,0.409,-0.227,-0.544,0.337]	[0.000,1.000,1.000,1.000,0.000,0.000]
19	valid	8.52	0.0013	[-0.546,0.345,-0.393,0.434,-0.268,0.411]	[0.382,0.461,-0.285,0.068,-0.646,0.372]	[1.000,0.000,1.000,1.000,0.000,0.000]
20	valid	8.52	0.0013	[-0.546,0.345,-0.393,0.434,-0.268,0.411]	[-0.474,-0.567,0.246,0.247,-0.463,-0.344]	[0.000,1.000,1.000,0.000,0.000,1.000]

Ordinary subset selection | seed 1 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	38.96	0.1971	[-0.519,0.422,0.113,0.519,-0.029,0.519]	[-0.054,-0.261,-0.025,0.815,0.006,0.514]	[0.000,1.000,1.000,0.000,1.000,0.000]
2	valid	11.50	0.0027	[-0.590,0.289,-0.323,0.414,-0.340,0.420]	[-0.753,0.185,0.380,0.083,0.492,-0.073]	[0.000,0.000,1.000,0.000,1.000,1.000]
3	valid	19.94	0.0103	[-0.446,0.165,-0.371,0.616,-0.208,0.462]	[0.144,0.471,0.582,-0.209,-0.507,-0.344]	[0.000,0.000,1.000,1.000,0.000,1.000]
4	valid	18.94	0.0162	[-0.402,0.344,-0.407,0.517,-0.113,0.524]	[0.671,0.475,-0.338,0.128,0.378,0.224]	[1.000,0.000,0.000,1.000,1.000,0.000]
5	valid	11.58	0.0018	[-0.384,0.329,-0.390,0.495,-0.312,0.501]	[-0.058,0.841,0.469,0.114,-0.194,-0.137]	[1.000,0.000,1.000,0.000,0.000,1.000]
6	valid	14.22	0.0040	[-0.346,0.296,-0.351,0.445,-0.519,0.451]	[0.680,-0.080,-0.007,-0.520,0.084,0.503]	[1.000,0.000,0.000,1.000,1.000,0.000]
7	valid	19.96	0.0081	[-0.660,0.194,-0.253,0.519,-0.258,0.357]	[0.110,-0.860,0.078,-0.437,0.148,-0.173]	[1.000,1.000,0.000,1.000,0.000,0.000]
8	valid	14.26	0.0039	[-0.599,0.347,-0.223,0.466,-0.392,0.316]	[0.455,0.286,0.502,0.400,-0.215,0.503]	[1.000,0.000,1.000,0.000,1.000,0.000]
9	valid	11.87	0.0031	[-0.592,0.343,-0.270,0.460,-0.387,0.312]	[-0.410,-0.016,-0.600,0.533,-0.107,0.420]	[0.000,1.000,0.000,0.000,1.000,1.000]
10	valid	9.49	0.0016	[-0.582,0.337,-0.323,0.452,-0.380,0.307]	[-0.446,-0.656,0.134,-0.312,-0.491,-0.117]	[0.000,1.000,1.000,1.000,0.000,0.000]
11	valid	7.69	0.0012	[-0.570,0.331,-0.372,0.444,-0.373,0.301]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,1.000,0.000,1.000,0.000]
12	valid	8.63	0.0017	[-0.583,0.295,-0.392,0.441,-0.340,0.332]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
13	valid	9.23	0.0019	[-0.587,0.282,-0.392,0.432,-0.339,0.347]	[0.562,-0.462,-0.473,0.091,-0.473,0.123]	[1.000,1.000,0.000,1.000,0.000,0.000]
14	valid	8.51	0.0013	[-0.584,0.300,-0.390,0.429,-0.337,0.345]	[-0.157,0.331,0.025,0.851,-0.329,0.181]	[1.000,0.000,1.000,0.000,0.000,1.000]
15	valid	6.62	0.0007	[-0.553,0.379,-0.363,0.405,-0.388,0.323]	[0.590,-0.332,0.430,0.510,0.303,0.075]	[1.000,1.000,1.000,0.000,0.000,0.000]
16	valid	5.54	0.0006	[-0.545,0.373,-0.398,0.398,-0.382,0.318]	[-0.013,0.610,-0.185,0.173,-0.744,0.105]	[1.000,0.000,0.000,1.000,0.000,1.000]
17	valid	5.96	0.0007	[-0.537,0.388,-0.421,0.383,-0.386,0.297]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
18	valid	5.96	0.0007	[-0.537,0.388,-0.421,0.383,-0.386,0.297]	[0.144,0.471,0.582,-0.209,-0.507,-0.344]	[0.000,0.000,1.000,1.000,0.000,1.000]
19	valid	5.96	0.0007	[-0.537,0.388,-0.421,0.383,-0.386,0.297]	[0.562,-0.462,-0.473,0.091,-0.473,0.123]	[1.000,1.000,0.000,1.000,0.000,0.000]
20	valid	5.96	0.0007	[-0.537,0.388,-0.421,0.383,-0.386,0.297]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]

Ordinary subset selection | seed 1 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	38.39	0.1301	[-0.210,0.398,-0.456,0.453,-0.436,-0.227]	[-0.110,0.309,-0.682,-0.173,0.630,0.033]	[0.000,0.000,0.000,1.000,1.000,1.000]
2	valid	25.09	0.0442	[-0.638,0.626,-0.503,0.041,-0.455,0.410]	[0.703,0.212,-0.262,0.152,-0.353,-0.494]	[1.000,0.000,0.000,1.000,0.000,1.000]
3	valid	21.73	0.0459	[-0.526,0.695,-0.603,0.506,-0.405,0.037]	[0.110,-0.860,0.078,-0.437,0.148,-0.173]	[1.000,1.000,0.000,1.000,0.000,0.000]
4	valid	14.78	0.0029	[-0.789,0.506,-0.423,0.342,-0.550,0.391]	[0.250,-0.124,0.112,-0.245,0.649,-0.654]	[1.000,0.000,0.000,0.000,1.000,1.000]
5	valid	8.62	0.0007	[-0.648,0.682,-0.661,0.508,-0.452,0.434]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
6	valid	9.47	0.0017	[-0.710,0.682,-0.698,0.466,-0.465,0.500]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
7	valid	4.81	0.0002	[-0.577,0.562,-0.620,0.533,-0.520,0.436]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
8	valid	5.95	0.0003	[-0.642,0.622,-0.698,0.576,-0.460,0.482]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
9	valid	7.58	0.0008	[-0.727,0.591,-0.673,0.487,-0.465,0.558]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
10	valid	6.78	0.0013	[-0.670,0.585,-0.689,0.494,-0.510,0.450]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
11	valid	6.44	0.0008	[-0.656,0.573,-0.743,0.539,-0.556,0.549]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
12	valid	6.02	0.0004	[-0.682,0.678,-0.659,0.575,-0.637,0.581]	[0.137,-0.573,0.372,-0.529,0.421,0.239]	[0.000,1.000,0.000,1.000,1.000,0.000]
13	valid	6.37	0.0012	[-0.651,0.593,-0.678,0.526,-0.624,0.531]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,1.000,0.000,1.000,0.000]
14	valid	9.37	0.0014	[-0.726,0.669,-0.673,0.531,-0.779,0.581]	[0.137,-0.573,0.372,-0.529,0.421,0.239]	[0.000,1.000,0.000,1.000,1.000,0.000]
15	valid	5.92	0.0004	[-0.720,0.697,-0.667,0.634,-0.700,0.636]	[-0.228,0.368,0.217,-0.793,-0.218,0.299]	[0.000,0.000,1.000,1.000,1.000,0.000]
16	valid	3.57	0.0001	[-0.739,0.660,-0.620,0.662,-0.633,0.562]	[-0.228,0.368,0.217,-0.793,-0.218,0.299]	[0.000,0.000,1.000,1.000,1.000,0.000]
17	valid	2.68	0.0002	[-0.741,0.645,-0.680,0.711,-0.641,0.629]	[0.673,-0.047,-0.566,0.224,-0.254,0.331]	[1.000,1.000,0.000,0.000,1.000,0.000]
18	valid	3.46	0.0001	[-0.765,0.704,-0.731,0.690,-0.682,0.602]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,1.000,0.000,1.000,0.000]
19	valid	1.75	0.0001	[-0.815,0.696,-0.744,0.770,-0.698,0.633]	[-0.720,0.239,0.286,0.256,-0.526,-0.006]	[0.000,1.000,1.000,0.000,0.000,1.000]
20	valid	1.74	0.0001	[-0.696,0.602,-0.621,0.656,-0.583,0.523]	[0.110,-0.860,0.078,-0.437,0.148,-0.173]	[0.000,1.000,0.000,1.000,0.000,1.000]

Ordinary subset selection | seed 1 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	49.96	0.1984	[-0.024,0.117,0.011,0.423,-0.003,0.242]	[-0.054,-0.261,-0.025,0.815,0.006,0.514]	[0.000,1.000,1.000,0.000,1.000,0.000]
2	valid	19.31	0.0063	[-0.138,0.211,-0.242,0.386,-0.163,0.233]	[-0.354,-0.292,-0.741,0.008,-0.489,-0.024]	[0.000,1.000,0.000,1.000,0.000,1.000]
3	valid	14.26	0.0040	[-0.256,0.178,-0.421,0.376,-0.325,0.297]	[-0.462,0.037,-0.577,-0.165,0.568,-0.320]	[0.000,0.000,0.000,1.000,1.000,1.000]
4	valid	8.67	0.0016	[-0.400,0.305,-0.331,0.416,-0.392,0.211]	[-0.611,0.612,-0.061,0.298,0.374,0.141]	[0.000,0.000,1.000,0.000,1.000,1.000]
5	valid	11.29	0.0044	[-0.362,0.313,-0.447,0.467,-0.401,0.197]	[-0.149,-0.313,-0.726,0.471,0.329,-0.154]	[0.000,1.000,0.000,0.000,1.000,1.000]
6	valid	14.58	0.0045	[-0.410,0.238,-0.402,0.575,-0.431,0.222]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
7	valid	21.25	0.0147	[-0.452,0.202,-0.501,0.682,-0.338,0.134]	[-0.410,-0.016,-0.600,0.533,-0.107,0.420]	[0.000,1.000,0.000,0.000,1.000,1.000]
8	valid	18.78	0.0182	[-0.494,0.191,-0.460,0.610,-0.437,0.149]	[-0.438,-0.103,-0.054,-0.001,0.801,-0.391]	[0.000,1.000,0.000,0.000,1.000,1.000]
9	valid	19.47	0.0222	[-0.534,0.215,-0.417,0.571,-0.514,0.119]	[0.490,0.520,-0.154,-0.215,0.644,-0.073]	[1.000,0.000,0.000,1.000,1.000,0.000]
10	valid	22.70	0.0213	[-0.618,0.175,-0.342,0.591,-0.542,0.121]	[-0.676,0.066,-0.050,0.440,0.480,0.334]	[0.000,1.000,1.000,0.000,1.000,0.000]
11	valid	19.95	0.0221	[-0.584,0.184,-0.425,0.563,-0.513,0.134]	[0.087,0.348,0.805,-0.110,-0.091,0.451]	[1.000,0.000,1.000,1.000,0.000,0.000]
12	valid	21.47	0.0203	[-0.604,0.154,-0.466,0.501,-0.559,0.131]	[-0.462,0.037,-0.577,-0.165,0.568,-0.320]	[0.000,0.000,0.000,1.000,1.000,1.000]
13	valid	24.77	0.0159	[-0.619,0.134,-0.410,0.416,-0.680,0.163]	[0.250,-0.124,0.112,-0.245,0.649,-0.654]	[1.000,0.000,0.000,0.000,1.000,1.000]
14	valid	22.93	0.0121	[-0.585,0.148,-0.379,0.496,-0.647,0.171]	[-0.106,0.510,0.045,0.750,0.123,-0.386]	[0.000,0.000,1.000,0.000,1.000,1.000]
15	valid	23.34	0.0123	[-0.553,0.132,-0.355,0.585,-0.626,0.179]	[0.124,-0.072,-0.090,-0.854,-0.201,-0.450]	[1.000,0.000,0.000,1.000,0.000,1.000]
16	valid	25.22	0.0175	[-0.608,0.114,-0.287,0.613,-0.622,0.190]	[-0.600,-0.024,-0.134,0.484,0.360,0.507]	[0.000,1.000,1.000,0.000,1.000,0.000]
17	valid	25.39	0.0313	[-0.665,0.095,-0.333,0.642,-0.546,0.154]	[-0.507,0.208,-0.667,0.392,-0.094,0.304]	[0.000,1.000,0.000,0.000,1.000,1.000]
18	valid	26.11	0.0385	[-0.705,0.088,-0.331,0.591,-0.574,0.149]	[-0.566,-0.227,0.409,-0.227,-0.544,0.337]	[0.000,1.000,1.000,1.000,0.000,0.000]
19	valid	28.25	0.0239	[-0.735,0.095,-0.272,0.526,-0.658,0.155]	[0.382,0.461,-0.285,0.068,-0.646,0.372]	[1.000,0.000,1.000,1.000,0.000,0.000]
20	valid	29.08	0.0246	[-0.759,0.103,-0.257,0.496,-0.676,0.153]	[-0.474,-0.567,0.246,0.247,-0.463,-0.344]	[0.000,1.000,1.000,0.000,0.000,1.000]

Ordinary subset selection | seed 2 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	16.00	0.0050	[-0.201,0.504,-0.505,0.386,-0.204,0.510]	[0.091,0.539,-0.134,-0.384,0.094,0.726]	[1.000,0.000,0.000,1.000,1.000,0.000]
2	valid	15.37	0.0098	[-0.169,0.326,-0.445,0.595,-0.451,0.331]	[0.127,-0.020,-0.673,0.298,-0.642,0.172]	[1.000,1.000,0.000,0.000,0.000,1.000]
3	valid	18.82	0.0088	[-0.491,0.233,-0.435,0.563,-0.370,0.248]	[-0.507,0.208,-0.667,0.392,-0.094,0.304]	[0.000,1.000,0.000,0.000,1.000,1.000]
4	valid	18.43	0.0088	[-0.489,0.248,-0.434,0.561,-0.369,0.247]	[-0.234,-0.246,-0.668,-0.642,0.156,0.050]	[0.000,1.000,0.000,1.000,1.000,0.000]
5	valid	19.78	0.0249	[-0.427,0.450,-0.375,0.492,-0.451,0.172]	[-0.157,0.331,0.025,0.851,-0.329,0.181]	[1.000,0.000,1.000,0.000,0.000,1.000]
6	valid	19.78	0.0249	[-0.427,0.450,-0.375,0.492,-0.451,0.172]	[0.605,-0.163,0.200,0.435,-0.421,0.448]	[1.000,1.000,1.000,0.000,0.000,0.000]
7	valid	14.10	0.0050	[-0.417,0.435,-0.363,0.487,-0.436,0.278]	[0.287,0.090,-0.636,0.166,0.235,0.650]	[1.000,1.000,0.000,0.000,1.000,0.000]
8	valid	14.10	0.0050	[-0.417,0.435,-0.363,0.487,-0.436,0.278]	[-0.157,0.331,0.025,0.851,-0.329,0.181]	[1.000,0.000,1.000,0.000,0.000,1.000]
9	valid	12.30	0.0039	[-0.399,0.491,-0.339,0.475,-0.377,0.342]	[0.455,0.286,0.502,0.400,-0.215,0.503]	[1.000,0.000,1.000,0.000,1.000,0.000]
10	valid	9.81	0.0020	[-0.386,0.463,-0.411,0.462,-0.364,0.349]	[0.054,0.210,0.474,-0.512,0.301,-0.612]	[0.000,0.000,1.000,1.000,0.000,1.000]
11	valid	9.81	0.0020	[-0.386,0.463,-0.411,0.462,-0.364,0.349]	[0.399,0.619,0.220,0.492,-0.304,-0.274]	[1.000,0.000,1.000,0.000,0.000,1.000]
12	valid	14.50	0.0030	[-0.312,0.407,-0.359,0.625,-0.355,0.301]	[0.360,0.319,0.652,-0.260,0.518,-0.080]	[0.000,0.000,1.000,1.000,1.000,0.000]
13	valid	14.50	0.0030	[-0.312,0.407,-0.359,0.625,-0.355,0.301]	[0.287,0.090,-0.636,0.166,0.235,0.650]	[1.000,1.000,0.000,0.000,1.000,0.000]
14	valid	14.50	0.0030	[-0.312,0.407,-0.359,0.625,-0.355,0.301]	[-0.462,0.037,-0.577,-0.165,0.568,-0.320]	[0.000,0.000,0.000,1.000,1.000,1.000]
15	valid	14.50	0.0030	[-0.312,0.407,-0.359,0.625,-0.355,0.301]	[0.012,-0.128,-0.829,-0.485,-0.067,0.237]	[1.000,1.000,0.000,1.000,0.000,0.000]
16	valid	14.50	0.0030	[-0.312,0.407,-0.359,0.625,-0.355,0.301]	[-0.157,0.331,0.025,0.851,-0.329,0.181]	[1.000,0.000,1.000,0.000,0.000,1.000]
17	valid	14.50	0.0030	[-0.312,0.407,-0.359,0.625,-0.355,0.301]	[-0.474,-0.567,0.246,0.247,-0.463,-0.344]	[0.000,1.000,1.000,0.000,0.000,1.000]
18	valid	14.50	0.0030	[-0.312,0.407,-0.359,0.625,-0.355,0.301]	[-0.107,0.581,-0.614,-0.344,0.323,-0.230]	[0.000,0.000,0.000,1.000,1.000,1.000]
19	valid	14.50	0.0030	[-0.312,0.407,-0.359,0.625,-0.355,0.301]	[0.230,-0.648,-0.058,0.014,-0.411,-0.595]	[1.000,1.000,0.000,0.000,0.000,1.000]
20	valid	14.50	0.0030	[-0.312,0.407,-0.359,0.625,-0.355,0.301]	[0.091,0.539,-0.134,-0.384,0.094,0.726]	[1.000,0.000,0.000,1.000,1.000,0.000]

Ordinary subset selection | seed 2 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	16.00	0.0050	[-0.201,0.504,-0.505,0.386,-0.204,0.510]	[0.091,0.539,-0.134,-0.384,0.094,0.726]	[1.000,0.000,0.000,1.000,1.000,0.000]
2	valid	21.41	0.0108	[-0.464,0.299,-0.176,0.535,-0.527,0.317]	[0.509,-0.134,-0.723,-0.311,0.323,0.001]	[1.000,0.000,0.000,1.000,1.000,0.000]
3	valid	9.43	0.0036	[-0.356,0.230,-0.466,0.410,-0.404,0.521]	[-0.169,-0.258,0.439,-0.640,-0.422,-0.354]	[0.000,0.000,1.000,1.000,0.000,1.000]
4	valid	16.02	0.0079	[-0.273,0.176,-0.357,0.644,-0.346,0.482]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[0.000,0.000,0.000,1.000,1.000,1.000]
5	valid	8.19	0.0006	[-0.261,0.325,-0.392,0.555,-0.318,0.513]	[0.671,0.475,-0.338,0.128,0.378,0.224]	[1.000,0.000,0.000,1.000,1.000,0.000]
6	valid	10.63	0.0031	[-0.238,0.297,-0.358,0.508,-0.499,0.469]	[0.680,-0.080,-0.007,-0.520,0.084,0.503]	[1.000,0.000,0.000,1.000,1.000,0.000]
7	valid	10.50	0.0024	[-0.360,0.265,-0.331,0.573,-0.400,0.450]	[-0.459,0.384,-0.073,0.327,0.227,0.692]	[0.000,1.000,1.000,0.000,1.000,0.000]
8	valid	8.66	0.0028	[-0.343,0.252,-0.436,0.546,-0.382,0.429]	[-0.270,0.085,-0.159,-0.836,-0.128,0.423]	[0.000,1.000,0.000,1.000,1.000,0.000]
9	valid	7.77	0.0022	[-0.372,0.320,-0.411,0.531,-0.402,0.383]	[0.446,-0.536,0.405,-0.246,0.114,0.525]	[1.000,1.000,1.000,0.000,0.000,0.000]
10	valid	5.08	0.0014	[-0.358,0.308,-0.395,0.510,-0.386,0.460]	[0.562,-0.462,-0.473,0.091,-0.473,0.123]	[1.000,1.000,0.000,1.000,0.000,0.000]
11	valid	5.57	0.0015	[-0.344,0.296,-0.470,0.490,-0.371,0.443]	[-0.446,-0.656,0.134,-0.312,-0.491,-0.117]	[0.000,1.000,1.000,1.000,0.000,0.000]
12	valid	4.19	0.0005	[-0.351,0.336,-0.441,0.483,-0.396,0.425]	[0.382,0.461,-0.285,0.068,-0.646,0.372]	[1.000,0.000,1.000,1.000,0.000,0.000]
13	valid	5.53	0.0012	[-0.305,0.318,-0.459,0.419,-0.433,0.482]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,0.000,0.000,1.000,1.000]
14	valid	5.26	0.0017	[-0.373,0.298,-0.430,0.469,-0.406,0.451]	[0.360,0.319,0.652,-0.260,0.518,-0.080]	[1.000,0.000,1.000,0.000,1.000,0.000]
15	valid	3.76	0.0001	[-0.368,0.334,-0.424,0.463,-0.400,0.445]	[0.455,0.286,0.502,0.400,-0.215,0.503]	[1.000,0.000,1.000,0.000,1.000,0.000]
16	valid	5.47	0.0005	[-0.380,0.378,-0.391,0.483,-0.397,0.411]	[0.703,0.212,-0.262,0.152,-0.353,-0.494]	[1.000,0.000,0.000,1.000,0.000,1.000]
17	valid	6.20	0.0010	[-0.374,0.372,-0.385,0.476,-0.427,0.405]	[-0.228,0.368,0.217,-0.793,-0.218,0.299]	[1.000,0.000,1.000,1.000,0.000,0.000]
18	valid	4.17	0.0001	[-0.364,0.362,-0.407,0.462,-0.415,0.431]	[0.440,-0.213,0.638,0.046,0.434,-0.405]	[0.000,0.000,1.000,0.000,1.000,1.000]
19	valid	4.91	0.0004	[-0.369,0.367,-0.388,0.484,-0.407,0.423]	[0.673,-0.047,-0.566,0.224,-0.254,0.331]	[1.000,1.000,0.000,0.000,1.000,0.000]
20	valid	5.07	0.0005	[-0.365,0.388,-0.384,0.480,-0.403,0.419]	[0.590,-0.332,0.430,0.510,0.303,0.075]	[1.000,1.000,1.000,0.000,0.000,0.000]

Ordinary subset selection | seed 2 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	32.51	0.0567	[-0.532,0.369,-0.026,0.167,-0.441,0.426]	[-0.186,0.078,-0.020,-0.271,-0.784,-0.521]	[0.000,0.000,1.000,1.000,0.000,1.000]
2	valid	29.71	0.0327	[-0.097,0.521,-0.073,0.293,-0.532,0.558]	[-0.228,0.368,0.217,-0.793,-0.218,0.299]	[1.000,0.000,1.000,1.000,0.000,0.000]
3	valid	10.09	0.0029	[-0.376,0.496,-0.552,0.489,-0.455,0.379]	[-0.507,0.208,-0.667,0.392,-0.094,0.304]	[0.000,1.000,0.000,0.000,1.000,1.000]
4	valid	9.86	0.0033	[-0.314,0.317,-0.616,0.611,-0.388,0.522]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,0.000,1.000,1.000,0.000,1.000]
5	valid	7.03	0.0027	[-0.455,0.393,-0.608,0.712,-0.604,0.601]	[0.796,0.423,-0.128,0.149,-0.334,-0.194]	[1.000,0.000,1.000,0.000,0.000,1.000]
6	valid	12.43	0.0071	[-0.382,0.249,-0.604,0.736,-0.508,0.555]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,0.000,1.000,1.000,0.000,1.000]
7	valid	10.77	0.0023	[-0.359,0.293,-0.583,0.683,-0.376,0.547]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,1.000,0.000,0.000,1.000]
8	valid	5.54	0.0006	[-0.468,0.434,-0.661,0.702,-0.461,0.702]	[0.382,0.461,-0.285,0.068,-0.646,0.372]	[1.000,0.000,1.000,1.000,0.000,0.000]
9	valid	6.28	0.0014	[-0.416,0.353,-0.525,0.627,-0.392,0.556]	[-0.507,0.208,-0.667,0.392,-0.094,0.304]	[0.000,1.000,0.000,0.000,1.000,1.000]
10	valid	5.92	0.0010	[-0.466,0.443,-0.656,0.746,-0.482,0.633]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
11	valid	5.30	0.0006	[-0.515,0.465,-0.701,0.737,-0.521,0.658]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
12	valid	6.64	0.0017	[-0.548,0.481,-0.706,0.783,-0.525,0.623]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
13	valid	5.69	0.0007	[-0.448,0.457,-0.648,0.770,-0.506,0.646]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,0.000,1.000,1.000,0.000,1.000]
14	valid	5.55	0.0010	[-0.362,0.357,-0.541,0.590,-0.421,0.508]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,0.000,0.000,1.000,1.000]
15	valid	2.42	0.0002	[-0.525,0.537,-0.677,0.761,-0.603,0.714]	[0.382,0.461,-0.285,0.068,-0.646,0.372]	[1.000,0.000,1.000,1.000,0.000,0.000]
16	valid	3.95	0.0010	[-0.495,0.439,-0.577,0.638,-0.463,0.612]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,1.000,0.000,0.000,1.000]
17	valid	4.75	0.0006	[-0.508,0.490,-0.632,0.783,-0.570,0.655]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,0.000,0.000,1.000,1.000]
18	valid	3.44	0.0002	[-0.529,0.507,-0.638,0.772,-0.605,0.716]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,0.000,0.000,1.000,1.000]
19	valid	2.63	0.0002	[-0.501,0.484,-0.591,0.672,-0.505,0.647]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
20	valid	3.97	0.0006	[-0.495,0.477,-0.591,0.727,-0.533,0.632]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]

Ordinary subset selection | seed 2 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	34.48	0.0227	[-0.044,0.284,-0.065,0.194,-0.045,0.411]	[0.091,0.539,-0.134,-0.384,0.094,0.726]	[1.000,0.000,0.000,1.000,1.000,0.000]
2	valid	12.88	0.0124	[-0.082,0.263,-0.317,0.296,-0.279,0.286]	[0.127,-0.020,-0.673,0.298,-0.642,0.172]	[1.000,1.000,0.000,0.000,0.000,1.000]
3	valid	25.80	0.0153	[-0.207,0.169,-0.565,0.405,-0.215,0.160]	[-0.507,0.208,-0.667,0.392,-0.094,0.304]	[0.000,1.000,0.000,0.000,1.000,1.000]
4	valid	32.32	0.0299	[-0.208,0.178,-0.739,0.534,-0.198,0.132]	[-0.234,-0.246,-0.668,-0.642,0.156,0.050]	[0.000,1.000,0.000,1.000,1.000,0.000]
5	valid	33.81	0.0388	[-0.160,0.199,-0.652,0.744,-0.219,0.095]	[-0.157,0.331,0.025,0.851,-0.329,0.181]	[1.000,0.000,1.000,0.000,0.000,1.000]
6	valid	31.67	0.0244	[-0.224,0.187,-0.608,0.814,-0.251,0.132]	[0.605,-0.163,0.200,0.435,-0.421,0.448]	[1.000,1.000,1.000,0.000,0.000,0.000]
7	valid	31.29	0.0298	[-0.227,0.145,-0.738,0.748,-0.244,0.182]	[0.287,0.090,-0.636,0.166,0.235,0.650]	[1.000,1.000,0.000,0.000,1.000,0.000]
8	valid	34.78	0.0308	[-0.184,0.151,-0.651,0.933,-0.248,0.145]	[-0.157,0.331,0.025,0.851,-0.329,0.181]	[1.000,0.000,1.000,0.000,0.000,1.000]
9	valid	35.90	0.0272	[-0.202,0.149,-0.717,0.958,-0.167,0.168]	[0.455,0.286,0.502,0.400,-0.215,0.503]	[1.000,0.000,1.000,0.000,1.000,0.000]
10	valid	38.07	0.0294	[-0.161,0.138,-0.751,1.023,-0.114,0.195]	[0.054,0.210,0.474,-0.512,0.301,-0.612]	[0.000,0.000,1.000,1.000,0.000,1.000]
11	valid	38.48	0.0287	[-0.165,0.159,-0.696,1.083,-0.113,0.186]	[0.399,0.619,0.220,0.492,-0.304,-0.274]	[1.000,0.000,1.000,0.000,0.000,1.000]
12	valid	39.50	0.0274	[-0.120,0.160,-0.803,1.049,-0.125,0.155]	[0.360,0.319,0.652,-0.260,0.518,-0.080]	[0.000,0.000,1.000,1.000,1.000,0.000]
13	valid	39.43	0.0299	[-0.118,0.132,-0.906,0.968,-0.120,0.178]	[0.287,0.090,-0.636,0.166,0.235,0.650]	[1.000,1.000,0.000,0.000,1.000,0.000]
14	valid	39.85	0.0303	[-0.125,0.113,-1.000,0.883,-0.133,0.175]	[-0.462,0.037,-0.577,-0.165,0.568,-0.320]	[0.000,0.000,0.000,1.000,1.000,1.000]
15	valid	42.01	0.0322	[-0.105,0.099,-1.090,0.865,-0.115,0.159]	[0.012,-0.128,-0.829,-0.485,-0.067,0.237]	[1.000,1.000,0.000,1.000,0.000,0.000]
16	valid	42.31	0.0333	[-0.091,0.099,-0.992,1.003,-0.114,0.136]	[-0.157,0.331,0.025,0.851,-0.329,0.181]	[1.000,0.000,1.000,0.000,0.000,1.000]
17	valid	41.49	0.0333	[-0.099,0.113,-0.976,0.987,-0.124,0.140]	[-0.474,-0.567,0.246,0.247,-0.463,-0.344]	[0.000,1.000,1.000,0.000,0.000,1.000]
18	valid	42.25	0.0336	[-0.087,0.119,-1.040,0.948,-0.119,0.129]	[-0.107,0.581,-0.614,-0.344,0.323,-0.230]	[0.000,0.000,0.000,1.000,1.000,1.000]
19	valid	40.66	0.0325	[-0.090,0.144,-1.014,0.911,-0.131,0.152]	[0.230,-0.648,-0.058,0.014,-0.411,-0.595]	[1.000,1.000,0.000,0.000,0.000,1.000]
20	valid	39.85	0.0291	[-0.085,0.156,-0.967,0.938,-0.124,0.174]	[0.091,0.539,-0.134,-0.384,0.094,0.726]	[1.000,0.000,0.000,1.000,1.000,0.000]

Ordinary subset selection | seed 3 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	34.60	0.1414	[-0.475,-0.235,-0.447,0.405,-0.471,0.366]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
2	valid	36.37	0.1494	[-0.511,-0.254,-0.323,0.498,-0.360,0.438]	[-0.308,0.274,0.274,0.341,-0.084,0.795]	[0.000,1.000,1.000,0.000,1.000,0.000]
3	valid	17.47	0.0297	[-0.490,0.083,-0.375,0.465,-0.512,0.366]	[0.012,-0.128,-0.829,-0.485,-0.067,0.237]	[1.000,1.000,0.000,1.000,0.000,0.000]
4	valid	12.91	0.0022	[-0.529,0.305,-0.343,0.548,-0.338,0.309]	[0.796,0.423,-0.128,0.149,-0.334,-0.194]	[1.000,0.000,1.000,0.000,0.000,1.000]
5	valid	15.96	0.0027	[-0.448,0.258,-0.290,0.464,-0.592,0.289]	[0.070,0.336,0.899,0.011,0.083,-0.260]	[0.000,0.000,1.000,0.000,1.000,1.000]
6	valid	14.04	0.0030	[-0.436,0.251,-0.282,0.451,-0.576,0.363]	[0.311,-0.695,-0.538,0.240,0.088,-0.257]	[1.000,1.000,0.000,0.000,0.000,1.000]
7	valid	14.04	0.0030	[-0.436,0.251,-0.282,0.451,-0.576,0.363]	[-0.474,-0.567,0.246,0.247,-0.463,-0.344]	[0.000,1.000,1.000,0.000,0.000,1.000]
8	valid	11.84	0.0044	[-0.442,0.393,-0.422,0.453,-0.463,0.230]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
9	valid	11.84	0.0044	[-0.442,0.393,-0.422,0.453,-0.463,0.230]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
10	valid	12.08	0.0044	[-0.445,0.401,-0.420,0.447,-0.462,0.226]	[0.703,0.212,-0.262,0.152,-0.353,-0.494]	[1.000,0.000,0.000,1.000,0.000,1.000]
11	valid	3.22	0.0003	[-0.421,0.380,-0.397,0.423,-0.437,0.390]	[0.562,-0.462,-0.473,0.091,-0.473,0.123]	[1.000,1.000,0.000,1.000,0.000,0.000]
12	valid	3.22	0.0003	[-0.421,0.380,-0.397,0.423,-0.437,0.390]	[0.094,0.765,0.432,-0.347,-0.201,-0.242]	[0.000,0.000,1.000,1.000,0.000,1.000]
13	valid	3.22	0.0003	[-0.421,0.380,-0.397,0.423,-0.437,0.390]	[0.230,-0.648,-0.058,0.014,-0.411,-0.595]	[1.000,1.000,0.000,0.000,0.000,1.000]
14	valid	6.91	0.0013	[-0.443,0.439,-0.385,0.403,-0.425,0.346]	[0.590,-0.332,0.430,0.510,0.303,0.075]	[1.000,1.000,1.000,0.000,0.000,0.000]
15	valid	6.91	0.0013	[-0.443,0.439,-0.385,0.403,-0.425,0.346]	[0.573,0.327,0.513,0.058,0.477,-0.266]	[1.000,0.000,1.000,0.000,1.000,0.000]
16	valid	6.91	0.0013	[-0.443,0.439,-0.385,0.403,-0.425,0.346]	[-0.446,-0.656,0.134,-0.312,-0.491,-0.117]	[0.000,1.000,1.000,1.000,0.000,0.000]
17	valid	6.91	0.0013	[-0.443,0.439,-0.385,0.403,-0.425,0.346]	[-0.600,-0.024,-0.134,0.484,0.360,0.507]	[0.000,1.000,1.000,0.000,1.000,0.000]
18	valid	6.91	0.0013	[-0.443,0.439,-0.385,0.403,-0.425,0.346]	[-0.110,0.309,-0.682,-0.173,0.630,0.033]	[0.000,0.000,0.000,1.000,1.000,1.000]
19	valid	6.91	0.0013	[-0.443,0.439,-0.385,0.403,-0.425,0.346]	[-0.094,-0.608,-0.324,-0.693,-0.014,-0.190]	[0.000,1.000,0.000,1.000,0.000,1.000]
20	valid	6.91	0.0013	[-0.443,0.439,-0.385,0.403,-0.425,0.346]	[0.622,-0.387,0.530,0.177,-0.383,0.065]	[1.000,1.000,1.000,0.000,0.000,0.000]

Ordinary subset selection | seed 3 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	34.60	0.1414	[-0.475,-0.235,-0.447,0.405,-0.471,0.366]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
2	valid	13.68	0.0049	[-0.448,0.287,-0.564,0.269,-0.456,0.343]	[0.399,0.619,0.220,0.492,-0.304,-0.274]	[1.000,0.000,1.000,0.000,0.000,1.000]
3	valid	18.09	0.0177	[-0.310,0.141,-0.540,0.518,-0.315,0.475]	[0.054,0.210,0.474,-0.512,0.301,-0.612]	[0.000,0.000,1.000,1.000,0.000,1.000]
4	valid	13.63	0.0019	[-0.279,0.371,-0.447,0.466,-0.289,0.533]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[0.000,1.000,0.000,1.000,0.000,1.000]
5	valid	15.11	0.0061	[-0.419,0.358,-0.324,0.478,-0.216,0.562]	[-0.228,0.368,0.217,-0.793,-0.218,0.299]	[0.000,0.000,1.000,1.000,1.000,0.000]
6	valid	15.54	0.0044	[-0.510,0.322,-0.257,0.456,-0.249,0.548]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
7	valid	5.97	0.0004	[-0.461,0.306,-0.380,0.412,-0.367,0.495]	[-0.637,-0.481,-0.339,-0.288,-0.353,0.200]	[0.000,1.000,0.000,1.000,0.000,1.000]
8	valid	6.15	0.0008	[-0.475,0.311,-0.344,0.395,-0.419,0.478]	[0.673,-0.047,-0.566,0.224,-0.254,0.331]	[1.000,1.000,0.000,1.000,0.000,0.000]
9	valid	5.71	0.0003	[-0.463,0.372,-0.336,0.386,-0.410,0.466]	[0.455,0.286,0.502,0.400,-0.215,0.503]	[1.000,0.000,1.000,0.000,1.000,0.000]
10	valid	6.92	0.0007	[-0.472,0.374,-0.325,0.363,-0.452,0.442]	[0.590,-0.332,0.430,0.510,0.303,0.075]	[1.000,0.000,1.000,0.000,1.000,0.000]
11	valid	6.27	0.0011	[-0.466,0.370,-0.321,0.390,-0.447,0.437]	[0.077,0.627,-0.555,0.059,0.536,0.048]	[1.000,0.000,0.000,0.000,1.000,1.000]
12	valid	3.38	0.0005	[-0.455,0.361,-0.380,0.381,-0.437,0.427]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
13	valid	2.75	0.0005	[-0.450,0.357,-0.403,0.377,-0.432,0.422]	[-0.446,-0.656,0.134,-0.312,-0.491,-0.117]	[0.000,1.000,1.000,1.000,0.000,0.000]
14	valid	3.09	0.0001	[-0.468,0.352,-0.400,0.373,-0.417,0.429]	[-0.410,-0.016,-0.600,0.533,-0.107,0.420]	[0.000,1.000,0.000,0.000,1.000,1.000]
15	valid	3.23	0.0005	[-0.466,0.351,-0.398,0.372,-0.426,0.427]	[0.440,-0.213,0.638,0.046,0.434,-0.405]	[1.000,0.000,1.000,0.000,1.000,0.000]
16	valid	3.23	0.0005	[-0.466,0.351,-0.398,0.372,-0.426,0.427]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
17	valid	3.23	0.0005	[-0.466,0.351,-0.398,0.372,-0.426,0.427]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
18	valid	3.23	0.0005	[-0.466,0.351,-0.398,0.372,-0.426,0.427]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
19	valid	3.23	0.0005	[-0.466,0.351,-0.398,0.372,-0.426,0.427]	[0.077,0.627,-0.555,0.059,0.536,0.048]	[1.000,0.000,0.000,0.000,1.000,1.000]
20	valid	3.23	0.0005	[-0.466,0.351,-0.398,0.372,-0.426,0.427]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[0.000,1.000,0.000,1.000,0.000,1.000]

Ordinary subset selection | seed 3 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	41.09	0.1357	[-0.292,0.349,-0.389,-0.230,-0.389,0.382]	[0.562,-0.462,-0.473,0.091,-0.473,0.123]	[1.000,1.000,0.000,1.000,0.000,0.000]
2	valid	18.52	0.0047	[-0.449,0.577,-0.341,0.207,-0.408,0.560]	[0.141,-0.170,0.112,0.333,0.056,0.908]	[1.000,1.000,1.000,0.000,0.000,0.000]
3	valid	19.72	0.0088	[-0.316,0.610,-0.680,0.446,-0.281,0.422]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
4	valid	12.23	0.0030	[-0.553,0.643,-0.624,0.434,-0.493,0.399]	[0.250,-0.124,0.112,-0.245,0.649,-0.654]	[1.000,0.000,0.000,0.000,1.000,1.000]
5	valid	10.90	0.0024	[-0.582,0.624,-0.724,0.537,-0.460,0.460]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
6	valid	10.27	0.0015	[-0.647,0.752,-0.806,0.573,-0.606,0.558]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
7	valid	6.95	0.0008	[-0.663,0.599,-0.754,0.594,-0.581,0.534]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
8	valid	6.81	0.0010	[-0.686,0.665,-0.736,0.583,-0.584,0.581]	[0.250,-0.124,0.112,-0.245,0.649,-0.654]	[1.000,0.000,0.000,0.000,1.000,1.000]
9	valid	8.94	0.0018	[-0.638,0.699,-0.792,0.550,-0.624,0.582]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[0.000,1.000,0.000,1.000,0.000,1.000]
10	valid	7.03	0.0011	[-0.636,0.646,-0.743,0.544,-0.575,0.681]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
11	valid	6.35	0.0006	[-0.668,0.666,-0.789,0.603,-0.618,0.654]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
12	valid	6.71	0.0008	[-0.702,0.707,-0.764,0.615,-0.590,0.649]	[-0.410,-0.016,-0.600,0.533,-0.107,0.420]	[0.000,1.000,0.000,0.000,1.000,1.000]
13	valid	5.28	0.0005	[-0.689,0.652,-0.651,0.558,-0.668,0.624]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,0.000,0.000,1.000,1.000]
14	valid	5.25	0.0009	[-0.717,0.652,-0.699,0.645,-0.803,0.644]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[0.000,1.000,0.000,1.000,0.000,1.000]
15	valid	4.56	0.0001	[-0.731,0.689,-0.701,0.609,-0.671,0.714]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
16	valid	5.51	0.0007	[-0.732,0.745,-0.720,0.648,-0.750,0.684]	[0.137,-0.573,0.372,-0.529,0.421,0.239]	[0.000,1.000,0.000,1.000,1.000,0.000]
17	valid	5.54	0.0004	[-0.630,0.626,-0.546,0.611,-0.655,0.575]	[-0.308,0.274,0.274,0.341,-0.084,0.795]	[0.000,1.000,1.000,0.000,1.000,0.000]
18	valid	4.97	0.0006	[-0.685,0.618,-0.687,0.587,-0.758,0.664]	[0.590,-0.332,0.430,0.510,0.303,0.075]	[1.000,0.000,1.000,0.000,1.000,0.000]
19	valid	3.62	0.0003	[-0.710,0.653,-0.686,0.638,-0.713,0.654]	[0.570,0.398,-0.261,-0.403,0.287,-0.451]	[1.000,0.000,0.000,1.000,0.000,1.000]
20	valid	3.35	0.0004	[-0.754,0.656,-0.709,0.670,-0.789,0.743]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,0.000,0.000,1.000,1.000]

Ordinary subset selection | seed 3 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	31.93	0.1110	[-0.287,-0.038,-0.087,0.391,-0.239,0.236]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
2	valid	36.21	0.1456	[-0.341,-0.099,-0.142,0.462,-0.170,0.496]	[-0.308,0.274,0.274,0.341,-0.084,0.795]	[0.000,1.000,1.000,0.000,1.000,0.000]
3	valid	33.01	0.1039	[-0.294,-0.063,-0.301,0.573,-0.158,0.509]	[0.012,-0.128,-0.829,-0.485,-0.067,0.237]	[1.000,1.000,0.000,1.000,0.000,0.000]
4	valid	28.34	0.0536	[-0.464,-0.000,-0.239,0.558,-0.194,0.512]	[0.796,0.423,-0.128,0.149,-0.334,-0.194]	[1.000,0.000,1.000,0.000,0.000,1.000]
5	valid	24.74	0.0423	[-0.411,0.031,-0.376,0.517,-0.190,0.541]	[0.070,0.336,0.899,0.011,0.083,-0.260]	[0.000,0.000,1.000,0.000,1.000,1.000]
6	valid	22.98	0.0316	[-0.429,0.097,-0.454,0.514,-0.151,0.543]	[0.311,-0.695,-0.538,0.240,0.088,-0.257]	[1.000,1.000,0.000,0.000,0.000,1.000]
7	valid	19.86	0.0142	[-0.482,0.142,-0.435,0.492,-0.183,0.555]	[-0.474,-0.567,0.246,0.247,-0.463,-0.344]	[0.000,1.000,1.000,0.000,0.000,1.000]
8	valid	23.86	0.0181	[-0.359,0.196,-0.545,0.659,-0.140,0.411]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
9	valid	26.15	0.0217	[-0.384,0.153,-0.486,0.755,-0.151,0.421]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
10	valid	21.76	0.0188	[-0.473,0.152,-0.488,0.627,-0.162,0.471]	[0.703,0.212,-0.262,0.152,-0.353,-0.494]	[1.000,0.000,0.000,1.000,0.000,1.000]
11	valid	19.32	0.0114	[-0.552,0.173,-0.543,0.513,-0.184,0.433]	[0.562,-0.462,-0.473,0.091,-0.473,0.123]	[1.000,1.000,0.000,1.000,0.000,0.000]
12	valid	18.68	0.0106	[-0.481,0.212,-0.578,0.529,-0.180,0.429]	[0.094,0.765,0.432,-0.347,-0.201,-0.242]	[0.000,0.000,1.000,1.000,0.000,1.000]
13	valid	15.78	0.0092	[-0.474,0.252,-0.529,0.475,-0.194,0.495]	[0.230,-0.648,-0.058,0.014,-0.411,-0.595]	[1.000,1.000,0.000,0.000,0.000,1.000]
14	valid	18.86	0.0119	[-0.530,0.252,-0.550,0.512,-0.143,0.438]	[0.590,-0.332,0.430,0.510,0.303,0.075]	[1.000,1.000,1.000,0.000,0.000,0.000]
15	valid	20.83	0.0141	[-0.603,0.257,-0.609,0.462,-0.158,0.341]	[0.573,0.327,0.513,0.058,0.477,-0.266]	[1.000,0.000,1.000,0.000,1.000,0.000]
16	valid	20.73	0.0141	[-0.639,0.296,-0.573,0.465,-0.171,0.291]	[-0.446,-0.656,0.134,-0.312,-0.491,-0.117]	[0.000,1.000,1.000,1.000,0.000,0.000]
17	valid	21.68	0.0145	[-0.715,0.262,-0.476,0.497,-0.174,0.314]	[-0.600,-0.024,-0.134,0.484,0.360,0.507]	[0.000,1.000,1.000,0.000,1.000,0.000]
18	valid	20.89	0.0159	[-0.673,0.265,-0.546,0.478,-0.197,0.281]	[-0.110,0.309,-0.682,-0.173,0.630,0.033]	[0.000,0.000,0.000,1.000,1.000,1.000]
19	valid	20.82	0.0142	[-0.622,0.290,-0.545,0.539,-0.177,0.269]	[-0.094,-0.608,-0.324,-0.693,-0.014,-0.190]	[0.000,1.000,0.000,1.000,0.000,1.000]
20	valid	22.68	0.0169	[-0.673,0.289,-0.571,0.497,-0.176,0.238]	[0.622,-0.387,0.530,0.177,-0.383,0.065]	[1.000,1.000,1.000,0.000,0.000,0.000]

Ordinary subset selection | seed 4 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	17.69	0.0046	[-0.275,0.494,-0.380,0.483,-0.483,0.263]	[0.135,0.489,0.691,0.326,-0.346,-0.201]	[1.000,0.000,1.000,0.000,0.000,1.000]
2	valid	6.71	0.0005	[-0.390,0.431,-0.420,0.398,-0.403,0.405]	[-0.462,-0.494,0.424,0.128,0.199,0.554]	[0.000,1.000,1.000,0.000,1.000,0.000]
3	valid	19.21	0.0170	[-0.603,0.394,-0.259,0.481,-0.399,0.152]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[1.000,1.000,0.000,1.000,0.000,0.000]
4	valid	19.21	0.0170	[-0.603,0.394,-0.259,0.481,-0.399,0.152]	[0.370,0.589,-0.242,-0.252,0.626,-0.038]	[1.000,0.000,0.000,1.000,1.000,0.000]
5	valid	28.53	0.0840	[-0.644,0.334,-0.000,0.469,-0.267,0.427]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
6	valid	28.53	0.0840	[-0.644,0.334,-0.000,0.469,-0.267,0.427]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[1.000,1.000,0.000,1.000,0.000,0.000]
7	valid	28.53	0.0840	[-0.644,0.334,-0.000,0.469,-0.267,0.427]	[0.425,0.367,0.097,-0.461,0.518,0.443]	[1.000,0.000,0.000,1.000,1.000,0.000]
8	valid	28.53	0.0840	[-0.644,0.334,-0.000,0.469,-0.267,0.427]	[-0.462,0.037,-0.577,-0.165,0.568,-0.320]	[0.000,0.000,0.000,1.000,1.000,1.000]
9	valid	28.53	0.0840	[-0.644,0.334,-0.000,0.469,-0.267,0.427]	[-0.720,0.239,0.286,0.256,-0.526,-0.006]	[0.000,1.000,1.000,0.000,0.000,1.000]
10	valid	24.34	0.0417	[-0.642,0.333,-0.079,0.468,-0.266,0.426]	[0.255,0.020,0.405,0.033,0.400,0.781]	[1.000,0.000,1.000,0.000,1.000,0.000]
11	valid	24.34	0.0417	[-0.642,0.333,-0.079,0.468,-0.266,0.426]	[0.294,-0.088,-0.068,0.098,0.920,0.214]	[1.000,1.000,0.000,0.000,1.000,0.000]
12	valid	16.20	0.0050	[-0.625,0.324,-0.243,0.455,-0.259,0.414]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
13	valid	16.20	0.0050	[-0.625,0.324,-0.243,0.455,-0.259,0.414]	[0.127,-0.020,-0.673,0.298,-0.642,0.172]	[1.000,1.000,0.000,0.000,0.000,1.000]
14	valid	16.20	0.0050	[-0.625,0.324,-0.243,0.455,-0.259,0.414]	[-0.738,0.601,-0.251,0.020,0.021,-0.174]	[0.000,0.000,0.000,1.000,1.000,1.000]
15	valid	16.20	0.0050	[-0.625,0.324,-0.243,0.455,-0.259,0.414]	[-0.186,0.078,-0.020,-0.271,-0.784,-0.521]	[0.000,0.000,1.000,1.000,0.000,1.000]
16	valid	16.20	0.0050	[-0.625,0.324,-0.243,0.455,-0.259,0.414]	[-0.026,-0.321,-0.030,0.145,-0.496,0.793]	[1.000,1.000,1.000,0.000,0.000,0.000]
17	valid	16.20	0.0050	[-0.625,0.324,-0.243,0.455,-0.259,0.414]	[0.490,0.520,-0.154,-0.215,0.644,-0.073]	[1.000,0.000,0.000,1.000,1.000,0.000]
18	valid	10.37	0.0024	[-0.592,0.306,-0.334,0.431,-0.324,0.392]	[-0.637,-0.481,-0.339,-0.288,-0.353,0.200]	[0.000,1.000,0.000,1.000,0.000,1.000]
19	valid	10.37	0.0024	[-0.592,0.306,-0.334,0.431,-0.324,0.392]	[-0.110,0.309,-0.682,-0.173,0.630,0.033]	[0.000,0.000,0.000,1.000,1.000,1.000]
20	valid	10.37	0.0024	[-0.592,0.306,-0.334,0.431,-0.324,0.392]	[-0.393,0.397,0.482,-0.374,-0.136,-0.545]	[0.000,0.000,1.000,1.000,0.000,1.000]

Ordinary subset selection | seed 4 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	17.69	0.0046	[-0.275,0.494,-0.380,0.483,-0.483,0.263]	[0.135,0.489,0.691,0.326,-0.346,-0.201]	[1.000,0.000,1.000,0.000,0.000,1.000]
2	valid	23.22	0.0623	[-0.508,0.396,-0.444,0.455,-0.426,-0.011]	[-0.410,-0.016,-0.600,0.533,-0.107,0.420]	[0.000,1.000,0.000,0.000,1.000,1.000]
3	valid	19.16	0.0131	[-0.609,0.242,-0.478,0.512,-0.153,0.241]	[0.230,-0.648,-0.058,0.014,-0.411,-0.595]	[1.000,1.000,0.000,0.000,0.000,1.000]
4	valid	16.58	0.0116	[-0.469,0.315,-0.637,0.394,-0.292,0.186]	[-0.013,0.610,-0.185,0.173,-0.744,0.105]	[1.000,0.000,0.000,1.000,0.000,1.000]
5	valid	13.17	0.0055	[-0.515,0.261,-0.580,0.420,-0.238,0.312]	[0.570,0.398,-0.261,-0.403,0.287,-0.451]	[1.000,0.000,0.000,1.000,0.000,1.000]
6	valid	15.40	0.0083	[-0.544,0.206,-0.515,0.432,-0.183,0.420]	[0.562,-0.462,-0.473,0.091,-0.473,0.123]	[1.000,1.000,0.000,1.000,0.000,0.000]
7	valid	12.78	0.0062	[-0.534,0.281,-0.505,0.424,-0.179,0.412]	[0.446,-0.536,0.405,-0.246,0.114,0.525]	[1.000,1.000,1.000,0.000,0.000,0.000]
8	valid	6.52	0.0019	[-0.504,0.379,-0.477,0.400,-0.253,0.389]	[0.703,0.212,-0.262,0.152,-0.353,-0.494]	[1.000,0.000,0.000,1.000,0.000,1.000]
9	valid	8.18	0.0037	[-0.449,0.377,-0.483,0.463,-0.239,0.388]	[0.796,0.423,-0.128,0.149,-0.334,-0.194]	[1.000,0.000,1.000,0.000,0.000,1.000]
10	valid	7.97	0.0043	[-0.494,0.367,-0.471,0.451,-0.233,0.378]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[1.000,1.000,0.000,1.000,0.000,0.000]
11	valid	5.87	0.0009	[-0.489,0.363,-0.466,0.446,-0.271,0.374]	[-0.157,0.331,0.025,0.851,-0.329,0.181]	[1.000,0.000,1.000,0.000,0.000,1.000]
12	valid	6.27	0.0014	[-0.477,0.416,-0.455,0.436,-0.265,0.365]	[-0.459,0.384,-0.073,0.327,0.227,0.692]	[0.000,0.000,1.000,1.000,1.000,0.000]
13	valid	5.50	0.0013	[-0.494,0.404,-0.442,0.423,-0.271,0.381]	[0.382,0.461,-0.285,0.068,-0.646,0.372]	[1.000,0.000,1.000,1.000,0.000,0.000]
14	valid	5.50	0.0013	[-0.494,0.404,-0.442,0.423,-0.271,0.381]	[-0.013,0.610,-0.185,0.173,-0.744,0.105]	[1.000,0.000,0.000,1.000,0.000,1.000]
15	valid	5.50	0.0013	[-0.494,0.404,-0.442,0.423,-0.271,0.381]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[1.000,1.000,0.000,1.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.796,0.423,-0.128,0.149,-0.334,-0.194]	[1.000,0.000,0.000,1.000,0.000,1.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[1.000,1.000,0.000,1.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.425,0.367,0.097,-0.461,0.518,0.443]	[1.000,0.000,0.000,1.000,1.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.462,0.037,-0.577,-0.165,0.568,-0.320]	[0.000,0.000,0.000,1.000,1.000,1.000]

Ordinary subset selection | seed 4 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	40.62	0.1670	[0.174,0.542,-0.468,0.292,-0.504,0.360]	[0.144,0.471,0.582,-0.209,-0.507,-0.344]	[0.000,0.000,1.000,1.000,0.000,1.000]
2	valid	17.57	0.0074	[-0.455,0.223,-0.636,0.620,-0.340,0.287]	[-0.507,0.208,-0.667,0.392,-0.094,0.304]	[0.000,1.000,0.000,0.000,1.000,1.000]
3	valid	8.35	0.0023	[-0.514,0.484,-0.565,0.643,-0.397,0.568]	[0.110,-0.860,0.078,-0.437,0.148,-0.173]	[0.000,1.000,0.000,1.000,0.000,1.000]
4	valid	6.29	0.0013	[-0.616,0.469,-0.607,0.696,-0.518,0.547]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
5	valid	6.03	0.0004	[-0.655,0.669,-0.672,0.694,-0.486,0.548]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
6	valid	6.23	0.0004	[-0.583,0.573,-0.638,0.532,-0.396,0.582]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
7	valid	3.30	0.0001	[-0.684,0.650,-0.680,0.637,-0.570,0.611]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
8	valid	4.65	0.0003	[-0.669,0.631,-0.737,0.676,-0.499,0.572]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
9	valid	3.79	0.0001	[-0.711,0.634,-0.699,0.591,-0.478,0.605]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
10	valid	2.19	0.0000	[-0.756,0.579,-0.757,0.630,-0.574,0.647]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[1.000,0.000,1.000,1.000,0.000,0.000]
11	valid	2.85	0.0002	[-0.696,0.554,-0.697,0.637,-0.493,0.606]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
12	valid	3.40	0.0001	[-0.709,0.579,-0.730,0.593,-0.481,0.603]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
13	valid	4.70	0.0006	[-0.707,0.533,-0.747,0.579,-0.476,0.508]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[1.000,1.000,0.000,1.000,0.000,0.000]
14	valid	4.85	0.0005	[-0.601,0.449,-0.653,0.531,-0.401,0.475]	[0.110,-0.860,0.078,-0.437,0.148,-0.173]	[0.000,1.000,0.000,1.000,0.000,1.000]
15	valid	1.39	0.0001	[-0.782,0.625,-0.730,0.693,-0.627,0.649]	[0.382,0.461,-0.285,0.068,-0.646,0.372]	[1.000,0.000,1.000,1.000,0.000,0.000]
16	valid	3.13	0.0002	[-0.746,0.556,-0.697,0.655,-0.485,0.604]	[0.332,-0.475,0.437,-0.543,0.245,-0.344]	[0.000,1.000,1.000,1.000,0.000,0.000]
17	valid	1.97	0.0001	[-0.688,0.583,-0.669,0.645,-0.556,0.584]	[0.382,0.461,-0.285,0.068,-0.646,0.372]	[1.000,0.000,1.000,1.000,0.000,0.000]
18	valid	1.13	0.0001	[-0.720,0.598,-0.672,0.632,-0.547,0.585]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,1.000,0.000,1.000,0.000]
19	valid	3.68	0.0009	[-0.724,0.607,-0.715,0.638,-0.656,0.561]	[-0.505,-0.657,0.267,0.263,0.310,-0.278]	[0.000,1.000,1.000,0.000,1.000,0.000]
20	valid	2.20	0.0004	[-0.788,0.612,-0.702,0.697,-0.550,0.606]	[0.137,-0.573,0.372,-0.529,0.421,0.239]	[0.000,1.000,1.000,1.000,0.000,0.000]

Ordinary subset selection | seed 4 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	29.85	0.0302	[-0.068,0.265,-0.405,0.169,-0.180,0.102]	[0.135,0.489,0.691,0.326,-0.346,-0.201]	[1.000,0.000,1.000,0.000,0.000,1.000]
2	valid	23.86	0.0095	[-0.216,0.441,-0.554,0.169,-0.206,0.291]	[-0.462,-0.494,0.424,0.128,0.199,0.554]	[0.000,1.000,1.000,0.000,1.000,0.000]
3	valid	28.43	0.0130	[-0.247,0.657,-0.538,0.236,-0.251,0.148]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[1.000,1.000,0.000,1.000,0.000,0.000]
4	valid	32.31	0.0275	[-0.264,0.798,-0.493,0.227,-0.337,0.106]	[0.370,0.589,-0.242,-0.252,0.626,-0.038]	[1.000,0.000,0.000,1.000,1.000,0.000]
5	valid	28.14	0.0319	[-0.284,0.600,-0.623,0.164,-0.474,0.133]	[0.266,-0.222,-0.520,-0.232,0.658,-0.350]	[1.000,0.000,0.000,0.000,1.000,1.000]
6	valid	31.62	0.0344	[-0.279,0.732,-0.573,0.182,-0.477,0.077]	[0.300,-0.661,-0.211,-0.430,-0.350,-0.349]	[1.000,1.000,0.000,1.000,0.000,0.000]
7	valid	31.37	0.0303	[-0.309,0.767,-0.426,0.211,-0.560,0.099]	[0.425,0.367,0.097,-0.461,0.518,0.443]	[1.000,0.000,0.000,1.000,1.000,0.000]
8	valid	29.25	0.0296	[-0.334,0.634,-0.494,0.190,-0.644,0.103]	[-0.462,0.037,-0.577,-0.165,0.568,-0.320]	[0.000,0.000,0.000,1.000,1.000,1.000]
9	valid	28.64	0.0353	[-0.412,0.499,-0.496,0.190,-0.723,0.091]	[-0.720,0.239,0.286,0.256,-0.526,-0.006]	[0.000,1.000,1.000,0.000,0.000,1.000]
10	valid	29.13	0.0345	[-0.407,0.448,-0.520,0.171,-0.757,0.116]	[0.255,0.020,0.405,0.033,0.400,0.781]	[1.000,0.000,1.000,0.000,1.000,0.000]
11	valid	34.05	0.0346	[-0.393,0.404,-0.466,0.155,-0.897,0.109]	[0.294,-0.088,-0.068,0.098,0.920,0.214]	[1.000,1.000,0.000,0.000,1.000,0.000]
12	valid	34.80	0.0245	[-0.425,0.337,-0.434,0.180,-0.939,0.116]	[0.487,-0.071,0.162,0.624,0.417,0.411]	[1.000,0.000,1.000,0.000,1.000,0.000]
13	valid	38.61	0.0322	[-0.377,0.286,-0.482,0.172,-1.028,0.090]	[0.127,-0.020,-0.673,0.298,-0.642,0.172]	[1.000,1.000,0.000,0.000,0.000,1.000]
14	valid	35.53	0.0361	[-0.448,0.324,-0.480,0.155,-0.943,0.088]	[-0.738,0.601,-0.251,0.020,0.021,-0.174]	[0.000,0.000,0.000,1.000,1.000,1.000]
15	valid	39.94	0.0318	[-0.418,0.291,-0.418,0.149,-1.071,0.092]	[-0.186,0.078,-0.020,-0.271,-0.784,-0.521]	[0.000,0.000,1.000,1.000,0.000,1.000]
16	valid	42.30	0.0250	[-0.378,0.294,-0.378,0.142,-1.141,0.109]	[-0.026,-0.321,-0.030,0.145,-0.496,0.793]	[1.000,1.000,1.000,0.000,0.000,0.000]
17	valid	44.87	0.0300	[-0.380,0.298,-0.334,0.129,-1.214,0.088]	[0.490,0.520,-0.154,-0.215,0.644,-0.073]	[1.000,0.000,0.000,1.000,1.000,0.000]
18	valid	44.43	0.0375	[-0.416,0.309,-0.328,0.124,-1.197,0.071]	[-0.637,-0.481,-0.339,-0.288,-0.353,0.200]	[0.000,1.000,0.000,1.000,0.000,1.000]
19	valid	46.51	0.0397	[-0.368,0.292,-0.352,0.112,-1.262,0.060]	[-0.110,0.309,-0.682,-0.173,0.630,0.033]	[0.000,0.000,0.000,1.000,1.000,1.000]
20	valid	44.19	0.0408	[-0.385,0.306,-0.382,0.117,-1.189,0.067]	[-0.393,0.397,0.482,-0.374,-0.136,-0.545]	[0.000,0.000,1.000,1.000,0.000,1.000]

Small budget-constrained selection | seed 0 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	25.59	0.0659	[-0.447,0.447,-0.447,0.447,-0.447,-0.038]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	16.89	0.0151	[-0.493,0.413,-0.413,0.413,-0.479,0.128]	[0.353,0.448,-0.509,-0.063,0.424,0.481]	[1.000,0.000,0.000,0.000,1.000,0.000]
3	valid	17.53	0.0161	[-0.491,0.411,-0.411,0.411,-0.489,0.117]	[0.407,-0.518,-0.600,-0.077,0.445,0.051]	[1.000,1.000,0.000,0.000,1.000,0.000]
4	valid	12.43	0.0095	[-0.475,0.399,-0.389,0.445,-0.473,0.205]	[-0.033,-0.479,0.012,-0.087,-0.857,0.164]	[0.000,1.000,0.000,1.000,0.000,0.000]
5	valid	22.32	0.0191	[-0.394,0.324,-0.611,0.448,-0.391,0.112]	[-0.365,0.210,0.374,-0.742,0.235,0.278]	[0.000,0.000,1.000,1.000,0.000,0.000]
6	valid	22.32	0.0191	[-0.394,0.324,-0.611,0.448,-0.391,0.112]	[-0.260,-0.012,0.314,0.902,0.141,-0.014]	[0.000,0.000,1.000,0.000,1.000,0.000]
7	valid	21.03	0.0184	[-0.382,0.397,-0.592,0.435,-0.380,0.109]	[0.122,-0.176,0.028,0.496,0.746,0.390]	[1.000,1.000,0.000,0.000,1.000,0.000]
8	valid	20.92	0.0184	[-0.381,0.407,-0.590,0.433,-0.378,0.108]	[-0.185,-0.096,0.017,-0.452,-0.847,-0.187]	[0.000,1.000,0.000,1.000,0.000,0.000]
9	valid	20.92	0.0184	[-0.381,0.407,-0.590,0.433,-0.378,0.108]	[0.673,0.153,0.372,0.242,0.571,-0.004]	[1.000,0.000,1.000,0.000,1.000,0.000]
10	valid	20.92	0.0184	[-0.381,0.407,-0.590,0.433,-0.378,0.108]	[-0.461,0.276,0.039,0.296,0.739,0.276]	[0.000,0.000,1.000,0.000,1.000,0.000]
11	valid	20.92	0.0184	[-0.381,0.407,-0.590,0.433,-0.378,0.108]	[-0.705,0.295,-0.275,-0.330,-0.445,-0.181]	[0.000,0.000,0.000,1.000,0.000,0.000]
12	valid	22.81	0.0228	[-0.330,0.611,-0.511,0.375,-0.328,0.094]	[-0.081,-0.319,0.193,-0.915,0.130,0.018]	[0.000,1.000,0.000,1.000,0.000,0.000]
13	valid	22.81	0.0228	[-0.330,0.611,-0.511,0.375,-0.328,0.094]	[0.080,-0.595,-0.463,-0.042,0.321,0.567]	[1.000,1.000,0.000,0.000,1.000,0.000]
14	valid	15.56	0.0088	[-0.320,0.577,-0.486,0.410,-0.318,0.243]	[-0.428,0.162,0.200,0.094,-0.682,-0.526]	[0.000,0.000,1.000,0.000,0.000,1.000]
15	valid	15.56	0.0088	[-0.320,0.577,-0.486,0.410,-0.318,0.243]	[-0.536,0.592,0.004,-0.576,-0.106,0.142]	[0.000,0.000,1.000,1.000,0.000,0.000]
16	valid	15.56	0.0088	[-0.320,0.577,-0.486,0.410,-0.318,0.243]	[-0.174,-0.275,-0.859,0.027,-0.296,-0.261]	[0.000,1.000,0.000,0.000,0.000,1.000]
17	valid	20.44	0.0143	[-0.227,0.507,-0.523,0.338,-0.525,0.163]	[-0.205,-0.189,0.668,-0.208,0.290,0.591]	[0.000,0.000,1.000,0.000,1.000,0.000]
18	valid	20.44	0.0143	[-0.227,0.507,-0.523,0.338,-0.525,0.163]	[-0.461,0.276,0.039,0.296,0.739,0.276]	[0.000,0.000,1.000,0.000,1.000,0.000]
19	valid	20.44	0.0143	[-0.227,0.507,-0.523,0.338,-0.525,0.163]	[0.138,0.648,-0.649,0.281,-0.230,0.085]	[1.000,0.000,0.000,0.000,0.000,0.000]
20	valid	20.44	0.0143	[-0.227,0.507,-0.523,0.338,-0.525,0.163]	[-0.357,-0.137,0.908,0.137,-0.054,-0.091]	[0.000,1.000,1.000,0.000,0.000,0.000]

Small budget-constrained selection | seed 0 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	25.59	0.0659	[-0.447,0.447,-0.447,0.447,-0.447,-0.038]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	4.17	0.0006	[-0.400,0.400,-0.400,0.400,-0.471,0.372]	[-0.113,-0.597,0.057,0.051,-0.151,0.776]	[0.000,1.000,1.000,0.000,0.000,0.000]
3	valid	22.86	0.0128	[-0.595,0.253,-0.253,0.253,-0.356,0.573]	[0.252,0.214,0.243,0.798,-0.352,-0.267]	[1.000,0.000,0.000,0.000,0.000,1.000]
4	valid	23.64	0.0266	[-0.471,0.415,-0.200,0.200,-0.316,0.652]	[0.307,-0.730,-0.027,0.323,-0.363,-0.369]	[0.000,1.000,0.000,0.000,0.000,1.000]
5	valid	21.54	0.0228	[-0.513,0.371,-0.327,0.230,-0.251,0.613]	[-0.365,0.210,0.374,-0.742,0.235,0.278]	[0.000,0.000,1.000,1.000,0.000,0.000]
6	valid	25.34	0.0353	[-0.546,0.336,-0.426,0.166,-0.200,0.582]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,0.000,1.000,0.000,0.000,1.000]
7	valid	20.51	0.0167	[-0.526,0.324,-0.410,0.197,-0.311,0.561]	[-0.126,-0.373,0.609,0.016,0.603,0.331]	[0.000,0.000,1.000,0.000,1.000,0.000]
8	valid	10.99	0.0016	[-0.431,0.501,-0.444,0.317,-0.313,0.408]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,1.000,0.000,1.000,0.000,0.000]
9	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,1.000,0.000,0.000,0.000,1.000]
10	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.407,-0.518,-0.600,-0.077,0.445,0.051]	[1.000,1.000,0.000,0.000,1.000,0.000]
11	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.033,-0.479,0.012,-0.087,-0.857,0.164]	[0.000,1.000,0.000,1.000,0.000,0.000]
12	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.365,0.210,0.374,-0.742,0.235,0.278]	[0.000,0.000,1.000,1.000,0.000,0.000]
13	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.260,-0.012,0.314,0.902,0.141,-0.014]	[0.000,0.000,1.000,0.000,1.000,0.000]
14	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.122,-0.176,0.028,0.496,0.746,0.390]	[1.000,1.000,0.000,0.000,1.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.185,-0.096,0.017,-0.452,-0.847,-0.187]	[0.000,1.000,0.000,1.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.673,0.153,0.372,0.242,0.571,-0.004]	[1.000,0.000,1.000,0.000,1.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.461,0.276,0.039,0.296,0.739,0.276]	[0.000,0.000,1.000,0.000,1.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.705,0.295,-0.275,-0.330,-0.445,-0.181]	[0.000,0.000,0.000,1.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.081,-0.319,0.193,-0.915,0.130,0.018]	[0.000,1.000,0.000,1.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.080,-0.595,-0.463,-0.042,0.321,0.567]	[1.000,1.000,0.000,0.000,1.000,0.000]

Small budget-constrained selection | seed 0 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	48.78	0.1259	[-0.373,0.140,-0.474,0.389,-0.289,-0.301]	[0.514,0.117,0.605,0.460,-0.095,-0.368]	[1.000,0.000,1.000,0.000,0.000,0.000]
2	valid	27.84	0.0335	[-0.504,0.633,-0.358,0.676,-0.244,0.023]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,1.000,0.000,1.000,0.000,0.000]
3	valid	22.62	0.0268	[-0.560,0.730,-0.442,0.681,-0.661,0.033]	[-0.205,-0.189,0.668,-0.208,0.290,0.591]	[0.000,0.000,1.000,0.000,1.000,0.000]
4	valid	13.99	0.0075	[-0.729,0.688,-0.445,0.624,-0.488,0.311]	[0.147,0.323,0.809,-0.074,0.069,0.457]	[1.000,0.000,1.000,1.000,0.000,0.000]
5	valid	8.41	0.0010	[-0.583,0.746,-0.397,0.564,-0.593,0.478]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,1.000,0.000,0.000,0.000,1.000]
6	valid	8.70	0.0007	[-0.532,0.815,-0.508,0.649,-0.502,0.566]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
7	valid	9.25	0.0011	[-0.437,0.725,-0.412,0.482,-0.598,0.485]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,1.000,0.000]
8	valid	6.49	0.0018	[-0.618,0.726,-0.445,0.640,-0.697,0.511]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,1.000,0.000,1.000,0.000,0.000]
9	valid	5.88	0.0007	[-0.670,0.775,-0.497,0.615,-0.711,0.577]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
10	valid	6.30	0.0019	[-0.539,0.699,-0.454,0.663,-0.654,0.473]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[1.000,1.000,0.000,1.000,0.000,0.000]
11	valid	6.96	0.0006	[-0.484,0.712,-0.430,0.581,-0.574,0.469]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
12	valid	8.78	0.0035	[-0.542,0.790,-0.484,0.609,-0.700,0.445]	[0.382,0.577,0.142,0.257,0.377,-0.542]	[1.000,0.000,1.000,0.000,1.000,0.000]
13	valid	9.43	0.0014	[-0.476,0.831,-0.492,0.594,-0.697,0.509]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
14	valid	7.70	0.0007	[-0.599,0.785,-0.531,0.533,-0.698,0.507]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[0.000,1.000,0.000,0.000,0.000,1.000]
15	valid	6.83	0.0007	[-0.531,0.800,-0.513,0.589,-0.674,0.553]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[0.000,1.000,0.000,0.000,0.000,1.000]
16	valid	7.23	0.0009	[-0.636,0.831,-0.509,0.601,-0.773,0.596]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[1.000,1.000,0.000,1.000,0.000,0.000]
17	valid	8.41	0.0010	[-0.527,0.836,-0.472,0.623,-0.747,0.585]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[0.000,1.000,0.000,0.000,0.000,1.000]
18	valid	8.27	0.0011	[-0.547,0.686,-0.421,0.450,-0.605,0.472]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[0.000,1.000,0.000,0.000,0.000,1.000]
19	valid	8.88	0.0007	[-0.478,0.756,-0.496,0.488,-0.647,0.484]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,1.000,0.000,1.000,0.000,0.000]
20	valid	8.34	0.0008	[-0.599,0.737,-0.455,0.495,-0.580,0.472]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[1.000,1.000,0.000,1.000,0.000,0.000]

Small budget-constrained selection | seed 0 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	53.54	0.1409	[-0.057,0.000,-0.228,0.002,-0.427,0.000]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	47.75	0.1713	[-0.224,0.222,-0.182,-0.028,-0.742,0.000]	[0.353,0.448,-0.509,-0.063,0.424,0.481]	[1.000,0.000,0.000,0.000,1.000,0.000]
3	valid	43.75	0.1379	[-0.288,0.327,-0.313,-0.020,-0.862,0.011]	[0.407,-0.518,-0.600,-0.077,0.445,0.051]	[1.000,1.000,0.000,0.000,1.000,0.000]
4	valid	48.42	0.1313	[-0.232,0.353,-0.243,-0.007,-1.126,0.008]	[-0.033,-0.479,0.012,-0.087,-0.857,0.164]	[0.000,1.000,0.000,1.000,0.000,0.000]
5	valid	41.30	0.0515	[-0.297,0.400,-0.312,0.069,-0.962,0.008]	[-0.365,0.210,0.374,-0.742,0.235,0.278]	[0.000,0.000,1.000,1.000,0.000,0.000]
6	valid	39.00	0.0436	[-0.314,0.369,-0.340,0.139,-0.952,0.008]	[-0.260,-0.012,0.314,0.902,0.141,-0.014]	[0.000,0.000,1.000,0.000,1.000,0.000]
7	valid	44.14	0.0454	[-0.280,0.338,-0.278,0.116,-1.161,0.032]	[0.122,-0.176,0.028,0.496,0.746,0.390]	[1.000,1.000,0.000,0.000,1.000,0.000]
8	valid	42.68	0.0418	[-0.272,0.356,-0.266,0.180,-1.127,0.013]	[-0.185,-0.096,0.017,-0.452,-0.847,-0.187]	[0.000,1.000,0.000,1.000,0.000,0.000]
9	valid	44.86	0.0446	[-0.317,0.313,-0.265,0.147,-1.219,0.010]	[0.673,0.153,0.372,0.242,0.571,-0.004]	[1.000,0.000,1.000,0.000,1.000,0.000]
10	valid	47.85	0.0446	[-0.314,0.286,-0.218,0.138,-1.350,0.008]	[-0.461,0.276,0.039,0.296,0.739,0.276]	[0.000,0.000,1.000,0.000,1.000,0.000]
11	valid	45.02	0.0446	[-0.395,0.302,-0.228,0.150,-1.249,0.007]	[-0.705,0.295,-0.275,-0.330,-0.445,-0.181]	[0.000,0.000,0.000,1.000,0.000,0.000]
12	valid	42.78	0.0407	[-0.401,0.331,-0.211,0.203,-1.185,0.007]	[-0.081,-0.319,0.193,-0.915,0.130,0.018]	[0.000,1.000,0.000,1.000,0.000,0.000]
13	valid	42.97	0.0446	[-0.363,0.383,-0.231,0.173,-1.200,0.006]	[0.080,-0.595,-0.463,-0.042,0.321,0.567]	[1.000,1.000,0.000,0.000,1.000,0.000]
14	valid	40.56	0.0421	[-0.425,0.390,-0.241,0.171,-1.127,0.023]	[-0.428,0.162,0.200,0.094,-0.682,-0.526]	[0.000,0.000,1.000,0.000,0.000,1.000]
15	valid	38.01	0.0453	[-0.474,0.447,-0.213,0.196,-1.040,0.020]	[-0.536,0.592,0.004,-0.576,-0.106,0.142]	[0.000,0.000,1.000,1.000,0.000,0.000]
16	valid	36.27	0.0422	[-0.475,0.460,-0.262,0.188,-0.990,0.023]	[-0.174,-0.275,-0.859,0.027,-0.296,-0.261]	[0.000,1.000,0.000,0.000,0.000,1.000]
17	valid	37.54	0.0435	[-0.478,0.401,-0.312,0.162,-1.027,0.022]	[-0.205,-0.189,0.668,-0.208,0.290,0.591]	[0.000,0.000,1.000,0.000,1.000,0.000]
18	valid	40.20	0.0435	[-0.475,0.376,-0.271,0.154,-1.115,0.019]	[-0.461,0.276,0.039,0.296,0.739,0.276]	[0.000,0.000,1.000,0.000,1.000,0.000]
19	valid	39.06	0.0436	[-0.480,0.362,-0.329,0.148,-1.074,0.019]	[0.138,0.648,-0.649,0.281,-0.230,0.085]	[1.000,0.000,0.000,0.000,0.000,0.000]
20	valid	38.09	0.0439	[-0.459,0.358,-0.394,0.146,-1.039,0.019]	[-0.357,-0.137,0.908,0.137,-0.054,-0.091]	[0.000,1.000,1.000,0.000,0.000,0.000]

Small budget-constrained selection | seed 1 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	28.53	0.0624	[-0.443,0.443,-0.443,0.443,-0.443,-0.142]	[0.514,0.117,0.605,0.460,-0.095,-0.368]	[1.000,0.000,1.000,0.000,0.000,0.000]
2	valid	28.53	0.0624	[-0.443,0.443,-0.443,0.443,-0.443,-0.142]	[0.384,-0.574,-0.279,-0.008,-0.622,-0.239]	[1.000,1.000,0.000,1.000,0.000,0.000]
3	valid	31.03	0.0615	[-0.656,0.364,-0.364,0.364,-0.402,-0.102]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
4	valid	31.03	0.0615	[-0.656,0.364,-0.364,0.364,-0.402,-0.102]	[0.549,0.000,0.537,-0.393,-0.240,-0.445]	[1.000,0.000,1.000,1.000,0.000,0.000]
5	valid	31.03	0.0615	[-0.656,0.364,-0.364,0.364,-0.402,-0.102]	[0.673,0.153,0.372,0.242,0.571,-0.004]	[1.000,0.000,1.000,0.000,1.000,0.000]
6	valid	31.03	0.0615	[-0.656,0.364,-0.364,0.364,-0.402,-0.102]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
7	valid	25.66	0.0199	[-0.492,0.303,-0.336,0.273,-0.692,0.027]	[0.147,0.323,0.809,-0.074,0.069,0.457]	[1.000,0.000,1.000,0.000,1.000,0.000]
8	valid	16.27	0.0091	[-0.454,0.347,-0.409,0.252,-0.638,0.187]	[-0.066,0.509,-0.733,-0.080,-0.352,0.264]	[0.000,0.000,0.000,1.000,0.000,0.000]
9	valid	17.24	0.0097	[-0.459,0.330,-0.387,0.277,-0.652,0.170]	[-0.117,-0.054,-0.081,-0.534,0.263,0.789]	[0.000,0.000,0.000,1.000,1.000,0.000]
10	valid	30.22	0.0188	[-0.467,0.183,-0.210,0.444,-0.711,0.035]	[-0.617,-0.050,-0.207,-0.045,0.735,0.179]	[0.000,0.000,0.000,1.000,1.000,0.000]
11	valid	23.93	0.0135	[-0.447,0.209,-0.256,0.445,-0.687,0.144]	[-0.428,0.162,0.200,0.094,-0.682,-0.526]	[0.000,0.000,1.000,0.000,0.000,1.000]
12	valid	23.93	0.0135	[-0.447,0.209,-0.256,0.445,-0.687,0.144]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
13	valid	23.93	0.0135	[-0.447,0.209,-0.256,0.445,-0.687,0.144]	[-0.543,0.565,-0.230,0.549,0.110,0.137]	[0.000,0.000,0.000,0.000,1.000,0.000]
14	valid	24.06	0.0135	[-0.444,0.224,-0.245,0.452,-0.685,0.138]	[0.307,-0.730,-0.027,0.323,-0.363,-0.369]	[1.000,1.000,0.000,0.000,0.000,0.000]
15	valid	24.06	0.0135	[-0.444,0.224,-0.245,0.452,-0.685,0.138]	[-0.043,0.600,-0.497,-0.486,0.392,0.031]	[0.000,0.000,0.000,1.000,1.000,0.000]
16	valid	24.06	0.0135	[-0.444,0.224,-0.245,0.452,-0.685,0.138]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,1.000,0.000]
17	valid	14.27	0.0027	[-0.516,0.254,-0.329,0.399,-0.543,0.326]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,0.000,1.000,0.000,0.000,1.000]
18	valid	14.27	0.0027	[-0.516,0.254,-0.329,0.399,-0.543,0.326]	[0.046,-0.506,0.027,0.687,-0.207,-0.476]	[0.000,1.000,0.000,0.000,0.000,1.000]
19	valid	13.51	0.0024	[-0.505,0.265,-0.333,0.405,-0.543,0.323]	[0.215,-0.587,-0.575,0.058,-0.038,-0.523]	[0.000,1.000,0.000,0.000,0.000,1.000]
20	valid	13.51	0.0024	[-0.505,0.265,-0.333,0.405,-0.543,0.323]	[0.029,0.208,0.555,-0.742,0.090,0.298]	[1.000,0.000,1.000,1.000,0.000,0.000]

Small budget-constrained selection | seed 1 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	28.53	0.0624	[-0.443,0.443,-0.443,0.443,-0.443,-0.142]	[0.514,0.117,0.605,0.460,-0.095,-0.368]	[1.000,0.000,1.000,0.000,0.000,0.000]
2	valid	16.09	0.0033	[-0.578,0.328,-0.532,0.328,-0.328,0.244]	[-0.066,0.509,-0.733,-0.080,-0.352,0.264]	[0.000,0.000,0.000,1.000,0.000,0.000]
3	valid	19.05	0.0099	[-0.468,0.586,-0.433,0.375,-0.278,0.177]	[-0.081,-0.319,0.193,-0.915,0.130,0.018]	[0.000,1.000,0.000,1.000,0.000,0.000]
4	valid	15.06	0.0106	[-0.365,0.528,-0.449,0.319,-0.517,0.133]	[-0.220,-0.033,0.564,-0.340,0.449,0.561]	[0.000,0.000,1.000,0.000,1.000,0.000]
5	valid	26.79	0.0490	[-0.127,0.482,-0.580,0.437,-0.472,-0.015]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]
6	valid	26.96	0.0193	[-0.097,0.426,-0.562,0.423,-0.561,0.000]	[0.147,0.323,0.809,-0.074,0.069,0.457]	[1.000,0.000,1.000,0.000,1.000,0.000]
7	valid	22.06	0.0148	[-0.143,0.421,-0.555,0.422,-0.558,0.073]	[-0.509,-0.595,-0.558,0.216,-0.169,-0.031]	[0.000,1.000,0.000,0.000,0.000,1.000]
8	valid	5.04	0.0004	[-0.315,0.440,-0.478,0.366,-0.485,0.331]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
9	valid	6.51	0.0009	[-0.286,0.402,-0.460,0.420,-0.513,0.325]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[1.000,1.000,0.000,1.000,0.000,0.000]
10	valid	14.99	0.0034	[-0.221,0.489,-0.608,0.337,-0.396,0.268]	[0.245,-0.141,0.121,0.841,0.283,0.344]	[1.000,0.000,1.000,0.000,1.000,0.000]
11	valid	14.45	0.0020	[-0.218,0.482,-0.600,0.333,-0.423,0.264]	[0.084,-0.630,0.281,-0.091,0.427,0.572]	[1.000,1.000,0.000,0.000,1.000,0.000]
12	valid	13.71	0.0012	[-0.234,0.480,-0.597,0.331,-0.421,0.268]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
13	valid	13.24	0.0012	[-0.229,0.470,-0.585,0.324,-0.459,0.262]	[-0.594,-0.378,0.498,0.219,0.416,-0.188]	[0.000,0.000,1.000,0.000,1.000,0.000]
14	valid	12.63	0.0012	[-0.233,0.467,-0.581,0.331,-0.455,0.272]	[-0.483,0.155,-0.698,0.316,0.188,-0.347]	[0.000,0.000,0.000,0.000,0.000,1.000]
15	valid	11.10	0.0017	[-0.254,0.457,-0.569,0.365,-0.446,0.267]	[-0.003,-0.100,0.152,-0.255,0.598,0.738]	[0.000,0.000,0.000,1.000,1.000,0.000]
16	valid	9.54	0.0017	[-0.267,0.444,-0.553,0.387,-0.446,0.279]	[0.424,-0.620,0.310,-0.475,0.258,0.219]	[1.000,1.000,0.000,1.000,0.000,0.000]
17	valid	9.19	0.0008	[-0.262,0.436,-0.542,0.379,-0.475,0.281]	[-0.058,0.235,0.010,-0.709,0.012,0.662]	[0.000,0.000,0.000,1.000,1.000,0.000]
18	valid	7.17	0.0012	[-0.283,0.414,-0.515,0.415,-0.474,0.295]	[-0.400,-0.307,0.044,-0.328,0.712,0.360]	[0.000,0.000,0.000,1.000,1.000,0.000]
19	valid	6.84	0.0007	[-0.281,0.411,-0.512,0.413,-0.471,0.312]	[0.545,-0.007,0.048,0.259,-0.790,-0.095]	[1.000,0.000,0.000,0.000,0.000,1.000]
20	valid	6.44	0.0019	[-0.293,0.391,-0.487,0.434,-0.492,0.305]	[0.766,-0.576,0.118,-0.140,-0.195,-0.104]	[1.000,1.000,0.000,1.000,0.000,0.000]

Small budget-constrained selection | seed 1 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	38.19	0.1658	[-0.350,-0.031,-0.478,0.023,-0.271,0.531]	[-0.428,0.162,0.200,0.094,-0.682,-0.526]	[0.000,0.000,1.000,0.000,0.000,1.000]
2	valid	10.89	0.0012	[-0.557,0.519,-0.477,0.373,-0.397,0.373]	[-0.501,0.782,0.124,0.315,0.096,-0.119]	[0.000,0.000,1.000,0.000,1.000,0.000]
3	valid	19.08	0.0229	[-0.580,0.316,-0.641,0.173,-0.679,0.618]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,0.000,1.000,0.000,0.000,1.000]
4	valid	17.27	0.0077	[-0.408,0.359,-0.689,0.632,-0.340,0.594]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]
5	valid	14.65	0.0065	[-0.379,0.366,-0.628,0.488,-0.408,0.628]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
6	valid	7.22	0.0012	[-0.487,0.550,-0.729,0.658,-0.562,0.533]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[1.000,1.000,0.000,1.000,0.000,0.000]
7	valid	13.56	0.0031	[-0.329,0.398,-0.704,0.662,-0.486,0.407]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[1.000,0.000,1.000,1.000,0.000,0.000]
8	valid	14.96	0.0068	[-0.411,0.551,-0.695,0.648,-0.334,0.450]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
9	valid	15.39	0.0078	[-0.262,0.459,-0.563,0.593,-0.344,0.391]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
10	valid	11.69	0.0019	[-0.344,0.596,-0.618,0.640,-0.473,0.430]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
11	valid	8.49	0.0019	[-0.467,0.603,-0.713,0.734,-0.592,0.501]	[0.396,-0.228,0.248,0.286,0.566,-0.573]	[1.000,0.000,1.000,0.000,1.000,0.000]
12	valid	9.54	0.0009	[-0.363,0.633,-0.704,0.630,-0.580,0.453]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
13	valid	9.72	0.0027	[-0.276,0.448,-0.500,0.536,-0.560,0.389]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,1.000,0.000]
14	valid	3.90	0.0003	[-0.496,0.538,-0.680,0.617,-0.649,0.448]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]
15	valid	4.63	0.0001	[-0.568,0.650,-0.713,0.665,-0.636,0.515]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
16	valid	5.06	0.0005	[-0.510,0.688,-0.755,0.665,-0.725,0.494]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
17	valid	8.02	0.0014	[-0.463,0.657,-0.759,0.766,-0.692,0.526]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
18	valid	3.60	0.0001	[-0.508,0.616,-0.715,0.634,-0.686,0.503]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
19	valid	6.26	0.0004	[-0.450,0.632,-0.735,0.649,-0.659,0.479]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]
20	valid	2.39	0.0002	[-0.560,0.609,-0.722,0.576,-0.664,0.470]	[0.545,-0.007,0.048,0.259,-0.790,-0.095]	[1.000,0.000,0.000,0.000,0.000,1.000]

Small budget-constrained selection | seed 1 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	41.06	0.1112	[-0.324,0.000,-0.397,0.283,-0.053,0.000]	[0.514,0.117,0.605,0.460,-0.095,-0.368]	[1.000,0.000,1.000,0.000,0.000,0.000]
2	valid	40.61	0.0949	[-0.492,0.216,-0.503,0.240,-0.045,-0.084]	[0.384,-0.574,-0.279,-0.008,-0.622,-0.239]	[1.000,1.000,0.000,1.000,0.000,0.000]
3	valid	32.26	0.0604	[-0.493,0.204,-0.535,0.226,-0.213,-0.040]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
4	valid	40.49	0.0912	[-0.636,0.153,-0.678,0.276,-0.160,-0.116]	[0.549,0.000,0.537,-0.393,-0.240,-0.445]	[1.000,0.000,1.000,1.000,0.000,0.000]
5	valid	41.20	0.0771	[-0.796,0.142,-0.691,0.218,-0.214,-0.092]	[0.673,0.153,0.372,0.242,0.571,-0.004]	[1.000,0.000,1.000,0.000,1.000,0.000]
6	valid	37.48	0.0612	[-0.780,0.197,-0.635,0.201,-0.315,-0.085]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
7	valid	37.19	0.0607	[-0.725,0.205,-0.808,0.174,-0.283,-0.044]	[0.147,0.323,0.809,-0.074,0.069,0.457]	[1.000,0.000,1.000,0.000,1.000,0.000]
8	valid	33.76	0.0611	[-0.698,0.280,-0.744,0.171,-0.339,-0.040]	[-0.066,0.509,-0.733,-0.080,-0.352,0.264]	[0.000,0.000,0.000,1.000,0.000,0.000]
9	valid	32.05	0.0596	[-0.692,0.249,-0.720,0.231,-0.370,-0.037]	[-0.117,-0.054,-0.081,-0.534,0.263,0.789]	[0.000,0.000,0.000,1.000,1.000,0.000]
10	valid	33.26	0.0592	[-0.773,0.208,-0.674,0.201,-0.438,-0.032]	[-0.617,-0.050,-0.207,-0.045,0.735,0.179]	[0.000,0.000,0.000,1.000,1.000,0.000]
11	valid	35.27	0.0539	[-0.874,0.203,-0.668,0.189,-0.387,-0.003]	[-0.428,0.162,0.200,0.094,-0.682,-0.526]	[0.000,0.000,1.000,0.000,0.000,1.000]
12	valid	33.90	0.0582	[-0.834,0.188,-0.655,0.172,-0.482,-0.003]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
13	valid	35.61	0.0533	[-0.938,0.218,-0.564,0.198,-0.439,-0.003]	[-0.543,0.565,-0.230,0.549,0.110,0.137]	[0.000,0.000,0.000,0.000,1.000,0.000]
14	valid	36.72	0.0525	[-0.976,0.266,-0.530,0.184,-0.408,-0.011]	[0.307,-0.730,-0.027,0.323,-0.363,-0.369]	[1.000,1.000,0.000,0.000,0.000,0.000]
15	valid	32.98	0.0523	[-0.873,0.304,-0.576,0.201,-0.424,-0.009]	[-0.043,0.600,-0.497,-0.486,0.392,0.031]	[0.000,0.000,0.000,1.000,1.000,0.000]
16	valid	33.92	0.0563	[-0.905,0.301,-0.520,0.169,-0.492,-0.008]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,1.000,0.000]
17	valid	33.40	0.0265	[-0.882,0.278,-0.567,0.169,-0.467,0.003]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,0.000,1.000,0.000,0.000,1.000]
18	valid	31.09	0.0244	[-0.838,0.343,-0.544,0.164,-0.499,0.013]	[0.046,-0.506,0.027,0.687,-0.207,-0.476]	[0.000,1.000,0.000,0.000,0.000,1.000]
19	valid	28.09	0.0251	[-0.721,0.404,-0.638,0.158,-0.467,0.022]	[0.215,-0.587,-0.575,0.058,-0.038,-0.523]	[0.000,1.000,0.000,0.000,0.000,1.000]
20	valid	27.64	0.0235	[-0.679,0.402,-0.704,0.185,-0.424,0.020]	[0.029,0.208,0.555,-0.742,0.090,0.298]	[1.000,0.000,1.000,1.000,0.000,0.000]

Small budget-constrained selection | seed 2 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	14.44	0.0135	[-0.444,0.443,-0.444,0.438,-0.444,0.141]	[0.353,0.892,0.177,0.173,0.002,0.134]	[1.000,0.000,1.000,0.000,1.000,0.000]
2	valid	19.96	0.0187	[-0.410,0.536,-0.457,0.404,-0.410,0.068]	[-0.357,-0.137,0.908,0.137,-0.054,-0.091]	[0.000,1.000,1.000,0.000,0.000,0.000]
3	valid	27.72	0.0239	[-0.199,0.494,-0.699,0.196,-0.199,0.387]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,0.000,1.000,0.000,0.000,1.000]
4	valid	27.64	0.0239	[-0.199,0.494,-0.699,0.199,-0.199,0.387]	[-0.536,0.592,0.004,-0.576,-0.106,0.142]	[0.000,0.000,1.000,1.000,0.000,0.000]
5	valid	28.25	0.0385	[-0.189,0.498,-0.690,0.189,-0.189,0.411]	[-0.226,-0.027,0.968,-0.032,-0.081,-0.065]	[0.000,0.000,1.000,0.000,0.000,1.000]
6	valid	25.28	0.0203	[-0.183,0.483,-0.669,0.304,-0.183,0.399]	[-0.289,-0.099,0.669,-0.415,-0.389,0.370]	[0.000,0.000,1.000,1.000,0.000,0.000]
7	valid	23.16	0.0196	[-0.150,0.417,-0.567,0.254,-0.552,0.335]	[0.084,-0.630,0.281,-0.091,0.427,0.572]	[1.000,1.000,0.000,0.000,1.000,0.000]
8	valid	23.16	0.0196	[-0.150,0.417,-0.567,0.254,-0.552,0.335]	[-0.226,-0.027,0.968,-0.032,-0.081,-0.065]	[0.000,0.000,1.000,0.000,0.000,1.000]
9	valid	23.86	0.0196	[-0.144,0.415,-0.578,0.247,-0.552,0.328]	[0.396,-0.228,0.248,0.286,0.566,-0.573]	[1.000,0.000,1.000,0.000,1.000,0.000]
10	valid	18.64	0.0216	[-0.262,0.429,-0.520,0.197,-0.452,0.483]	[-0.501,0.782,0.124,0.315,0.096,-0.119]	[0.000,0.000,1.000,0.000,0.000,1.000]
11	valid	19.83	0.0255	[-0.233,0.450,-0.529,0.195,-0.450,0.472]	[0.502,-0.302,0.198,-0.080,0.768,-0.147]	[1.000,1.000,0.000,0.000,1.000,0.000]
12	valid	19.83	0.0255	[-0.233,0.450,-0.529,0.195,-0.450,0.472]	[-0.332,0.375,0.215,0.555,0.601,0.184]	[0.000,0.000,1.000,0.000,1.000,0.000]
13	valid	19.83	0.0255	[-0.233,0.450,-0.529,0.195,-0.450,0.472]	[0.084,-0.630,0.281,-0.091,0.427,0.572]	[1.000,1.000,0.000,0.000,1.000,0.000]
14	valid	19.83	0.0255	[-0.233,0.450,-0.529,0.195,-0.450,0.472]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
15	valid	19.83	0.0255	[-0.233,0.450,-0.529,0.195,-0.450,0.472]	[0.183,-0.196,0.420,0.081,0.836,-0.216]	[1.000,0.000,1.000,0.000,1.000,0.000]
16	valid	19.83	0.0255	[-0.233,0.450,-0.529,0.195,-0.450,0.472]	[-0.226,-0.027,0.968,-0.032,-0.081,-0.065]	[0.000,0.000,1.000,0.000,0.000,1.000]
17	valid	19.83	0.0255	[-0.233,0.450,-0.529,0.195,-0.450,0.472]	[0.029,0.208,0.555,-0.742,0.090,0.298]	[1.000,0.000,1.000,1.000,0.000,0.000]
18	valid	19.83	0.0255	[-0.233,0.450,-0.529,0.195,-0.450,0.472]	[-0.450,-0.212,-0.516,0.464,-0.323,-0.409]	[0.000,1.000,0.000,0.000,0.000,1.000]
19	valid	19.83	0.0255	[-0.233,0.450,-0.529,0.195,-0.450,0.472]	[-0.277,-0.229,-0.711,-0.110,0.444,0.395]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	19.83	0.0255	[-0.233,0.450,-0.529,0.195,-0.450,0.472]	[0.353,0.892,0.177,0.173,0.002,0.134]	[1.000,0.000,1.000,0.000,1.000,0.000]

Small budget-constrained selection | seed 2 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	14.44	0.0135	[-0.444,0.443,-0.444,0.438,-0.444,0.141]	[0.353,0.892,0.177,0.173,0.002,0.134]	[1.000,0.000,1.000,0.000,1.000,0.000]
2	valid	9.49	0.0021	[-0.381,0.391,-0.381,0.376,-0.381,0.520]	[-0.039,-0.468,0.013,-0.267,-0.117,-0.833]	[0.000,1.000,0.000,0.000,0.000,1.000]
3	valid	20.44	0.0026	[-0.266,0.274,-0.266,0.622,-0.408,0.479]	[-0.400,-0.307,0.044,-0.328,0.712,0.360]	[0.000,0.000,0.000,1.000,1.000,0.000]
4	valid	23.97	0.0123	[-0.282,0.223,-0.217,0.506,-0.644,0.390]	[0.177,0.180,0.756,-0.012,0.405,-0.448]	[1.000,0.000,1.000,0.000,1.000,0.000]
5	valid	19.30	0.0118	[-0.256,0.208,-0.458,0.464,-0.584,0.354]	[0.396,-0.228,0.248,0.286,0.566,-0.573]	[1.000,0.000,1.000,0.000,1.000,0.000]
6	valid	12.38	0.0044	[-0.478,0.202,-0.443,0.449,-0.464,0.343]	[0.514,0.117,0.605,0.460,-0.095,-0.368]	[1.000,0.000,1.000,0.000,0.000,0.000]
7	valid	10.70	0.0072	[-0.349,0.345,-0.480,0.488,-0.459,0.282]	[0.084,-0.630,0.281,-0.091,0.427,0.572]	[1.000,1.000,0.000,0.000,1.000,0.000]
8	valid	8.97	0.0075	[-0.489,0.321,-0.447,0.454,-0.427,0.263]	[0.307,-0.730,-0.027,0.323,-0.363,-0.369]	[1.000,1.000,0.000,0.000,0.000,0.000]
9	valid	9.94	0.0008	[-0.576,0.287,-0.396,0.402,-0.377,0.353]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,0.000,1.000,0.000,0.000,1.000]
10	valid	11.81	0.0011	[-0.595,0.282,-0.396,0.356,-0.340,0.409]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,0.000,1.000]
11	valid	11.14	0.0007	[-0.576,0.273,-0.383,0.427,-0.329,0.396]	[-0.055,-0.644,0.309,-0.348,-0.604,-0.036]	[0.000,1.000,0.000,1.000,0.000,0.000]
12	valid	12.32	0.0020	[-0.595,0.278,-0.340,0.381,-0.363,0.419]	[-0.220,-0.033,0.564,-0.340,0.449,0.561]	[0.000,0.000,1.000,0.000,1.000,0.000]
13	valid	8.74	0.0005	[-0.561,0.325,-0.367,0.392,-0.359,0.403]	[-0.357,-0.137,0.908,0.137,-0.054,-0.091]	[0.000,1.000,1.000,0.000,0.000,0.000]
14	valid	22.59	0.0011	[-0.752,0.259,-0.292,0.312,-0.286,0.321]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[1.000,1.000,0.000,1.000,0.000,0.000]
15	valid	11.80	0.0006	[-0.592,0.297,-0.321,0.420,-0.363,0.386]	[-0.274,-0.015,-0.261,-0.243,-0.862,-0.234]	[0.000,1.000,0.000,1.000,0.000,0.000]
16	valid	6.53	0.0000	[-0.541,0.378,-0.392,0.399,-0.343,0.365]	[0.245,-0.141,0.121,0.841,0.283,0.344]	[1.000,1.000,0.000,0.000,1.000,0.000]
17	valid	5.46	0.0004	[-0.512,0.360,-0.373,0.430,-0.369,0.386]	[-0.594,-0.378,0.498,0.219,0.416,-0.188]	[0.000,0.000,1.000,0.000,1.000,0.000]
18	valid	3.81	0.0001	[-0.495,0.383,-0.382,0.415,-0.379,0.383]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
19	valid	4.38	0.0001	[-0.507,0.388,-0.382,0.401,-0.378,0.378]	[-0.617,-0.050,-0.207,-0.045,0.735,0.179]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	1.83	0.0001	[-0.453,0.395,-0.445,0.410,-0.374,0.363]	[0.382,0.577,0.142,0.257,0.377,-0.542]	[1.000,0.000,1.000,0.000,1.000,0.000]

Small budget-constrained selection | seed 2 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	55.92	0.3120	[0.039,0.515,-0.038,-0.182,-0.410,0.476]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
2	valid	28.44	0.0886	[-0.373,0.374,-0.479,-0.012,-0.650,0.499]	[0.396,-0.228,0.248,0.286,0.566,-0.573]	[1.000,0.000,1.000,0.000,1.000,0.000]
3	valid	17.93	0.0176	[-0.443,0.419,-0.463,0.200,-0.661,0.444]	[0.245,-0.141,0.121,0.841,0.283,0.344]	[1.000,1.000,0.000,0.000,1.000,0.000]
4	valid	8.90	0.0021	[-0.628,0.466,-0.512,0.422,-0.619,0.395]	[0.252,0.214,0.243,0.798,-0.352,-0.267]	[1.000,0.000,1.000,0.000,0.000,0.000]
5	valid	11.78	0.0042	[-0.546,0.579,-0.599,0.350,-0.701,0.517]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
6	valid	6.17	0.0007	[-0.485,0.557,-0.597,0.597,-0.464,0.464]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]
7	valid	7.13	0.0010	[-0.555,0.690,-0.764,0.651,-0.585,0.556]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]
8	valid	6.14	0.0003	[-0.578,0.677,-0.737,0.614,-0.681,0.589]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
9	valid	5.62	0.0005	[-0.569,0.588,-0.696,0.545,-0.491,0.526]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[0.000,1.000,0.000,0.000,0.000,1.000]
10	valid	5.03	0.0007	[-0.645,0.676,-0.748,0.598,-0.602,0.673]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
11	valid	4.92	0.0004	[-0.669,0.664,-0.754,0.570,-0.667,0.586]	[0.252,0.214,0.243,0.798,-0.352,-0.267]	[1.000,0.000,1.000,0.000,0.000,0.000]
12	valid	5.41	0.0005	[-0.634,0.649,-0.707,0.554,-0.515,0.528]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[1.000,1.000,0.000,1.000,0.000,0.000]
13	valid	8.16	0.0017	[-0.624,0.710,-0.775,0.537,-0.636,0.490]	[0.382,0.577,0.142,0.257,0.377,-0.542]	[1.000,0.000,1.000,0.000,1.000,0.000]
14	valid	6.04	0.0010	[-0.634,0.610,-0.745,0.559,-0.676,0.521]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]
15	valid	3.19	0.0003	[-0.637,0.607,-0.708,0.595,-0.607,0.556]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
16	valid	5.32	0.0006	[-0.662,0.703,-0.760,0.603,-0.700,0.574]	[0.382,0.577,0.142,0.257,0.377,-0.542]	[1.000,0.000,1.000,0.000,1.000,0.000]
17	valid	5.44	0.0008	[-0.584,0.660,-0.723,0.664,-0.623,0.537]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[1.000,1.000,0.000,1.000,0.000,0.000]
18	valid	8.38	0.0017	[-0.511,0.638,-0.746,0.579,-0.619,0.491]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
19	valid	6.14	0.0005	[-0.695,0.665,-0.746,0.553,-0.527,0.531]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,0.000,1.000]
20	valid	5.71	0.0005	[-0.660,0.739,-0.778,0.620,-0.621,0.569]	[0.382,0.577,0.142,0.257,0.377,-0.542]	[1.000,0.000,1.000,0.000,1.000,0.000]

Small budget-constrained selection | seed 2 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	51.64	0.0923	[-0.150,0.433,-0.074,0.000,-0.001,0.055]	[0.353,0.892,0.177,0.173,0.002,0.134]	[1.000,0.000,1.000,0.000,1.000,0.000]
2	valid	43.73	0.0428	[-0.138,0.457,-0.351,0.036,-0.015,0.051]	[-0.357,-0.137,0.908,0.137,-0.054,-0.091]	[0.000,1.000,1.000,0.000,0.000,0.000]
3	valid	43.69	0.1672	[-0.112,0.347,-0.444,-0.052,-0.015,0.259]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,0.000,1.000,0.000,0.000,1.000]
4	valid	33.87	0.0406	[-0.250,0.560,-0.390,0.095,-0.039,0.227]	[-0.536,0.592,0.004,-0.576,-0.106,0.142]	[0.000,0.000,1.000,1.000,0.000,0.000]
5	valid	35.26	0.0405	[-0.263,0.500,-0.592,0.082,-0.035,0.216]	[-0.226,-0.027,0.968,-0.032,-0.081,-0.065]	[0.000,0.000,1.000,0.000,0.000,1.000]
6	valid	35.83	0.0149	[-0.289,0.406,-0.793,0.134,-0.086,0.187]	[-0.289,-0.099,0.669,-0.415,-0.389,0.370]	[0.000,0.000,1.000,1.000,0.000,0.000]
7	valid	32.16	0.0207	[-0.295,0.589,-0.637,0.115,-0.145,0.179]	[0.084,-0.630,0.281,-0.091,0.427,0.572]	[1.000,1.000,0.000,0.000,1.000,0.000]
8	valid	36.40	0.0203	[-0.291,0.520,-0.821,0.100,-0.129,0.166]	[-0.226,-0.027,0.968,-0.032,-0.081,-0.065]	[0.000,0.000,1.000,0.000,0.000,1.000]
9	valid	37.58	0.0206	[-0.358,0.424,-0.895,0.094,-0.201,0.084]	[0.396,-0.228,0.248,0.286,0.566,-0.573]	[1.000,0.000,1.000,0.000,1.000,0.000]
10	valid	36.56	0.0278	[-0.401,0.532,-0.844,0.084,-0.171,0.084]	[-0.501,0.782,0.124,0.315,0.096,-0.119]	[0.000,0.000,1.000,0.000,0.000,1.000]
11	valid	33.34	0.0332	[-0.466,0.567,-0.735,0.080,-0.236,0.069]	[0.502,-0.302,0.198,-0.080,0.768,-0.147]	[1.000,1.000,0.000,0.000,1.000,0.000]
12	valid	31.34	0.0363	[-0.476,0.591,-0.706,0.112,-0.284,0.060]	[-0.332,0.375,0.215,0.555,0.601,0.184]	[0.000,0.000,1.000,0.000,1.000,0.000]
13	valid	32.16	0.0390	[-0.461,0.730,-0.580,0.098,-0.325,0.056]	[0.084,-0.630,0.281,-0.091,0.427,0.572]	[1.000,1.000,0.000,0.000,1.000,0.000]
14	valid	31.14	0.0406	[-0.447,0.691,-0.577,0.090,-0.407,0.053]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
15	valid	30.64	0.0391	[-0.441,0.604,-0.612,0.084,-0.494,0.041]	[0.183,-0.196,0.420,0.081,0.836,-0.216]	[1.000,0.000,1.000,0.000,1.000,0.000]
16	valid	31.95	0.0399	[-0.434,0.556,-0.730,0.076,-0.458,0.040]	[-0.226,-0.027,0.968,-0.032,-0.081,-0.065]	[0.000,0.000,1.000,0.000,0.000,1.000]
17	valid	33.03	0.0406	[-0.405,0.551,-0.811,0.101,-0.411,0.037]	[0.029,0.208,0.555,-0.742,0.090,0.298]	[1.000,0.000,1.000,1.000,0.000,0.000]
18	valid	33.79	0.0331	[-0.430,0.523,-0.884,0.114,-0.353,0.049]	[-0.450,-0.212,-0.516,0.464,-0.323,-0.409]	[0.000,1.000,0.000,0.000,0.000,1.000]
19	valid	36.35	0.0416	[-0.411,0.491,-0.971,0.096,-0.357,0.042]	[-0.277,-0.229,-0.711,-0.110,0.444,0.395]	[0.000,1.000,0.000,0.000,1.000,0.000]
20	valid	36.05	0.0438	[-0.414,0.567,-0.936,0.088,-0.329,0.042]	[0.353,0.892,0.177,0.173,0.002,0.134]	[1.000,0.000,1.000,0.000,1.000,0.000]

Small budget-constrained selection | seed 3 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	29.06	0.0795	[-0.447,0.447,-0.447,0.447,-0.447,-0.038]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	24.85	0.0332	[-0.379,0.551,-0.379,0.379,-0.514,0.028]	[-0.004,-0.478,0.183,0.145,0.592,0.606]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	29.17	0.0376	[-0.291,0.528,-0.567,0.291,-0.480,0.000]	[0.183,-0.196,0.420,0.081,0.836,-0.216]	[1.000,0.000,1.000,0.000,1.000,0.000]
4	valid	29.23	0.0755	[-0.289,0.525,-0.564,0.310,-0.476,-0.006]	[-0.185,-0.096,0.017,-0.452,-0.847,-0.187]	[0.000,1.000,0.000,1.000,0.000,0.000]
5	valid	29.36	0.0764	[-0.277,0.550,-0.562,0.299,-0.464,-0.003]	[0.469,-0.200,0.402,0.296,0.103,0.693]	[1.000,1.000,1.000,0.000,0.000,0.000]
6	valid	29.32	0.0747	[-0.250,0.507,-0.572,0.402,-0.439,-0.000]	[-0.365,0.210,0.374,-0.742,0.235,0.278]	[0.000,0.000,1.000,1.000,0.000,0.000]
7	valid	29.32	0.0747	[-0.250,0.507,-0.572,0.402,-0.439,-0.000]	[0.029,0.208,0.555,-0.742,0.090,0.298]	[1.000,0.000,1.000,1.000,0.000,0.000]
8	valid	29.00	0.0359	[-0.250,0.507,-0.572,0.402,-0.439,0.006]	[0.080,-0.595,-0.463,-0.042,0.321,0.567]	[1.000,1.000,0.000,0.000,1.000,0.000]
9	valid	29.00	0.0359	[-0.250,0.507,-0.572,0.402,-0.439,0.006]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
10	valid	13.29	0.0063	[-0.229,0.464,-0.523,0.401,-0.402,0.369]	[-0.039,-0.468,0.013,-0.267,-0.117,-0.833]	[0.000,1.000,0.000,0.000,0.000,1.000]
11	valid	11.98	0.0013	[-0.344,0.364,-0.410,0.422,-0.315,0.551]	[0.382,0.577,0.142,0.257,0.377,-0.542]	[1.000,0.000,0.000,0.000,0.000,1.000]
12	valid	11.98	0.0013	[-0.344,0.364,-0.410,0.422,-0.315,0.551]	[0.036,-0.388,-0.217,0.361,-0.715,-0.400]	[0.000,1.000,0.000,0.000,0.000,1.000]
13	valid	10.90	0.0009	[-0.339,0.397,-0.404,0.416,-0.310,0.543]	[-0.277,-0.229,-0.711,-0.110,0.444,0.395]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	10.90	0.0009	[-0.339,0.397,-0.404,0.416,-0.310,0.543]	[0.122,0.400,0.712,-0.503,-0.243,0.082]	[1.000,0.000,1.000,1.000,0.000,0.000]
15	valid	10.90	0.0009	[-0.339,0.397,-0.404,0.416,-0.310,0.543]	[0.138,0.648,-0.649,0.281,-0.230,0.085]	[1.000,0.000,0.000,0.000,0.000,0.000]
16	valid	10.64	0.0021	[-0.306,0.360,-0.412,0.385,-0.387,0.556]	[-0.010,-0.599,0.097,-0.205,0.355,0.681]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	10.66	0.0021	[-0.306,0.360,-0.412,0.385,-0.387,0.556]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,0.000,1.000]
18	valid	10.66	0.0021	[-0.306,0.360,-0.412,0.385,-0.387,0.556]	[0.369,0.161,-0.500,-0.728,-0.154,-0.186]	[1.000,0.000,0.000,1.000,0.000,0.000]
19	valid	10.96	0.0019	[-0.297,0.354,-0.407,0.406,-0.391,0.551]	[-0.197,-0.321,0.480,-0.640,-0.361,-0.297]	[0.000,0.000,1.000,1.000,0.000,0.000]
20	valid	10.54	0.0019	[-0.296,0.364,-0.407,0.404,-0.389,0.548]	[0.145,-0.215,-0.062,0.001,-0.905,-0.331]	[0.000,1.000,0.000,0.000,0.000,1.000]

Small budget-constrained selection | seed 3 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	29.06	0.0795	[-0.447,0.447,-0.447,0.447,-0.447,-0.038]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	7.08	0.0026	[-0.400,0.400,-0.400,0.400,-0.471,0.372]	[-0.113,-0.597,0.057,0.051,-0.151,0.776]	[0.000,1.000,1.000,0.000,0.000,0.000]
3	valid	23.28	0.0202	[-0.246,0.246,-0.609,0.246,-0.349,0.571]	[0.252,0.214,0.243,0.798,-0.352,-0.267]	[0.000,0.000,1.000,0.000,0.000,1.000]
4	valid	17.10	0.0188	[-0.195,0.429,-0.483,0.195,-0.550,0.452]	[0.084,-0.630,0.281,-0.091,0.427,0.572]	[1.000,1.000,0.000,0.000,1.000,0.000]
5	valid	13.63	0.0086	[-0.183,0.403,-0.454,0.384,-0.518,0.426]	[-0.708,-0.466,-0.409,-0.295,0.108,0.123]	[0.000,1.000,0.000,1.000,0.000,0.000]
6	valid	17.22	0.0119	[-0.160,0.352,-0.397,0.336,-0.453,0.613]	[-0.448,0.057,-0.834,-0.145,-0.249,-0.131]	[0.000,0.000,0.000,0.000,0.000,1.000]
7	valid	30.02	0.0163	[-0.102,0.225,-0.254,0.514,-0.290,0.725]	[-0.705,0.295,-0.275,-0.330,-0.445,-0.181]	[0.000,0.000,0.000,1.000,0.000,0.000]
8	valid	27.59	0.0152	[-0.094,0.207,-0.233,0.472,-0.479,0.665]	[-0.504,0.029,-0.836,-0.136,0.098,-0.137]	[0.000,0.000,0.000,1.000,1.000,0.000]
9	valid	24.07	0.0131	[-0.091,0.313,-0.226,0.458,-0.465,0.646]	[0.245,-0.141,0.121,0.841,0.283,0.344]	[1.000,1.000,0.000,0.000,1.000,0.000]
10	valid	13.13	0.0025	[-0.214,0.398,-0.340,0.411,-0.418,0.580]	[0.382,0.577,0.142,0.257,0.377,-0.542]	[1.000,0.000,0.000,0.000,0.000,1.000]
11	valid	12.06	0.0026	[-0.204,0.486,-0.324,0.392,-0.398,0.553]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,1.000,0.000,0.000,0.000,1.000]
12	valid	12.61	0.0066	[-0.214,0.571,-0.439,0.339,-0.345,0.448]	[-0.197,-0.321,0.480,-0.640,-0.361,-0.297]	[0.000,0.000,1.000,1.000,0.000,0.000]
13	valid	10.54	0.0028	[-0.264,0.564,-0.433,0.335,-0.341,0.443]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
14	valid	8.87	0.0018	[-0.318,0.554,-0.426,0.329,-0.335,0.435]	[0.384,-0.574,-0.279,-0.008,-0.622,-0.239]	[1.000,1.000,0.000,1.000,0.000,0.000]
15	valid	5.26	0.0009	[-0.323,0.515,-0.396,0.306,-0.421,0.450]	[0.177,0.180,0.756,-0.012,0.405,-0.448]	[1.000,0.000,1.000,0.000,1.000,0.000]
16	valid	6.85	0.0005	[-0.349,0.538,-0.420,0.314,-0.372,0.419]	[0.514,0.117,0.605,0.460,-0.095,-0.368]	[1.000,0.000,1.000,0.000,0.000,0.000]
17	valid	5.69	0.0000	[-0.349,0.523,-0.413,0.336,-0.374,0.427]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,1.000,0.000,1.000,0.000,0.000]
18	valid	4.06	0.0000	[-0.350,0.492,-0.399,0.367,-0.378,0.446]	[0.766,-0.576,0.118,-0.140,-0.195,-0.104]	[1.000,1.000,0.000,1.000,0.000,0.000]
19	valid	3.87	0.0000	[-0.350,0.480,-0.394,0.378,-0.380,0.453]	[-0.055,-0.644,0.309,-0.348,-0.604,-0.036]	[0.000,1.000,0.000,1.000,0.000,0.000]
20	valid	2.16	0.0000	[-0.342,0.469,-0.385,0.370,-0.426,0.443]	[0.147,0.323,0.809,-0.074,0.069,0.457]	[1.000,0.000,1.000,0.000,1.000,0.000]

Small budget-constrained selection | seed 3 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	8.19	0.0010	[-0.464,0.409,-0.373,0.378,-0.374,0.522]	[0.513,0.253,-0.346,-0.132,-0.312,0.662]	[1.000,0.000,0.000,1.000,0.000,0.000]
2	valid	21.62	0.0261	[-0.254,0.432,-0.193,0.219,-0.229,0.625]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
3	valid	8.60	0.0011	[-0.412,0.511,-0.288,0.462,-0.523,0.607]	[0.117,-0.020,-0.714,0.431,0.536,-0.047]	[1.000,1.000,0.000,0.000,1.000,0.000]
4	valid	15.36	0.0132	[-0.496,0.721,-0.390,0.169,-0.635,0.500]	[0.396,-0.228,0.248,0.286,0.566,-0.573]	[1.000,1.000,0.000,0.000,1.000,0.000]
5	valid	13.89	0.0056	[-0.565,0.695,-0.213,0.560,-0.472,0.539]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,1.000,0.000,1.000,0.000,0.000]
6	valid	9.98	0.0037	[-0.356,0.683,-0.415,0.463,-0.558,0.394]	[-0.205,-0.189,0.668,-0.208,0.290,0.591]	[0.000,0.000,1.000,0.000,1.000,0.000]
7	valid	6.03	0.0011	[-0.510,0.734,-0.391,0.517,-0.561,0.606]	[0.086,-0.258,0.294,-0.376,0.176,-0.817]	[0.000,1.000,0.000,0.000,0.000,1.000]
8	valid	6.54	0.0019	[-0.508,0.740,-0.376,0.528,-0.624,0.601]	[0.177,0.180,0.756,-0.012,0.405,-0.448]	[1.000,0.000,1.000,0.000,1.000,0.000]
9	valid	6.94	0.0010	[-0.524,0.769,-0.411,0.582,-0.569,0.588]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,1.000,0.000,1.000,0.000,0.000]
10	valid	8.69	0.0043	[-0.646,0.772,-0.357,0.635,-0.656,0.684]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,0.000,1.000]
11	valid	8.49	0.0015	[-0.430,0.665,-0.352,0.584,-0.522,0.534]	[-0.205,-0.189,0.668,-0.208,0.290,0.591]	[0.000,0.000,1.000,0.000,1.000,0.000]
12	valid	6.11	0.0012	[-0.533,0.702,-0.410,0.607,-0.590,0.678]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,0.000,1.000]
13	valid	9.58	0.0033	[-0.655,0.672,-0.323,0.532,-0.624,0.706]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[0.000,1.000,0.000,0.000,0.000,1.000]
14	valid	7.59	0.0015	[-0.632,0.662,-0.387,0.473,-0.576,0.699]	[0.549,0.000,0.537,-0.393,-0.240,-0.445]	[1.000,0.000,1.000,1.000,0.000,0.000]
15	valid	6.07	0.0010	[-0.465,0.623,-0.410,0.560,-0.483,0.566]	[-0.197,-0.321,0.480,-0.640,-0.361,-0.297]	[0.000,0.000,1.000,1.000,0.000,0.000]
16	valid	4.38	0.0004	[-0.665,0.709,-0.501,0.590,-0.653,0.723]	[0.384,-0.574,-0.279,-0.008,-0.622,-0.239]	[1.000,1.000,0.000,1.000,0.000,0.000]
17	valid	1.56	0.0000	[-0.608,0.755,-0.564,0.618,-0.701,0.714]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,1.000,0.000,1.000,0.000,0.000]
18	valid	4.78	0.0006	[-0.625,0.688,-0.499,0.624,-0.651,0.755]	[-0.594,-0.378,0.498,0.219,0.416,-0.188]	[0.000,1.000,0.000,0.000,1.000,0.000]
19	valid	6.80	0.0010	[-0.715,0.707,-0.485,0.654,-0.630,0.702]	[0.514,0.117,0.605,0.460,-0.095,-0.368]	[1.000,0.000,1.000,0.000,0.000,0.000]
20	valid	7.26	0.0004	[-0.691,0.677,-0.442,0.498,-0.572,0.660]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[0.000,1.000,0.000,0.000,0.000,1.000]

Small budget-constrained selection | seed 3 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	53.73	0.1549	[-0.057,0.000,-0.228,0.002,-0.427,0.000]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
2	valid	52.29	0.0711	[-0.049,0.182,-0.115,0.054,-0.805,0.000]	[-0.004,-0.478,0.183,0.145,0.592,0.606]	[0.000,1.000,0.000,0.000,1.000,0.000]
3	valid	57.33	0.1138	[-0.069,0.106,-0.169,0.042,-1.150,-0.035]	[0.183,-0.196,0.420,0.081,0.836,-0.216]	[1.000,0.000,1.000,0.000,1.000,0.000]
4	valid	58.75	0.1053	[-0.073,0.092,-0.126,0.082,-1.446,-0.027]	[-0.185,-0.096,0.017,-0.452,-0.847,-0.187]	[0.000,1.000,0.000,1.000,0.000,0.000]
5	valid	53.29	0.1002	[-0.167,0.130,-0.218,0.140,-1.290,-0.026]	[0.469,-0.200,0.402,0.296,0.103,0.693]	[1.000,1.000,1.000,0.000,0.000,0.000]
6	valid	47.77	0.0951	[-0.217,0.155,-0.278,0.244,-1.119,-0.026]	[-0.365,0.210,0.374,-0.742,0.235,0.278]	[0.000,0.000,1.000,1.000,0.000,0.000]
7	valid	43.62	0.0937	[-0.206,0.168,-0.357,0.351,-0.996,-0.024]	[0.029,0.208,0.555,-0.742,0.090,0.298]	[1.000,0.000,1.000,1.000,0.000,0.000]
8	valid	43.16	0.0885	[-0.188,0.226,-0.412,0.294,-1.038,-0.021]	[0.080,-0.595,-0.463,-0.042,0.321,0.567]	[1.000,1.000,0.000,0.000,1.000,0.000]
9	valid	46.48	0.0874	[-0.167,0.251,-0.344,0.245,-1.210,-0.017]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
10	valid	44.79	0.0469	[-0.169,0.301,-0.338,0.217,-1.197,0.019]	[-0.039,-0.468,0.013,-0.267,-0.117,-0.833]	[0.000,1.000,0.000,0.000,0.000,1.000]
11	valid	39.40	0.0435	[-0.216,0.301,-0.311,0.217,-0.970,0.056]	[0.382,0.577,0.142,0.257,0.377,-0.542]	[1.000,0.000,0.000,0.000,0.000,1.000]
12	valid	34.60	0.0373	[-0.207,0.400,-0.362,0.285,-0.957,0.095]	[0.036,-0.388,-0.217,0.361,-0.715,-0.400]	[0.000,1.000,0.000,0.000,0.000,1.000]
13	valid	36.36	0.0383	[-0.205,0.384,-0.423,0.238,-0.998,0.083]	[-0.277,-0.229,-0.711,-0.110,0.444,0.395]	[0.000,1.000,0.000,0.000,1.000,0.000]
14	valid	33.81	0.0386	[-0.195,0.406,-0.503,0.264,-0.904,0.079]	[0.122,0.400,0.712,-0.503,-0.243,0.082]	[1.000,0.000,1.000,1.000,0.000,0.000]
15	valid	31.90	0.0425	[-0.197,0.501,-0.477,0.250,-0.857,0.079]	[0.138,0.648,-0.649,0.281,-0.230,0.085]	[1.000,0.000,0.000,0.000,0.000,0.000]
16	valid	33.74	0.0419	[-0.180,0.587,-0.417,0.208,-0.905,0.072]	[-0.010,-0.599,0.097,-0.205,0.355,0.681]	[0.000,1.000,0.000,0.000,1.000,0.000]
17	valid	29.08	0.0353	[-0.206,0.587,-0.417,0.196,-0.721,0.100]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,0.000,1.000]
18	valid	27.76	0.0326	[-0.234,0.619,-0.417,0.251,-0.721,0.092]	[0.369,0.161,-0.500,-0.728,-0.154,-0.186]	[1.000,0.000,0.000,1.000,0.000,0.000]
19	valid	28.24	0.0310	[-0.238,0.520,-0.468,0.301,-0.775,0.087]	[-0.197,-0.321,0.480,-0.640,-0.361,-0.297]	[0.000,0.000,1.000,1.000,0.000,0.000]
20	valid	27.35	0.0284	[-0.206,0.578,-0.469,0.289,-0.744,0.110]	[0.145,-0.215,-0.062,0.001,-0.905,-0.331]	[0.000,1.000,0.000,0.000,0.000,1.000]

Small budget-constrained selection | seed 4 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	50.56	0.2518	[-0.572,-0.125,-0.572,0.572,-0.052,-0.000]	[0.623,-0.021,0.047,-0.764,-0.140,-0.079]	[1.000,0.000,1.000,1.000,0.000,0.000]
2	valid	25.82	0.0338	[-0.412,0.331,-0.606,0.422,-0.412,0.075]	[-0.461,0.276,0.039,0.296,0.739,0.276]	[0.000,0.000,1.000,0.000,1.000,0.000]
3	valid	35.70	0.0484	[-0.219,0.461,-0.453,0.696,-0.219,0.040]	[-0.055,-0.644,0.309,-0.348,-0.604,-0.036]	[0.000,1.000,0.000,1.000,0.000,0.000]
4	valid	35.01	0.0839	[-0.212,0.446,-0.438,0.673,-0.332,-0.000]	[-0.783,-0.543,0.041,0.254,0.160,0.003]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	20.73	0.0150	[-0.152,0.485,-0.390,0.550,-0.257,0.471]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
6	valid	20.73	0.0150	[-0.152,0.485,-0.390,0.550,-0.257,0.471]	[-0.055,-0.644,0.309,-0.348,-0.604,-0.036]	[0.000,1.000,0.000,1.000,0.000,0.000]
7	valid	20.73	0.0150	[-0.152,0.485,-0.390,0.550,-0.257,0.471]	[0.087,0.723,0.584,-0.285,-0.016,-0.218]	[1.000,0.000,1.000,1.000,0.000,0.000]
8	valid	15.32	0.0090	[-0.432,0.459,-0.343,0.465,-0.212,0.474]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
9	valid	15.32	0.0090	[-0.432,0.459,-0.343,0.465,-0.212,0.474]	[-0.734,0.164,0.539,0.097,-0.129,-0.343]	[0.000,0.000,1.000,0.000,0.000,1.000]
10	valid	15.32	0.0090	[-0.432,0.459,-0.343,0.465,-0.212,0.474]	[0.085,-0.109,-0.474,-0.669,-0.538,0.138]	[1.000,1.000,0.000,1.000,0.000,0.000]
11	valid	15.32	0.0090	[-0.432,0.459,-0.343,0.465,-0.212,0.474]	[-0.196,-0.693,0.094,-0.424,0.015,0.541]	[0.000,1.000,0.000,1.000,0.000,0.000]
12	valid	15.32	0.0090	[-0.432,0.459,-0.343,0.465,-0.212,0.474]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
13	valid	15.32	0.0090	[-0.432,0.459,-0.343,0.465,-0.212,0.474]	[-0.357,-0.137,0.908,0.137,-0.054,-0.091]	[0.000,1.000,1.000,0.000,0.000,0.000]
14	valid	15.32	0.0090	[-0.432,0.459,-0.343,0.465,-0.212,0.474]	[0.219,0.018,-0.325,-0.316,0.017,0.864]	[1.000,0.000,0.000,1.000,0.000,0.000]
15	valid	15.32	0.0090	[-0.432,0.459,-0.343,0.465,-0.212,0.474]	[-0.473,-0.617,-0.120,-0.542,-0.277,-0.108]	[0.000,1.000,0.000,1.000,0.000,0.000]
16	valid	15.32	0.0090	[-0.432,0.459,-0.343,0.465,-0.212,0.474]	[0.578,-0.104,-0.273,0.747,0.117,-0.091]	[1.000,1.000,0.000,0.000,1.000,0.000]
17	valid	15.12	0.0090	[-0.431,0.458,-0.342,0.465,-0.216,0.473]	[-0.117,-0.054,-0.081,-0.534,0.263,0.789]	[0.000,0.000,0.000,1.000,1.000,0.000]
18	valid	15.12	0.0090	[-0.431,0.458,-0.342,0.465,-0.216,0.473]	[0.286,0.579,0.097,0.210,0.481,0.547]	[1.000,0.000,1.000,0.000,1.000,0.000]
19	valid	15.12	0.0090	[-0.431,0.458,-0.342,0.465,-0.216,0.473]	[0.369,0.161,-0.500,-0.728,-0.154,-0.186]	[1.000,0.000,0.000,1.000,0.000,0.000]
20	valid	16.96	0.0083	[-0.384,0.430,-0.467,0.498,-0.184,0.410]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]

Small budget-constrained selection | seed 4 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	50.56	0.2518	[-0.572,-0.125,-0.572,0.572,-0.052,-0.000]	[0.623,-0.021,0.047,-0.764,-0.140,-0.079]	[1.000,0.000,1.000,1.000,0.000,0.000]
2	valid	27.31	0.0283	[-0.415,0.415,-0.646,0.427,-0.124,0.202]	[-0.369,-0.499,0.271,0.164,-0.327,0.638]	[0.000,1.000,1.000,0.000,0.000,0.000]
3	valid	40.60	0.0700	[-0.245,0.530,-0.513,0.627,-0.000,0.048]	[0.424,-0.620,0.310,-0.475,0.258,0.219]	[1.000,1.000,0.000,1.000,0.000,0.000]
4	valid	25.24	0.0256	[-0.232,0.502,-0.485,0.593,-0.232,0.232]	[-0.641,0.061,-0.123,0.280,-0.235,0.660]	[0.000,0.000,0.000,0.000,0.000,0.000]
5	valid	18.00	0.0109	[-0.254,0.443,-0.428,0.524,-0.205,0.489]	[0.335,0.216,-0.187,-0.157,0.603,-0.647]	[1.000,0.000,0.000,0.000,0.000,1.000]
6	valid	16.08	0.0108	[-0.362,0.380,-0.559,0.472,-0.250,0.356]	[0.396,-0.228,0.248,0.286,0.566,-0.573]	[1.000,0.000,1.000,0.000,1.000,0.000]
7	valid	18.89	0.0152	[-0.342,0.359,-0.528,0.554,-0.236,0.336]	[0.766,-0.576,0.118,-0.140,-0.195,-0.104]	[1.000,1.000,0.000,1.000,0.000,0.000]
8	valid	18.58	0.0150	[-0.428,0.345,-0.507,0.533,-0.227,0.323]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
9	valid	15.94	0.0146	[-0.407,0.410,-0.483,0.506,-0.249,0.338]	[0.502,-0.302,0.198,-0.080,0.768,-0.147]	[1.000,1.000,0.000,0.000,1.000,0.000]
10	valid	9.64	0.0015	[-0.381,0.384,-0.452,0.474,-0.328,0.412]	[0.307,-0.730,-0.027,0.323,-0.363,-0.369]	[0.000,1.000,0.000,0.000,0.000,1.000]
11	valid	16.23	0.0130	[-0.360,0.398,-0.529,0.498,-0.245,0.354]	[0.245,-0.141,0.121,0.841,0.283,0.344]	[1.000,0.000,1.000,0.000,1.000,0.000]
12	valid	14.42	0.0094	[-0.355,0.392,-0.522,0.491,-0.291,0.349]	[-0.126,-0.373,0.609,0.016,0.603,0.331]	[0.000,0.000,1.000,0.000,1.000,0.000]
13	valid	15.25	0.0127	[-0.322,0.430,-0.535,0.477,-0.291,0.336]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
14	valid	15.12	0.0127	[-0.331,0.420,-0.522,0.492,-0.284,0.344]	[-0.055,-0.644,0.309,-0.348,-0.604,-0.036]	[0.000,1.000,0.000,1.000,0.000,0.000]
15	valid	14.39	0.0127	[-0.329,0.417,-0.519,0.489,-0.305,0.342]	[-0.010,-0.599,0.097,-0.205,0.355,0.681]	[0.000,1.000,0.000,0.000,1.000,0.000]
16	valid	13.46	0.0076	[-0.325,0.412,-0.513,0.483,-0.302,0.369]	[0.382,0.577,0.142,0.257,0.377,-0.542]	[1.000,0.000,0.000,0.000,0.000,1.000]
17	valid	15.46	0.0114	[-0.307,0.443,-0.529,0.479,-0.284,0.345]	[-0.617,-0.050,-0.207,-0.045,0.735,0.179]	[0.000,1.000,0.000,0.000,1.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.245,-0.141,0.121,0.841,0.283,0.344]	[1.000,1.000,0.000,0.000,1.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000,0.000,0.000]	[-0.055,-0.644,0.309,-0.348,-0.604,-0.036]	[0.000,1.000,1.000,0.000,0.000,0.000]

Small budget-constrained selection | seed 4 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	44.05	0.1708	[0.024,0.607,-0.093,0.021,-0.189,0.661]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[0.000,1.000,0.000,0.000,0.000,1.000]
2	valid	39.87	0.1920	[-0.486,0.537,0.250,0.296,-0.338,0.670]	[0.545,-0.007,0.048,0.259,-0.790,-0.095]	[1.000,0.000,0.000,0.000,0.000,1.000]
3	valid	12.76	0.0059	[-0.482,0.515,-0.286,0.314,-0.577,0.628]	[0.578,-0.104,-0.273,0.747,0.117,-0.091]	[1.000,1.000,0.000,0.000,1.000,0.000]
4	valid	11.81	0.0025	[-0.450,0.286,-0.581,0.458,-0.383,0.626]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]
5	valid	6.69	0.0005	[-0.675,0.500,-0.570,0.547,-0.531,0.660]	[0.384,-0.574,-0.279,-0.008,-0.622,-0.239]	[1.000,1.000,0.000,1.000,0.000,0.000]
6	valid	11.28	0.0046	[-0.586,0.344,-0.689,0.434,-0.568,0.511]	[0.396,-0.228,0.248,0.286,0.566,-0.573]	[1.000,0.000,1.000,0.000,1.000,0.000]
7	valid	8.14	0.0023	[-0.502,0.517,-0.740,0.568,-0.547,0.575]	[0.384,-0.574,-0.279,-0.008,-0.622,-0.239]	[1.000,1.000,0.000,1.000,0.000,0.000]
8	valid	8.78	0.0016	[-0.481,0.618,-0.650,0.382,-0.500,0.538]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
9	valid	7.76	0.0022	[-0.610,0.524,-0.746,0.501,-0.536,0.598]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]
10	valid	4.54	0.0005	[-0.659,0.541,-0.688,0.593,-0.620,0.682]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]
11	valid	7.27	0.0008	[-0.606,0.569,-0.711,0.624,-0.499,0.688]	[0.382,0.577,0.142,0.257,0.377,-0.542]	[1.000,0.000,0.000,0.000,0.000,1.000]
12	valid	5.27	0.0000	[-0.578,0.535,-0.601,0.554,-0.492,0.663]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
13	valid	6.03	0.0000	[-0.654,0.565,-0.716,0.556,-0.537,0.696]	[0.396,-0.228,0.248,0.286,0.566,-0.573]	[1.000,0.000,1.000,0.000,1.000,0.000]
14	valid	5.53	0.0005	[-0.626,0.519,-0.687,0.596,-0.560,0.686]	[0.177,0.180,0.756,-0.012,0.405,-0.448]	[1.000,0.000,1.000,0.000,1.000,0.000]
15	valid	4.86	0.0007	[-0.688,0.616,-0.779,0.605,-0.634,0.725]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
16	valid	6.62	0.0007	[-0.573,0.704,-0.790,0.634,-0.605,0.697]	[0.252,0.214,0.243,0.798,-0.352,-0.267]	[0.000,0.000,1.000,0.000,0.000,1.000]
17	valid	4.46	0.0004	[-0.602,0.679,-0.787,0.641,-0.652,0.757]	[0.177,0.180,0.756,-0.012,0.405,-0.448]	[1.000,0.000,1.000,0.000,1.000,0.000]
18	valid	4.72	0.0006	[-0.500,0.575,-0.671,0.610,-0.605,0.677]	[0.177,0.180,0.756,-0.012,0.405,-0.448]	[1.000,0.000,1.000,0.000,1.000,0.000]
19	valid	6.07	0.0012	[-0.547,0.653,-0.715,0.628,-0.585,0.653]	[0.384,-0.574,-0.279,-0.008,-0.622,-0.239]	[1.000,1.000,0.000,1.000,0.000,0.000]
20	valid	3.75	0.0008	[-0.665,0.634,-0.760,0.696,-0.710,0.757]	[0.081,-0.613,0.131,-0.395,-0.446,-0.496]	[0.000,1.000,0.000,0.000,0.000,1.000]

Small budget-constrained selection | seed 4 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	60.52	0.3569	[-0.323,-0.010,-0.022,0.417,0.000,-0.037]	[0.623,-0.021,0.047,-0.764,-0.140,-0.079]	[1.000,0.000,1.000,1.000,0.000,0.000]
2	valid	48.62	0.1140	[-0.486,0.072,-0.030,0.501,-0.231,-0.031]	[-0.461,0.276,0.039,0.296,0.739,0.276]	[0.000,0.000,1.000,0.000,1.000,0.000]
3	valid	45.77	0.2517	[-0.427,0.226,0.045,0.591,-0.400,-0.026]	[-0.055,-0.644,0.309,-0.348,-0.604,-0.036]	[0.000,1.000,0.000,1.000,0.000,0.000]
4	valid	44.82	0.2504	[-0.607,0.295,0.042,0.577,-0.369,-0.021]	[-0.783,-0.543,0.041,0.254,0.160,0.003]	[0.000,1.000,0.000,0.000,1.000,0.000]
5	valid	40.28	0.1918	[-0.531,0.362,0.070,0.462,-0.369,0.072]	[0.126,-0.175,0.132,-0.207,-0.832,-0.448]	[0.000,1.000,0.000,0.000,0.000,1.000]
6	valid	40.64	0.1918	[-0.471,0.486,0.098,0.501,-0.481,0.062]	[-0.055,-0.644,0.309,-0.348,-0.604,-0.036]	[0.000,1.000,0.000,1.000,0.000,0.000]
7	valid	40.14	0.1873	[-0.441,0.642,0.036,0.522,-0.429,0.036]	[0.087,0.723,0.584,-0.285,-0.016,-0.218]	[1.000,0.000,1.000,1.000,0.000,0.000]
8	valid	38.83	0.1864	[-0.424,0.597,0.024,0.475,-0.577,0.034]	[0.068,0.631,-0.136,-0.042,0.736,0.187]	[1.000,0.000,0.000,0.000,1.000,0.000]
9	valid	36.64	0.0617	[-0.546,0.586,-0.011,0.452,-0.525,0.052]	[-0.734,0.164,0.539,0.097,-0.129,-0.343]	[0.000,0.000,1.000,0.000,0.000,1.000]
10	valid	35.90	0.0615	[-0.520,0.564,-0.038,0.573,-0.479,0.056]	[0.085,-0.109,-0.474,-0.669,-0.538,0.138]	[1.000,1.000,0.000,1.000,0.000,0.000]
11	valid	37.86	0.0655	[-0.495,0.669,-0.028,0.603,-0.416,0.049]	[-0.196,-0.693,0.094,-0.424,0.015,0.541]	[0.000,1.000,0.000,1.000,0.000,0.000]
12	valid	36.14	0.0646	[-0.485,0.626,-0.044,0.565,-0.522,0.046]	[-0.110,0.538,-0.421,0.003,0.711,-0.123]	[0.000,0.000,0.000,0.000,1.000,0.000]
13	valid	35.36	0.0642	[-0.470,0.633,-0.074,0.571,-0.515,0.045]	[-0.357,-0.137,0.908,0.137,-0.054,-0.091]	[0.000,1.000,1.000,0.000,0.000,0.000]
14	valid	36.65	0.0659	[-0.510,0.578,-0.067,0.673,-0.457,0.040]	[0.219,0.018,-0.325,-0.316,0.017,0.864]	[1.000,0.000,0.000,1.000,0.000,0.000]
15	valid	37.81	0.0673	[-0.515,0.620,-0.060,0.700,-0.425,0.033]	[-0.473,-0.617,-0.120,-0.542,-0.277,-0.108]	[0.000,1.000,0.000,1.000,0.000,0.000]
16	valid	37.49	0.0671	[-0.612,0.599,-0.066,0.646,-0.413,0.028]	[0.578,-0.104,-0.273,0.747,0.117,-0.091]	[1.000,1.000,0.000,0.000,1.000,0.000]
17	valid	38.47	0.0671	[-0.586,0.530,-0.063,0.747,-0.422,0.025]	[-0.117,-0.054,-0.081,-0.534,0.263,0.789]	[0.000,0.000,0.000,1.000,1.000,0.000]
18	valid	37.01	0.0634	[-0.584,0.595,-0.059,0.662,-0.456,0.036]	[0.286,0.579,0.097,0.210,0.481,0.547]	[1.000,0.000,1.000,0.000,1.000,0.000]
19	valid	38.45	0.0671	[-0.595,0.540,-0.065,0.756,-0.414,0.029]	[0.369,0.161,-0.500,-0.728,-0.154,-0.186]	[1.000,0.000,0.000,1.000,0.000,0.000]
20	valid	41.56	0.0668	[-0.585,0.452,-0.076,0.913,-0.348,0.028]	[0.044,-0.334,0.374,-0.528,0.318,0.605]	[1.000,0.000,1.000,1.000,0.000,0.000]

Stronger parameter noise | seed 0 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	43.02	0.0789	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	35.20	0.0659	[-0.134,0.235,-0.485,0.832]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
3	valid	35.20	0.0659	[-0.134,0.235,-0.485,0.832]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
4	valid	33.49	0.0613	[-0.399,-0.058,-0.578,0.709]	[-0.618,0.483,-0.526,0.328]	[0.000,1.000,0.000,0.000]
5	valid	33.49	0.0613	[-0.399,-0.058,-0.578,0.709]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
6	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
7	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[0.863,0.029,0.095,0.495]	[1.000,0.000,0.000,0.000]
8	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
9	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
10	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
11	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[-0.805,0.381,-0.140,-0.432]	[0.000,0.000,0.000,1.000]
12	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[-0.261,0.493,-0.095,-0.824]	[0.000,0.000,0.000,1.000]
13	valid	16.60	0.0177	[-0.489,0.309,-0.453,0.678]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
14	valid	16.60	0.0177	[-0.489,0.309,-0.453,0.678]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
15	valid	16.59	0.0177	[-0.486,0.313,-0.453,0.679]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
16	valid	16.32	0.0177	[-0.495,0.313,-0.449,0.674]	[0.872,0.150,0.014,-0.466]	[1.000,0.000,0.000,0.000]
17	valid	16.25	0.0178	[-0.482,0.330,-0.448,0.677]	[-0.602,-0.667,0.317,0.303]	[0.000,1.000,0.000,0.000]
18	valid	16.25	0.0178	[-0.482,0.330,-0.448,0.677]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
19	valid	16.25	0.0178	[-0.482,0.330,-0.448,0.677]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
20	valid	16.25	0.0178	[-0.482,0.330,-0.448,0.677]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]

Stronger parameter noise | seed 0 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	43.02	0.0789	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	22.12	0.0432	[-0.550,0.068,-0.629,0.545]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
3	valid	29.36	0.0666	[-0.811,0.016,-0.462,0.360]	[0.405,0.823,-0.005,-0.399]	[1.000,0.000,0.000,0.000]
4	valid	29.80	0.0605	[-0.658,-0.023,-0.697,0.283]	[0.473,-0.294,0.808,0.193]	[0.000,0.000,1.000,0.000]
5	valid	7.00	0.0041	[-0.621,0.382,-0.580,0.363]	[-0.237,-0.863,0.169,-0.413]	[0.000,1.000,0.000,0.000]
6	valid	9.00	0.0129	[-0.522,0.558,-0.534,0.362]	[-0.007,-0.683,-0.056,-0.728]	[0.000,1.000,0.000,0.000]
7	valid	9.21	0.0097	[-0.521,0.561,-0.467,0.442]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
8	valid	10.09	0.0117	[-0.579,0.535,-0.418,0.452]	[0.540,0.560,0.062,-0.625]	[1.000,0.000,0.000,0.000]
9	valid	7.55	0.0062	[-0.565,0.522,-0.463,0.441]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
10	valid	5.32	0.0039	[-0.549,0.507,-0.507,0.429]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
11	valid	6.28	0.0056	[-0.550,0.507,-0.479,0.459]	[-0.162,-0.596,-0.390,-0.683]	[0.000,0.000,0.000,1.000]
12	valid	6.28	0.0056	[-0.550,0.507,-0.479,0.459]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
13	valid	6.28	0.0056	[-0.550,0.507,-0.479,0.459]	[-0.162,-0.596,-0.390,-0.683]	[0.000,0.000,0.000,1.000]
14	valid	6.28	0.0056	[-0.550,0.507,-0.479,0.459]	[0.540,0.560,0.062,-0.625]	[1.000,0.000,0.000,0.000]
15	valid	6.28	0.0056	[-0.550,0.507,-0.479,0.459]	[-0.007,-0.683,-0.056,-0.728]	[0.000,1.000,0.000,0.000]
16	valid	6.28	0.0056	[-0.550,0.507,-0.479,0.459]	[-0.007,-0.683,-0.056,-0.728]	[0.000,1.000,0.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.007,-0.683,-0.056,-0.728]	[0.000,0.000,0.000,1.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]

Stronger parameter noise | seed 0 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	57.86	0.3149	[-0.001,0.056,-0.505,-0.074]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
2	valid	55.47	0.3261	[-0.529,0.062,-0.516,-0.327]	[0.787,0.491,0.145,-0.344]	[1.000,0.000,0.000,0.000]
3	valid	6.85	0.0043	[-0.500,0.391,-0.590,0.332]	[-0.376,0.796,-0.152,-0.450]	[0.000,0.000,0.000,1.000]
4	valid	24.86	0.0529	[-0.629,0.019,-0.546,0.418]	[-0.242,0.238,-0.136,0.931]	[0.000,1.000,0.000,0.000]
5	valid	8.97	0.0110	[-0.550,0.273,-0.580,0.434]	[0.226,-0.935,0.057,-0.266]	[0.000,1.000,0.000,0.000]
6	valid	10.06	0.0061	[-0.441,0.388,-0.638,0.331]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
7	valid	7.73	0.0066	[-0.615,0.523,-0.683,0.367]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
8	valid	9.09	0.0047	[-0.608,0.392,-0.545,0.302]	[0.563,-0.674,0.433,-0.203]	[1.000,0.000,0.000,0.000]
9	valid	9.39	0.0050	[-0.724,0.445,-0.553,0.374]	[0.500,-0.762,0.411,-0.016]	[1.000,0.000,0.000,0.000]
10	valid	10.37	0.0074	[-0.684,0.470,-0.659,0.314]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
11	valid	8.85	0.0066	[-0.481,0.341,-0.423,0.241]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
12	valid	9.12	0.0074	[-0.579,0.403,-0.549,0.286]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
13	valid	2.72	0.0009	[-0.641,0.447,-0.650,0.533]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
14	valid	7.22	0.0040	[-0.607,0.396,-0.453,0.390]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
15	valid	6.88	0.0040	[-0.604,0.420,-0.456,0.392]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
16	valid	5.40	0.0029	[-0.687,0.495,-0.558,0.460]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
17	valid	4.07	0.0002	[-0.720,0.498,-0.615,0.559]	[-0.794,-0.426,0.031,-0.432]	[0.000,0.000,0.000,1.000]
18	valid	6.03	0.0029	[-0.541,0.365,-0.419,0.394]	[0.543,0.120,0.645,-0.524]	[1.000,0.000,0.000,0.000]
19	valid	6.14	0.0023	[-0.550,0.383,-0.449,0.339]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
20	valid	6.18	0.0032	[-0.676,0.464,-0.522,0.460]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 0 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	50.40	0.2082	[-0.082,0.000,-0.515,0.000]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	37.73	0.0841	[-0.082,0.000,-0.289,0.337]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
3	valid	56.76	0.2107	[0.086,0.000,-0.255,0.646]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
4	valid	53.89	0.1894	[-0.108,-0.152,-0.251,0.634]	[-0.618,0.483,-0.526,0.328]	[0.000,1.000,0.000,0.000]
5	valid	47.89	0.1853	[-0.101,-0.143,-0.456,0.563]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
6	valid	44.49	0.1316	[-0.265,-0.143,-0.300,0.563]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
7	valid	40.84	0.1104	[-0.438,-0.137,-0.270,0.540]	[0.863,0.029,0.095,0.495]	[1.000,0.000,0.000,0.000]
8	valid	47.52	0.1609	[-0.386,-0.185,-0.238,0.710]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
9	valid	39.69	0.0962	[-0.505,-0.073,-0.230,0.686]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
10	valid	41.33	0.1079	[-0.668,-0.068,-0.144,0.638]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
11	valid	40.80	0.1133	[-0.602,0.004,-0.130,0.792]	[-0.805,0.381,-0.140,-0.432]	[0.000,0.000,0.000,1.000]
12	valid	43.08	0.1091	[-0.530,0.041,-0.114,0.941]	[-0.261,0.493,-0.095,-0.824]	[0.000,0.000,0.000,1.000]
13	valid	41.42	0.0820	[-0.513,0.104,-0.114,0.938]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
14	valid	39.28	0.0871	[-0.636,0.097,-0.106,0.861]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
15	valid	38.07	0.0880	[-0.574,0.158,-0.106,0.861]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
16	valid	35.65	0.0845	[-0.735,0.156,-0.105,0.739]	[0.872,0.150,0.014,-0.466]	[1.000,0.000,0.000,0.000]
17	valid	35.02	0.0958	[-0.722,0.218,-0.078,0.726]	[-0.602,-0.667,0.317,0.303]	[0.000,1.000,0.000,0.000]
18	valid	37.17	0.1099	[-0.843,0.200,-0.042,0.665]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
19	valid	38.63	0.1158	[-0.960,0.182,-0.038,0.600]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
20	valid	36.27	0.1105	[-0.934,0.235,-0.059,0.584]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]

Stronger parameter noise | seed 1 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	76.92	0.5929	[-0.947,-0.172,0.128,-0.239]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	22.52	0.0207	[-0.796,0.424,-0.336,0.273]	[-0.520,-0.406,-0.727,0.188]	[0.000,1.000,0.000,0.000]
3	valid	34.13	0.1197	[-0.720,0.479,-0.498,-0.065]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
4	valid	34.13	0.1197	[-0.720,0.479,-0.498,-0.065]	[0.660,-0.284,-0.415,0.558]	[1.000,0.000,0.000,0.000]
5	valid	41.22	0.1977	[-0.738,0.524,-0.388,-0.174]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
6	valid	38.45	0.1572	[-0.751,0.492,-0.421,-0.126]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
7	valid	38.45	0.1572	[-0.751,0.492,-0.421,-0.126]	[-0.704,-0.560,-0.219,-0.377]	[0.000,1.000,0.000,0.000]
8	valid	38.51	0.1596	[-0.652,0.711,-0.253,-0.076]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
9	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
10	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.501,0.494,0.669,-0.241]	[0.000,0.000,1.000,0.000]
11	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
12	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
13	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.613,-0.730,0.233,-0.193]	[0.000,1.000,0.000,0.000]
14	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.794,-0.426,0.031,-0.432]	[0.000,1.000,0.000,0.000]
15	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.238,-0.766,0.324,0.502]	[0.000,1.000,0.000,0.000]
16	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
17	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
18	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[-0.462,0.422,0.201,0.754]	[0.000,0.000,1.000,0.000]
19	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.787,0.491,0.145,-0.344]	[1.000,0.000,0.000,0.000]
20	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]

Stronger parameter noise | seed 1 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	76.92	0.5929	[-0.947,-0.172,0.128,-0.239]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	20.22	0.0170	[-0.757,0.423,-0.432,0.247]	[-0.686,0.426,-0.583,-0.092]	[0.000,0.000,0.000,1.000]
3	valid	28.37	0.0521	[-0.767,0.190,-0.220,0.572]	[0.291,-0.106,0.323,-0.894]	[0.000,0.000,0.000,1.000]
4	valid	19.94	0.0223	[-0.694,0.459,-0.199,0.518]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
5	valid	21.74	0.0360	[-0.385,0.387,-0.778,0.312]	[0.500,-0.762,0.411,-0.016]	[0.000,0.000,1.000,0.000]
6	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
7	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.520,-0.406,-0.727,0.188]	[0.000,1.000,0.000,0.000]
8	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
9	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.660,-0.284,-0.415,0.558]	[1.000,0.000,0.000,0.000]
10	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
11	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
12	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.704,-0.560,-0.219,-0.377]	[0.000,1.000,0.000,0.000]
13	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
14	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.501,0.494,0.669,-0.241]	[0.000,0.000,1.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.613,-0.730,0.233,-0.193]	[0.000,1.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.794,-0.426,0.031,-0.432]	[0.000,1.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.238,-0.766,0.324,0.502]	[0.000,1.000,0.000,0.000]

Stronger parameter noise | seed 1 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	63.66	0.3826	[0.076,0.556,-0.142,-0.100]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
2	valid	34.47	0.0771	[-0.011,0.656,-0.295,0.485]	[-0.099,-0.037,-0.437,-0.893]	[0.000,0.000,0.000,1.000]
3	valid	22.61	0.0579	[-0.617,0.614,-0.127,0.528]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
4	valid	14.90	0.0154	[-0.612,0.692,-0.296,0.544]	[0.198,-0.025,-0.905,-0.376]	[0.000,0.000,0.000,1.000]
5	valid	9.64	0.0080	[-0.575,0.665,-0.691,0.427]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
6	valid	5.44	0.0042	[-0.570,0.585,-0.693,0.625]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
7	valid	7.92	0.0042	[-0.506,0.673,-0.634,0.498]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
8	valid	7.20	0.0050	[-0.536,0.569,-0.438,0.398]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
9	valid	9.71	0.0104	[-0.591,0.681,-0.453,0.438]	[0.543,0.120,0.645,-0.524]	[1.000,0.000,0.000,0.000]
10	valid	3.84	0.0008	[-0.642,0.636,-0.586,0.514]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
11	valid	6.26	0.0025	[-0.665,0.708,-0.552,0.521]	[0.540,0.560,0.062,-0.625]	[1.000,0.000,0.000,0.000]
12	valid	9.25	0.0078	[-0.650,0.657,-0.564,0.401]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
13	valid	5.37	0.0017	[-0.687,0.692,-0.584,0.524]	[-0.543,0.615,0.311,-0.480]	[0.000,0.000,0.000,1.000]
14	valid	9.06	0.0068	[-0.597,0.650,-0.527,0.395]	[-0.162,-0.596,-0.390,-0.683]	[0.000,1.000,0.000,0.000]
15	valid	7.24	0.0052	[-0.605,0.621,-0.532,0.420]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
16	valid	6.49	0.0046	[-0.603,0.652,-0.536,0.458]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
17	valid	9.38	0.0046	[-0.548,0.682,-0.493,0.425]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
18	valid	6.20	0.0047	[-0.642,0.680,-0.559,0.488]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
19	valid	5.03	0.0016	[-0.616,0.637,-0.555,0.482]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
20	valid	3.77	0.0012	[-0.564,0.570,-0.508,0.461]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 1 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.48	0.4395	[-0.501,-0.241,0.000,0.000]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	49.51	0.2497	[-0.488,-0.060,-0.319,0.000]	[-0.520,-0.406,-0.727,0.188]	[0.000,1.000,0.000,0.000]
3	valid	46.48	0.2320	[-0.428,-0.020,-0.608,0.000]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
4	valid	39.57	0.0942	[-0.667,-0.017,-0.522,0.136]	[0.660,-0.284,-0.415,0.558]	[1.000,0.000,0.000,0.000]
5	valid	45.95	0.1211	[-0.954,-0.015,-0.322,0.121]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
6	valid	38.48	0.0841	[-0.879,0.092,-0.469,0.111]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
7	valid	37.18	0.1016	[-0.863,0.201,-0.460,0.043]	[-0.704,-0.560,-0.219,-0.377]	[0.000,1.000,0.000,0.000]
8	valid	30.10	0.0713	[-0.676,0.330,-0.460,0.043]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
9	valid	24.55	0.0415	[-0.652,0.330,-0.460,0.131]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
10	valid	27.42	0.0572	[-0.626,0.316,-0.626,0.096]	[-0.501,0.494,0.669,-0.241]	[0.000,0.000,1.000,0.000]
11	valid	31.04	0.0692	[-0.780,0.267,-0.568,0.087]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
12	valid	34.42	0.1031	[-0.717,0.245,-0.708,0.023]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
13	valid	31.42	0.0993	[-0.710,0.340,-0.638,0.023]	[-0.613,-0.730,0.233,-0.193]	[0.000,1.000,0.000,0.000]
14	valid	29.19	0.0984	[-0.678,0.438,-0.598,0.022]	[-0.794,-0.426,0.031,-0.432]	[0.000,1.000,0.000,0.000]
15	valid	28.20	0.0897	[-0.655,0.558,-0.515,0.021]	[-0.238,-0.766,0.324,0.502]	[0.000,1.000,0.000,0.000]
16	valid	31.44	0.1022	[-0.814,0.429,-0.494,0.021]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
17	valid	35.70	0.1109	[-0.960,0.396,-0.403,0.019]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
18	valid	33.60	0.0792	[-0.943,0.389,-0.424,0.057]	[-0.462,0.422,0.201,0.754]	[0.000,0.000,1.000,0.000]
19	valid	37.82	0.0860	[-1.069,0.349,-0.364,0.052]	[0.787,0.491,0.145,-0.344]	[1.000,0.000,0.000,0.000]
20	valid	41.52	0.1062	[-1.185,0.301,-0.322,0.046]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]

Stronger parameter noise | seed 2 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	60.45	0.2583	[0.286,0.002,-0.914,0.286]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
2	valid	33.13	0.0438	[-0.367,0.447,-0.808,0.114]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
3	valid	36.69	0.0718	[-0.623,0.229,-0.740,0.106]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
4	valid	32.93	0.0774	[-0.570,0.460,-0.677,0.067]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
5	valid	33.37	0.0774	[-0.579,0.455,-0.674,0.060]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
6	valid	33.37	0.0774	[-0.579,0.455,-0.674,0.060]	[-0.686,0.426,-0.583,-0.092]	[0.000,0.000,0.000,1.000]
7	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
8	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
9	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.376,0.796,-0.152,-0.450]	[0.000,0.000,0.000,1.000]
10	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.670,-0.109,0.690,-0.251]	[0.000,0.000,1.000,0.000]
11	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.300,0.937,0.153,-0.090]	[0.000,0.000,1.000,0.000]
12	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.237,-0.863,0.169,-0.413]	[0.000,1.000,0.000,0.000]
13	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
14	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
15	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
16	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
17	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
18	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.194,0.637,0.742,0.079]	[0.000,0.000,1.000,0.000]
19	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
20	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 2 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	60.45	0.2583	[0.286,0.002,-0.914,0.286]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
2	valid	46.23	0.1513	[0.225,0.677,-0.664,0.225]	[-0.005,-0.305,-0.938,0.166]	[0.000,1.000,0.000,0.000]
3	valid	21.37	0.0416	[-0.094,0.482,-0.654,0.575]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
4	valid	22.78	0.0417	[-0.443,0.360,-0.756,0.320]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
5	valid	22.21	0.0300	[-0.407,0.615,-0.630,0.245]	[-0.602,-0.667,0.317,0.303]	[0.000,1.000,0.000,0.000]
6	valid	29.09	0.0463	[-0.601,0.484,-0.623,0.130]	[0.769,-0.202,0.494,0.352]	[1.000,0.000,0.000,0.000]
7	valid	29.17	0.0456	[-0.530,0.636,-0.549,0.114]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
8	valid	33.60	0.0932	[-0.605,0.606,-0.515,0.044]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
9	valid	29.15	0.0462	[-0.601,0.602,-0.511,0.123]	[-0.272,0.576,0.069,-0.768]	[0.000,0.000,0.000,1.000]
10	valid	28.53	0.0419	[-0.571,0.572,-0.575,0.125]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
11	valid	7.19	0.0074	[-0.393,0.394,-0.550,0.623]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
12	valid	7.87	0.0074	[-0.387,0.388,-0.563,0.619]	[0.765,0.309,0.563,-0.053]	[0.000,0.000,1.000,0.000]
13	valid	8.05	0.0087	[-0.402,0.380,-0.567,0.610]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
14	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
15	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
16	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
17	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
18	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
19	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.563,-0.674,0.433,-0.203]	[1.000,0.000,0.000,0.000]

Stronger parameter noise | seed 2 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	86.15	0.6042	[-0.558,-0.287,0.049,-0.054]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	74.60	0.4336	[-0.627,-0.123,-0.311,-0.271]	[-0.100,-0.189,0.951,0.225]	[0.000,0.000,1.000,0.000]
3	valid	36.73	0.1208	[-0.711,0.520,-0.456,0.023]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
4	valid	14.40	0.0125	[-0.690,0.553,-0.585,0.461]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
5	valid	17.24	0.0235	[-0.645,0.437,-0.429,0.397]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
6	valid	13.12	0.0103	[-0.762,0.663,-0.582,0.553]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
7	valid	14.69	0.0167	[-0.611,0.561,-0.502,0.402]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
8	valid	11.73	0.0078	[-0.709,0.637,-0.647,0.542]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
9	valid	6.56	0.0022	[-0.625,0.582,-0.550,0.631]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
10	valid	7.47	0.0042	[-0.623,0.649,-0.553,0.609]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
11	valid	4.33	0.0011	[-0.598,0.614,-0.590,0.660]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
12	valid	4.58	0.0022	[-0.576,0.537,-0.567,0.643]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
13	valid	3.21	0.0004	[-0.537,0.559,-0.568,0.625]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
14	valid	3.30	0.0008	[-0.542,0.553,-0.589,0.635]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
15	valid	5.16	0.0016	[-0.549,0.652,-0.559,0.624]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
16	valid	3.93	0.0008	[-0.551,0.569,-0.536,0.626]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
17	valid	3.55	0.0009	[-0.482,0.498,-0.486,0.552]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
18	valid	4.65	0.0011	[-0.570,0.706,-0.604,0.683]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
19	valid	4.06	0.0011	[-0.644,0.682,-0.651,0.721]	[-0.794,-0.426,0.031,-0.432]	[0.000,0.000,0.000,1.000]
20	valid	3.90	0.0006	[-0.501,0.559,-0.535,0.574]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 2 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	79.48	0.4224	[0.314,0.000,-0.490,0.000]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
2	valid	56.39	0.1684	[-0.005,0.064,-0.490,0.000]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
3	valid	44.38	0.1426	[-0.260,0.064,-0.333,0.000]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
4	valid	36.95	0.1091	[-0.172,0.292,-0.333,0.000]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
5	valid	41.30	0.1352	[-0.398,0.148,-0.333,0.000]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
6	valid	33.03	0.0811	[-0.385,0.335,-0.321,0.036]	[-0.686,0.426,-0.583,-0.092]	[0.000,0.000,0.000,1.000]
7	valid	19.26	0.0242	[-0.380,0.330,-0.289,0.192]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
8	valid	32.25	0.0399	[-0.572,0.203,-0.275,0.182]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
9	valid	24.28	0.0392	[-0.540,0.192,-0.311,0.316]	[-0.376,0.796,-0.152,-0.450]	[0.000,0.000,0.000,1.000]
10	valid	25.95	0.0399	[-0.516,0.183,-0.458,0.252]	[-0.670,-0.109,0.690,-0.251]	[0.000,0.000,1.000,0.000]
11	valid	32.52	0.0434	[-0.485,0.172,-0.614,0.169]	[-0.300,0.937,0.153,-0.090]	[0.000,0.000,1.000,0.000]
12	valid	27.86	0.0452	[-0.477,0.282,-0.567,0.167]	[-0.237,-0.863,0.169,-0.413]	[0.000,1.000,0.000,0.000]
13	valid	24.32	0.0324	[-0.474,0.409,-0.468,0.166]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
14	valid	31.86	0.0545	[-0.448,0.386,-0.602,0.077]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
15	valid	26.86	0.0378	[-0.439,0.379,-0.601,0.156]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
16	valid	30.16	0.0603	[-0.567,0.292,-0.581,0.151]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
17	valid	33.74	0.0669	[-0.687,0.262,-0.540,0.140]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
18	valid	33.83	0.0656	[-0.611,0.249,-0.663,0.133]	[0.194,0.637,0.742,0.079]	[0.000,0.000,1.000,0.000]
19	valid	35.73	0.0657	[-0.562,0.229,-0.777,0.128]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
20	valid	39.09	0.0659	[-0.448,0.217,-0.925,0.121]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 3 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	46.07	0.0913	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	29.99	0.0323	[-0.134,0.694,-0.683,0.181]	[-0.005,-0.305,-0.938,0.166]	[0.000,1.000,0.000,0.000]
3	valid	11.94	0.0127	[-0.266,0.578,-0.542,0.549]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
4	valid	26.37	0.0398	[-0.098,0.455,-0.447,0.764]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
5	valid	30.01	0.0615	[-0.014,0.426,-0.555,0.714]	[0.434,0.565,0.692,-0.118]	[0.000,0.000,1.000,0.000]
6	valid	30.01	0.0615	[-0.014,0.426,-0.555,0.714]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
7	valid	18.58	0.0308	[-0.381,0.337,-0.644,0.571]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
8	valid	10.45	0.0051	[-0.333,0.505,-0.537,0.588]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
9	valid	10.45	0.0051	[-0.333,0.505,-0.537,0.588]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
10	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.263,0.146,0.859,-0.414]	[0.000,0.000,1.000,0.000]
11	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[0.077,0.882,-0.212,0.415]	[1.000,0.000,0.000,0.000]
12	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.018,0.197,-0.819,-0.539]	[0.000,0.000,0.000,1.000]
13	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
14	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
15	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
16	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[0.328,0.107,-0.455,-0.821]	[0.000,0.000,0.000,1.000]
17	valid	11.53	0.0110	[-0.263,0.621,-0.485,0.557]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
18	valid	11.86	0.0119	[-0.258,0.610,-0.512,0.547]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
19	valid	11.86	0.0119	[-0.258,0.610,-0.512,0.547]	[-0.013,0.056,-0.998,-0.022]	[0.000,0.000,0.000,1.000]
20	valid	11.86	0.0119	[-0.258,0.610,-0.512,0.547]	[-0.457,0.529,0.535,0.474]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 3 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	46.07	0.0913	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	32.05	0.0826	[-0.550,0.068,-0.629,0.545]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
3	valid	38.85	0.1016	[-0.349,-0.000,-0.472,0.810]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
4	valid	41.52	0.1056	[-0.254,-0.023,-0.755,0.604]	[0.765,0.309,0.563,-0.053]	[0.000,0.000,1.000,0.000]
5	valid	30.35	0.0620	[-0.294,0.175,-0.736,0.584]	[0.077,0.882,-0.212,0.415]	[1.000,0.000,0.000,0.000]
6	valid	22.02	0.0380	[-0.317,0.324,-0.701,0.551]	[0.226,-0.935,0.057,-0.266]	[0.000,1.000,0.000,0.000]
7	valid	14.29	0.0153	[-0.415,0.421,-0.642,0.488]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
8	valid	15.15	0.0240	[-0.442,0.395,-0.643,0.485]	[0.769,-0.202,0.494,0.352]	[1.000,0.000,0.000,0.000]
9	valid	13.89	0.0176	[-0.437,0.420,-0.635,0.479]	[-0.100,-0.640,0.394,0.652]	[0.000,1.000,0.000,0.000]
10	valid	14.36	0.0202	[-0.450,0.408,-0.635,0.477]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
11	valid	13.65	0.0176	[-0.447,0.422,-0.631,0.474]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
12	valid	13.65	0.0176	[-0.447,0.422,-0.631,0.474]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
13	valid	13.65	0.0176	[-0.447,0.422,-0.631,0.474]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
14	valid	13.65	0.0176	[-0.447,0.422,-0.631,0.474]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
15	valid	13.65	0.0176	[-0.447,0.422,-0.631,0.474]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.263,0.146,0.859,-0.414]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 3 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	52.60	0.0960	[-0.051,0.089,-0.611,0.048]	[-0.220,0.017,0.903,-0.368]	[0.000,0.000,1.000,0.000]
2	valid	61.78	0.3659	[-0.657,0.113,-0.518,-0.334]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
3	valid	41.01	0.2047	[-0.574,0.538,-0.488,-0.166]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
4	valid	18.51	0.0196	[-0.587,0.598,-0.604,0.234]	[-0.444,0.444,-0.465,-0.624]	[0.000,0.000,0.000,1.000]
5	valid	11.46	0.0150	[-0.700,0.591,-0.452,0.489]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
6	valid	9.03	0.0046	[-0.637,0.685,-0.530,0.442]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]
7	valid	12.89	0.0138	[-0.758,0.575,-0.597,0.501]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
8	valid	5.64	0.0030	[-0.527,0.604,-0.620,0.582]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
9	valid	11.28	0.0113	[-0.650,0.638,-0.817,0.593]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
10	valid	7.21	0.0041	[-0.595,0.650,-0.678,0.551]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
11	valid	9.74	0.0047	[-0.474,0.661,-0.738,0.561]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
12	valid	5.73	0.0027	[-0.560,0.722,-0.725,0.665]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
13	valid	7.50	0.0031	[-0.485,0.676,-0.661,0.513]	[-0.007,-0.683,-0.056,-0.728]	[0.000,1.000,0.000,0.000]
14	valid	7.32	0.0031	[-0.455,0.605,-0.610,0.481]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
15	valid	6.08	0.0019	[-0.598,0.797,-0.755,0.619]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
16	valid	5.01	0.0016	[-0.526,0.731,-0.659,0.575]	[0.317,0.914,0.253,-0.015]	[0.000,0.000,1.000,0.000]
17	valid	6.79	0.0019	[-0.496,0.745,-0.681,0.558]	[0.514,-0.350,-0.042,0.782]	[0.000,1.000,0.000,0.000]
18	valid	9.07	0.0035	[-0.417,0.669,-0.649,0.493]	[0.765,0.309,0.563,-0.053]	[0.000,0.000,1.000,0.000]
19	valid	6.94	0.0020	[-0.509,0.784,-0.724,0.606]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
20	valid	9.77	0.0039	[-0.416,0.690,-0.664,0.488]	[0.514,-0.350,-0.042,0.782]	[0.000,1.000,0.000,0.000]

Stronger parameter noise | seed 3 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	58.59	0.2668	[-0.082,0.000,-0.515,0.000]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	30.67	0.0596	[-0.077,0.314,-0.480,0.161]	[-0.005,-0.305,-0.938,0.166]	[0.000,1.000,0.000,0.000]
3	valid	26.12	0.0557	[-0.070,0.286,-0.460,0.417]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
4	valid	37.37	0.0934	[-0.063,0.151,-0.413,0.684]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
5	valid	38.13	0.0896	[-0.057,0.138,-0.638,0.572]	[0.434,0.565,0.692,-0.118]	[0.000,0.000,1.000,0.000]
6	valid	43.18	0.1017	[-0.050,0.120,-0.858,0.473]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
7	valid	40.20	0.1002	[-0.155,0.106,-0.844,0.465]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
8	valid	32.83	0.0577	[-0.155,0.214,-0.731,0.465]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
9	valid	38.95	0.0751	[-0.155,0.191,-0.920,0.413]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
10	valid	45.85	0.0878	[-0.135,0.167,-1.113,0.302]	[-0.263,0.146,0.859,-0.414]	[0.000,0.000,1.000,0.000]
11	valid	43.07	0.0918	[-0.141,0.157,-1.051,0.404]	[0.077,0.882,-0.212,0.415]	[1.000,0.000,0.000,0.000]
12	valid	40.84	0.0922	[-0.134,0.147,-0.983,0.516]	[-0.018,0.197,-0.819,-0.539]	[0.000,0.000,0.000,1.000]
13	valid	44.89	0.1019	[-0.115,0.127,-1.118,0.455]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
14	valid	49.25	0.1103	[-0.100,0.110,-1.273,0.356]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
15	valid	48.17	0.1047	[-0.145,0.109,-1.252,0.351]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
16	valid	46.75	0.1184	[-0.121,0.105,-1.200,0.437]	[0.328,0.107,-0.455,-0.821]	[0.000,0.000,0.000,1.000]
17	valid	46.22	0.0969	[-0.121,0.148,-1.200,0.389]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
18	valid	50.23	0.1029	[-0.107,0.131,-1.347,0.301]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
19	valid	48.25	0.1052	[-0.123,0.125,-1.287,0.366]	[-0.013,0.056,-0.998,-0.022]	[0.000,0.000,0.000,1.000]
20	valid	50.37	0.1036	[-0.133,0.107,-1.371,0.312]	[-0.457,0.529,0.535,0.474]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 4 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	35.26	0.1033	[-0.421,0.527,0.098,0.732]	[-0.477,0.831,0.084,-0.272]	[0.000,0.000,0.000,1.000]
2	valid	53.94	0.2301	[-0.487,0.511,0.449,0.548]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
3	valid	47.90	0.2100	[-0.303,0.830,0.311,0.350]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
4	valid	47.90	0.2100	[-0.303,0.830,0.311,0.350]	[-0.259,-0.531,-0.093,0.801]	[0.000,1.000,0.000,0.000]
5	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[0.291,-0.106,0.323,-0.894]	[0.000,0.000,0.000,1.000]
6	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
7	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[-0.406,-0.902,-0.036,0.142]	[0.000,1.000,0.000,0.000]
8	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
9	valid	10.11	0.0057	[-0.402,0.662,-0.299,0.558]	[-0.100,-0.189,0.951,0.225]	[0.000,0.000,1.000,0.000]
10	valid	10.11	0.0057	[-0.402,0.662,-0.299,0.558]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
11	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
12	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
13	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
14	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.266,0.572,0.740,0.233]	[0.000,0.000,1.000,0.000]
15	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.489,0.191,-0.234,0.818]	[1.000,0.000,0.000,0.000]
16	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.503,0.277,-0.818,-0.031]	[0.000,0.000,0.000,1.000]
17	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
18	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.157,0.765,-0.245,-0.575]	[0.000,0.000,0.000,1.000]
19	valid	6.20	0.0040	[-0.394,0.584,-0.414,0.576]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
20	valid	6.36	0.0031	[-0.392,0.570,-0.434,0.578]	[-0.064,-0.505,0.844,0.167]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 4 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	35.26	0.1033	[-0.421,0.527,0.098,0.732]	[-0.477,0.831,0.084,-0.272]	[0.000,0.000,0.000,1.000]
2	valid	44.26	0.1876	[-0.294,0.833,0.252,0.394]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
3	valid	19.63	0.0289	[-0.400,0.784,-0.159,0.447]	[-0.457,0.529,0.535,0.474]	[0.000,0.000,1.000,0.000]
4	valid	28.77	0.0777	[-0.627,0.733,-0.072,0.255]	[0.814,-0.418,0.255,0.313]	[1.000,0.000,0.000,0.000]
5	valid	9.79	0.0069	[-0.520,0.609,-0.490,0.345]	[0.223,0.325,0.735,-0.551]	[0.000,0.000,1.000,0.000]
6	valid	22.58	0.0269	[-0.520,0.786,-0.275,0.193]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
7	valid	15.20	0.0140	[-0.493,0.745,-0.342,0.292]	[0.340,0.096,-0.269,-0.896]	[0.000,0.000,0.000,1.000]
8	valid	13.35	0.0133	[-0.471,0.711,-0.442,0.279]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
9	valid	14.59	0.0139	[-0.430,0.650,-0.568,0.265]	[-0.543,0.615,0.311,-0.480]	[0.000,0.000,1.000,0.000]
10	valid	12.69	0.0115	[-0.381,0.721,-0.491,0.306]	[-0.612,0.267,-0.382,0.639]	[0.000,0.000,1.000,0.000]
11	valid	12.42	0.0115	[-0.397,0.716,-0.488,0.304]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
12	valid	12.71	0.0115	[-0.390,0.716,-0.495,0.301]	[0.317,0.914,0.253,-0.015]	[0.000,0.000,1.000,0.000]
13	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
14	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.259,-0.531,-0.093,0.801]	[0.000,1.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.291,-0.106,0.323,-0.894]	[0.000,0.000,0.000,1.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.406,-0.902,-0.036,0.142]	[0.000,1.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.100,-0.189,0.951,0.225]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 4 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	38.00	0.1218	[0.061,0.514,-0.083,0.417]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
2	valid	35.34	0.0798	[0.133,0.531,-0.517,0.418]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
3	valid	7.66	0.0021	[-0.387,0.662,-0.597,0.483]	[0.850,0.267,-0.190,0.413]	[1.000,0.000,0.000,0.000]
4	valid	5.72	0.0033	[-0.470,0.619,-0.412,0.594]	[-0.612,0.267,-0.382,0.639]	[0.000,0.000,1.000,0.000]
5	valid	8.11	0.0072	[-0.527,0.528,-0.385,0.502]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
6	valid	7.46	0.0058	[-0.494,0.535,-0.362,0.402]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
7	valid	6.83	0.0038	[-0.616,0.658,-0.462,0.545]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
8	valid	10.74	0.0126	[-0.730,0.684,-0.477,0.533]	[-0.612,0.267,-0.382,0.639]	[0.000,0.000,1.000,0.000]
9	valid	8.63	0.0076	[-0.592,0.615,-0.395,0.568]	[-0.612,0.267,-0.382,0.639]	[0.000,0.000,1.000,0.000]
10	valid	9.10	0.0051	[-0.402,0.622,-0.393,0.646]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,0.000,1.000]
11	valid	4.76	0.0032	[-0.509,0.686,-0.469,0.642]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]
12	valid	3.79	0.0022	[-0.513,0.818,-0.544,0.684]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
13	valid	2.40	0.0008	[-0.485,0.670,-0.485,0.585]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
14	valid	0.84	0.0004	[-0.457,0.666,-0.472,0.523]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
15	valid	2.39	0.0012	[-0.506,0.733,-0.495,0.607]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
16	valid	1.79	0.0005	[-0.458,0.648,-0.472,0.552]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
17	valid	1.72	0.0010	[-0.380,0.521,-0.363,0.428]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,0.000,1.000]
18	valid	2.82	0.0014	[-0.499,0.645,-0.455,0.554]	[0.543,0.120,0.645,-0.524]	[1.000,0.000,0.000,0.000]
19	valid	2.50	0.0014	[-0.487,0.651,-0.447,0.544]	[-0.612,0.267,-0.382,0.639]	[0.000,0.000,1.000,0.000]
20	valid	2.31	0.0014	[-0.458,0.631,-0.432,0.527]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]

Stronger parameter noise | seed 4 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	52.58	0.1957	[-0.499,0.000,0.000,0.256]	[-0.477,0.831,0.084,-0.272]	[0.000,0.000,0.000,1.000]
2	valid	47.29	0.1468	[-0.844,0.246,0.000,0.193]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
3	valid	34.52	0.1021	[-0.601,0.534,0.000,0.193]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
4	valid	33.59	0.0914	[-0.510,0.786,-0.024,0.164]	[-0.259,-0.531,-0.093,0.801]	[0.000,1.000,0.000,0.000]
5	valid	28.65	0.0810	[-0.423,0.771,-0.023,0.321]	[0.291,-0.106,0.323,-0.894]	[0.000,0.000,0.000,1.000]
6	valid	37.08	0.0930	[-0.367,1.022,-0.020,0.172]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
7	valid	41.09	0.1025	[-0.297,1.214,-0.019,0.139]	[-0.406,-0.902,-0.036,0.142]	[0.000,1.000,0.000,0.000]
8	valid	40.94	0.1119	[-0.287,1.176,0.046,0.218]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
9	valid	38.13	0.0991	[-0.287,1.100,-0.015,0.218]	[-0.100,-0.189,0.951,0.225]	[0.000,0.000,1.000,0.000]
10	valid	42.02	0.0991	[-0.226,1.312,-0.013,0.189]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
11	valid	41.69	0.0989	[-0.152,1.305,-0.013,0.270]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
12	valid	40.13	0.0815	[-0.162,1.292,-0.059,0.268]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
13	valid	42.35	0.0781	[-0.134,1.408,-0.072,0.221]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
14	valid	40.93	0.0575	[-0.149,1.382,-0.112,0.216]	[-0.266,0.572,0.740,0.233]	[0.000,0.000,1.000,0.000]
15	valid	39.23	0.0588	[-0.199,1.341,-0.130,0.210]	[0.489,0.191,-0.234,0.818]	[1.000,0.000,0.000,0.000]
16	valid	37.82	0.0667	[-0.256,1.301,-0.126,0.207]	[-0.503,0.277,-0.818,-0.031]	[0.000,0.000,0.000,1.000]
17	valid	36.40	0.0655	[-0.243,1.269,-0.123,0.265]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
18	valid	35.07	0.0638	[-0.220,1.234,-0.120,0.332]	[0.157,0.765,-0.245,-0.575]	[0.000,0.000,0.000,1.000]
19	valid	34.84	0.0560	[-0.220,1.234,-0.161,0.291]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
20	valid	31.14	0.0349	[-0.220,1.079,-0.209,0.291]	[-0.064,-0.505,0.844,0.167]	[0.000,0.000,1.000,0.000]

Query-dependent parameter noise | seed 0 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	43.02	0.0789	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	35.20	0.0659	[-0.134,0.235,-0.485,0.832]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
3	valid	35.20	0.0659	[-0.134,0.235,-0.485,0.832]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
4	valid	33.49	0.0613	[-0.399,-0.058,-0.578,0.709]	[-0.618,0.483,-0.526,0.328]	[0.000,1.000,0.000,0.000]
5	valid	33.49	0.0613	[-0.399,-0.058,-0.578,0.709]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
6	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
7	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[0.863,0.029,0.095,0.495]	[1.000,0.000,0.000,0.000]
8	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
9	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
10	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
11	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[-0.805,0.381,-0.140,-0.432]	[0.000,0.000,0.000,1.000]
12	valid	31.30	0.0744	[-0.585,0.015,-0.362,0.726]	[-0.261,0.493,-0.095,-0.824]	[0.000,0.000,0.000,1.000]
13	valid	16.60	0.0177	[-0.489,0.309,-0.453,0.678]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
14	valid	16.60	0.0177	[-0.489,0.309,-0.453,0.678]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
15	valid	16.59	0.0177	[-0.486,0.313,-0.453,0.679]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
16	valid	16.32	0.0177	[-0.495,0.313,-0.449,0.674]	[0.872,0.150,0.014,-0.466]	[1.000,0.000,0.000,0.000]
17	valid	16.25	0.0178	[-0.482,0.330,-0.448,0.677]	[-0.602,-0.667,0.317,0.303]	[0.000,1.000,0.000,0.000]
18	valid	16.25	0.0178	[-0.482,0.330,-0.448,0.677]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
19	valid	16.25	0.0178	[-0.482,0.330,-0.448,0.677]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
20	valid	16.25	0.0178	[-0.482,0.330,-0.448,0.677]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]

Query-dependent parameter noise | seed 0 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	43.02	0.0789	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	22.12	0.0432	[-0.550,0.068,-0.629,0.545]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
3	valid	29.36	0.0666	[-0.811,0.016,-0.462,0.360]	[0.405,0.823,-0.005,-0.399]	[1.000,0.000,0.000,0.000]
4	valid	29.80	0.0605	[-0.658,-0.023,-0.697,0.283]	[0.473,-0.294,0.808,0.193]	[0.000,0.000,1.000,0.000]
5	valid	7.00	0.0041	[-0.621,0.382,-0.580,0.363]	[-0.237,-0.863,0.169,-0.413]	[0.000,1.000,0.000,0.000]
6	valid	7.58	0.0027	[-0.630,0.353,-0.496,0.483]	[-0.007,-0.683,-0.056,-0.728]	[0.000,0.000,0.000,1.000]
7	valid	7.66	0.0051	[-0.610,0.413,-0.453,0.502]	[0.259,-0.495,-0.732,0.389]	[0.000,1.000,0.000,0.000]
8	valid	10.68	0.0148	[-0.581,0.484,-0.397,0.521]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
9	valid	9.50	0.0144	[-0.617,0.455,-0.414,0.490]	[0.563,-0.674,0.433,-0.203]	[1.000,0.000,0.000,0.000]
10	valid	10.95	0.0184	[-0.616,0.481,-0.390,0.486]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
11	valid	9.56	0.0148	[-0.610,0.476,-0.412,0.481]	[0.434,0.565,0.692,-0.118]	[0.000,0.000,1.000,0.000]
12	valid	9.86	0.0157	[-0.603,0.471,-0.407,0.499]	[-0.794,-0.426,0.031,-0.432]	[0.000,0.000,0.000,1.000]
13	valid	10.48	0.0176	[-0.592,0.462,-0.400,0.524]	[-0.700,0.090,0.547,-0.451]	[0.000,0.000,0.000,1.000]
14	valid	9.48	0.0125	[-0.570,0.455,-0.423,0.538]	[0.223,0.325,0.735,-0.551]	[0.000,0.000,1.000,0.000]
15	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.700,0.090,0.547,-0.451]	[0.000,0.000,1.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.618,0.483,-0.526,0.328]	[0.000,1.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.863,0.029,0.095,0.495]	[1.000,0.000,0.000,0.000]

Query-dependent parameter noise | seed 0 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	57.86	0.3149	[-0.001,0.056,-0.505,-0.074]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
2	valid	55.47	0.3261	[-0.529,0.062,-0.516,-0.327]	[0.787,0.491,0.145,-0.344]	[1.000,0.000,0.000,0.000]
3	valid	6.85	0.0043	[-0.500,0.391,-0.590,0.332]	[-0.376,0.796,-0.152,-0.450]	[0.000,0.000,0.000,1.000]
4	valid	11.54	0.0115	[-0.425,0.538,-0.434,0.453]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
5	valid	4.71	0.0029	[-0.607,0.594,-0.637,0.544]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
6	valid	9.42	0.0057	[-0.502,0.487,-0.781,0.529]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
7	valid	5.07	0.0032	[-0.613,0.508,-0.716,0.448]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
8	valid	6.92	0.0038	[-0.654,0.411,-0.723,0.445]	[0.563,-0.674,0.433,-0.203]	[1.000,0.000,0.000,0.000]
9	valid	9.04	0.0109	[-0.695,0.337,-0.687,0.572]	[0.500,-0.762,0.411,-0.016]	[1.000,0.000,0.000,0.000]
10	valid	13.35	0.0134	[-0.697,0.241,-0.659,0.482]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
11	valid	9.65	0.0062	[-0.705,0.319,-0.614,0.477]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
12	valid	10.71	0.0048	[-0.645,0.401,-0.600,0.293]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
13	valid	3.01	0.0004	[-0.700,0.486,-0.644,0.504]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
14	valid	10.97	0.0080	[-0.547,0.347,-0.513,0.243]	[0.500,-0.762,0.411,-0.016]	[1.000,0.000,0.000,0.000]
15	valid	6.19	0.0031	[-0.747,0.464,-0.701,0.460]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
16	valid	4.05	0.0009	[-0.648,0.410,-0.622,0.492]	[-0.543,0.615,0.311,-0.480]	[0.000,0.000,0.000,1.000]
17	valid	5.30	0.0010	[-0.471,0.288,-0.405,0.363]	[0.377,-0.706,-0.186,-0.571]	[0.000,0.000,0.000,1.000]
18	valid	4.20	0.0009	[-0.699,0.442,-0.648,0.550]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
19	valid	3.55	0.0009	[-0.564,0.369,-0.525,0.443]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
20	valid	2.93	0.0009	[-0.593,0.403,-0.552,0.465]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]

Query-dependent parameter noise | seed 0 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	50.40	0.2082	[-0.082,0.000,-0.515,0.000]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	37.73	0.0841	[-0.082,0.000,-0.289,0.337]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
3	valid	56.76	0.2107	[0.086,0.000,-0.255,0.646]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
4	valid	53.89	0.1894	[-0.108,-0.152,-0.251,0.634]	[-0.618,0.483,-0.526,0.328]	[0.000,1.000,0.000,0.000]
5	valid	47.89	0.1853	[-0.101,-0.143,-0.456,0.563]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
6	valid	44.49	0.1316	[-0.265,-0.143,-0.300,0.563]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
7	valid	40.84	0.1104	[-0.438,-0.137,-0.270,0.540]	[0.863,0.029,0.095,0.495]	[1.000,0.000,0.000,0.000]
8	valid	47.52	0.1609	[-0.386,-0.185,-0.238,0.710]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
9	valid	39.69	0.0962	[-0.505,-0.073,-0.230,0.686]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
10	valid	41.33	0.1079	[-0.668,-0.068,-0.144,0.638]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
11	valid	40.80	0.1133	[-0.602,0.004,-0.130,0.792]	[-0.805,0.381,-0.140,-0.432]	[0.000,0.000,0.000,1.000]
12	valid	43.08	0.1091	[-0.530,0.041,-0.114,0.941]	[-0.261,0.493,-0.095,-0.824]	[0.000,0.000,0.000,1.000]
13	valid	41.42	0.0820	[-0.513,0.104,-0.114,0.938]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
14	valid	39.28	0.0871	[-0.636,0.097,-0.106,0.861]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
15	valid	38.07	0.0880	[-0.574,0.158,-0.106,0.861]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
16	valid	35.65	0.0845	[-0.735,0.156,-0.105,0.739]	[0.872,0.150,0.014,-0.466]	[1.000,0.000,0.000,0.000]
17	valid	35.02	0.0958	[-0.722,0.218,-0.078,0.726]	[-0.602,-0.667,0.317,0.303]	[0.000,1.000,0.000,0.000]
18	valid	37.17	0.1099	[-0.843,0.200,-0.042,0.665]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
19	valid	38.63	0.1158	[-0.960,0.182,-0.038,0.600]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
20	valid	36.27	0.1105	[-0.934,0.235,-0.059,0.584]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]

Query-dependent parameter noise | seed 1 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	76.92	0.5929	[-0.947,-0.172,0.128,-0.239]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	22.52	0.0207	[-0.796,0.424,-0.336,0.273]	[-0.520,-0.406,-0.727,0.188]	[0.000,1.000,0.000,0.000]
3	valid	34.13	0.1197	[-0.720,0.479,-0.498,-0.065]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
4	valid	34.13	0.1197	[-0.720,0.479,-0.498,-0.065]	[0.660,-0.284,-0.415,0.558]	[1.000,0.000,0.000,0.000]
5	valid	41.22	0.1977	[-0.738,0.524,-0.388,-0.174]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
6	valid	38.45	0.1572	[-0.751,0.492,-0.421,-0.126]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
7	valid	38.45	0.1572	[-0.751,0.492,-0.421,-0.126]	[-0.704,-0.560,-0.219,-0.377]	[0.000,1.000,0.000,0.000]
8	valid	38.51	0.1596	[-0.652,0.711,-0.253,-0.076]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
9	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
10	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.501,0.494,0.669,-0.241]	[0.000,0.000,1.000,0.000]
11	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
12	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
13	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.613,-0.730,0.233,-0.193]	[0.000,1.000,0.000,0.000]
14	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.794,-0.426,0.031,-0.432]	[0.000,1.000,0.000,0.000]
15	valid	14.95	0.0135	[-0.566,0.603,-0.503,0.252]	[-0.238,-0.766,0.324,0.502]	[0.000,1.000,0.000,0.000]
16	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
17	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
18	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[-0.462,0.422,0.201,0.754]	[0.000,0.000,1.000,0.000]
19	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.787,0.491,0.145,-0.344]	[1.000,0.000,0.000,0.000]
20	valid	20.84	0.0257	[-0.676,0.603,-0.372,0.202]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]

Query-dependent parameter noise | seed 1 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	76.92	0.5929	[-0.947,-0.172,0.128,-0.239]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	20.22	0.0170	[-0.757,0.423,-0.432,0.247]	[-0.686,0.426,-0.583,-0.092]	[0.000,0.000,0.000,1.000]
3	valid	28.37	0.0521	[-0.767,0.190,-0.220,0.572]	[0.291,-0.106,0.323,-0.894]	[0.000,0.000,0.000,1.000]
4	valid	19.94	0.0223	[-0.694,0.459,-0.199,0.518]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
5	valid	22.33	0.0452	[-0.602,0.657,-0.154,0.426]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
6	valid	11.50	0.0129	[-0.568,0.619,-0.364,0.402]	[-0.501,0.494,0.669,-0.241]	[0.000,0.000,1.000,0.000]
7	valid	9.57	0.0131	[-0.548,0.523,-0.345,0.554]	[0.534,0.374,0.019,-0.758]	[0.000,0.000,0.000,1.000]
8	valid	2.18	0.0005	[-0.510,0.503,-0.471,0.515]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]
9	valid	7.30	0.0097	[-0.598,0.476,-0.398,0.507]	[0.405,0.823,-0.005,-0.399]	[1.000,0.000,0.000,0.000]
10	valid	9.49	0.0109	[-0.595,0.430,-0.386,0.558]	[0.540,0.560,0.062,-0.625]	[0.000,0.000,0.000,1.000]
11	valid	10.80	0.0123	[-0.614,0.425,-0.367,0.555]	[0.509,-0.697,-0.096,-0.496]	[1.000,0.000,0.000,0.000]
12	valid	10.26	0.0109	[-0.611,0.423,-0.378,0.552]	[0.473,-0.294,0.808,0.193]	[0.000,0.000,1.000,0.000]
13	valid	9.30	0.0109	[-0.605,0.419,-0.401,0.546]	[0.434,0.565,0.692,-0.118]	[0.000,0.000,1.000,0.000]
14	valid	8.31	0.0065	[-0.597,0.413,-0.426,0.539]	[0.223,0.325,0.735,-0.551]	[0.000,0.000,1.000,0.000]
15	valid	8.03	0.0065	[-0.594,0.412,-0.435,0.537]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
16	valid	7.38	0.0038	[-0.586,0.406,-0.459,0.529]	[-0.700,0.090,0.547,-0.451]	[0.000,0.000,1.000,0.000]
17	valid	7.16	0.0055	[-0.596,0.428,-0.436,0.521]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
18	valid	7.16	0.0055	[-0.596,0.428,-0.436,0.521]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
19	valid	7.16	0.0055	[-0.596,0.428,-0.436,0.521]	[-0.700,0.090,0.547,-0.451]	[0.000,0.000,1.000,0.000]
20	valid	7.16	0.0055	[-0.596,0.428,-0.436,0.521]	[0.540,0.560,0.062,-0.625]	[0.000,0.000,0.000,1.000]

Query-dependent parameter noise | seed 1 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	63.66	0.3826	[0.076,0.556,-0.142,-0.100]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
2	valid	34.47	0.0771	[-0.011,0.656,-0.295,0.485]	[-0.099,-0.037,-0.437,-0.893]	[0.000,0.000,0.000,1.000]
3	valid	22.61	0.0579	[-0.617,0.614,-0.127,0.528]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
4	valid	14.90	0.0154	[-0.612,0.692,-0.296,0.544]	[0.198,-0.025,-0.905,-0.376]	[0.000,0.000,0.000,1.000]
5	valid	9.64	0.0080	[-0.575,0.665,-0.691,0.427]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
6	valid	5.44	0.0042	[-0.570,0.585,-0.693,0.625]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
7	valid	7.92	0.0042	[-0.506,0.673,-0.634,0.498]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
8	valid	7.20	0.0050	[-0.536,0.569,-0.438,0.398]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
9	valid	2.70	0.0006	[-0.636,0.645,-0.648,0.563]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
10	valid	5.38	0.0018	[-0.592,0.661,-0.633,0.504]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
11	valid	8.17	0.0057	[-0.591,0.679,-0.611,0.435]	[0.540,0.560,0.062,-0.625]	[1.000,0.000,0.000,0.000]
12	valid	7.92	0.0045	[-0.540,0.620,-0.543,0.401]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
13	valid	5.57	0.0018	[-0.641,0.623,-0.671,0.497]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
14	valid	9.41	0.0085	[-0.666,0.664,-0.689,0.421]	[0.534,0.374,0.019,-0.758]	[1.000,0.000,0.000,0.000]
15	valid	5.32	0.0013	[-0.525,0.529,-0.538,0.409]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
16	valid	7.74	0.0040	[-0.397,0.449,-0.396,0.293]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
17	valid	6.71	0.0044	[-0.384,0.436,-0.408,0.306]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
18	valid	5.96	0.0043	[-0.376,0.436,-0.376,0.316]	[0.540,0.560,0.062,-0.625]	[0.000,0.000,0.000,1.000]
19	valid	4.99	0.0017	[-0.375,0.406,-0.397,0.315]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
20	valid	4.38	0.0013	[-0.344,0.361,-0.352,0.286]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]

Query-dependent parameter noise | seed 1 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	75.48	0.4395	[-0.501,-0.241,0.000,0.000]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	49.51	0.2497	[-0.488,-0.060,-0.319,0.000]	[-0.520,-0.406,-0.727,0.188]	[0.000,1.000,0.000,0.000]
3	valid	46.48	0.2320	[-0.428,-0.020,-0.608,0.000]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
4	valid	39.57	0.0942	[-0.667,-0.017,-0.522,0.136]	[0.660,-0.284,-0.415,0.558]	[1.000,0.000,0.000,0.000]
5	valid	45.95	0.1211	[-0.954,-0.015,-0.322,0.121]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
6	valid	38.48	0.0841	[-0.879,0.092,-0.469,0.111]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
7	valid	37.18	0.1016	[-0.863,0.201,-0.460,0.043]	[-0.704,-0.560,-0.219,-0.377]	[0.000,1.000,0.000,0.000]
8	valid	30.10	0.0713	[-0.676,0.330,-0.460,0.043]	[0.500,-0.762,0.411,-0.016]	[0.000,1.000,0.000,0.000]
9	valid	24.55	0.0415	[-0.652,0.330,-0.460,0.131]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
10	valid	27.42	0.0572	[-0.626,0.316,-0.626,0.096]	[-0.501,0.494,0.669,-0.241]	[0.000,0.000,1.000,0.000]
11	valid	31.04	0.0692	[-0.780,0.267,-0.568,0.087]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
12	valid	34.42	0.1031	[-0.717,0.245,-0.708,0.023]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
13	valid	31.42	0.0993	[-0.710,0.340,-0.638,0.023]	[-0.613,-0.730,0.233,-0.193]	[0.000,1.000,0.000,0.000]
14	valid	29.19	0.0984	[-0.678,0.438,-0.598,0.022]	[-0.794,-0.426,0.031,-0.432]	[0.000,1.000,0.000,0.000]
15	valid	28.20	0.0897	[-0.655,0.558,-0.515,0.021]	[-0.238,-0.766,0.324,0.502]	[0.000,1.000,0.000,0.000]
16	valid	31.44	0.1022	[-0.814,0.429,-0.494,0.021]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
17	valid	35.70	0.1109	[-0.960,0.396,-0.403,0.019]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
18	valid	33.60	0.0792	[-0.943,0.389,-0.424,0.057]	[-0.462,0.422,0.201,0.754]	[0.000,0.000,1.000,0.000]
19	valid	37.82	0.0860	[-1.069,0.349,-0.364,0.052]	[0.787,0.491,0.145,-0.344]	[1.000,0.000,0.000,0.000]
20	valid	41.52	0.1062	[-1.185,0.301,-0.322,0.046]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]

Query-dependent parameter noise | seed 2 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	60.45	0.2583	[0.286,0.002,-0.914,0.286]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
2	valid	33.13	0.0438	[-0.367,0.447,-0.808,0.114]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
3	valid	36.69	0.0718	[-0.623,0.229,-0.740,0.106]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
4	valid	32.93	0.0774	[-0.570,0.460,-0.677,0.067]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
5	valid	33.37	0.0774	[-0.579,0.455,-0.674,0.060]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
6	valid	33.37	0.0774	[-0.579,0.455,-0.674,0.060]	[-0.686,0.426,-0.583,-0.092]	[0.000,0.000,0.000,1.000]
7	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
8	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
9	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.376,0.796,-0.152,-0.450]	[0.000,0.000,0.000,1.000]
10	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.670,-0.109,0.690,-0.251]	[0.000,0.000,1.000,0.000]
11	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.300,0.937,0.153,-0.090]	[0.000,0.000,1.000,0.000]
12	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.237,-0.863,0.169,-0.413]	[0.000,1.000,0.000,0.000]
13	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
14	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
15	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
16	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
17	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
18	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.194,0.637,0.742,0.079]	[0.000,0.000,1.000,0.000]
19	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
20	valid	21.28	0.0300	[-0.559,0.526,-0.589,0.253]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]

Query-dependent parameter noise | seed 2 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	60.45	0.2583	[0.286,0.002,-0.914,0.286]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
2	valid	46.23	0.1513	[0.225,0.677,-0.664,0.225]	[-0.005,-0.305,-0.938,0.166]	[0.000,1.000,0.000,0.000]
3	valid	21.37	0.0416	[-0.094,0.482,-0.654,0.575]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
4	valid	22.78	0.0417	[-0.443,0.360,-0.756,0.320]	[0.976,-0.186,-0.095,-0.056]	[1.000,0.000,0.000,0.000]
5	valid	22.21	0.0300	[-0.407,0.615,-0.630,0.245]	[-0.602,-0.667,0.317,0.303]	[0.000,1.000,0.000,0.000]
6	valid	29.09	0.0463	[-0.601,0.484,-0.623,0.130]	[0.769,-0.202,0.494,0.352]	[1.000,0.000,0.000,0.000]
7	valid	29.17	0.0456	[-0.530,0.636,-0.549,0.114]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
8	valid	33.60	0.0932	[-0.605,0.606,-0.515,0.044]	[0.651,-0.542,-0.469,-0.251]	[1.000,0.000,0.000,0.000]
9	valid	29.15	0.0462	[-0.601,0.602,-0.511,0.123]	[-0.272,0.576,0.069,-0.768]	[0.000,0.000,0.000,1.000]
10	valid	28.53	0.0419	[-0.571,0.572,-0.575,0.125]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
11	valid	7.19	0.0074	[-0.393,0.394,-0.550,0.623]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
12	valid	7.87	0.0074	[-0.387,0.388,-0.563,0.619]	[0.765,0.309,0.563,-0.053]	[0.000,0.000,1.000,0.000]
13	valid	8.05	0.0087	[-0.402,0.380,-0.567,0.610]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
14	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
15	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
16	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
17	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
18	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
19	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
20	valid	8.54	0.0087	[-0.398,0.375,-0.576,0.607]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]

Query-dependent parameter noise | seed 2 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	86.15	0.6042	[-0.558,-0.287,0.049,-0.054]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
2	valid	74.60	0.4336	[-0.627,-0.123,-0.311,-0.271]	[-0.100,-0.189,0.951,0.225]	[0.000,0.000,1.000,0.000]
3	valid	36.73	0.1208	[-0.711,0.520,-0.456,0.023]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
4	valid	14.40	0.0125	[-0.690,0.553,-0.585,0.461]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
5	valid	17.24	0.0235	[-0.645,0.437,-0.429,0.397]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
6	valid	13.12	0.0103	[-0.762,0.663,-0.582,0.553]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
7	valid	14.69	0.0167	[-0.611,0.561,-0.502,0.402]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
8	valid	11.73	0.0078	[-0.709,0.637,-0.647,0.542]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
9	valid	6.56	0.0022	[-0.625,0.582,-0.550,0.631]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
10	valid	7.47	0.0042	[-0.623,0.649,-0.553,0.609]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
11	valid	4.33	0.0011	[-0.598,0.614,-0.590,0.660]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
12	valid	4.58	0.0022	[-0.576,0.537,-0.567,0.643]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
13	valid	3.21	0.0004	[-0.537,0.559,-0.568,0.625]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
14	valid	3.30	0.0008	[-0.542,0.553,-0.589,0.635]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
15	valid	5.16	0.0016	[-0.549,0.652,-0.559,0.624]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
16	valid	3.93	0.0008	[-0.551,0.569,-0.536,0.626]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
17	valid	3.55	0.0009	[-0.482,0.498,-0.486,0.552]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
18	valid	4.65	0.0011	[-0.570,0.706,-0.604,0.683]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
19	valid	4.06	0.0011	[-0.644,0.682,-0.651,0.721]	[-0.794,-0.426,0.031,-0.432]	[0.000,0.000,0.000,1.000]
20	valid	3.90	0.0006	[-0.501,0.559,-0.535,0.574]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]

Query-dependent parameter noise | seed 2 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	79.48	0.4224	[0.314,0.000,-0.490,0.000]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
2	valid	56.39	0.1684	[-0.005,0.064,-0.490,0.000]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
3	valid	44.38	0.1426	[-0.260,0.064,-0.333,0.000]	[0.884,-0.005,0.412,0.221]	[1.000,0.000,0.000,0.000]
4	valid	36.95	0.1091	[-0.172,0.292,-0.333,0.000]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
5	valid	41.30	0.1352	[-0.398,0.148,-0.333,0.000]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
6	valid	33.03	0.0811	[-0.385,0.335,-0.321,0.036]	[-0.686,0.426,-0.583,-0.092]	[0.000,0.000,0.000,1.000]
7	valid	19.26	0.0242	[-0.380,0.330,-0.289,0.192]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
8	valid	32.25	0.0399	[-0.572,0.203,-0.275,0.182]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
9	valid	24.28	0.0392	[-0.540,0.192,-0.311,0.316]	[-0.376,0.796,-0.152,-0.450]	[0.000,0.000,0.000,1.000]
10	valid	25.95	0.0399	[-0.516,0.183,-0.458,0.252]	[-0.670,-0.109,0.690,-0.251]	[0.000,0.000,1.000,0.000]
11	valid	32.52	0.0434	[-0.485,0.172,-0.614,0.169]	[-0.300,0.937,0.153,-0.090]	[0.000,0.000,1.000,0.000]
12	valid	27.86	0.0452	[-0.477,0.282,-0.567,0.167]	[-0.237,-0.863,0.169,-0.413]	[0.000,1.000,0.000,0.000]
13	valid	24.32	0.0324	[-0.474,0.409,-0.468,0.166]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
14	valid	31.86	0.0545	[-0.448,0.386,-0.602,0.077]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
15	valid	26.86	0.0378	[-0.439,0.379,-0.601,0.156]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
16	valid	30.16	0.0603	[-0.567,0.292,-0.581,0.151]	[0.514,-0.350,-0.042,0.782]	[1.000,0.000,0.000,0.000]
17	valid	33.74	0.0669	[-0.687,0.262,-0.540,0.140]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
18	valid	33.83	0.0656	[-0.611,0.249,-0.663,0.133]	[0.194,0.637,0.742,0.079]	[0.000,0.000,1.000,0.000]
19	valid	35.73	0.0657	[-0.562,0.229,-0.777,0.128]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
20	valid	39.09	0.0659	[-0.448,0.217,-0.925,0.121]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]

Query-dependent parameter noise | seed 3 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	46.07	0.0913	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	29.99	0.0323	[-0.134,0.694,-0.683,0.181]	[-0.005,-0.305,-0.938,0.166]	[0.000,1.000,0.000,0.000]
3	valid	11.94	0.0127	[-0.266,0.578,-0.542,0.549]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
4	valid	26.37	0.0398	[-0.098,0.455,-0.447,0.764]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
5	valid	30.01	0.0615	[-0.014,0.426,-0.555,0.714]	[0.434,0.565,0.692,-0.118]	[0.000,0.000,1.000,0.000]
6	valid	30.01	0.0615	[-0.014,0.426,-0.555,0.714]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
7	valid	18.58	0.0308	[-0.381,0.337,-0.644,0.571]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
8	valid	10.45	0.0051	[-0.333,0.505,-0.537,0.588]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
9	valid	10.45	0.0051	[-0.333,0.505,-0.537,0.588]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
10	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.263,0.146,0.859,-0.414]	[0.000,0.000,1.000,0.000]
11	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[0.077,0.882,-0.212,0.415]	[1.000,0.000,0.000,0.000]
12	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.018,0.197,-0.819,-0.539]	[0.000,0.000,0.000,1.000]
13	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
14	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
15	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
16	valid	11.54	0.0071	[-0.319,0.501,-0.568,0.570]	[0.328,0.107,-0.455,-0.821]	[0.000,0.000,0.000,1.000]
17	valid	11.53	0.0110	[-0.263,0.621,-0.485,0.557]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
18	valid	11.86	0.0119	[-0.258,0.610,-0.512,0.547]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
19	valid	11.86	0.0119	[-0.258,0.610,-0.512,0.547]	[-0.013,0.056,-0.998,-0.022]	[0.000,0.000,0.000,1.000]
20	valid	11.86	0.0119	[-0.258,0.610,-0.512,0.547]	[-0.457,0.529,0.535,0.474]	[0.000,0.000,1.000,0.000]

Query-dependent parameter noise | seed 3 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	46.07	0.0913	[-0.054,0.393,-0.917,0.046]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	32.05	0.0826	[-0.550,0.068,-0.629,0.545]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
3	valid	38.85	0.1016	[-0.349,-0.000,-0.472,0.810]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
4	valid	41.52	0.1056	[-0.254,-0.023,-0.755,0.604]	[0.765,0.309,0.563,-0.053]	[0.000,0.000,1.000,0.000]
5	valid	30.35	0.0620	[-0.294,0.175,-0.736,0.584]	[0.077,0.882,-0.212,0.415]	[1.000,0.000,0.000,0.000]
6	valid	22.02	0.0380	[-0.317,0.324,-0.701,0.551]	[0.226,-0.935,0.057,-0.266]	[0.000,1.000,0.000,0.000]
7	valid	14.29	0.0153	[-0.415,0.421,-0.642,0.488]	[-0.242,0.238,-0.136,0.931]	[0.000,0.000,1.000,0.000]
8	valid	15.15	0.0240	[-0.442,0.395,-0.643,0.485]	[0.769,-0.202,0.494,0.352]	[1.000,0.000,0.000,0.000]
9	valid	13.89	0.0176	[-0.437,0.420,-0.635,0.479]	[-0.100,-0.640,0.394,0.652]	[0.000,1.000,0.000,0.000]
10	valid	14.36	0.0202	[-0.450,0.408,-0.635,0.477]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
11	valid	13.65	0.0176	[-0.447,0.422,-0.631,0.474]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
12	valid	13.65	0.0176	[-0.447,0.422,-0.631,0.474]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
13	valid	13.65	0.0176	[-0.447,0.422,-0.631,0.474]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
14	valid	13.65	0.0176	[-0.447,0.422,-0.631,0.474]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
15	valid	13.65	0.0176	[-0.447,0.422,-0.631,0.474]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
16	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.765,0.309,0.563,-0.053]	[1.000,0.000,0.000,0.000]
17	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
18	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[-0.263,0.146,0.859,-0.414]	[0.000,0.000,1.000,0.000]

Query-dependent parameter noise | seed 3 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	52.60	0.0960	[-0.051,0.089,-0.611,0.048]	[-0.220,0.017,0.903,-0.368]	[0.000,0.000,1.000,0.000]
2	valid	61.78	0.3659	[-0.657,0.113,-0.518,-0.334]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
3	valid	41.01	0.2047	[-0.574,0.538,-0.488,-0.166]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
4	valid	18.51	0.0196	[-0.587,0.598,-0.604,0.234]	[-0.444,0.444,-0.465,-0.624]	[0.000,0.000,0.000,1.000]
5	valid	11.46	0.0150	[-0.700,0.591,-0.452,0.489]	[-0.190,-0.302,0.443,-0.822]	[0.000,0.000,0.000,1.000]
6	valid	9.03	0.0046	[-0.637,0.685,-0.530,0.442]	[-0.618,0.483,-0.526,0.328]	[0.000,0.000,0.000,1.000]
7	valid	12.89	0.0138	[-0.758,0.575,-0.597,0.501]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
8	valid	5.64	0.0030	[-0.527,0.604,-0.620,0.582]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
9	valid	11.28	0.0113	[-0.650,0.638,-0.817,0.593]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
10	valid	7.21	0.0041	[-0.595,0.650,-0.678,0.551]	[0.563,-0.674,0.433,-0.203]	[0.000,1.000,0.000,0.000]
11	valid	9.74	0.0047	[-0.474,0.661,-0.738,0.561]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
12	valid	5.73	0.0027	[-0.560,0.722,-0.725,0.665]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
13	valid	7.50	0.0031	[-0.485,0.676,-0.661,0.513]	[-0.007,-0.683,-0.056,-0.728]	[0.000,1.000,0.000,0.000]
14	valid	7.32	0.0031	[-0.455,0.605,-0.610,0.481]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
15	valid	6.08	0.0019	[-0.598,0.797,-0.755,0.619]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
16	valid	6.00	0.0028	[-0.473,0.591,-0.583,0.487]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
17	valid	4.04	0.0016	[-0.658,0.828,-0.757,0.688]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
18	valid	3.08	0.0003	[-0.540,0.716,-0.591,0.558]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
19	valid	4.87	0.0011	[-0.549,0.770,-0.668,0.583]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,1.000,0.000]
20	valid	4.71	0.0011	[-0.498,0.711,-0.622,0.556]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]

Query-dependent parameter noise | seed 3 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	58.59	0.2668	[-0.082,0.000,-0.515,0.000]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
2	valid	30.67	0.0596	[-0.077,0.314,-0.480,0.161]	[-0.005,-0.305,-0.938,0.166]	[0.000,1.000,0.000,0.000]
3	valid	26.12	0.0557	[-0.070,0.286,-0.460,0.417]	[-0.624,0.413,-0.042,-0.663]	[0.000,0.000,0.000,1.000]
4	valid	37.37	0.0934	[-0.063,0.151,-0.413,0.684]	[-0.224,-0.355,-0.625,-0.659]	[0.000,0.000,0.000,1.000]
5	valid	38.13	0.0896	[-0.057,0.138,-0.638,0.572]	[0.434,0.565,0.692,-0.118]	[0.000,0.000,1.000,0.000]
6	valid	43.18	0.1017	[-0.050,0.120,-0.858,0.473]	[0.018,0.321,0.942,-0.100]	[0.000,0.000,1.000,0.000]
7	valid	40.20	0.1002	[-0.155,0.106,-0.844,0.465]	[0.680,-0.072,-0.634,0.361]	[1.000,0.000,0.000,0.000]
8	valid	32.83	0.0577	[-0.155,0.214,-0.731,0.465]	[0.083,-0.850,0.327,0.405]	[0.000,1.000,0.000,0.000]
9	valid	38.95	0.0751	[-0.155,0.191,-0.920,0.413]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
10	valid	45.85	0.0878	[-0.135,0.167,-1.113,0.302]	[-0.263,0.146,0.859,-0.414]	[0.000,0.000,1.000,0.000]
11	valid	43.07	0.0918	[-0.141,0.157,-1.051,0.404]	[0.077,0.882,-0.212,0.415]	[1.000,0.000,0.000,0.000]
12	valid	40.84	0.0922	[-0.134,0.147,-0.983,0.516]	[-0.018,0.197,-0.819,-0.539]	[0.000,0.000,0.000,1.000]
13	valid	44.89	0.1019	[-0.115,0.127,-1.118,0.455]	[-0.518,0.672,0.526,0.049]	[0.000,0.000,1.000,0.000]
14	valid	49.25	0.1103	[-0.100,0.110,-1.273,0.356]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
15	valid	48.17	0.1047	[-0.145,0.109,-1.252,0.351]	[0.974,0.024,-0.154,0.162]	[1.000,0.000,0.000,0.000]
16	valid	46.75	0.1184	[-0.121,0.105,-1.200,0.437]	[0.328,0.107,-0.455,-0.821]	[0.000,0.000,0.000,1.000]
17	valid	46.22	0.0969	[-0.121,0.148,-1.200,0.389]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
18	valid	50.23	0.1029	[-0.107,0.131,-1.347,0.301]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
19	valid	48.25	0.1052	[-0.123,0.125,-1.287,0.366]	[-0.013,0.056,-0.998,-0.022]	[0.000,0.000,0.000,1.000]
20	valid	50.37	0.1036	[-0.133,0.107,-1.371,0.312]	[-0.457,0.529,0.535,0.474]	[0.000,0.000,1.000,0.000]

Query-dependent parameter noise | seed 4 | Pedro

t	status	angle	regret	theta_hat	S	observed Y
1	valid	35.26	0.1033	[-0.421,0.527,0.098,0.732]	[-0.477,0.831,0.084,-0.272]	[0.000,0.000,0.000,1.000]
2	valid	53.94	0.2301	[-0.487,0.511,0.449,0.548]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
3	valid	47.90	0.2100	[-0.303,0.830,0.311,0.350]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
4	valid	47.90	0.2100	[-0.303,0.830,0.311,0.350]	[-0.259,-0.531,-0.093,0.801]	[0.000,1.000,0.000,0.000]
5	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[0.291,-0.106,0.323,-0.894]	[0.000,0.000,0.000,1.000]
6	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
7	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[-0.406,-0.902,-0.036,0.142]	[0.000,1.000,0.000,0.000]
8	valid	44.55	0.1815	[-0.250,0.817,0.257,0.451]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
9	valid	10.11	0.0057	[-0.402,0.662,-0.299,0.558]	[-0.100,-0.189,0.951,0.225]	[0.000,0.000,1.000,0.000]
10	valid	10.11	0.0057	[-0.402,0.662,-0.299,0.558]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
11	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
12	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
13	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
14	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.266,0.572,0.740,0.233]	[0.000,0.000,1.000,0.000]
15	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.489,0.191,-0.234,0.818]	[1.000,0.000,0.000,0.000]
16	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[-0.503,0.277,-0.818,-0.031]	[0.000,0.000,0.000,1.000]
17	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
18	valid	12.10	0.0085	[-0.370,0.663,-0.282,0.586]	[0.157,0.765,-0.245,-0.575]	[0.000,0.000,0.000,1.000]
19	valid	6.20	0.0040	[-0.394,0.584,-0.414,0.576]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
20	valid	6.36	0.0031	[-0.392,0.570,-0.434,0.578]	[-0.064,-0.505,0.844,0.167]	[0.000,0.000,1.000,0.000]

Query-dependent parameter noise | seed 4 | Genious

t	status	angle	regret	theta_hat	S	observed Y
1	valid	35.26	0.1033	[-0.421,0.527,0.098,0.732]	[-0.477,0.831,0.084,-0.272]	[0.000,0.000,0.000,1.000]
2	valid	44.26	0.1876	[-0.294,0.833,0.252,0.394]	[0.509,-0.697,-0.096,-0.496]	[0.000,1.000,0.000,0.000]
3	valid	19.63	0.0289	[-0.400,0.784,-0.159,0.447]	[-0.457,0.529,0.535,0.474]	[0.000,0.000,1.000,0.000]
4	valid	28.77	0.0777	[-0.627,0.733,-0.072,0.255]	[0.814,-0.418,0.255,0.313]	[1.000,0.000,0.000,0.000]
5	valid	9.79	0.0069	[-0.520,0.609,-0.490,0.345]	[0.223,0.325,0.735,-0.551]	[0.000,0.000,1.000,0.000]
6	valid	22.58	0.0269	[-0.520,0.786,-0.275,0.193]	[0.651,-0.542,-0.469,-0.251]	[0.000,1.000,0.000,0.000]
7	valid	15.20	0.0140	[-0.493,0.745,-0.342,0.292]	[0.340,0.096,-0.269,-0.896]	[0.000,0.000,0.000,1.000]
8	valid	13.35	0.0133	[-0.471,0.711,-0.442,0.279]	[0.496,0.017,0.713,0.495]	[0.000,0.000,1.000,0.000]
9	valid	7.43	0.0027	[-0.453,0.684,-0.425,0.382]	[-0.543,0.615,0.311,-0.480]	[0.000,0.000,0.000,1.000]
10	valid	1.74	0.0000	[-0.417,0.630,-0.456,0.470]	[0.540,0.560,0.062,-0.625]	[0.000,0.000,0.000,1.000]
11	valid	12.95	0.0095	[-0.311,0.499,-0.524,0.617]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
12	valid	10.00	0.0065	[-0.329,0.539,-0.502,0.591]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
13	valid	8.18	0.0035	[-0.380,0.528,-0.491,0.579]	[0.553,0.736,0.366,0.133]	[1.000,0.000,0.000,0.000]
14	valid	3.99	0.0014	[-0.419,0.570,-0.457,0.539]	[0.317,0.914,0.253,-0.015]	[1.000,0.000,0.000,0.000]
15	valid	2.60	0.0008	[-0.404,0.613,-0.440,0.518]	[-0.007,-0.683,-0.056,-0.728]	[0.000,1.000,0.000,0.000]
16	valid	2.78	0.0002	[-0.403,0.599,-0.456,0.520]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
17	valid	2.75	0.0008	[-0.392,0.616,-0.457,0.508]	[-0.162,-0.596,-0.390,-0.683]	[0.000,1.000,0.000,0.000]
18	valid	3.71	0.0008	[-0.392,0.592,-0.485,0.511]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,1.000,0.000]
19	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
20	degenerate_cone	-	-	[0.000,0.000,0.000,0.000]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]

Query-dependent parameter noise | seed 4 | Score Base

t	status	angle	regret	theta_hat	S	observed Y
1	valid	38.00	0.1218	[0.061,0.514,-0.083,0.417]	[0.124,-0.331,-0.404,0.843]	[0.000,1.000,0.000,0.000]
2	valid	35.34	0.0798	[0.133,0.531,-0.517,0.418]	[-0.288,-0.189,0.880,-0.327]	[0.000,0.000,1.000,0.000]
3	valid	7.66	0.0021	[-0.387,0.662,-0.597,0.483]	[0.850,0.267,-0.190,0.413]	[1.000,0.000,0.000,0.000]
4	valid	5.72	0.0033	[-0.470,0.619,-0.412,0.594]	[-0.612,0.267,-0.382,0.639]	[0.000,0.000,1.000,0.000]
5	valid	8.11	0.0072	[-0.527,0.528,-0.385,0.502]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
6	valid	7.46	0.0058	[-0.494,0.535,-0.362,0.402]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
7	valid	6.83	0.0038	[-0.616,0.658,-0.462,0.545]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
8	valid	10.74	0.0126	[-0.730,0.684,-0.477,0.533]	[-0.612,0.267,-0.382,0.639]	[0.000,0.000,1.000,0.000]
9	valid	8.63	0.0076	[-0.592,0.615,-0.395,0.568]	[-0.612,0.267,-0.382,0.639]	[0.000,0.000,1.000,0.000]
10	valid	4.44	0.0017	[-0.481,0.753,-0.507,0.487]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
11	valid	4.09	0.0016	[-0.375,0.613,-0.438,0.525]	[0.579,0.143,0.579,-0.555]	[0.000,0.000,0.000,1.000]
12	valid	3.19	0.0008	[-0.432,0.642,-0.476,0.568]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
13	valid	5.01	0.0032	[-0.401,0.558,-0.387,0.529]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,0.000,1.000]
14	valid	3.16	0.0014	[-0.461,0.674,-0.479,0.594]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
15	valid	4.78	0.0019	[-0.428,0.716,-0.509,0.623]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]
16	valid	3.85	0.0016	[-0.503,0.629,-0.448,0.566]	[0.712,-0.384,0.323,-0.491]	[1.000,0.000,0.000,0.000]
17	valid	4.38	0.0023	[-0.474,0.611,-0.456,0.584]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,0.000,1.000]
18	valid	2.13	0.0008	[-0.487,0.714,-0.516,0.606]	[0.543,0.120,0.645,-0.524]	[0.000,0.000,1.000,0.000]
19	valid	2.95	0.0020	[-0.490,0.671,-0.466,0.586]	[0.579,0.143,0.579,-0.555]	[1.000,0.000,0.000,0.000]
20	valid	3.75	0.0010	[-0.431,0.531,-0.400,0.487]	[-0.612,0.267,-0.382,0.639]	[0.000,1.000,0.000,0.000]

Query-dependent parameter noise | seed 4 | Online SAMD

t	status	angle	regret	theta_hat	S	observed Y
1	valid	52.58	0.1957	[-0.499,0.000,0.000,0.256]	[-0.477,0.831,0.084,-0.272]	[0.000,0.000,0.000,1.000]
2	valid	47.29	0.1468	[-0.844,0.246,0.000,0.193]	[0.286,0.290,0.195,0.892]	[1.000,0.000,0.000,0.000]
3	valid	34.52	0.1021	[-0.601,0.534,0.000,0.193]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
4	valid	33.59	0.0914	[-0.510,0.786,-0.024,0.164]	[-0.259,-0.531,-0.093,0.801]	[0.000,1.000,0.000,0.000]
5	valid	28.65	0.0810	[-0.423,0.771,-0.023,0.321]	[0.291,-0.106,0.323,-0.894]	[0.000,0.000,0.000,1.000]
6	valid	37.08	0.0930	[-0.367,1.022,-0.020,0.172]	[0.377,-0.706,-0.186,-0.571]	[0.000,1.000,0.000,0.000]
7	valid	41.09	0.1025	[-0.297,1.214,-0.019,0.139]	[-0.406,-0.902,-0.036,0.142]	[0.000,1.000,0.000,0.000]
8	valid	40.94	0.1119	[-0.287,1.176,0.046,0.218]	[-0.345,0.105,0.680,-0.638]	[0.000,0.000,0.000,1.000]
9	valid	38.13	0.0991	[-0.287,1.100,-0.015,0.218]	[-0.100,-0.189,0.951,0.225]	[0.000,0.000,1.000,0.000]
10	valid	42.02	0.0991	[-0.226,1.312,-0.013,0.189]	[0.216,-0.862,0.395,0.233]	[0.000,1.000,0.000,0.000]
11	valid	41.69	0.0989	[-0.152,1.305,-0.013,0.270]	[0.405,0.823,-0.005,-0.399]	[0.000,0.000,0.000,1.000]
12	valid	40.13	0.0815	[-0.162,1.292,-0.059,0.268]	[-0.125,0.724,0.669,0.117]	[0.000,0.000,1.000,0.000]
13	valid	42.35	0.0781	[-0.134,1.408,-0.072,0.221]	[-0.965,-0.209,-0.127,0.096]	[0.000,1.000,0.000,0.000]
14	valid	40.93	0.0575	[-0.149,1.382,-0.112,0.216]	[-0.266,0.572,0.740,0.233]	[0.000,0.000,1.000,0.000]
15	valid	39.23	0.0588	[-0.199,1.341,-0.130,0.210]	[0.489,0.191,-0.234,0.818]	[1.000,0.000,0.000,0.000]
16	valid	37.82	0.0667	[-0.256,1.301,-0.126,0.207]	[-0.503,0.277,-0.818,-0.031]	[0.000,0.000,0.000,1.000]
17	valid	36.40	0.0655	[-0.243,1.269,-0.123,0.265]	[0.056,0.815,-0.158,-0.555]	[0.000,0.000,0.000,1.000]
18	valid	35.07	0.0638	[-0.220,1.234,-0.120,0.332]	[0.157,0.765,-0.245,-0.575]	[0.000,0.000,0.000,1.000]
19	valid	34.84	0.0560	[-0.220,1.234,-0.161,0.291]	[-0.282,0.705,0.573,-0.308]	[0.000,0.000,1.000,0.000]
20	valid	31.14	0.0349	[-0.220,1.079,-0.209,0.291]	[-0.064,-0.505,0.844,0.167]	[0.000,0.000,1.000,0.000]